

INPUT SUMMARY

FLUID PACKAGE: Basis-1(Peng-Robinson)

Property Package Type: PengRob

Component List - 1: Benzene /Propene /Propane /Cumene /14-iP-BZ /

Reaction Set: Set-1

Reaction 'Rxn-1':

Reactants: Benzene, Stoich Coeff -1 / Propene, Stoich Coeff -1 / Cumene,
Stoich Coeff 1 /

Basis Data: Basis = Molar Conc'n / Component = Benzene / Phase = LiquidPhase /
RateUnits = kgmole/m³-s /

Kinetic Coefficients: Fwd Freq Factor = 650000 / Fwd Activation Energy = 52.56
/

Reaction 'Rxn-2':

Reactants: Cumene, Stoich Coeff -1 / Propene, Stoich Coeff -1 / 14-iP-BZ,
Stoich Coeff 1 /

Basis Data: Basis = Molar Conc'n / Component = Cumene / Phase = LiquidPhase /
RateUnits = kgmole/m³-s /

Kinetic Coefficients: Fwd Freq Factor = 450000 / Fwd Activation Energy = 55 /

Reaction Set: Set-2

Reaction 'Rxn-4':

Reactants: Benzene, Stoich Coeff -1 / 14-iP-BZ, Stoich Coeff -1 / Cumene,
Stoich Coeff 2 /

Basis Data: Component = 14-iP-BZ / Phase = Overall /

FLOWSHEET: Main

Fluid Package: Basis-1

STREAM: Fresh Propylene (Material Stream)

Temperature = 25 C

Pressure = 1500 kPa

Mass Flow = 537999.998 kg/h

Composition Basis (In Mole Fractions):Benzene = 0/ Propene = 0.95/ Propane = 0.05/ Cumene = 0/ 14-iP-BZ = 0/

UNIT OPERATION: P-1 (Pump)

Feed Stream = Total_Benzene

Product Stream = Benzene HP

Energy Stream = Q-P1

AdiabaticEfficiency = 75 %

CurveCollectionName = CC-0

SelectedCurveCollection = True

UseMultiphasePump = False

PlotAllCollections = False

MalfunctionType = 17

Active = 0

variable1 = 100

MalfunctionType = 18

Active = 0

MalfunctionType = 19

Active = 0

NPSHRequiredAP = Turton Method

UnitsforSuctionSpecificSpeed = SI

UseNewCurveInterpolation = False

UNIT OPERATION: P-2 (Pump)

Feed Stream = Fresh Propylene

Product Stream = Propylene HP

Energy Stream = Q-P2

AdiabaticEfficiency = 75 %

CurveCollectionName = CC-0

SelectedCurveCollection = True

UseMultiphasePump = False

PlotAllCollections = False

MalfunctionType = 17

Active = 0

variable1 = 100

MalfunctionType = 18

Active = 0

MalfunctionType = 19

Active = 0

NPSHRequiredAP = Turton Method

UnitsforSuctionSpecificSpeed = SI

UseNewCurveInterpolation = False

STREAM: Benzene HP (Material Stream)

Pressure = 3500 kPa

STREAM: Q-P1 (Energy Stream)

STREAM: Propylene HP (Material Stream)

Pressure = 3500 kPa

STREAM: Q-P2 (Energy Stream)

UNIT OPERATION: MIX-100 (Mixer)

Feed Stream = Benzene HP

Feed Stream = Propylene HP

Product Stream = Mixed Feed

UseTrivialSolution = True

STREAM: Mixed Feed (Material Stream)

UNIT OPERATION: E-100 (Heater)

Feed Stream = Mixed Feed

Product Stream = Reactor Feed

Energy Stream = Q-Heater

Pressure Drop = 50 kPa

STREAM: Reactor Feed (Material Stream)

Temperature = 150 C

STREAM: Q-Heater (Energy Stream)

UNIT OPERATION: R-1 (Plug Flow Reactor)

Feed Stream = Reactor Feed

Product Stream = Alkylation Effluents

Energy Stream = Q-R1

Reaction Set=Set-1

Delta P = 50 kPa

NumberOfSegments = 100

TubeLength = 8 m

TubeDiameter = 5 m

VoidFraction = 0.45

STREAM: Alkylation Effluents (Material Stream)

Temperature = 150 C

STREAM: Q-R1 (Energy Stream)

UNIT OPERATION: VLV-100 (Valve)

Feed Stream = Alkylation Effluents

Product Stream = C1-Feed

ValveManufacturer = FISHER

ValveManufacturerType = 0

C1 = 33.4664011

Cv = 943.578019 USGPM(60F,1psi)

RigorousSizingMethod = True

UseXtTable = False

RigorousFlowCalc = True

useImplicitISAModel = False

MalfunctionType = 8

Active = 0

MalfunctionType = 9

Active = 0

MalfunctionType = 10

Active = 0

MalfunctionType = 11

Active = 0

MalfunctionType = 12

Active = 0

MalfunctionType = 13

Active = 0

STREAM: C1-Feed (Material Stream)

Pressure = 1500 kPa

UNIT OPERATION: T-100 (Absorber)

TwoLiquidCheck = 2 Liquid Check

FixedDampingFactor = 0.08

ShowEffDiagFlag = True

STREAM: Reflux-1 (Material Stream)

Temperature = 41.8970712 C

Pressure = 1450 kPa

Molar Flow = 19994.058 kgmole/h

Composition Basis (In Mole Fractions):Benzene = 4.76837052e-06/ Propene =
0.0412530217/ Propane = 0.95874221/ Cumene = 3.02448032e-11/ 14-iP-BZ =
3.83060209e-15/

STREAM: C1-Top (Material Stream)

STREAM: C1-Bot (Material Stream)

UNIT OPERATION: E-101 (Heater)

Feed Stream = C1-Bot

Product Stream = Heated_Bot

Energy Stream = Q-Reboiler

Pressure Drop = 0 kPa

STREAM: Q-Reboiler (Energy Stream)

UNIT OPERATION: V-100 (Separator)

Feed Stream = Heated_Bot

Vapour Product = Boilup-New

Liquid Product = C2-Feed

STREAM: C2-Feed (Material Stream)

UNIT OPERATION: E-102 (Cooler)

Feed Stream = C1-Top

Product Stream = Cooled-New

Energy Stream = Q-Condenser

Pressure Drop = 0 kPa

STREAM: Q-Condenser (Energy Stream)

UNIT OPERATION: P-100 (Pump)

Feed Stream = Reflux-New

Product Stream = Reflux-Temp

Energy Stream = Q-Reflux

AdiabaticEfficiency = 75 %

CurveCollectionName = CC-0

SelectedCurveCollection = True

UseMultiphasePump = False

PlotAllCollections = False

MalfunctionType = 17

Active = 0

variable1 = 100

MalfunctionType = 18

Active = 0

MalfunctionType = 19

Active = 0

NPSHRequiredAP = Turton Method

UnitsforSuctionSpecificSpeed = SI

UseNewCurveInterpolation = False

STREAM: Q-Reflux (Energy Stream)

UNIT OPERATION: V-101 (Separator)

Feed Stream = Cooled-New

Vapour Product = Fuel-Gas

Liquid Product = Reflux-New

STREAM: Fuel-Gas (Material Stream)

STREAM: Reflux-Temp (Material Stream)

Vapour Fraction = 0

Pressure = 1450 kPa

STREAM: Boilup-New (Material Stream)

Molar Flow = 13000 kgmole/h

STREAM: Cooled-New (Material Stream)

Vapour Fraction = 0

STREAM: Reflux-New (Material Stream)

STREAM: Heated_Bot (Material Stream)

UNIT OPERATION: VLV-101 (Valve)

Feed Stream = C2-Feed_New

Product Stream = C2-Feed-LP

ValveManufacturer = FISHER

ValveManufacturerType = 0

C1 = 33.4664011

RigorousSizingMethod = True

UseXtTable = False

RigorousFlowCalc = True
useImplicitLSAModel = False
MalfunctionType = 8
Active = 0
MalfunctionType = 9
Active = 0
MalfunctionType = 10
Active = 0
MalfunctionType = 11
Active = 0
MalfunctionType = 12
Active = 0
MalfunctionType = 13
Active = 0

STREAM: C2-Feed-LP (Material Stream)

Pressure = 120 kPa

UNIT OPERATION: T-101 (Absorber)

TwoLiquidCheck = 2 Liquid Check

FixedDampingFactor = 0.08

TargetType = 1

Phase = Liquid

ShowEffDiagFlag = True

Specification Name = Max_BenzeneSpecConvergedStatus = InactiveSpecification Value =

0.001Specification Name = Reflux RatioSpecConvergedStatus = InactiveSpecification
Value = 1.5

STREAM: Reflux-2 (Material Stream)

Temperature = 50.0236337 C

Pressure = 150 kPa

Molar Flow = 8327.2388 kgmole/h

Composition Basis (In Mole Fractions):Benzene = 0.693891975/ Propene =
0.000372218346/ Propane = 0.035378081/ Cumene = 0.270118107/ 14-iP-BZ =
0.000239618443/

STREAM: Boilup-2 (Material Stream)

Temperature = 172.65195 C

Pressure = 120 kPa

Molar Flow = 8500 kgmole/h

Composition Basis (In Mole Fractions):Benzene = 4.31673923e-06/ Propene =
2.45454556e-26/ Propane = 1.0749396e-23/ Cumene = 0.879351507/ 14-iP-BZ =
0.120644176/

STREAM: C2-Top (Material Stream)

STREAM: C2-Bot (Material Stream)

UNIT OPERATION: E-103 (Cooler)

Feed Stream = C2-Top

Product Stream = C2-Cooled

Energy Stream = Q-Cond2

Pressure Drop = 0 kPa

STREAM: C2-Cooled (Material Stream)

Temperature = 50 C

STREAM: Q-Cond2 (Energy Stream)

UNIT OPERATION: V-102 (Separator)

Feed Stream = C2-Cooled

Vapour Product = C2-Gas

Liquid Product = C2-Liquid

STREAM: C2-Gas (Material Stream)

Vapour Fraction = 1

STREAM: C2-Liquid (Material Stream)

Vapour Fraction = 0

UNIT OPERATION: P-101 (Pump)

Feed Stream = C2-Liquid

Product Stream = C2-PumpOut

Energy Stream = Q-Pump2

Delta P = 50 kPa

AdiabaticEfficiency = 75 %

CurveCollectionName = CC-0

SelectedCurveCollection = True

UseMultiphasePump = False

PlotAllCollections = False

MalfunctionType = 17

Active = 0

variable1 = 100

MalfunctionType = 18

Active = 0

MalfunctionType = 19

Active = 0

NPSHRequiredAP = Turton Method

UnitsforSuctionSpecificSpeed = SI

UseNewCurveInterpolation = False

STREAM: C2-PumpOut (Material Stream)

STREAM: Q-Pump2 (Energy Stream)

UNIT OPERATION: TEE-100 (Tee)

Feed Stream = C2-PumpOut

Product Stream = Reflux-New2
Product Stream = Benzene Ovrhd

STREAM: Reflux-New2 (Material Stream)

Vapour Fraction = 0

STREAM: Benzene Ovrhd (Material Stream)

UNIT OPERATION: E-104 (Heater)

Feed Stream = C2-Bot

Product Stream = C2-Heated_Bot

Energy Stream = Q-Reb2

Pressure Drop = 0 kPa

STREAM: C2-Heated_Bot (Material Stream)

STREAM: Q-Reb2 (Energy Stream)

UNIT OPERATION: V-103 (Separator)

Feed Stream = C2-Heated_Bot

Vapour Product = Boilup-New2

Liquid Product = C2-Bottoms

STREAM: Boilup-New2 (Material Stream)

Molar Flow = 8500 kgmole/h

STREAM: C2-Bottoms (Material Stream)

UNIT OPERATION: T-102 (Absorber)

TwoLiquidCheck = 2 Liquid Check

FixedDampingFactor = 0.08

ShowEffDiagFlag = True

STREAM: Reflux-3 (Material Stream)

Temperature = 50.0193173 C

Pressure = 150 kPa

Molar Flow = 1374.67401 kgmole/h

Composition Basis (In Mole Fractions):Benzene = 3.35548494e-06/ Propene =
1.90048947e-27/ Propane = 9.11070296e-25/ Cumene = 0.99595934/ 14-iP-BZ =
0.00403730496/

STREAM: Boilup-3 (Material Stream)

Temperature = 176.581534 C

Pressure = 120 kPa

Molar Flow = 5000 kgmole/h

Composition Basis (In Mole Fractions):Benzene = 4.78956119e-09/ Propene = 0/
Propane = 0/ Cumene = 0.832516847/ 14-iP-BZ = 0.167483148/

STREAM: C3-Top (Material Stream)

STREAM: C3-Bot (Material Stream)

UNIT OPERATION: E-105 (Cooler)

Feed Stream = C3-Top

Product Stream = C3-Cooled

Energy Stream = Q-Cond3

Pressure Drop = 0 kPa

STREAM: C3-Cooled (Material Stream)

Temperature = 50 C

STREAM: Q-Cond3 (Energy Stream)

UNIT OPERATION: V-104 (Separator)

Feed Stream = C3-Cooled

Vapour Product = C3-Gas

Liquid Product = C3-Liquid

STREAM: C3-Gas (Material Stream)

Vapour Fraction = 1

STREAM: C3-Liquid (Material Stream)

Vapour Fraction = 0

UNIT OPERATION: P-102 (Pump)

Feed Stream = C3-Liquid

Product Stream = C3-PumpOut

Energy Stream = Q-Pump3

Delta P = 50 kPa

AdiabaticEfficiency = 75 %

CurveCollectionName = CC-0

SelectedCurveCollection = True

UseMultiphasePump = False

PlotAllCollections = False

MalfunctionType = 17

Active = 0

variable1 = 100

MalfunctionType = 18

Active = 0

MalfunctionType = 19

Active = 0

NPSHRequiredAP = Turton Method

UnitsforSuctionSpecificSpeed = SI

UseNewCurveInterpolation = False

STREAM: C3-PumpOut (Material Stream)

STREAM: Q-Pump3 (Energy Stream)

UNIT OPERATION: TEE-101 (Tee)

Feed Stream = C3-PumpOut

Product Stream = CumeneProduct Stream = Reflux_new3

STREAM: Cumene (Material Stream)

UNIT OPERATION: E-106 (Heater)

Feed Stream = C3-Bot

Product Stream = C3-Heated_Bot

Energy Stream = Q-reb3

Pressure Drop = 0 kPa

UNIT OPERATION: V-105 (Separator)

Feed Stream = C3-Heated_Bot

Vapour Product = Boilup-New3

Liquid Product = DIPB_To_R2

STREAM: C3-Heated_Bot (Material Stream)

STREAM: Q-reb3 (Energy Stream)

STREAM: Boilup-New3 (Material Stream)

Molar Flow = 5000 kgmole/h

STREAM: DIPB_To_R2 (Material Stream)

UNIT OPERATION: MIX-101 (Mixer)

Feed Stream = DIPB_To_R2

Feed Stream = Transalkylation Recycle

Product Stream = Trans_Mixed

UseTrivialSolution = True

STREAM: Trans_Mixed (Material Stream)

UNIT OPERATION: P-103 (Pump)

Feed Stream = Cooled-Trans_Mixed

Product Stream = P6-Out

Energy Stream = Q-P6

AdiabaticEfficiency = 75 %

CurveCollectionName = CC-0

SelectedCurveCollection = True

UseMultiphasePump = False

PlotAllCollections = False

MalfunctionType = 17

Active = 0

variable1 = 100

MalfunctionType = 18

Active = 0

MalfunctionType = 19

Active = 0

NPSHRequiredAP = Turton Method

UnitsforSuctionSpecificSpeed = SI

UseNewCurveInterpolation = False

STREAM: Q-P6 (Energy Stream)

STREAM: P6-Out (Material Stream)

Pressure = 2500 kPa

UNIT OPERATION: E-107 (Cooler)

Feed Stream = Trans_Mixed

Product Stream = Cooled-Trans_Mixed

Energy Stream = Q-107

Pressure Drop = 0 kPa

STREAM: Q-107 (Energy Stream)

STREAM: Cooled-Trans_Mixed (Material Stream)

Temperature = 80 C

UNIT OPERATION: E-108 (Heater)

Feed Stream = P6-Out

Product Stream = Trans Feed

Energy Stream = Q-E9

Pressure Drop = 0 kPa

STREAM: Q-E9 (Energy Stream)

STREAM: Trans Feed (Material Stream)

Temperature = 210 C

UNIT OPERATION: CRV-100 (Conversion Reactor)

Feed Stream = Trans Feed

Vapour Product = Trans_Vap

Liquid Product = Trans_Effluents

Reaction Set=Set-2

Diameter = 2 m

Length = 6 m

ReactionName = Rxn-4

STREAM: Trans_Vap (Material Stream)

STREAM: Trans_Effluents (Material Stream)

UNIT OPERATION: MIX-102 (Mixer)

Feed Stream = C2-Feed

Feed Stream = Trans_Recycled

Product Stream = C2-Feed_New

UseTrivialSolution = True

STREAM: C2-Feed_New (Material Stream)

UNIT OPERATION: RCY-1 (Recycle)

Inlet Stream = Trans_Effluents

Output Stream = Trans_Recycled

IsIgnored = True

FlowSensitivity = 100

Acceleration = 1

STREAM: Trans_Recycled (Material Stream)

Temperature = 210.823168 C

Pressure = 2500 kPa

Molar Flow = 12796.4671 kgmole/h

Composition Basis (In Mole Fractions):Benzene = 0.19866248/ Propene =
2.7012528e-14/ Propane = 0.0299030736/ Cumene = 0.649775331/ 14-iP-BZ =
0.121659116/

UNIT OPERATION: TEE-102 (Tee)

Feed Stream = Benzene Ovrhd

Product Stream = Transalkylation Recycle
Product Stream = Alkylation Recycle

UNIT OPERATION: MIX-103 (Mixer)

Feed Stream = Alkylation Recycled

Feed Stream = Fresh Benzene

Product Stream = Total_Benzene

UseTrivialSolution = True

STREAM: Total_Benzene (Material Stream)

STREAM: Fresh Benzene (Material Stream)

Temperature = 25 C

Pressure = 1500 kPa

Mass Flow = 1000000 kg/h

Composition Basis (In Mass Fractions):Benzene = 1/ Propene = 0/ Propane = 0/

Cumene = 0/ 14-iP-BZ = 0/

STREAM: Transalkylation Recycle (Material Stream)

STREAM: Alkylation Recycle (Material Stream)

UNIT OPERATION: RCY-2 (Recycle)

Inlet Stream = Alkylation Recycle

Output Stream = Alkylation Recycled

FlowSensitivity = 100

Acceleration = 1

STREAM: Alkylation Recycled (Material Stream)

Temperature = 50.0236415 C

Pressure = 150 kPa

Molar Flow = 9715.40914 kgmole/h

Composition Basis (In Mole Fractions):Benzene = 0.694823002/ Propene =
0.000372074979/ Propane = 0.0353496433/ Cumene = 0.269216821/ 14-iP-BZ =
0.000238459424/

UNIT OPERATION: RCY-3 (Recycle)

Inlet Stream = Boilup-New

Output Stream = Boil-Up

STREAM: Boil-Up (Material Stream)

Temperature = 219.255877 C

Pressure = 1500 kPa

Molar Flow = 13000 kgmole/h

Composition Basis (In Mole Fractions):Benzene = 0.592086806/ Propene =
0.00399028252/ Propane = 0.259175607/ Cumene = 0.120356196/ 14-iP-BZ =

0.0243911082/

UNIT OPERATION: RCY-4 (Recycle)

Inlet Stream = Reflux-Temp

Output Stream = Reflux-1

UNIT OPERATION: RCY-5 (Recycle)

Inlet Stream = Boilup-New2

Output Stream = Boilup-2

UNIT OPERATION: RCY-6 (Recycle)

Inlet Stream = Reflux-New2

Output Stream = Reflux-2

STREAM: Reflux_new3 (Material Stream)

Vapour Fraction = 0

UNIT OPERATION: RCY-7 (Recycle)

Inlet Stream = Boilup-New3

Output Stream = Boilup-3

UNIT OPERATION: RCY-8 (Recycle)

Inlet Stream = Reflux_new3

Output Stream = Reflux-3

FLOWSHEET: COL1 (OWNER: T-100)

Fluid Package: Basis-1

UNIT OPERATION: Main Tower (Tower)

StageNumber = 5 (Feed)/ NumberOfColumnStages = 10

WHtSpeced = 1

RateHoldup = 0.0883572188

StgNumber = 0

StgNumber = 1

StgNumber = 2

StgNumber = 3

StgNumber = 4

StgNumber = 5

StgNumber = 6

StgNumber = 7

StgNumber = 8

StgNumber = 9

HasTPSAROption = True

STREAM: Reflux-1 (Material Stream)

STREAM: C1-Top (Material Stream)

STREAM: C1-Bot (Material Stream)

STREAM: C1-Feed (Material Stream)

STREAM: Boil-Up (Material Stream)

FLOWSHEET: COL2 (OWNER: T-101)

Fluid Package: Basis-1

UNIT OPERATION: Main Tower (Tower)

StageNumber = 6 (Feed)/ NumberOfColumnStages = 22

WHtSpeced = 1

RateHoldup = 0.0883572188

StgNumber = 0

StgNumber = 1

StgNumber = 2

StgNumber = 3

StgNumber = 4

StgNumber = 5

StgNumber = 6

StgNumber = 7

StgNumber = 8

StgNumber = 9

StgNumber = 10

StgNumber = 11

StgNumber = 12

StgNumber = 13

StgNumber = 14

StgNumber = 15

StgNumber = 16

StgNumber = 17

StgNumber = 18

StgNumber = 19

StgNumber = 20

StgNumber = 21

HasTPSAROption = True

STREAM: Reflux-2 (Material Stream)

STREAM: Boilup-2 (Material Stream)

STREAM: C2-Top (Material Stream)

STREAM: C2-Bot (Material Stream)

STREAM: C2-Feed-LP (Material Stream)

FLWSHEET: COL3 (OWNER: T-102)

Fluid Package: Basis-1

UNIT OPERATION: Main Tower (Tower)

StageNumber = 10 (Feed)/ NumberOfColumnStages = 20

WHtSpeced = 1

RateHoldup = 0.0883572188

StgNumber = 0

StgNumber = 1

StgNumber = 2

StgNumber = 3

StgNumber = 4

StgNumber = 5

StgNumber = 6

StgNumber = 7

StgNumber = 8

StgNumber = 9

StgNumber = 10

StgNumber = 11

StgNumber = 12

StgNumber = 13

StgNumber = 14

StgNumber = 15

StgNumber = 16

StgNumber = 17

StgNumber = 18

StgNumber = 19

HasTPSAROption = True

STREAM: Reflux-3 (Material Stream)

STREAM: Boilup-3 (Material Stream)

STREAM: C3-Top (Material Stream)

STREAM: C3-Bot (Material Stream)

STREAM: C2-Bottoms (Material Stream)

##

OUTPUT SUMMARY

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Company Name Not Available Case Name: Final 2 (2).xml

Bedford, MA

USA Unit Set: SI10

Date/Time: Tue Sep 8 22:18:12 2026

Basis-1 (Fluid Package): Component List

Fluid Package: Basis-1

COMPONENT LIST

Component List - 1 [HYSYS Databanks]

COMPONENT	TYPE	MOLECULAR WEIGHT	BOILING PT (C)	IDEAL LIQ DENSITY (kg/m3)	CRITICAL TEMP (C)
Benzene	Pure	78.11	80.09	882.2	288.9
Propene	Pure	42.08	-47.75	521.0	91.85
Propane	Pure	44.10	-42.10	506.7	96.75
Cumene	Pure	120.2	152.4	864.8	358.0
14-iP-BZ	Pure	162.3	210.5	859.8	415.9

(Continued..) Component List - 1 [HYSYS Databanks]

COMPONENT	CRITICAL PRES (kPa)	CRITICAL VOL (m3/kgmole)	ACENTRICITY	HEAT OF FORM (kJ/kgmole)
Benzene	4924	0.2600	0.2150	8.298e+004
Propene	4620	0.1810	0.1480	2.043e+004
Propane	4257	0.2000	0.1524	-1.039e+005
Cumene	3209	0.4276	0.3377	3940
14-iP-BZ	2450	0.5980	0.3900	-7.651e+004

Case (Simulation Case): Mass and Energy Balance, Utility Balance, Process CO2 Emissions

Simulation Case: Case

OVERALL MASS BALANCE

In Stream	Count	Mass Flow	Out Stream	Count	Mass Flow
		(kg/h)			(kg/h)

Fresh Propylene	Yes	5.380e+005	Fuel-Gas	Yes	0.0000
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Fresh Benzene	Yes	1.000e+006	C2-Gas	Yes	4.896e+004
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C3-Gas	Yes	0.0000
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Cumene	Yes	3.870e+005
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Trans_Vap	Yes	0.0000
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Total In MassFlow (kg/h) 1.538e+006 Total Out MassFlow (kg/h) 4.359e+005

Mass Imbalance (kg/h) -1.102e+006 Rel Mass Imbalance Pct (%) -71.66

OVERALL ENERGY BALANCE

InStream	Count	Energy Flow	OutStream	Count	Energy Flow
		(kJ/h)			(kJ/h)

Q-P1	Yes	9.681e+06	Q-R1	Yes	1.185e+09
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Fresh Propylene	Yes	-1.864e+07	Q-Condenser	Yes	2.674e+08
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Q-P2	Yes	2.828e+06	Fuel-Gas	Yes	-0.000e-01
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Q-Heater	Yes	5.723e+08	Q-Cond2	Yes	1.164e+09
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Q-Reboiler	Yes	6.565e+08	C2-Gas	Yes	-4.592e+07
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Q-Reflux	Yes	0.000e-01	Q-Cond3	Yes	2.855e+08
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Q-Pump2	Yes	1.945e+05	C3-Gas	Yes	0.000e-01
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Q-Reb2	Yes	3.380e+08	Cumene	Yes	-1.129e+08
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Q-Pump3 Yes 4.404e+04 Q-107 Yes 2.980e+08

Q-reb3 Yes 2.041e+08 Trans_Vap Yes 0.000e-01

Q-P6 Yes 9.964e+06

Q-E9 Yes 7.134e+08

Fresh Benzene Yes 6.433e+08

Total In EnergyFlow (kJ/h) 3.132e+009 Total Out EnergyFlow (kJ/h) 3.041e+009

Energy Imbalance (kJ/h) -9.116e+007 Rel Energy Imbalance Pct (%) -2.91

OVERALL UTILITY BALANCE

Utility Name	Usage Info	Energy Flow	Mass Flow	Cost
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Hot Utility Summary	Cold Utility Summary
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Utility Flow ---	Utility Flow ---
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Utility Cost ---	Utility Cost ---
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Carbon Emiss. ---	Carbon Emiss. ---
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Carbon Fees ---	Carbon Fees ---
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PROCESS CO2 EMISSIONS

Inlet Stream	Count	IFPP (1995)	IFPP (2007)	EPA (2009)
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	(kg/h)	(kg/h)	(kg/h)	
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Fresh Propylene	Yes	0.000e-01	0.000e-01	0.000e-01
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Fresh Benzene	Yes	0.000e-01	0.000e-01	0.000e-01
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Total from Inlets	---	---	---	
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Total Carbon Fees	---	---	---	
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from Inlets (Cost/hr)

Outlet Stream	Count	IFPP (1995)	IFPP (2007)	EPA (2009)
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		(kg/h)	(kg/h)	(kg/h)
Fuel-Gas	Yes	0.000e-01	0.000e-01	0.000e-01
C2-Gas	Yes	0.000e-01	0.000e-01	0.000e-01
C3-Gas	Yes	0.000e-01	0.000e-01	0.000e-01
Cumene	Yes	0.000e-01	0.000e-01	0.000e-01
Trans_Vap	Yes	0.000e-01	0.000e-01	0.000e-01
Total from Outlets		---	---	---
Total Carbon Fees		---	---	---
from Outlets (Cost/hr)				

All Material Streams (Case (Main)+ Templates): Conditions, Composition, K Value, Acid Gas Performance, Package Properties, Attachments

Material Stream: Fresh Propylene Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	25.00	25.00
Pressure: (kPa)	1500	1500
Molar Flow (kgmole/h)	1.275e+004	1.275e+004
Mass Flow (kg/h)	5.380e+005	5.380e+005
Std Ideal Liq VolFlow (m3/h)	1034	1034

Molar Enthalpy (kJ/kgmole) -1.461e+03 -1.461e+03

Molar Entropy (kJ/kgmole-C) 2.370e+01 2.370e+01

Heat Flow (kJ/h) -1.864e+07 -1.864e+07

Liq VolFlow @Std Cond (m3/h) 1030 1030

COMPOSITION

Overall Phase

Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
Propene	1.212e+004	0.9500	5.099e+005	0.9477	978.7	0.9463	
Propane	637.7	0.0500	2.812e+004	0.0523	55.50	0.0537	
Cumene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
14-iP-BZ	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
Total	1.275e+004	1.0000	5.380e+005	1.0000	1034	1.0000	

Liquid Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
Propene	1.212e+004	0.9500	5.099e+005	0.9477	978.7	0.9463	
Propane	637.7	0.0500	2.812e+004	0.0523	55.50	0.0537	
Cumene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
14-iP-BZ	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
Total	1.275e+004	1.0000	5.380e+005	1.0000	1034	1.0000	

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO PRODUCT FROM LOGICAL CONNECTION

Pump: P-2

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Benzene HP

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction 0.0000 1.0000

Temperature: (C)	38.71	38.71
Pressure: (kPa)	3500	3500
Molar Flow (kgmole/h)	2.252e+004	2.252e+004
Mass Flow (kg/h)	1.857e+006	1.857e+006
Std Ideal Liq VolFlow (m3/h)	2125	2125
Molar Enthalpy (kJ/kgmole)	3.921e+04	3.921e+04
Molar Entropy (kJ/kgmole-C)	-1.215e+02	-1.215e+02
Heat Flow (kJ/h)	8.828e+08	8.828e+08
Liq VolFlow @Std Cond (m3/h)	2110	2110

COMPOSITION

Overall Phase Vapour Fraction 0.0000

COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.955e+004	0.8683	1.527e+006	0.8223	1731	0.8145
Propene	3.615	0.0002	152.1	0.0001	0.2920	0.0001
Propane	343.4	0.0153	1.514e+004	0.0082	29.89	0.0141
Cumene	2616	0.1162	3.144e+005	0.1693	363.5	0.1710
14-iP-BZ	2.317	0.0001	375.9	0.0002	0.4373	0.0002
Total	2.252e+004	1.0000	1.857e+006	1.0000	2125	1.0000

Liquid Phase Phase Fraction 1.000

COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
------------	-----------	-----------	-----------	-----------	-------------	-------------

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.955e+004	0.8683	1.527e+006	0.8223	1731	0.8145

Propene	3.615	0.0002	152.1	0.0001	0.2920	0.0001
Propane	343.4	0.0153	1.514e+004	0.0082	29.89	0.0141
Cumene	2616	0.1162	3.144e+005	0.1693	363.5	0.1710
14-iP-BZ	2.317	0.0001	375.9	0.0002	0.4373	0.0002
Total	2.252e+004	1.0000	1.857e+006	1.0000	2125	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Mixer: MIX-100	Pump: P-1	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Propylene HP

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	27.38	27.38
Pressure: (kPa)	3500	3500
Molar Flow (kgmole/h)	1.275e+004	1.275e+004
Mass Flow (kg/h)	5.380e+005	5.380e+005
Std Ideal Liq VolFlow (m3/h)	1034	1034
Molar Enthalpy (kJ/kgmole)	-1.239e+03	-1.239e+03
Molar Entropy (kJ/kgmole-C)	2.391e+01	2.391e+01
Heat Flow (kJ/h)	-1.581e+07	-1.581e+07
Liq VolFlow @Std Cond (m3/h)	1030	1030

COMPOSITION

Overall Phase	Vapour Fraction 0.0000
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propene	1.212e+004	0.9500	5.099e+005	0.9477	978.7	0.9463
Propane	637.7	0.0500	2.812e+004	0.0523	55.50	0.0537
Cumene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
14-iP-BZ	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Total	1.275e+004	1.0000	5.380e+005	1.0000	1034	1.0000

Liquid Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propene	1.212e+004	0.9500	5.099e+005	0.9477	978.7	0.9463
Propane	637.7	0.0500	2.812e+004	0.0523	55.50	0.0537
Cumene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
14-iP-BZ	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Total	1.275e+004	1.0000	5.380e+005	1.0000	1034	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Mixer: MIX-100	Pump: P-2	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Mixed Feed

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	36.71	36.71
Pressure: (kPa)	3500	3500
Molar Flow (kgmole/h)	3.527e+004	3.527e+004
Mass Flow (kg/h)	2.395e+006	2.395e+006
Std Ideal Liq VolFlow (m3/h)	3160	3160
Molar Enthalpy (kJ/kgmole)	2.458e+04	2.458e+04
Molar Entropy (kJ/kgmole-C)	-6.432e+01	-6.432e+01
Heat Flow (kJ/h)	8.670e+08	8.670e+08
Liq VolFlow @Std Cond (m3/h)	3080	3080

COMPOSITION

Overall Phase

Vapour Fraction 0.0000

COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW
	LIQVOL FRAC				

(kgmole/h)

(kg/h)

(m3/h)

Benzene	1.955e+004	0.5543	1.527e+006	0.6376	1731	0.5479
Propene	1.212e+004	0.3436	5.100e+005	0.2129	979.0	0.3099
Propane	981.2	0.0278	4.327e+004	0.0181	85.39	0.0270
Cumene	2616	0.0742	3.144e+005	0.1312	363.5	0.1151
14-iP-BZ	2.317	0.0001	375.9	0.0002	0.4373	0.0001
Total	3.527e+004	1.0000	2.395e+006	1.0000	3160	1.0000
Liquid Phase					Phase Fraction	1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.955e+004	0.5543	1.527e+006	0.6376	1731	0.5479
Propene	1.212e+004	0.3436	5.100e+005	0.2129	979.0	0.3099
Propane	981.2	0.0278	4.327e+004	0.0181	85.39	0.0270
Cumene	2616	0.0742	3.144e+005	0.1312	363.5	0.1151
14-iP-BZ	2.317	0.0001	375.9	0.0002	0.4373	0.0001
Total	3.527e+004	1.0000	2.395e+006	1.0000	3160	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Heater: E-100	Mixer: MIX-100	
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UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Reactor Feed	Fluid Package: Basis-1
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Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	150.0	150.0
Pressure: (kPa)	3450	3450
Molar Flow (kgmole/h)	3.527e+004	3.527e+004
Mass Flow (kg/h)	2.395e+006	2.395e+006
Std Ideal Liq VolFlow (m3/h)	3160	3160
Molar Enthalpy (kJ/kgmole)	4.081e+04	4.081e+04
Molar Entropy (kJ/kgmole-C)	-2.006e+01	-2.006e+01
Heat Flow (kJ/h)	1.439e+09	1.439e+09
Liq VolFlow @Std Cond (m3/h)	3080	3080

COMPOSITION

Overall Phase

Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.955e+004	0.5543	1.527e+006	0.6376	1731	0.5479
Propene	1.212e+004	0.3436	5.100e+005	0.2129	979.0	0.3099
Propane	981.2	0.0278	4.327e+004	0.0181	85.39	0.0270
Cumene	2616	0.0742	3.144e+005	0.1312	363.5	0.1151
14-iP-BZ	2.317	0.0001	375.9	0.0002	0.4373	0.0001
Total	3.527e+004	1.0000	2.395e+006	1.0000	3160	1.0000

Liquid Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.955e+004	0.5543	1.527e+006	0.6376	1731	0.5479
Propene	1.212e+004	0.3436	5.100e+005	0.2129	979.0	0.3099
Propane	981.2	0.0278	4.327e+004	0.0181	85.39	0.0270
Cumene	2616	0.0742	3.144e+005	0.1312	363.5	0.1151
14-iP-BZ	2.317	0.0001	375.9	0.0002	0.4373	0.0001
Total	3.527e+004	1.0000	2.395e+006	1.0000	3160	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---

Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Plug Flow Reactor: R-1 Heater: E-100

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Alkylolation Effluents	Fluid Package: Basis-1
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Property Package: Peng-Robinson

CONDITIONS

	OVERALL	LIQUID PH.	VAPOUR PH.
Vapour / Phase Fraction	0.0000	1.0000	0.0000
Temperature: (C)	150.0	150.0	150.0
Pressure: (kPa)	3400	3400	3400
Molar Flow (kgmole/h)	2.315e+004	2.315e+004	0.0000
Mass Flow (kg/h)	2.395e+006	2.395e+006	0.0000
Std Ideal Liq VolFlow (m3/h)	2789	2789	0.0000

Molar Enthalpy (kJ/kgmole) 1.100e+04 1.100e+04 1.093e+04
Molar Entropy (kJ/kgmole-C) -1.605e+01 -1.605e+01 -1.594e+01
Heat Flow (kJ/h) 2.547e+08 2.547e+08 0.000e-01
Liq VolFlow @Std Cond (m3/h) 2735 2735 0.0000

COMPOSITION

Overall Phase Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.118e+004	0.4830	8.734e+005	0.3646	990.0	0.3549
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	981.2	0.0424	4.327e+004	0.0181	85.39	0.0306
Cumene	7238	0.3126	8.699e+005	0.3632	1006	0.3606
14-iP-BZ	3751	0.1620	6.087e+005	0.2541	708.0	0.2538
Total	2.315e+004	1.0000	2.395e+006	1.0000	2789	1.0000

Liquid Phase Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.118e+004	0.4830	8.734e+005	0.3646	990.0	0.3549
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	981.2	0.0424	4.327e+004	0.0181	85.39	0.0306
Cumene	7238	0.3126	8.699e+005	0.3632	1006	0.3606
14-iP-BZ	3751	0.1620	6.087e+005	0.2541	708.0	0.2538
Total	2.315e+004	1.0000	2.395e+006	1.0000	2789	1.0000

Vapour Phase

Phase Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	0.0000	0.4826	0.0000	0.3644	0.0000	0.3546
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	0.0000	0.0428	0.0000	0.0182	0.0000	0.0309
Cumene	0.0000	0.3126	0.0000	0.3632	0.0000	0.3606
14-iP-BZ	0.0000	0.1620	0.0000	0.2542	0.0000	0.2538
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.9992	0.9992	---
Propene	0.0000	0.0000	---
Propane	1.010	1.010	---
Cumene	1.000	1.000	---
14-iP-BZ	1.000	1.000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Valve: VLV-100	Plug Flow Reactor: R-1	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C1-Feed

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	150.4	150.4
Pressure: (kPa)	1500	1500
Molar Flow (kgmole/h)	2.315e+004	2.315e+004
Mass Flow (kg/h)	2.395e+006	2.395e+006
Std Ideal Liq VolFlow (m3/h)	2789	2789
Molar Enthalpy (kJ/kgmole)	1.100e+04	1.100e+04
Molar Entropy (kJ/kgmole-C)	-1.541e+01	-1.541e+01
Heat Flow (kJ/h)	2.547e+08	2.547e+08
Liq VolFlow @Std Cond (m3/h)	2735	2735

COMPOSITION

Overall Phase

Vapour Fraction 0.0000

COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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(kgmole/h)	(kg/h)	(m3/h)
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Benzene	1.118e+004	0.4830	8.734e+005	0.3646	990.0	0.3549
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	981.2	0.0424	4.327e+004	0.0181	85.39	0.0306
Cumene	7238	0.3126	8.699e+005	0.3632	1006	0.3606
14-iP-BZ	3751	0.1620	6.087e+005	0.2541	708.0	0.2538
Total	2.315e+004	1.0000	2.395e+006	1.0000	2789	1.0000
Liquid Phase	Phase Fraction 1.000					

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.118e+004	0.4830	8.734e+005	0.3646	990.0	0.3549
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	981.2	0.0424	4.327e+004	0.0181	85.39	0.0306
Cumene	7238	0.3126	8.699e+005	0.3632	1006	0.3606
14-iP-BZ	3751	0.1620	6.087e+005	0.2541	708.0	0.2538
Total	2.315e+004	1.0000	2.395e+006	1.0000	2789	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Absorber: T-100	Valve: VLV-100	
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UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Reflux-1

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	41.90	41.90
Pressure: (kPa)	1450	1450
Molar Flow (kgmole/h)	1.999e+004	1.999e+004
Mass Flow (kg/h)	8.800e+005	8.800e+005
Std Ideal Liq VolFlow (m3/h)	1735	1735
Molar Enthalpy (kJ/kgmole)	-1.127e+05	-1.127e+05
Molar Entropy (kJ/kgmole-C)	9.602e+01	9.602e+01
Heat Flow (kJ/h)	-2.253e+09	-2.253e+09
Liq VolFlow @Std Cond (m3/h)	1731	1731

COMPOSITION

Overall Phase

Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	9.534e-002	0.0000	7.447	0.0000	8.441e-003	0.0000	
Propene	824.8	0.0413	3.471e+004	0.0394	66.63	0.0384	
Propane	1.917e+004	0.9587	8.453e+005	0.9606	1668	0.9616	
Cumene	6.047e-007	0.0000	7.268e-005	0.0000	8.405e-008	0.0000	
14-iP-BZ	7.659e-011	0.0000	1.243e-008	0.0000	1.446e-011	0.0000	
Total	1.999e+004	1.0000	8.800e+005	1.0000	1735	1.0000	

Liquid Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	9.534e-002	0.0000	7.447	0.0000	8.441e-003	0.0000	
Propene	824.8	0.0413	3.471e+004	0.0394	66.63	0.0384	
Propane	1.917e+004	0.9587	8.453e+005	0.9606	1668	0.9616	
Cumene	6.047e-007	0.0000	7.268e-005	0.0000	8.405e-008	0.0000	
14-iP-BZ	7.659e-011	0.0000	1.243e-008	0.0000	1.446e-011	0.0000	
Total	1.999e+004	1.0000	8.800e+005	1.0000	1735	1.0000	

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---

Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Absorber: T-100	Recycle: RCY-4	
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UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C1-Top	Fluid Package: Basis-1
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Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH.

Vapour / Phase Fraction	1.0000	1.0000
Temperature: (C)	41.98	41.98
Pressure: (kPa)	1450	1450
Molar Flow (kgmole/h)	2.000e+004	2.000e+004
Mass Flow (kg/h)	8.804e+005	8.804e+005
Std Ideal Liq VolFlow (m3/h)	1736	1736

Molar Enthalpy (kJ/kgmole) -9.939e+04 -9.939e+04

Molar Entropy (kJ/kgmole-C) 1.385e+02 1.385e+02

Heat Flow (kJ/h) -1.988e+09 -1.988e+09

Liq VolFlow @Std Cond (m3/h) 1732 1732

COMPOSITION

Overall Phase

Vapour Fraction 1.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	8.936e-002	0.0000	6.980	0.0000	7.912e-003	0.0000	
Propene	810.7	0.0405	3.412e+004	0.0387	65.49	0.0377	
Propane	1.919e+004	0.9595	8.463e+005	0.9612	1670	0.9623	
Cumene	5.545e-007	0.0000	6.665e-005	0.0000	7.708e-008	0.0000	
14-iP-BZ	7.045e-011	0.0000	1.143e-008	0.0000	1.330e-011	0.0000	
Total	2.000e+004	1.0000	8.804e+005	1.0000	1736	1.0000	

Vapour Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	8.936e-002	0.0000	6.980	0.0000	7.912e-003	0.0000	
Propene	810.7	0.0405	3.412e+004	0.0387	65.49	0.0377	
Propane	1.919e+004	0.9595	8.463e+005	0.9612	1670	0.9623	
Cumene	5.545e-007	0.0000	6.665e-005	0.0000	7.708e-008	0.0000	
14-iP-BZ	7.045e-011	0.0000	1.143e-008	0.0000	1.330e-011	0.0000	
Total	2.000e+004	1.0000	8.804e+005	1.0000	1736	1.0000	

K VALUE

COMPONENTS		MIXED	LIGHT	HEAVY
Benzene	---	---	---	
Propene	---	---	---	
Propane	---	---	---	
Cumene	---	---	---	
14-iP-BZ	---	---	---	

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Cooler: E-102	Absorber: T-100	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C1-Bot Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction 0.0000 1.0000

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.888e+004	0.5223	1.475e+006	0.4355	1672	0.4133

Propene	65.97	0.0018	2776	0.0008	5.328	0.0013
Propane	4327	0.1197	1.908e+005	0.0563	376.6	0.0931
Cumene	8802	0.2435	1.058e+006	0.3124	1223	0.3025
14-iP-BZ	4068	0.1126	6.602e+005	0.1950	767.9	0.1898
Total	3.614e+004	1.0000	3.386e+006	1.0000	4045	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Heater: E-101	Absorber: T-100	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C2-Feed

Fluid Package: Basis-1

CONDITIONS

Vapour / Phase Fraction	0.0000	0.0000	1.0000
Temperature: (C)	219.3	219.3	219.3
Pressure: (kPa)	1500	1500	1500
Molar Flow (kgmole/h)	2.314e+004	0.0000	2.314e+004
Mass Flow (kg/h)	2.395e+006	0.0000	2.395e+006
Std Ideal Liq VolFlow (m3/h)	2788	0.0000	2788
Molar Enthalpy (kJ/kgmole)	2.794e+04	4.205e+04	2.794e+04
Molar Entropy (kJ/kgmole-C)	2.118e+01	6.127e+01	2.118e+01
Heat Flow (kJ/h)	6.467e+08	0.000e-01	6.467e+08
Liq VolFlow @Std Cond (m3/h)	2734	0.0000	2734

Overall Phase Vapour Fraction 0.0000

	(kgmole/h)		(kg/h)		(m3/h)	
Benzene	1.118e+004	0.4833	8.736e+005	0.3648	990.3	0.3551
Propene	14.15	0.0006	595.6	0.0002	1.143	0.0004
Propane	958.3	0.0414	4.226e+004	0.0176	83.41	0.0299
Cumene	7235	0.3126	8.696e+005	0.3631	1006	0.3606
14-iP-BZ	3751	0.1621	6.086e+005	0.2542	707.9	0.2539
Total	2.314e+004	1.0000	2.395e+006	1.0000	2788	1.0000

Vapour Phase

Phase Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	0.0000	0.5918	0.0000	0.6060	0.0000	0.5422
Propene	0.0000	0.0040	0.0000	0.0022	0.0000	0.0033
Propane	0.0000	0.2591	0.0000	0.1498	0.0000	0.2333
Cumene	0.0000	0.1206	0.0000	0.1900	0.0000	0.1734
14-iP-BZ	0.0000	0.0245	0.0000	0.0520	0.0000	0.0478
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000

Liquid Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.118e+004	0.4833	8.736e+005	0.3648	990.3	0.3551
Propene	14.15	0.0006	595.6	0.0002	1.143	0.0004
Propane	958.3	0.0414	4.226e+004	0.0176	83.41	0.0299
Cumene	7235	0.3126	8.696e+005	0.3631	1006	0.3606
14-iP-BZ	3751	0.1621	6.086e+005	0.2542	707.9	0.2539
Total	2.314e+004	1.0000	2.395e+006	1.0000	2788	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	1.225	1.225	---
Propene	6.515	6.515	---
Propane	6.257	6.257	---

Cumene	0.3857	0.3857	---
14-iP-BZ	0.1509	0.1509	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Mixer: MIX-102	Separator: V-100	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Fuel-Gas	Fluid Package: Basis-1
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Property Package: Peng-Robinson

CONDITIONS

	OVERALL	LIQUID PH.	VAPOUR PH.
Vapour / Phase Fraction	1.0000	0.0000	1.0000
Temperature: (C)	41.91	41.91	41.91
Pressure: (kPa)	1450	1450	1450
Molar Flow (kgmole/h)	0.0000	0.0000	0.0000
Mass Flow (kg/h)	0.0000	0.0000	0.0000
Std Ideal Liq VolFlow (m3/h)	0.0000	0.0000	0.0000
Molar Enthalpy (kJ/kgmole)	-9.852e+04	-1.128e+05	-9.852e+04

Molar Entropy (kJ/kgmole-C) 1.381e+02 9.606e+01 1.381e+02

Heat Flow (kJ/h) -0.000e-01 0.000e-01 0.000e-01

Liq VolFlow @Std Cond (m3/h) 0.0000 0.0000 0.0000

COMPOSITION

Overall Phase

Vapour Fraction 1.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propene	0.0000	0.0476	0.0000	0.0455	0.0000	0.0443
Propane	0.0000	0.9524	0.0000	0.9545	0.0000	0.9557
Cumene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
14-iP-BZ	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000

Liquid Phase

Phase Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propene	0.0000	0.0405	0.0000	0.0387	0.0000	0.0377
Propane	0.0000	0.9595	0.0000	0.9612	0.0000	0.9623
Cumene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
14-iP-BZ	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000

Vapour Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propene	0.0000	0.0476	0.0000	0.0455	0.0000	0.0443
Propane	0.0000	0.9524	0.0000	0.9545	0.0000	0.9557
Cumene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
14-iP-BZ	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	8.040e-002	8.040e-002	---
Propene	1.174	1.174	---
Propane	0.9926	0.9926	---
Cumene	7.082e-003	7.082e-003	---
14-iP-BZ	1.237e-003	1.237e-003	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Separator: V-101

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Reflux-Temp

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	41.91	41.91
Pressure: (kPa)	1450	1450
Molar Flow (kgmole/h)	2.000e+004	2.000e+004
Mass Flow (kg/h)	8.804e+005	8.804e+005
Std Ideal Liq VolFlow (m3/h)	1736	1736
Molar Enthalpy (kJ/kgmole)	-1.128e+05	-1.128e+05
Molar Entropy (kJ/kgmole-C)	9.606e+01	9.606e+01
Heat Flow (kJ/h)	-2.256e+09	-2.256e+09
Liq VolFlow @Std Cond (m3/h)	1732	1732

COMPOSITION

Overall Phase	Vapour Fraction 0.0000
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COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	8.936e-002	0.0000	6.980	0.0000	7.912e-003	0.0000

Propene	810.7	0.0405	3.412e+004	0.0387	65.49	0.0377
Propane	1.919e+004	0.9595	8.463e+005	0.9612	1670	0.9623
Cumene	5.545e-007	0.0000	6.665e-005	0.0000	7.708e-008	0.0000
14-iP-BZ	7.045e-011	0.0000	1.143e-008	0.0000	1.330e-011	0.0000
Total	2.000e+004	1.0000	8.804e+005	1.0000	1736	1.0000
Liquid Phase	Phase Fraction 1.000					

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)	
Benzene	8.936e-002	0.0000	6.980	0.0000	7.912e-003	0.0000
Propene	810.7	0.0405	3.412e+004	0.0387	65.49	0.0377
Propane	1.919e+004	0.9595	8.463e+005	0.9612	1670	0.9623
Cumene	5.545e-007	0.0000	6.665e-005	0.0000	7.708e-008	0.0000
14-iP-BZ	7.045e-011	0.0000	1.143e-008	0.0000	1.330e-011	0.0000
Total	2.000e+004	1.0000	8.804e+005	1.0000	1736	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Recycle: RCY-4 Pump: P-100

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Boilup-New

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

	OVERALL	VAPOUR PH.	LIQUID PH.
Vapour / Phase Fraction	1.0000	1.0000	0.0000
Temperature: (C)	219.3	219.3	219.3
Pressure: (kPa)	1500	1500	1500
Molar Flow (kgmole/h)	1.300e+004	1.300e+004	0.0000
Mass Flow (kg/h)	9.917e+005	9.917e+005	0.0000
Std Ideal Liq VolFlow (m3/h)	1256	1256	0.0000
Molar Enthalpy (kJ/kgmole)	4.205e+04	4.205e+04	2.794e+04
Molar Entropy (kJ/kgmole-C)	6.127e+01	6.127e+01	2.118e+01
Heat Flow (kJ/h)	5.467e+08	5.467e+08	0.000e-01
Liq VolFlow @Std Cond (m3/h)	1228	1228	0.0000

COMPOSITION

Overall Phase

Vapour Fraction 1.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	7694	0.5918	6.010e+005	0.6060	681.2	0.5422
Propene	51.81	0.0040	2180	0.0022	4.185	0.0033
Propane	3369	0.2591	1.485e+005	0.1498	293.2	0.2333
Cumene	1568	0.1206	1.884e+005	0.1900	217.9	0.1734
14-iP-BZ	317.9	0.0245	5.159e+004	0.0520	60.00	0.0478
Total	1.300e+004	1.0000	9.917e+005	1.0000	1256	1.0000

Vapour Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	7694	0.5918	6.010e+005	0.6060	681.2	0.5422
Propene	51.81	0.0040	2180	0.0022	4.185	0.0033
Propane	3369	0.2591	1.485e+005	0.1498	293.2	0.2333
Cumene	1568	0.1206	1.884e+005	0.1900	217.9	0.1734
14-iP-BZ	317.9	0.0245	5.159e+004	0.0520	60.00	0.0478
Total	1.300e+004	1.0000	9.917e+005	1.0000	1256	1.0000

Liquid Phase

Phase Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	0.0000	0.4833	0.0000	0.3648	0.0000	0.3551

Propene	0.0000	0.0006	0.0000	0.0002	0.0000	0.0004
Propane	0.0000	0.0414	0.0000	0.0176	0.0000	0.0299
Cumene	0.0000	0.3126	0.0000	0.3631	0.0000	0.3606
14-iP-BZ	0.0000	0.1621	0.0000	0.2542	0.0000	0.2539
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	1.225	1.225	---
Propene	6.515	6.515	---
Propane	6.257	6.257	---
Cumene	0.3857	0.3857	---
14-iP-BZ	0.1509	0.1509	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Recycle: RCY-3	Separator: V-100	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Cooled-New

Fluid Package: Basis-1

	(kgmole/h)		(kg/h)		(m3/h)	
Benzene	8.936e-002	0.0000	6.980	0.0000	7.912e-003	0.0000
Propene	810.7	0.0405	3.412e+004	0.0387	65.49	0.0377
Propane	1.919e+004	0.9595	8.463e+005	0.9612	1670	0.9623
Cumene	5.545e-007	0.0000	6.665e-005	0.0000	7.708e-008	0.0000
14-iP-BZ	7.045e-011	0.0000	1.143e-008	0.0000	1.330e-011	0.0000
Total	2.000e+004	1.0000	8.804e+005	1.0000	1736	1.0000

Liquid Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	8.936e-002	0.0000	6.980	0.0000	7.912e-003	0.0000
Propene	810.7	0.0405	3.412e+004	0.0387	65.49	0.0377
Propane	1.919e+004	0.9595	8.463e+005	0.9612	1670	0.9623
Cumene	5.545e-007	0.0000	6.665e-005	0.0000	7.708e-008	0.0000
14-iP-BZ	7.045e-011	0.0000	1.143e-008	0.0000	1.330e-011	0.0000
Total	2.000e+004	1.0000	8.804e+005	1.0000	1736	1.0000

Vapour Phase

Phase Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propene	0.0000	0.0476	0.0000	0.0455	0.0000	0.0443
Propane	0.0000	0.9524	0.0000	0.9545	0.0000	0.9557
Cumene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
14-iP-BZ	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	8.040e-002	8.040e-002	---
Propene	1.174	1.174	---
Propane	0.9926	0.9926	---

Cumene	7.082e-003	7.082e-003	---
14-iP-BZ	1.237e-003	1.237e-003	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Separator: V-101	Cooler: E-102	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Reflux-New	Fluid Package: Basis-1
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Property Package: Peng-Robinson

CONDITIONS

	OVERALL	LIQUID PH.	VAPOUR PH.
Vapour / Phase Fraction	0.0000	1.0000	0.0000
Temperature: (C)	41.91	41.91	41.91
Pressure: (kPa)	1450	1450	1450
Molar Flow (kgmole/h)	2.000e+004	2.000e+004	0.0000
Mass Flow (kg/h)	8.804e+005	8.804e+005	0.0000
Std Ideal Liq VolFlow (m3/h)	1736	1736	0.0000
Molar Enthalpy (kJ/kgmole)	-1.128e+05	-1.128e+05	-9.852e+04

Molar Entropy (kJ/kgmole-C) 9.606e+01 9.606e+01 1.381e+02

Heat Flow (kJ/h) -2.256e+09 -2.256e+09 0.000e-01

Liq VolFlow @Std Cond (m3/h) 1732 1732 0.0000

COMPOSITION

Overall Phase Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	8.936e-002	0.0000	6.980	0.0000	7.912e-003	0.0000	
Propene	810.7	0.0405	3.412e+004	0.0387	65.49	0.0377	
Propane	1.919e+004	0.9595	8.463e+005	0.9612	1670	0.9623	
Cumene	5.545e-007	0.0000	6.665e-005	0.0000	7.708e-008	0.0000	
14-iP-BZ	7.045e-011	0.0000	1.143e-008	0.0000	1.330e-011	0.0000	
Total	2.000e+004	1.0000	8.804e+005	1.0000	1736	1.0000	

Liquid Phase Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	8.936e-002	0.0000	6.980	0.0000	7.912e-003	0.0000	
Propene	810.7	0.0405	3.412e+004	0.0387	65.49	0.0377	
Propane	1.919e+004	0.9595	8.463e+005	0.9612	1670	0.9623	
Cumene	5.545e-007	0.0000	6.665e-005	0.0000	7.708e-008	0.0000	
14-iP-BZ	7.045e-011	0.0000	1.143e-008	0.0000	1.330e-011	0.0000	
Total	2.000e+004	1.0000	8.804e+005	1.0000	1736	1.0000	

Vapour Phase Phase Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propene	0.0000	0.0476	0.0000	0.0455	0.0000	0.0443
Propane	0.0000	0.9524	0.0000	0.9545	0.0000	0.9557
Cumene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
14-iP-BZ	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	8.040e-002	8.040e-002	---
Propene	1.174	1.174	---
Propane	0.9926	0.9926	---
Cumene	7.082e-003	7.082e-003	---
14-iP-BZ	1.237e-003	1.237e-003	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Pump: P-100	Separator: V-101	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Heated_Bot

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

	OVERALL	VAPOUR PH.	LIQUID PH.
Vapour / Phase Fraction	0.3597	0.3597	0.6403
Temperature: (C)	219.3	219.3	219.3
Pressure: (kPa)	1500	1500	1500
Molar Flow (kgmole/h)	3.614e+004	1.300e+004	2.314e+004
Mass Flow (kg/h)	3.386e+006	9.918e+005	2.395e+006
Std Ideal Liq VolFlow (m3/h)	4045	1257	2788
Molar Enthalpy (kJ/kgmole)	3.302e+04	4.205e+04	2.794e+04
Molar Entropy (kJ/kgmole-C)	3.560e+01	6.127e+01	2.118e+01
Heat Flow (kJ/h)	1.193e+09	5.467e+08	6.467e+08
Liq VolFlow @Std Cond (m3/h)	3954	1228	2734

COMPOSITION

Overall Phase	Vapour Fraction 0.3597
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COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.888e+004	0.5223	1.475e+006	0.4355	1672	0.4133

Propene	65.97	0.0018	2776	0.0008	5.328	0.0013
Propane	4327	0.1197	1.908e+005	0.0563	376.6	0.0931
Cumene	8802	0.2435	1.058e+006	0.3124	1223	0.3025
14-iP-BZ	4068	0.1126	6.602e+005	0.1950	767.9	0.1898
Total	3.614e+004	1.0000	3.386e+006	1.0000	4045	1.0000
Vapour Phase			Phase Fraction 0.3597			

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)	
Benzene	7695	0.5918	6.010e+005	0.6060	681.3	0.5422
Propene	51.81	0.0040	2180	0.0022	4.185	0.0033
Propane	3369	0.2591	1.486e+005	0.1498	293.2	0.2333
Cumene	1568	0.1206	1.884e+005	0.1900	217.9	0.1734
14-iP-BZ	317.9	0.0245	5.159e+004	0.0520	60.01	0.0478
Total	1.300e+004	1.0000	9.918e+005	1.0000	1257	1.0000
Liquid Phase			Phase Fraction 0.6403			

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)	
Benzene	1.118e+004	0.4833	8.736e+005	0.3648	990.3	0.3552
Propene	14.15	0.0006	595.6	0.0002	1.143	0.0004
Propane	958.3	0.0414	4.226e+004	0.0176	83.40	0.0299
Cumene	7235	0.3126	8.696e+005	0.3631	1006	0.3606
14-iP-BZ	3750	0.1621	6.086e+005	0.2542	707.9	0.2539
Total	2.314e+004	1.0000	2.395e+006	1.0000	2788	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	1.225	1.225	---
Propene	6.515	6.515	---
Propane	6.257	6.257	---
Cumene	0.3857	0.3857	---
14-iP-BZ	0.1509	0.1509	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Separator: V-100	Heater: E-101	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C2-Feed-LP Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

	OVERALL	VAPOUR PH.	LIQUID PH.
Vapour / Phase Fraction	0.6057	0.6057	0.3943
Temperature: (C)	141.8	141.8	141.8

Pressure: (kPa)	120.0	120.0	120.0
Molar Flow (kgmole/h)	3.594e+004	2.177e+004	1.417e+004
Mass Flow (kg/h)	3.862e+006	2.059e+006	1.804e+006
Std Ideal Liq VolFlow (m3/h)	4496	2409	2088
Molar Enthalpy (kJ/kgmole)	2.116e+04	5.212e+04	-2.639e+04
Molar Entropy (kJ/kgmole-C)	3.975e+01	5.357e+01	1.851e+01
Heat Flow (kJ/h)	7.606e+08	1.135e+09	-3.741e+08
Liq VolFlow @Std Cond (m3/h)	4418	2368	2068

COMPOSITION

Overall Phase	Vapour Fraction 0.6057
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.373e+004	0.3819	1.072e+006	0.2776	1215	0.2703
Propene	14.15	0.0004	595.6	0.0002	1.143	0.0003
Propane	1341	0.0373	5.913e+004	0.0153	116.7	0.0260
Cumene	1.555e+004	0.4327	1.869e+006	0.4839	2161	0.4807
14-iP-BZ	5308	0.1477	8.613e+005	0.2230	1002	0.2228
Total	3.594e+004	1.0000	3.862e+006	1.0000	4496	1.0000

Vapour Phase	Phase Fraction 0.6057
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.181e+004	0.5424	9.223e+005	0.4480	1045	0.4341
Propene	13.98	0.0006	588.2	0.0003	1.129	0.0005

Propane	1323	0.0608	5.832e+004	0.0283	115.1	0.0478
Cumene	7651	0.3515	9.196e+005	0.4467	1063	0.4415
14-iP-BZ	972.6	0.0447	1.578e+005	0.0767	183.6	0.0762
Total	2.177e+004	1.0000	2.059e+006	1.0000	2409	1.0000
Liquid Phase	Phase Fraction 0.3943					

COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1919	0.1354	1.499e+005	0.0831	169.9	0.0814
Propene	0.1775	0.0000	7.468	0.0000	1.433e-002	0.0000
Propane	18.36	0.0013	809.7	0.0004	1.598	0.0008
Cumene	7899	0.5574	9.495e+005	0.5264	1098	0.5259
14-iP-BZ	4335	0.3059	7.034e+005	0.3900	818.2	0.3919
Total	1.417e+004	1.0000	1.804e+006	1.0000	2088	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	4.006	4.006	---
Propene	51.28	51.28	---
Propane	46.89	46.89	---
Cumene	0.6305	0.6305	---
14-iP-BZ	0.1461	0.1461	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Absorber: T-101	Valve: VLV-101	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Reflux-2

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	50.02	50.02
Pressure: (kPa)	150.0	150.0
Molar Flow (kgmole/h)	8327	8327
Mass Flow (kg/h)	7.351e+005	7.351e+005
Std Ideal Liq VolFlow (m3/h)	850.5	850.5
Molar Enthalpy (kJ/kgmole)	2.358e+04	2.358e+04
Molar Entropy (kJ/kgmole-C)	-9.317e+01	-9.317e+01
Heat Flow (kJ/h)	1.963e+08	1.963e+08
Liq VolFlow @Std Cond (m3/h)	839.9	839.9

COMPOSITION

Overall Phase

Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	5778	0.6939	4.513e+005	0.6139	511.6	0.6015
Propene	3.100	0.0004	130.4	0.0002	0.2504	0.0003
Propane	294.6	0.0354	1.299e+004	0.0177	25.64	0.0301
Cumene	2249	0.2701	2.704e+005	0.3678	312.6	0.3676
14-iP-BZ	1.995	0.0002	323.8	0.0004	0.3766	0.0004
Total	8327	1.0000	7.351e+005	1.0000	850.5	1.0000
Liquid Phase					Phase Fraction	1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	5778	0.6939	4.513e+005	0.6139	511.6	0.6015
Propene	3.100	0.0004	130.4	0.0002	0.2504	0.0003
Propane	294.6	0.0354	1.299e+004	0.0177	25.64	0.0301
Cumene	2249	0.2701	2.704e+005	0.3678	312.6	0.3676
14-iP-BZ	1.995	0.0002	323.8	0.0004	0.3766	0.0004
Total	8327	1.0000	7.351e+005	1.0000	850.5	1.0000
K VALUE						

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---

14-iP-BZ	0.0000	0.0000	---
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UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Absorber: T-101	Recycle: RCY-6	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Boilup-2	Fluid Package: Basis-1
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Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH.

Vapour / Phase Fraction	1.0000	1.0000
Temperature: (C)	172.7	172.7
Pressure: (kPa)	120.0	120.0
Molar Flow (kgmole/h)	8500	8500
Mass Flow (kg/h)	1.065e+006	1.065e+006
Std Ideal Liq VolFlow (m3/h)	1232	1232
Molar Enthalpy (kJ/kgmole)	2.310e+04	2.310e+04
Molar Entropy (kJ/kgmole-C)	1.339e+02	1.339e+02

Heat Flow (kJ/h) 1.963e+08 1.963e+08

Liq VolFlow @Std Cond (m3/h) 1229 1229

COMPOSITION

Overall Phase Vapour Fraction 1.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	3.669e-002	0.0000	2.866	0.0000	3.249e-003	0.0000	
Propene	2.086e-022	0.0000	8.780e-021	0.0000	1.685e-023	0.0000	
Propane	9.137e-020	0.0000	4.029e-018	0.0000	7.952e-021	0.0000	
Cumene	7474	0.8794	8.984e+005	0.8437	1039	0.8430	
14-iP-BZ	1025	0.1206	1.664e+005	0.1563	193.5	0.1570	
Total	8500	1.0000	1.065e+006	1.0000	1232	1.0000	

Vapour Phase Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	3.669e-002	0.0000	2.866	0.0000	3.249e-003	0.0000	
Propene	2.086e-022	0.0000	8.780e-021	0.0000	1.685e-023	0.0000	
Propane	9.137e-020	0.0000	4.029e-018	0.0000	7.952e-021	0.0000	
Cumene	7474	0.8794	8.984e+005	0.8437	1039	0.8430	
14-iP-BZ	1025	0.1206	1.664e+005	0.1563	193.5	0.1570	
Total	8500	1.0000	1.065e+006	1.0000	1232	1.0000	

K VALUE

COMPONENTS		MIXED	LIGHT	HEAVY
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Benzene	---	---	---	
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Propene	---	---	---	
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Propane	---	---	---	
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Cumene	---	---	---	
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14-iP-BZ	---	---	---	
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UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Absorber: T-101	Recycle: RCY-5	
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UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C2-Top

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH.

Vapour / Phase Fraction	1.0000	1.0000
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Temperature: (C)	115.6	115.6
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Pressure: (kPa)	100.0	100.0
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Molar Flow (kgmole/h) 2.866e+004 2.866e+004

Mass Flow (kg/h) 2.498e+006 2.498e+006

Std Ideal Liq VolFlow (m3/h) 2914 2914

Molar Enthalpy (kJ/kgmole) 6.185e+04 6.185e+04

Molar Entropy (kJ/kgmole-C) 2.507e+01 2.507e+01

Heat Flow (kJ/h) 1.773e+09 1.773e+09

Liq VolFlow @Std Cond (m3/h) 2874 2874

COMPOSITION

Overall Phase Vapour Fraction 1.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.950e+004	0.6806	1.524e+006	0.6098	1727	0.5927	
Propene	17.25	0.0006	726.1	0.0003	1.394	0.0005	
Propane	1636	0.0571	7.212e+004	0.0289	142.3	0.0489	
Cumene	7494	0.2615	9.008e+005	0.3606	1042	0.3575	
14-iP-BZ	6.632	0.0002	1076	0.0004	1.252	0.0004	
Total	2.866e+004	1.0000	2.498e+006	1.0000	2914	1.0000	

Vapour Phase Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.950e+004	0.6806	1.524e+006	0.6098	1727	0.5927	
Propene	17.25	0.0006	726.1	0.0003	1.394	0.0005	
Propane	1636	0.0571	7.212e+004	0.0289	142.3	0.0489	

Cumene	7494	0.2615	9.008e+005	0.3606	1042	0.3575
14-iP-BZ	6.632	0.0002	1076	0.0004	1.252	0.0004
Total	2.866e+004	1.0000	2.498e+006	1.0000	2914	1.0000

K VALUE

COMPONENTS		MIXED	LIGHT	HEAVY
Benzene	---	---	---	
Propene	---	---	---	
Propane	---	---	---	
Cumene	---	---	---	
14-iP-BZ	---	---	---	

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Cooler: E-103	Absorber: T-101	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C2-Bot	Fluid Package: Basis-1
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Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction 0.0000 1.0000

Temperature: (C) 169.5 169.5

Pressure: (kPa) 120.0 120.0

Molar Flow (kgmole/h) 2.411e+004 2.411e+004

Mass Flow (kg/h) 3.164e+006 3.164e+006

Std Ideal Liq VolFlow (m3/h) 3666 3666

Molar Enthalpy (kJ/kgmole) -2.569e+04 -2.569e+04

Molar Entropy (kJ/kgmole-C) 5.065e+01 5.065e+01

Heat Flow (kJ/h) -6.194e+08 -6.194e+08

Liq VolFlow @Std Cond (m3/h) 3653 3653

COMPOSITION

Overall Phase Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	5.063e-002	0.0000	3.954	0.0000	4.483e-003	0.0000	
Propene	2.173e-022	0.0000	9.143e-021	0.0000	1.755e-023	0.0000	
Propane	9.538e-020	0.0000	4.206e-018	0.0000	8.301e-021	0.0000	
Cumene	1.778e+004	0.7375	2.137e+006	0.6754	2471	0.6742	
14-iP-BZ	6328	0.2625	1.027e+006	0.3246	1194	0.3258	
Total	2.411e+004	1.0000	3.164e+006	1.0000	3666	1.0000	

Liquid Phase Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	5.063e-002	0.0000	3.954	0.0000	4.483e-003	0.0000	
Propene	2.173e-022	0.0000	9.143e-021	0.0000	1.755e-023	0.0000	
Propane	9.538e-020	0.0000	4.206e-018	0.0000	8.301e-021	0.0000	
Cumene	1.778e+004	0.7375	2.137e+006	0.6754	2471	0.6742	
14-iP-BZ	6328	0.2625	1.027e+006	0.3246	1194	0.3258	
Total	2.411e+004	1.0000	3.164e+006	1.0000	3666	1.0000	

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Heater: E-104	Absorber: T-101	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C2-Cooled

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

	OVERALL	VAPOUR PH.	LIQUID PH.
Vapour / Phase Fraction	0.0318	0.0318	0.9682
Temperature: (C)	50.00	50.00	50.00
Pressure: (kPa)	100.0	100.0	100.0
Molar Flow (kgmole/h)	2.866e+004	911.1	2.775e+004
Mass Flow (kg/h)	2.498e+006	4.896e+004	2.449e+006
Std Ideal Liq VolFlow (m3/h)	2914	79.98	2834
Molar Enthalpy (kJ/kgmole)	2.124e+04	-5.041e+04	2.359e+04
Molar Entropy (kJ/kgmole-C)	-8.659e+01	1.152e+02	-9.321e+01
Heat Flow (kJ/h)	6.088e+08	-4.592e+07	6.547e+08
Liq VolFlow @Std Cond (m3/h)	2874	77.86	2798

COMPOSITION

Overall Phase	Vapour Fraction 0.0318
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COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.950e+004	0.6806	1.524e+006	0.6098	1727	0.5927
Propene	17.25	0.0006	726.1	0.0003	1.394	0.0005

Propane	1636	0.0571	7.212e+004	0.0289	142.3	0.0489
Cumene	7494	0.2615	9.008e+005	0.3606	1042	0.3575
14-iP-BZ	6.632	0.0002	1076	0.0004	1.252	0.0004
Total	2.866e+004	1.0000	2.498e+006	1.0000	2914	1.0000
Vapour Phase			Phase Fraction 3.179e-002			

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	243.1	0.2668	1.898e+004	0.3878	21.52	0.2691
Propene	6.927	0.0076	291.5	0.0060	0.5596	0.0070
Propane	654.2	0.7180	2.885e+004	0.5893	56.93	0.7119
Cumene	6.924	0.0076	832.2	0.0170	0.9624	0.0120
14-iP-BZ	1.005e-003	0.0000	0.1631	0.0000	1.897e-004	0.0000
Total	911.1	1.0000	4.896e+004	1.0000	79.98	1.0000
Liquid Phase			Phase Fraction 0.9682			

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.926e+004	0.6942	1.505e+006	0.6143	1705	0.6019
Propene	10.33	0.0004	434.6	0.0002	0.8342	0.0003
Propane	981.4	0.0354	4.328e+004	0.0177	85.41	0.0301
Cumene	7487	0.2698	8.999e+005	0.3674	1041	0.3673
14-iP-BZ	6.631	0.0002	1076	0.0004	1.252	0.0004
Total	2.775e+004	1.0000	2.449e+006	1.0000	2834	1.0000
K VALUE						

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.3843	0.3843	---
Propene	20.43	20.43	---
Propane	20.30	20.30	---
Cumene	2.816e-002	2.816e-002	---
14-iP-BZ	4.616e-003	4.616e-003	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Separator: V-102	Cooler: E-103	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C2-Gas Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

	OVERALL	VAPOUR PH.	LIQUID PH.
Vapour / Phase Fraction	1.0000	1.0000	0.0000
Temperature: (C)	50.00	50.00	50.00
Pressure: (kPa)	100.0	100.0	100.0

Molar Flow (kgmole/h)	911.1	911.1	0.0000
Mass Flow (kg/h)	4.896e+004	4.896e+004	0.0000
Std Ideal Liq VolFlow (m3/h)	79.98	79.98	0.0000
Molar Enthalpy (kJ/kgmole)	-5.041e+04	-5.041e+04	2.359e+04
Molar Entropy (kJ/kgmole-C)	1.152e+02	1.152e+02	-9.321e+01
Heat Flow (kJ/h)	-4.592e+07	-4.592e+07	0.000e-01
Liq VolFlow @Std Cond (m3/h)	77.86	77.86	0.0000

COMPOSITION

Overall Phase	Vapour Fraction 1.0000
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	243.1	0.2668	1.898e+004	0.3878	21.52	0.2691	
Propene	6.927	0.0076	291.5	0.0060	0.5596	0.0070	
Propane	654.2	0.7180	2.885e+004	0.5893	56.93	0.7119	
Cumene	6.924	0.0076	832.2	0.0170	0.9624	0.0120	
14-iP-BZ	1.005e-003	0.0000	0.1631	0.0000	1.897e-004	0.0000	
Total	911.1	1.0000	4.896e+004	1.0000	79.98	1.0000	

Vapour Phase	Phase Fraction 1.000
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	243.1	0.2668	1.898e+004	0.3878	21.52	0.2691	
Propene	6.927	0.0076	291.5	0.0060	0.5596	0.0070	
Propane	654.2	0.7180	2.885e+004	0.5893	56.93	0.7119	

Cumene	6.924	0.0076	832.2	0.0170	0.9624	0.0120
14-iP-BZ	1.005e-003	0.0000	0.1631	0.0000	1.897e-004	0.0000
Total	911.1	1.0000	4.896e+004	1.0000	79.98	1.0000
Liquid Phase				Phase Fraction	0.0000	

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	0.0000	0.6942	0.0000	0.6143	0.0000	0.6019
Propene	0.0000	0.0004	0.0000	0.0002	0.0000	0.0003
Propane	0.0000	0.0354	0.0000	0.0177	0.0000	0.0301
Cumene	0.0000	0.2698	0.0000	0.3674	0.0000	0.3673
14-iP-BZ	0.0000	0.0002	0.0000	0.0004	0.0000	0.0004
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.3843	0.3843	---
Propene	20.43	20.43	---
Propane	20.30	20.30	---
Cumene	2.816e-002	2.816e-002	---
14-iP-BZ	4.616e-003	4.616e-003	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Separator: V-102		

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C2-Liquid

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

	OVERALL	VAPOUR PH.	LIQUID PH.
Vapour / Phase Fraction	0.0000	0.0000	1.0000
Temperature: (C)	50.00	50.00	50.00
Pressure: (kPa)	100.0	100.0	100.0
Molar Flow (kgmole/h)	2.775e+004	0.0000	2.775e+004
Mass Flow (kg/h)	2.449e+006	0.0000	2.449e+006
Std Ideal Liq VolFlow (m3/h)	2834	0.0000	2834
Molar Enthalpy (kJ/kgmole)	2.359e+04	-5.041e+04	2.359e+04
Molar Entropy (kJ/kgmole-C)	-9.321e+01	1.152e+02	-9.321e+01
Heat Flow (kJ/h)	6.547e+08	0.000e-01	6.547e+08
Liq VolFlow @Std Cond (m3/h)	2798	0.0000	2798

COMPOSITION

Overall Phase

Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.926e+004	0.6942	1.505e+006	0.6143	1705	0.6019	
Propene	10.33	0.0004	434.6	0.0002	0.8342	0.0003	
Propane	981.4	0.0354	4.328e+004	0.0177	85.41	0.0301	
Cumene	7487	0.2698	8.999e+005	0.3674	1041	0.3673	
14-iP-BZ	6.631	0.0002	1076	0.0004	1.252	0.0004	
Total	2.775e+004	1.0000	2.449e+006	1.0000	2834	1.0000	
Vapour Phase					Phase Fraction	0.0000	

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	0.0000	0.2668	0.0000	0.3878	0.0000	0.2691	
Propene	0.0000	0.0076	0.0000	0.0060	0.0000	0.0070	
Propane	0.0000	0.7180	0.0000	0.5893	0.0000	0.7119	
Cumene	0.0000	0.0076	0.0000	0.0170	0.0000	0.0120	
14-iP-BZ	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000	
Liquid Phase					Phase Fraction	1.000	

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.926e+004	0.6942	1.505e+006	0.6143	1705	0.6019	
Propene	10.33	0.0004	434.6	0.0002	0.8342	0.0003	
Propane	981.4	0.0354	4.328e+004	0.0177	85.41	0.0301	

Cumene	7487	0.2698	8.999e+005	0.3674	1041	0.3673
14-iP-BZ	6.631	0.0002	1076	0.0004	1.252	0.0004
Total	2.775e+004	1.0000	2.449e+006	1.0000	2834	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.3843	0.3843	---
Propene	20.43	20.43	---
Propane	20.30	20.30	---
Cumene	2.816e-002	2.816e-002	---
14-iP-BZ	4.616e-003	4.616e-003	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Pump: P-101	Separator: V-102	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C2-PumpOut	Fluid Package: Basis-1
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Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction 0.0000 1.0000
 Temperature: (C) 50.02 50.02
 Pressure: (kPa) 150.0 150.0
 Molar Flow (kgmole/h) 2.775e+004 2.775e+004
 Mass Flow (kg/h) 2.449e+006 2.449e+006
 Std Ideal Liq VolFlow (m3/h) 2834 2834
 Molar Enthalpy (kJ/kgmole) 2.360e+04 2.360e+04
 Molar Entropy (kJ/kgmole-C) -9.321e+01 -9.321e+01
 Heat Flow (kJ/h) 6.549e+08 6.549e+08
 Liq VolFlow @Std Cond (m3/h) 2798 2798

COMPOSITION

Overall Phase Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.926e+004	0.6942	1.505e+006	0.6143	1705	0.6019	
Propene	10.33	0.0004	434.6	0.0002	0.8342	0.0003	
Propane	981.4	0.0354	4.328e+004	0.0177	85.41	0.0301	
Cumene	7487	0.2698	8.999e+005	0.3674	1041	0.3673	
14-iP-BZ	6.631	0.0002	1076	0.0004	1.252	0.0004	
Total	2.775e+004	1.0000	2.449e+006	1.0000	2834	1.0000	
Liquid Phase							Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.926e+004	0.6942	1.505e+006	0.6143	1705	0.6019	
Propene	10.33	0.0004	434.6	0.0002	0.8342	0.0003	
Propane	981.4	0.0354	4.328e+004	0.0177	85.41	0.0301	
Cumene	7487	0.2698	8.999e+005	0.3674	1041	0.3673	
14-iP-BZ	6.631	0.0002	1076	0.0004	1.252	0.0004	
Total	2.775e+004	1.0000	2.449e+006	1.0000	2834	1.0000	

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Tee: TEE-100	Pump: P-101	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Reflux-New2

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	50.02	50.02
Pressure: (kPa)	150.0	150.0
Molar Flow (kgmole/h)	8324	8324
Mass Flow (kg/h)	7.348e+005	7.348e+005
Std Ideal Liq VolFlow (m3/h)	850.1	850.1
Molar Enthalpy (kJ/kgmole)	2.360e+04	2.360e+04
Molar Entropy (kJ/kgmole-C)	-9.321e+01	-9.321e+01
Heat Flow (kJ/h)	1.965e+08	1.965e+08
Liq VolFlow @Std Cond (m3/h)	839.5	839.5

COMPOSITION

Overall Phase	Vapour Fraction 0.0000
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)	
Benzene	5779	0.6942	4.514e+005	0.6143	511.6	0.6019
Propene	3.098	0.0004	130.4	0.0002	0.2503	0.0003

Propane	294.4	0.0354	1.298e+004	0.0177	25.62	0.0301
Cumene	2246	0.2698	2.700e+005	0.3674	312.2	0.3673
14-iP-BZ	1.989	0.0002	322.8	0.0004	0.3755	0.0004
Total	8324	1.0000	7.348e+005	1.0000	850.1	1.0000
Liquid Phase					Phase Fraction	1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	5779	0.6942	4.514e+005	0.6143	511.6	0.6019
Propene	3.098	0.0004	130.4	0.0002	0.2503	0.0003
Propane	294.4	0.0354	1.298e+004	0.0177	25.62	0.0301
Cumene	2246	0.2698	2.700e+005	0.3674	312.2	0.3673
14-iP-BZ	1.989	0.0002	322.8	0.0004	0.3755	0.0004
Total	8324	1.0000	7.348e+005	1.0000	850.1	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Recycle: RCY-6	Tee: TEE-100	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Benzene Ovrhd

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	50.02	50.02
Pressure: (kPa)	150.0	150.0
Molar Flow (kgmole/h)	1.942e+004	1.942e+004
Mass Flow (kg/h)	1.714e+006	1.714e+006
Std Ideal Liq VolFlow (m3/h)	1984	1984
Molar Enthalpy (kJ/kgmole)	2.360e+04	2.360e+04
Molar Entropy (kJ/kgmole-C)	-9.321e+01	-9.321e+01
Heat Flow (kJ/h)	4.584e+08	4.584e+08
Liq VolFlow @Std Cond (m3/h)	1959	1959

COMPOSITION

Overall Phase

Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.348e+004	0.6942	1.053e+006	0.6143	1194	0.6019
Propene	7.229	0.0004	304.2	0.0002	0.5839	0.0003
Propane	687.0	0.0354	3.029e+004	0.0177	59.79	0.0301
Cumene	5241	0.2698	6.299e+005	0.3674	728.5	0.3673
14-iP-BZ	4.642	0.0002	753.3	0.0004	0.8761	0.0004
Total	1.942e+004	1.0000	1.714e+006	1.0000	1984	1.0000

Liquid Phase Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.348e+004	0.6942	1.053e+006	0.6143	1194	0.6019
Propene	7.229	0.0004	304.2	0.0002	0.5839	0.0003
Propane	687.0	0.0354	3.029e+004	0.0177	59.79	0.0301
Cumene	5241	0.2698	6.299e+005	0.3674	728.5	0.3673
14-iP-BZ	4.642	0.0002	753.3	0.0004	0.8761	0.0004
Total	1.942e+004	1.0000	1.714e+006	1.0000	1984	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---

14-iP-BZ 0.0000 0.0000 ---

UNIT OPERATIONS

FEED TO PRODUCT FROM LOGICAL CONNECTION

Tee: TEE-102 Tee: TEE-100

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C2-Heated_Bot Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH. LIQUID PH.

Vapour / Phase Fraction 0.3526 0.3526 0.6474

Temperature: (C) 172.7 172.7 172.7

Pressure: (kPa) 120.0 120.0 120.0

Molar Flow (kgmole/h) 2.411e+004 8501 1.561e+004

Mass Flow (kg/h) 3.164e+006 1.065e+006 2.099e+006

Std Ideal Liq VolFlow (m3/h) 3666 1233 2433

Molar Enthalpy (kJ/kgmole) -1.167e+04 2.309e+04 -3.061e+04

Molar Entropy (kJ/kgmole-C) 8.221e+01 1.339e+02 5.406e+01

Liq VolFlow @Std Cond (m3/h)	3653	1229	2424
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COMPOSITION

Overall Phase

Vapour Fraction 0.3526

COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)		(kg/h)		(m3/h)	
Benzene	5.063e-002	0.0000	3.954	0.0000	4.483e-003	0.0000
Propene	2.173e-022	0.0000	9.143e-021	0.0000	1.755e-023	0.0000
Propane	9.538e-020	0.0000	4.206e-018	0.0000	8.301e-021	0.0000
Cumene	1.778e+004	0.7375	2.137e+006	0.6754	2471	0.6742
14-iP-BZ	6328	0.2625	1.027e+006	0.3246	1194	0.3258
Total	2.411e+004	1.0000	3.164e+006	1.0000	3666	1.0000

Vapour Phase

Phase Fraction 0.3526

COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)		(kg/h)		(m3/h)	
Benzene	3.988e-002	0.0000	3.115	0.0000	3.531e-003	0.0000
Propene	2.112e-022	0.0000	8.888e-021	0.0000	1.706e-023	0.0000
Propane	9.247e-020	0.0000	4.078e-018	0.0000	8.048e-021	0.0000
Cumene	7474	0.8792	8.983e+005	0.8435	1039	0.8428
14-iP-BZ	1027	0.1208	1.666e+005	0.1565	193.8	0.1572
Total	8501	1.0000	1.065e+006	1.0000	1233	1.0000

Liquid Phase

Phase Fraction 0.6474

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.075e-002	0.0000	0.8396	0.0000	9.518e-004	0.0000	
Propene	6.078e-024	0.0000	2.558e-022	0.0000	4.910e-025	0.0000	
Propane	2.909e-021	0.0000	1.283e-019	0.0000	2.532e-022	0.0000	
Cumene	1.031e+004	0.6603	1.239e+006	0.5901	1432	0.5887	
14-iP-BZ	5302	0.3397	8.603e+005	0.4099	1001	0.4113	
Total	1.561e+004	1.0000	2.099e+006	1.0000	2433	1.0000	

K VALUE

COMPONENTS		MIXED	LIGHT	HEAVY
Benzene	6.811	6.811	---	
Propene	63.80	63.80	---	
Propane	58.37	58.37	---	
Cumene	1.331	1.331	---	
14-iP-BZ	0.3556	0.3556	---	

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Separator: V-103	Heater: E-104	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Boilup-New2

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

	OVERALL	VAPOUR PH.	LIQUID PH.
Vapour / Phase Fraction	1.0000	1.0000	0.0000
Temperature: (C)	172.7	172.7	172.7
Pressure: (kPa)	120.0	120.0	120.0
Molar Flow (kgmole/h)	8500	8500	0.0000
Mass Flow (kg/h)	1.065e+006	1.065e+006	0.0000
Std Ideal Liq VolFlow (m3/h)	1232	1232	0.0000
Molar Enthalpy (kJ/kgmole)	2.309e+04	2.309e+04	-3.061e+04
Molar Entropy (kJ/kgmole-C)	1.339e+02	1.339e+02	5.406e+01
Heat Flow (kJ/h)	1.963e+08	1.963e+08	0.000e-01
Liq VolFlow @Std Cond (m3/h)	1229	1229	0.0000

COMPOSITION

Overall Phase	Vapour Fraction 1.0000
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COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	3.988e-002	0.0000	3.115	0.0000	3.531e-003	0.0000
Propene	2.113e-022	0.0000	8.890e-021	0.0000	1.706e-023	0.0000

Propane	9.249e-020	0.0000	4.079e-018	0.0000	8.050e-021	0.0000
Cumene	7473	0.8792	8.983e+005	0.8436	1039	0.8428
14-iP-BZ	1027	0.1208	1.666e+005	0.1564	193.8	0.1572
Total	8500	1.0000	1.065e+006	1.0000	1232	1.0000
Vapour Phase	Phase Fraction 1.000					

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)	
Benzene	3.988e-002	0.0000	3.115	0.0000	3.531e-003	0.0000
Propene	2.113e-022	0.0000	8.890e-021	0.0000	1.706e-023	0.0000
Propane	9.249e-020	0.0000	4.079e-018	0.0000	8.050e-021	0.0000
Cumene	7473	0.8792	8.983e+005	0.8436	1039	0.8428
14-iP-BZ	1027	0.1208	1.666e+005	0.1564	193.8	0.1572
Total	8500	1.0000	1.065e+006	1.0000	1232	1.0000
Liquid Phase	Phase Fraction 0.0000					

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)	
Benzene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Cumene	0.0000	0.6603	0.0000	0.5902	0.0000	0.5888
14-iP-BZ	0.0000	0.3397	0.0000	0.4098	0.0000	0.4112
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	6.811	6.811	---
Propene	63.80	63.80	---
Propane	58.37	58.37	---
Cumene	1.331	1.331	---
14-iP-BZ	0.3556	0.3556	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Recycle: RCY-5	Separator: V-103	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C2-Bottoms Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

	OVERALL	VAPOUR PH.	LIQUID PH.
Vapour / Phase Fraction	0.0000	0.0000	1.0000
Temperature: (C)	172.7	172.7	172.7
Pressure: (kPa)	120.0	120.0	120.0

Cumene	0.0000	0.8792	0.0000	0.8436	0.0000	0.8428
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14-iP-BZ	0.0000	0.1208	0.0000	0.1564	0.0000	0.1572
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Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000
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Liquid Phase					Phase Fraction	1.000
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COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)		(kg/h)		(m3/h)	
Benzene	1.075e-002	0.0000	0.8399	0.0000	9.520e-004	0.0000
Propene	6.081e-024	0.0000	2.559e-022	0.0000	4.912e-025	0.0000
Propane	2.910e-021	0.0000	1.283e-019	0.0000	2.532e-022	0.0000
Cumene	1.031e+004	0.6603	1.239e+006	0.5902	1433	0.5888
14-iP-BZ	5301	0.3397	8.603e+005	0.4098	1001	0.4112
Total	1.561e+004	1.0000	2.099e+006	1.0000	2433	1.0000

K VALUE

COMPONENTS		MIXED	LIGHT	HEAVY
Benzene	6.811	6.811	---	
Propene	63.80	63.80	---	
Propane	58.37	58.37	---	
Cumene	1.331	1.331	---	
14-iP-BZ	0.3556	0.3556	---	

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Absorber: T-102	Separator: V-103	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Reflux-3

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	50.02	50.02
Pressure: (kPa)	150.0	150.0
Molar Flow (kgmole/h)	1375	1375
Mass Flow (kg/h)	1.655e+005	1.655e+005
Std Ideal Liq VolFlow (m3/h)	191.3	191.3
Molar Enthalpy (kJ/kgmole)	-3.516e+04	-3.516e+04
Molar Entropy (kJ/kgmole-C)	-3.253e+01	-3.253e+01
Heat Flow (kJ/h)	-4.833e+07	-4.833e+07
Liq VolFlow @Std Cond (m3/h)	191.0	191.0

COMPOSITION

Overall Phase

Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	4.613e-003	0.0000	0.3603	0.0000	4.084e-004	0.0000	
Propene	2.613e-024	0.0000	1.099e-022	0.0000	2.110e-025	0.0000	
Propane	1.252e-021	0.0000	5.523e-020	0.0000	1.090e-022	0.0000	
Cumene	1369	0.9960	1.646e+005	0.9946	190.3	0.9945	
14-iP-BZ	5.550	0.0040	900.6	0.0054	1.048	0.0055	
Total	1375	1.0000	1.655e+005	1.0000	191.3	1.0000	
Liquid Phase					Phase Fraction	1.000	

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	4.613e-003	0.0000	0.3603	0.0000	4.084e-004	0.0000	
Propene	2.613e-024	0.0000	1.099e-022	0.0000	2.110e-025	0.0000	
Propane	1.252e-021	0.0000	5.523e-020	0.0000	1.090e-022	0.0000	
Cumene	1369	0.9960	1.646e+005	0.9946	190.3	0.9945	
14-iP-BZ	5.550	0.0040	900.6	0.0054	1.048	0.0055	
Total	1375	1.0000	1.655e+005	1.0000	191.3	1.0000	

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Absorber: T-102	Recycle: RCY-8	
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UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Boilup-3

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH.

Vapour / Phase Fraction	1.0000	1.0000
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Temperature: (C)	176.6	176.6
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Pressure: (kPa)	120.0	120.0
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Molar Flow (kgmole/h)	5000	5000
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Mass Flow (kg/h)	6.362e+005	6.362e+005
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Std Ideal Liq VolFlow (m3/h)	736.6	736.6
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Molar Enthalpy (kJ/kgmole)	2.082e+04	2.082e+04
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Molar Entropy (kJ/kgmole-C)	1.374e+02	1.374e+02
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Heat Flow (kJ/h)	1.041e+08	1.041e+08
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Liq VolFlow @Std Cond (m3/h) 734.3 734.3

COMPOSITION

Overall Phase

Vapour Fraction 1.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	2.395e-005	0.0000	1.871e-003	0.0000	2.120e-006	0.0000	
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
Propane	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
Cumene	4163	0.8325	5.003e+005	0.7864	578.6	0.7854	
14-iP-BZ	837.4	0.1675	1.359e+005	0.2136	158.1	0.2146	
Total	5000	1.0000	6.362e+005	1.0000	736.6	1.0000	

Vapour Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	2.395e-005	0.0000	1.871e-003	0.0000	2.120e-006	0.0000	
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
Propane	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
Cumene	4163	0.8325	5.003e+005	0.7864	578.6	0.7854	
14-iP-BZ	837.4	0.1675	1.359e+005	0.2136	158.1	0.2146	
Total	5000	1.0000	6.362e+005	1.0000	736.6	1.0000	

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
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Benzene	---	---	---
Propene	---	---	---
Propane	---	---	---
Cumene	---	---	---
14-iP-BZ	---	---	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Absorber: T-102	Recycle: RCY-7	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C3-Top Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH.

Vapour / Phase Fraction	1.0000	1.0000
Temperature: (C)	153.5	153.5
Pressure: (kPa)	100.0	100.0
Molar Flow (kgmole/h)	4594	4594

Mass Flow (kg/h) 5.528e+005 5.528e+005

Std Ideal Liq VolFlow (m3/h) 639.3 639.3

Molar Enthalpy (kJ/kgmole) 2.703e+04 2.703e+04

Molar Entropy (kJ/kgmole-C) 1.213e+02 1.213e+02

Heat Flow (kJ/h) 1.242e+08 1.242e+08

Liq VolFlow @Std Cond (m3/h) 638.1 638.1

COMPOSITION

Overall Phase Vapour Fraction 1.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.537e-002	0.0000	1.201	0.0000	1.361e-003	0.0000	
Propene	8.681e-024	0.0000	3.653e-022	0.0000	7.012e-025	0.0000	
Propane	4.162e-021	0.0000	1.835e-019	0.0000	3.622e-022	0.0000	
Cumene	4577	0.9963	5.501e+005	0.9950	636.1	0.9950	
14-iP-BZ	16.87	0.0037	2737	0.0050	3.184	0.0050	
Total	4594	1.0000	5.528e+005	1.0000	639.3	1.0000	

Vapour Phase Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.537e-002	0.0000	1.201	0.0000	1.361e-003	0.0000	
Propene	8.681e-024	0.0000	3.653e-022	0.0000	7.012e-025	0.0000	
Propane	4.162e-021	0.0000	1.835e-019	0.0000	3.622e-022	0.0000	
Cumene	4577	0.9963	5.501e+005	0.9950	636.1	0.9950	

14-iP-BZ	16.87	0.0037	2737	0.0050	3.184	0.0050
Total	4594	1.0000	5.528e+005	1.0000	639.3	1.0000

K VALUE

COMPONENTS		MIXED	LIGHT	HEAVY
Benzene	---	---	---	
Propene	---	---	---	
Propane	---	---	---	
Cumene	---	---	---	
14-iP-BZ	---	---	---	

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Cooler: E-105	Absorber: T-102	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C3-Bot

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction 0.0000 1.0000

Temperature: (C) 173.2 173.2

Pressure: (kPa) 120.0 120.0

Molar Flow (kgmole/h) 1.739e+004 1.739e+004

Mass Flow (kg/h) 2.348e+006 2.348e+006

Std Ideal Liq VolFlow (m3/h) 2722 2722

Molar Enthalpy (kJ/kgmole) -3.140e+04 -3.140e+04

Molar Entropy (kJ/kgmole-C) 5.463e+01 5.463e+01

Heat Flow (kJ/h) -5.461e+08 -5.461e+08

Liq VolFlow @Std Cond (m3/h) 2712 2712

COMPOSITION

Overall Phase Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.566e-005	0.0000	1.223e-003	0.0000	1.386e-006	0.0000	
Propene	1.209e-026	0.0000	5.088e-025	0.0000	9.766e-028	0.0000	
Propane	1.236e-026	0.0000	5.451e-025	0.0000	1.076e-027	0.0000	
Cumene	1.126e+004	0.6476	1.354e+006	0.5765	1565	0.5751	
14-iP-BZ	6127	0.3524	9.943e+005	0.4235	1156	0.4249	
Total	1.739e+004	1.0000	2.348e+006	1.0000	2722	1.0000	

Liquid Phase Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.566e-005	0.0000	1.223e-003	0.0000	1.386e-006	0.0000
Propene	1.209e-026	0.0000	5.088e-025	0.0000	9.766e-028	0.0000
Propane	1.236e-026	0.0000	5.451e-025	0.0000	1.076e-027	0.0000
Cumene	1.126e+004	0.6476	1.354e+006	0.5765	1565	0.5751
14-iP-BZ	6127	0.3524	9.943e+005	0.4235	1156	0.4249
Total	1.739e+004	1.0000	2.348e+006	1.0000	2722	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Heater: E-106	Absorber: T-102	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C3-Cooled

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	50.00	50.00
Pressure: (kPa)	100.0	100.0
Molar Flow (kgmole/h)	4594	4594
Mass Flow (kg/h)	5.528e+005	5.528e+005
Std Ideal Liq VolFlow (m3/h)	639.3	639.3
Molar Enthalpy (kJ/kgmole)	-3.513e+04	-3.513e+04
Molar Entropy (kJ/kgmole-C)	-3.255e+01	-3.255e+01
Heat Flow (kJ/h)	-1.614e+08	-1.614e+08
Liq VolFlow @Std Cond (m3/h)	638.1	638.1

COMPOSITION

Overall Phase	Vapour Fraction 0.0000
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COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.537e-002	0.0000	1.201	0.0000	1.361e-003	0.0000
Propene	8.681e-024	0.0000	3.653e-022	0.0000	7.012e-025	0.0000
Propane	4.162e-021	0.0000	1.835e-019	0.0000	3.622e-022	0.0000
Cumene	4577	0.9963	5.501e+005	0.9950	636.1	0.9950

14-iP-BZ	16.87	0.0037	2737	0.0050	3.184	0.0050
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Total	4594	1.0000	5.528e+005	1.0000	639.3	1.0000
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Liquid Phase	Phase Fraction	1.000
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COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)		(kg/h)		(m3/h)	
Benzene	1.537e-002	0.0000	1.201	0.0000	1.361e-003	0.0000
Propene	8.681e-024	0.0000	3.653e-022	0.0000	7.012e-025	0.0000
Propane	4.162e-021	0.0000	1.835e-019	0.0000	3.622e-022	0.0000
Cumene	4577	0.9963	5.501e+005	0.9950	636.1	0.9950
14-iP-BZ	16.87	0.0037	2737	0.0050	3.184	0.0050
Total	4594	1.0000	5.528e+005	1.0000	639.3	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Separator: V-104	Cooler: E-105	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C3-Gas

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

	OVERALL	LIQUID PH.	VAPOUR PH.
Vapour / Phase Fraction	1.0000	0.0000	1.0000
Temperature: (C)	50.00	50.00	50.00
Pressure: (kPa)	100.0	100.0	100.0
Molar Flow (kgmole/h)	0.0000	0.0000	0.0000
Mass Flow (kg/h)	0.0000	0.0000	0.0000
Std Ideal Liq VolFlow (m3/h)	0.0000	0.0000	0.0000
Molar Enthalpy (kJ/kgmole)	7.288e+03	-3.513e+04	7.288e+03
Molar Entropy (kJ/kgmole-C)	6.782e+01	-3.255e+01	6.782e+01
Heat Flow (kJ/h)	0.000e-01	0.000e-01	0.000e-01
Liq VolFlow @Std Cond (m3/h)	0.0000	0.0000	0.0000

COMPOSITION

Overall Phase

Vapour Fraction 1.0000

Cumene	0.0000	0.9994	0.0000	0.9992	0.0000	0.9992
14-iP-BZ	0.0000	0.0006	0.0000	0.0008	0.0000	0.0008
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000

K VALUE

COMPONENTS		MIXED	LIGHT	HEAVY
Benzene	11.64	11.64	---	
Propene	29.51	29.51	---	
Propane	29.51	29.51	---	
Cumene	1.003	1.003	---	
14-iP-BZ	0.1539	0.1539	---	

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
	Separator: V-104	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C3-Liquid	Fluid Package: Basis-1
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Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH. VAPOUR PH.

Vapour / Phase Fraction	0.0000	1.0000	0.0000
Temperature: (C)	50.00	50.00	50.00
Pressure: (kPa)	100.0	100.0	100.0
Molar Flow (kgmole/h)	4594	4594	0.0000
Mass Flow (kg/h)	5.528e+005	5.528e+005	0.0000
Std Ideal Liq VolFlow (m3/h)	639.3	639.3	0.0000
Molar Enthalpy (kJ/kgmole)	-3.513e+04	-3.513e+04	7.288e+03
Molar Entropy (kJ/kgmole-C)	-3.255e+01	-3.255e+01	6.782e+01
Heat Flow (kJ/h)	-1.614e+08	-1.614e+08	0.000e-01
Liq VolFlow @Std Cond (m3/h)	638.1	638.1	0.0000

COMPOSITION

Overall Phase	Vapour Fraction 0.0000
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.537e-002	0.0000	1.201	0.0000	1.361e-003	0.0000
Propene	8.681e-024	0.0000	3.653e-022	0.0000	7.012e-025	0.0000
Propane	4.162e-021	0.0000	1.835e-019	0.0000	3.622e-022	0.0000
Cumene	4577	0.9963	5.501e+005	0.9950	636.1	0.9950
14-iP-BZ	16.87	0.0037	2737	0.0050	3.184	0.0050
Total	4594	1.0000	5.528e+005	1.0000	639.3	1.0000
Liquid Phase				Phase Fraction 1.000		

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.537e-002	0.0000	1.201	0.0000	1.361e-003	0.0000	
Propene	8.681e-024	0.0000	3.653e-022	0.0000	7.012e-025	0.0000	
Propane	4.162e-021	0.0000	1.835e-019	0.0000	3.622e-022	0.0000	
Cumene	4577	0.9963	5.501e+005	0.9950	636.1	0.9950	
14-iP-BZ	16.87	0.0037	2737	0.0050	3.184	0.0050	
Total	4594	1.0000	5.528e+005	1.0000	639.3	1.0000	
Vapour Phase					Phase Fraction	0.0000	

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
Propane	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
Cumene	0.0000	0.9994	0.0000	0.9992	0.0000	0.9992	
14-iP-BZ	0.0000	0.0006	0.0000	0.0008	0.0000	0.0008	
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000	

K VALUE

COMPONENTS		MIXED	LIGHT	HEAVY
Benzene	11.64	11.64	---	
Propene	29.51	29.51	---	
Propane	29.51	29.51	---	
Cumene	1.003	1.003	---	
14-iP-BZ	0.1539	0.1539	---	

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Pump: P-102	Separator: V-104	
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UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C3-PumpOut

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
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Temperature: (C)	50.02	50.02
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Pressure: (kPa)	150.0	150.0
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Molar Flow (kgmole/h)	4594	4594
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Mass Flow (kg/h)	5.528e+005	5.528e+005
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Std Ideal Liq VolFlow (m3/h)	639.3	639.3
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Molar Enthalpy (kJ/kgmole)	-3.512e+04	-3.512e+04
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Molar Entropy (kJ/kgmole-C)	-3.254e+01	-3.254e+01
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Heat Flow (kJ/h)	-1.613e+08	-1.613e+08
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Liq VolFlow @Std Cond (m3/h) 638.1 638.1

COMPOSITION

Overall Phase

Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.537e-002	0.0000	1.201	0.0000	1.361e-003	0.0000	
Propene	8.681e-024	0.0000	3.653e-022	0.0000	7.012e-025	0.0000	
Propane	4.162e-021	0.0000	1.835e-019	0.0000	3.622e-022	0.0000	
Cumene	4577	0.9963	5.501e+005	0.9950	636.1	0.9950	
14-iP-BZ	16.87	0.0037	2737	0.0050	3.184	0.0050	
Total	4594	1.0000	5.528e+005	1.0000	639.3	1.0000	

Liquid Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.537e-002	0.0000	1.201	0.0000	1.361e-003	0.0000	
Propene	8.681e-024	0.0000	3.653e-022	0.0000	7.012e-025	0.0000	
Propane	4.162e-021	0.0000	1.835e-019	0.0000	3.622e-022	0.0000	
Cumene	4577	0.9963	5.501e+005	0.9950	636.1	0.9950	
14-iP-BZ	16.87	0.0037	2737	0.0050	3.184	0.0050	
Total	4594	1.0000	5.528e+005	1.0000	639.3	1.0000	

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
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Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Tee: TEE-101	Pump: P-102	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Cumene Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	50.02	50.02
Pressure: (kPa)	150.0	150.0
Molar Flow (kgmole/h)	3216	3216

Mass Flow (kg/h) 3.870e+005 3.870e+005
 Std Ideal Liq VolFlow (m3/h) 447.5 447.5
 Molar Enthalpy (kJ/kgmole) -3.512e+04 -3.512e+04
 Molar Entropy (kJ/kgmole-C) -3.254e+01 -3.254e+01
 Heat Flow (kJ/h) -1.129e+08 -1.129e+08
 Liq VolFlow @Std Cond (m3/h) 446.6 446.6

COMPOSITION

Overall Phase Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
 LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.076e-002	0.0000	0.8406	0.0000	9.528e-004	0.0000	
Propene	6.077e-024	0.0000	2.557e-022	0.0000	4.909e-025	0.0000	
Propane	2.914e-021	0.0000	1.285e-019	0.0000	2.536e-022	0.0000	
Cumene	3204	0.9963	3.851e+005	0.9950	445.3	0.9950	
14-iP-BZ	11.81	0.0037	1916	0.0050	2.229	0.0050	
Total	3216	1.0000	3.870e+005	1.0000	447.5	1.0000	

Liquid Phase Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
 LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.076e-002	0.0000	0.8406	0.0000	9.528e-004	0.0000	
Propene	6.077e-024	0.0000	2.557e-022	0.0000	4.909e-025	0.0000	
Propane	2.914e-021	0.0000	1.285e-019	0.0000	2.536e-022	0.0000	
Cumene	3204	0.9963	3.851e+005	0.9950	445.3	0.9950	

14-iP-BZ	11.81	0.0037	1916	0.0050	2.229	0.0050
Total	3216	1.0000	3.870e+005	1.0000	447.5	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
	Tee: TEE-101	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C3-Heated_Bot	Fluid Package: Basis-1
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Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH. LIQUID PH.

Vapour / Phase Fraction	0.2876	0.2876	0.7124
Temperature: (C)	176.6	176.6	176.6
Pressure: (kPa)	120.0	120.0	120.0
Molar Flow (kgmole/h)	1.739e+004	5000	1.239e+004
Mass Flow (kg/h)	2.348e+006	6.363e+005	1.712e+006
Std Ideal Liq VolFlow (m3/h)	2722	736.7	1985
Molar Enthalpy (kJ/kgmole)	-1.967e+04	2.082e+04	-3.601e+04
Molar Entropy (kJ/kgmole-C)	8.083e+01	1.374e+02	5.801e+01
Heat Flow (kJ/h)	-3.420e+08	1.041e+08	-4.461e+08
Liq VolFlow @Std Cond (m3/h)	2712	734.4	1978

COMPOSITION

Overall Phase	Vapour Fraction 0.2876
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.566e-005	0.0000	1.223e-003	0.0000	1.386e-006	0.0000
Propene	1.209e-026	0.0000	5.088e-025	0.0000	9.766e-028	0.0000
Propane	1.236e-026	0.0000	5.451e-025	0.0000	1.076e-027	0.0000
Cumene	1.126e+004	0.6476	1.354e+006	0.5765	1565	0.5751
14-iP-BZ	6127	0.3524	9.943e+005	0.4235	1156	0.4249
Total	1.739e+004	1.0000	2.348e+006	1.0000	2722	1.0000

Vapour Phase	Phase Fraction 0.2876
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.168e-005	0.0000	9.120e-004	0.0000	1.034e-006	0.0000
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Cumene	4162	0.8324	5.003e+005	0.7863	578.6	0.7853
14-iP-BZ	837.8	0.1676	1.360e+005	0.2137	158.1	0.2147
Total	5000	1.0000	6.363e+005	1.0000	736.7	1.0000
Liquid Phase					Phase Fraction	0.7124

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	3.980e-006	0.0000	3.109e-004	0.0000	3.524e-007	0.0000
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Cumene	7099	0.5730	8.533e+005	0.4985	986.7	0.4971
14-iP-BZ	5290	0.4270	8.584e+005	0.5015	998.4	0.5029
Total	1.239e+004	1.0000	1.712e+006	1.0000	1985	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	7.269	7.269	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	1.453	1.453	---
14-iP-BZ	0.3924	0.3924	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Separator: V-105	Heater: E-106	
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UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Boilup-New3

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL	VAPOUR PH.	LIQUID PH.
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Vapour / Phase Fraction	1.0000	1.0000	0.0000
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Temperature: (C)	176.6	176.6	176.6
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Pressure: (kPa)	120.0	120.0	120.0
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Molar Flow (kgmole/h)	5000	5000	0.0000
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Mass Flow (kg/h)	6.362e+005	6.362e+005	0.0000
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Std Ideal Liq VolFlow (m3/h)	736.6	736.6	0.0000
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Molar Enthalpy (kJ/kgmole)	2.082e+04	2.082e+04	-3.601e+04
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Molar Entropy (kJ/kgmole-C)	1.374e+02	1.374e+02	5.801e+01
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Heat Flow (kJ/h)	1.041e+08	1.041e+08	0.000e-01
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Liq VolFlow @Std Cond (m3/h)	734.3	734.3	0.0000
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COMPOSITION

Overall Phase

Vapour Fraction 1.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.168e-005	0.0000	9.121e-004	0.0000	1.034e-006	0.0000	
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
Propane	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
Cumene	4162	0.8325	5.003e+005	0.7863	578.5	0.7854	
14-iP-BZ	837.7	0.1675	1.359e+005	0.2137	158.1	0.2146	
Total	5000	1.0000	6.362e+005	1.0000	736.6	1.0000	

Vapour Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.168e-005	0.0000	9.121e-004	0.0000	1.034e-006	0.0000	
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
Propane	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	
Cumene	4162	0.8325	5.003e+005	0.7863	578.5	0.7854	
14-iP-BZ	837.7	0.1675	1.359e+005	0.2137	158.1	0.2146	
Total	5000	1.0000	6.362e+005	1.0000	736.6	1.0000	

Liquid Phase

Phase Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Cumene	0.0000	0.5731	0.0000	0.4985	0.0000	0.4971
14-iP-BZ	0.0000	0.4269	0.0000	0.5015	0.0000	0.5029
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	7.269	7.269	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	1.453	1.453	---
14-iP-BZ	0.3924	0.3924	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Recycle: RCY-7	Separator: V-105	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: DIPB_To_R2

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH. LIQUID PH.

Vapour / Phase Fraction	0.0000	0.0000	1.0000
Temperature: (C)	176.6	176.6	176.6
Pressure: (kPa)	120.0	120.0	120.0
Molar Flow (kgmole/h)	1.239e+004	0.0000	1.239e+004
Mass Flow (kg/h)	1.712e+006	0.0000	1.712e+006
Std Ideal Liq VolFlow (m3/h) 1985	0.0000	1985	
Molar Enthalpy (kJ/kgmole)	-3.601e+04	2.082e+04	-3.601e+04
Molar Entropy (kJ/kgmole-C)	5.801e+01	1.374e+02	5.801e+01
Heat Flow (kJ/h)	-4.461e+08	0.000e-01	-4.461e+08
Liq VolFlow @Std Cond (m3/h) 1978	0.0000	1978	

COMPOSITION

Overall Phase	Vapour Fraction 0.0000
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	3.981e-006	0.0000	3.109e-004	0.0000	3.524e-007	0.0000
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Cumene	7100	0.5731	8.533e+005	0.4985	986.8	0.4971

14-iP-BZ	5289	0.4269	8.583e+005	0.5015	998.3	0.5029
Total	1.239e+004	1.0000	1.712e+006	1.0000	1985	1.0000
Vapour Phase			Phase Fraction 0.0000			

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Cumene	0.0000	0.8325	0.0000	0.7863	0.0000	0.7854
14-iP-BZ	0.0000	0.1675	0.0000	0.2137	0.0000	0.2146
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000
Liquid Phase			Phase Fraction 1.000			

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	3.981e-006	0.0000	3.109e-004	0.0000	3.524e-007	0.0000
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Cumene	7100	0.5731	8.533e+005	0.4985	986.8	0.4971
14-iP-BZ	5289	0.4269	8.583e+005	0.5015	998.3	0.5029
Total	1.239e+004	1.0000	1.712e+006	1.0000	1985	1.0000
K VALUE						

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	7.269	7.269	---

Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	1.453	1.453	---
14-iP-BZ	0.3924	0.3924	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Mixer: MIX-101	Separator: V-105	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Trans_Mixed Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

	OVERALL	VAPOUR PH.	LIQUID PH.
Vapour / Phase Fraction	0.1226	0.1226	0.8774
Temperature: (C)	123.5	123.5	123.5
Pressure: (kPa)	120.0	120.0	120.0
Molar Flow (kgmole/h)	2.210e+004	2709	1.939e+004
Mass Flow (kg/h)	2.569e+006	2.269e+005	2.342e+006

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1870	0.6904	1.461e+005	0.6438	165.6	0.6139
Propene	3.104	0.0011	130.6	0.0006	0.2508	0.0009
Propane	291.4	0.1076	1.285e+004	0.0566	25.36	0.0940
Cumene	485.4	0.1792	5.835e+004	0.2571	67.47	0.2502
14-iP-BZ	58.56	0.0216	9502	0.0419	11.05	0.0410

Total 2709 1.0000 2.269e+005 1.0000 269.7 1.0000

Liquid Phase

Phase Fraction 0.8774

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)	
Benzene	4871	0.2512	3.805e+005	0.1625	431.3	0.1593
Propene	0.5102	0.0000	21.47	0.0000	4.121e-002	0.0000
Propane	52.11	0.0027	2298	0.0010	4.535	0.0017
Cumene	9235	0.4762	1.110e+006	0.4739	1284	0.4741
14-iP-BZ	5233	0.2699	8.492e+005	0.3626	987.7	0.3649
Total	1.939e+004	1.0000	2.342e+006	1.0000	2707	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	2.748	2.748	---
Propene	43.56	43.56	---
Propane	40.03	40.03	---
Cumene	0.3763	0.3763	---
14-iP-BZ	8.011e-002	8.011e-002	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Cooler: E-107	Mixer: MIX-101	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: P6-Out

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	81.00	81.00
Pressure: (kPa)	2500	2500
Molar Flow (kgmole/h)	2.210e+004	2.210e+004
Mass Flow (kg/h)	2.569e+006	2.569e+006
Std Ideal Liq VolFlow (m3/h)	2977	2977
Molar Enthalpy (kJ/kgmole)	-2.285e+04	-2.285e+04
Molar Entropy (kJ/kgmole-C)	-3.738e+01	-3.738e+01
Heat Flow (kJ/h)	-5.049e+08	-5.049e+08
Liq VolFlow @Std Cond (m3/h)	2930	2930

COMPOSITION

Overall Phase

Vapour Fraction 0.0000

COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	6742	0.3050	5.266e+005	0.2050	596.9	0.2005
Propene	3.614	0.0002	152.1	0.0001	0.2920	0.0001
Propane	343.5	0.0155	1.515e+004	0.0059	29.89	0.0100
Cumene	9720	0.4398	1.168e+006	0.4548	1351	0.4538
14-iP-BZ	5292	0.2394	8.587e+005	0.3343	998.8	0.3355
Total	2.210e+004	1.0000	2.569e+006	1.0000	2977	1.0000
Liquid Phase					Phase Fraction	1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	6742	0.3050	5.266e+005	0.2050	596.9	0.2005
Propene	3.614	0.0002	152.1	0.0001	0.2920	0.0001
Propane	343.5	0.0155	1.515e+004	0.0059	29.89	0.0100
Cumene	9720	0.4398	1.168e+006	0.4548	1351	0.4538
14-iP-BZ	5292	0.2394	8.587e+005	0.3343	998.8	0.3355
Total	2.210e+004	1.0000	2.569e+006	1.0000	2977	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Heater: E-108	Pump: P-103	
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UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Cooled-Trans_Mixed	Fluid Package: Basis-1
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Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	80.00	80.00
Pressure: (kPa)	120.0	120.0
Molar Flow (kgmole/h)	2.210e+004	2.210e+004
Mass Flow (kg/h)	2.569e+006	2.569e+006
Std Ideal Liq VolFlow (m3/h)	2977	2977
Molar Enthalpy (kJ/kgmole)	-2.330e+04	-2.330e+04
Molar Entropy (kJ/kgmole-C)	-3.767e+01	-3.767e+01
Heat Flow (kJ/h)	-5.149e+08	-5.149e+08
Liq VolFlow @Std Cond (m3/h)	2930	2930

COMPOSITION

Overall Phase

Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	6742	0.3050	5.266e+005	0.2050	596.9	0.2005
Propene	3.614	0.0002	152.1	0.0001	0.2920	0.0001
Propane	343.5	0.0155	1.515e+004	0.0059	29.89	0.0100
Cumene	9720	0.4398	1.168e+006	0.4548	1351	0.4538
14-iP-BZ	5292	0.2394	8.587e+005	0.3343	998.8	0.3355
Total	2.210e+004	1.0000	2.569e+006	1.0000	2977	1.0000

Liquid Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	6742	0.3050	5.266e+005	0.2050	596.9	0.2005
Propene	3.614	0.0002	152.1	0.0001	0.2920	0.0001
Propane	343.5	0.0155	1.515e+004	0.0059	29.89	0.0100
Cumene	9720	0.4398	1.168e+006	0.4548	1351	0.4538
14-iP-BZ	5292	0.2394	8.587e+005	0.3343	998.8	0.3355
Total	2.210e+004	1.0000	2.569e+006	1.0000	2977	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---

Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Pump: P-103	Cooler: E-107	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Trans Feed	Fluid Package: Basis-1
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Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	210.0	210.0
Pressure: (kPa)	2500	2500
Molar Flow (kgmole/h)	2.210e+004	2.210e+004
Mass Flow (kg/h)	2.569e+006	2.569e+006

Std Ideal Liq VolFlow (m3/h) 2977 2977

Molar Enthalpy (kJ/kgmole) 9.434e+03 9.434e+03

Molar Entropy (kJ/kgmole-C) 3.983e+01 3.983e+01

Heat Flow (kJ/h) 2.085e+08 2.085e+08

Liq VolFlow @Std Cond (m3/h) 2930 2930

COMPOSITION

Overall Phase Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	6742	0.3050	5.266e+005	0.2050	596.9	0.2005
Propene	3.614	0.0002	152.1	0.0001	0.2920	0.0001
Propane	343.5	0.0155	1.515e+004	0.0059	29.89	0.0100
Cumene	9720	0.4398	1.168e+006	0.4548	1351	0.4538
14-iP-BZ	5292	0.2394	8.587e+005	0.3343	998.8	0.3355
Total	2.210e+004	1.0000	2.569e+006	1.0000	2977	1.0000

Liquid Phase Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	6742	0.3050	5.266e+005	0.2050	596.9	0.2005
Propene	3.614	0.0002	152.1	0.0001	0.2920	0.0001
Propane	343.5	0.0155	1.515e+004	0.0059	29.89	0.0100
Cumene	9720	0.4398	1.168e+006	0.4548	1351	0.4538
14-iP-BZ	5292	0.2394	8.587e+005	0.3343	998.8	0.3355

Total 2.210e+004 1.0000 2.569e+006 1.0000 2977 1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Conversion Reactor: CRV-100 Heater: E-108

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Trans_Vap

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH. VAPOUR PH.

Vapour / Phase Fraction	1.0000	0.0000	1.0000
Temperature: (C)	210.6	210.6	210.6
Pressure: (kPa)	2500	2500	2500
Molar Flow (kgmole/h)	0.0000	0.0000	0.0000
Mass Flow (kg/h)	0.0000	0.0000	0.0000
Std Ideal Liq VolFlow (m3/h)	0.0000	0.0000	0.0000
Molar Enthalpy (kJ/kgmole)	3.149e+04	9.434e+03	3.149e+04
Molar Entropy (kJ/kgmole-C)	7.973e+01	5.518e+01	7.973e+01
Heat Flow (kJ/h)	0.000e-01	0.000e-01	0.000e-01
Liq VolFlow @Std Cond (m3/h)	0.0000	0.0000	0.0000

COMPOSITION

Overall Phase	Vapour Fraction 1.0000
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COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	0.0000	0.3025	0.0000	0.2383	0.0000	0.2255
Propene	0.0000	0.0014	0.0000	0.0006	0.0000	0.0009
Propane	0.0000	0.1265	0.0000	0.0563	0.0000	0.0927
Cumene	0.0000	0.5356	0.0000	0.6492	0.0000	0.6268
14-iP-BZ	0.0000	0.0340	0.0000	0.0557	0.0000	0.0541
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000

Liquid Phase	Phase Fraction 0.0000
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COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)	(kg/h)	(m3/h)
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Benzene	0.0000	0.1614	0.0000	0.1084	0.0000	0.1060
Propene	0.0000	0.0002	0.0000	0.0001	0.0000	0.0001
Propane	0.0000	0.0155	0.0000	0.0059	0.0000	0.0100
Cumene	0.0000	0.7271	0.0000	0.7519	0.0000	0.7498
14-iP-BZ	0.0000	0.0958	0.0000	0.1337	0.0000	0.1341
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000

Vapour Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	0.0000	0.3025	0.0000	0.2383	0.0000	0.2255
Propene	0.0000	0.0014	0.0000	0.0006	0.0000	0.0009
Propane	0.0000	0.1265	0.0000	0.0563	0.0000	0.0927
Cumene	0.0000	0.5356	0.0000	0.6492	0.0000	0.6268
14-iP-BZ	0.0000	0.0340	0.0000	0.0557	0.0000	0.0541
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	1.874	1.874	---
Propene	8.464	8.464	---
Propane	8.139	8.139	---
Cumene	0.7366	0.7366	---
14-iP-BZ	0.3553	0.3553	---

UNIT OPERATIONS

FEED TO PRODUCT FROM LOGICAL CONNECTION

Conversion Reactor: CRV-100

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Trans_Effluents

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH. VAPOUR PH.

Vapour / Phase Fraction 0.0000 1.0000 0.0000

Temperature: (C) 210.6 210.6 210.6

Pressure: (kPa) 2500 2500 2500

Molar Flow (kgmole/h) 2.210e+004 2.210e+004 0.0000

Mass Flow (kg/h) 2.569e+006 2.569e+006 0.0000

Std Ideal Liq VolFlow (m3/h) 2979 2979 0.0000

Molar Enthalpy (kJ/kgmole) 9.434e+03 9.434e+03 3.149e+04

Molar Entropy (kJ/kgmole-C) 5.518e+01 5.518e+01 7.973e+01

Heat Flow (kJ/h) 2.085e+08 2.085e+08 0.000e-01

Liq VolFlow @Std Cond (m3/h) 2951 2951 0.0000

COMPOSITION

Overall Phase

Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	3567	0.1614	2.786e+005	0.1084	315.8	0.1060
Propene	3.614	0.0002	152.1	0.0001	0.2920	0.0001
Propane	343.5	0.0155	1.515e+004	0.0059	29.89	0.0100
Cumene	1.607e+004	0.7271	1.932e+006	0.7519	2234	0.7498
14-iP-BZ	2117	0.0958	3.435e+005	0.1337	399.5	0.1341
Total	2.210e+004	1.0000	2.569e+006	1.0000	2979	1.0000

Liquid Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	3567	0.1614	2.786e+005	0.1084	315.8	0.1060
Propene	3.614	0.0002	152.1	0.0001	0.2920	0.0001
Propane	343.5	0.0155	1.515e+004	0.0059	29.89	0.0100
Cumene	1.607e+004	0.7271	1.932e+006	0.7519	2234	0.7498
14-iP-BZ	2117	0.0958	3.435e+005	0.1337	399.5	0.1341
Total	2.210e+004	1.0000	2.569e+006	1.0000	2979	1.0000

Vapour Phase

Phase Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)
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Benzene	0.0000	0.3025	0.0000	0.2383	0.0000	0.2255
Propene	0.0000	0.0014	0.0000	0.0006	0.0000	0.0009
Propane	0.0000	0.1265	0.0000	0.0563	0.0000	0.0927
Cumene	0.0000	0.5356	0.0000	0.6492	0.0000	0.6268
14-iP-BZ	0.0000	0.0340	0.0000	0.0557	0.0000	0.0541
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	1.874	1.874	---
Propene	8.464	8.464	---
Propane	8.139	8.139	---
Cumene	0.7366	0.7366	---
14-iP-BZ	0.3553	0.3553	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Recycle: RCY-1	Conversion Reactor: CRV-100	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C2-Feed_New

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	216.2	216.2
Pressure: (kPa)	1500	1500
Molar Flow (kgmole/h)	3.594e+004	3.594e+004
Mass Flow (kg/h)	3.862e+006	3.862e+006
Std Ideal Liq VolFlow (m3/h)	4496	4496
Molar Enthalpy (kJ/kgmole)	2.116e+04	2.116e+04
Molar Entropy (kJ/kgmole-C)	3.309e+01	3.309e+01
Heat Flow (kJ/h)	7.606e+08	7.606e+08
Liq VolFlow @Std Cond (m3/h)	4418	4418

COMPOSITION

Overall Phase	Vapour Fraction 0.0000
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.373e+004	0.3819	1.072e+006	0.2776	1215	0.2703
Propene	14.15	0.0004	595.6	0.0002	1.143	0.0003
Propane	1341	0.0373	5.913e+004	0.0153	116.7	0.0260
Cumene	1.555e+004	0.4327	1.869e+006	0.4839	2161	0.4807
14-iP-BZ	5308	0.1477	8.613e+005	0.2230	1002	0.2228

Total 3.594e+004 1.0000 3.862e+006 1.0000 4496 1.0000

Liquid Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.373e+004	0.3819	1.072e+006	0.2776	1215	0.2703	
Propene	14.15	0.0004	595.6	0.0002	1.143	0.0003	
Propane	1341	0.0373	5.913e+004	0.0153	116.7	0.0260	
Cumene	1.555e+004	0.4327	1.869e+006	0.4839	2161	0.4807	
14-iP-BZ	5308	0.1477	8.613e+005	0.2230	1002	0.2228	
Total	3.594e+004	1.0000	3.862e+006	1.0000	4496	1.0000	

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Valve: VLV-101	Mixer: MIX-102	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Trans_Recycled

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	210.8	210.8
Pressure: (kPa)	2500	2500
Molar Flow (kgmole/h)	1.280e+004	1.280e+004
Mass Flow (kg/h)	1.467e+006	1.467e+006
Std Ideal Liq VolFlow (m3/h)	1708	1708
Molar Enthalpy (kJ/kgmole)	8.897e+03	8.897e+03
Molar Entropy (kJ/kgmole-C)	5.292e+01	5.292e+01
Heat Flow (kJ/h)	1.139e+08	1.139e+08
Liq VolFlow @Std Cond (m3/h)	1687	1687

COMPOSITION

Overall Phase

Vapour Fraction 0.0000

COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	2542	0.1987	1.986e+005	0.1353	225.1	0.1318
Propene	3.457e-010	0.0000	1.455e-008	0.0000	2.792e-011	0.0000
Propane	382.7	0.0299	1.687e+004	0.0115	33.30	0.0195
Cumene	8315	0.6498	9.994e+005	0.6810	1156	0.6767
14-iP-BZ	1557	0.1217	2.526e+005	0.1722	293.8	0.1720
Total	1.280e+004	1.0000	1.467e+006	1.0000	1708	1.0000
Liquid Phase					Phase Fraction	1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	2542	0.1987	1.986e+005	0.1353	225.1	0.1318
Propene	3.457e-010	0.0000	1.455e-008	0.0000	2.792e-011	0.0000
Propane	382.7	0.0299	1.687e+004	0.0115	33.30	0.0195
Cumene	8315	0.6498	9.994e+005	0.6810	1156	0.6767
14-iP-BZ	1557	0.1217	2.526e+005	0.1722	293.8	0.1720
Total	1.280e+004	1.0000	1.467e+006	1.0000	1708	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Mixer: MIX-102	Recycle: RCY-1	
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UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Total_Benzene

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
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Temperature: (C)	37.11	37.11
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Pressure: (kPa)	150.0	150.0
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Molar Flow (kgmole/h)	2.252e+004	2.252e+004
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Mass Flow (kg/h)	1.857e+006	1.857e+006
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Std Ideal Liq VolFlow (m3/h)	2125	2125
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Molar Enthalpy (kJ/kgmole)	3.878e+04	3.878e+04
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Molar Entropy (kJ/kgmole-C)	-1.218e+02	-1.218e+02
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Heat Flow (kJ/h)	8.732e+08	8.732e+08
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Liq VolFlow @Std Cond (m3/h)	2110	2110
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COMPOSITION

Overall Phase

Vapour Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.955e+004	0.8683	1.527e+006	0.8223	1731	0.8145	
Propene	3.615	0.0002	152.1	0.0001	0.2920	0.0001	
Propane	343.4	0.0153	1.514e+004	0.0082	29.89	0.0141	
Cumene	2616	0.1162	3.144e+005	0.1693	363.5	0.1710	
14-iP-BZ	2.317	0.0001	375.9	0.0002	0.4373	0.0002	
Total	2.252e+004	1.0000	1.857e+006	1.0000	2125	1.0000	

Liquid Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.955e+004	0.8683	1.527e+006	0.8223	1731	0.8145	
Propene	3.615	0.0002	152.1	0.0001	0.2920	0.0001	
Propane	343.4	0.0153	1.514e+004	0.0082	29.89	0.0141	
Cumene	2616	0.1162	3.144e+005	0.1693	363.5	0.1710	
14-iP-BZ	2.317	0.0001	375.9	0.0002	0.4373	0.0002	
Total	2.252e+004	1.0000	1.857e+006	1.0000	2125	1.0000	

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---

Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Pump: P-1	Mixer: MIX-103	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Fresh Benzene

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	25.00	25.00
Pressure: (kPa)	1500	1500
Molar Flow (kgmole/h)	1.280e+004	1.280e+004
Mass Flow (kg/h)	1.000e+006	1.000e+006

Total 1.280e+004 1.0000 1.000e+006 1.0000 1134 1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Mixer: MIX-103

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Transalkylation Recycle Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

(kgmole/h) (kg/h) (m³/h)

Benzene	6742	0.6942	5.266e+005	0.6143	596.9	0.6019
Propene	3.614	0.0004	152.1	0.0002	0.2920	0.0003
Propane	343.5	0.0354	1.515e+004	0.0177	29.89	0.0301
Cumene	2621	0.2698	3.150e+005	0.3674	364.2	0.3673
14-iP-BZ	2.321	0.0002	376.6	0.0004	0.4381	0.0004
Total	9712	1.0000	8.572e+005	1.0000	991.8	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Mixer: MIX-101	Tee: TEE-102	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Alkylation Recycle

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	50.02	50.02
Pressure: (kPa)	150.0	150.0
Molar Flow (kgmole/h)	9712	9712
Mass Flow (kg/h)	8.572e+005	8.572e+005
Std Ideal Liq VolFlow (m3/h)	991.8	991.8
Molar Enthalpy (kJ/kgmole)	2.360e+04	2.360e+04
Molar Entropy (kJ/kgmole-C)	-9.321e+01	-9.321e+01
Heat Flow (kJ/h)	2.292e+08	2.292e+08
Liq VolFlow @Std Cond (m3/h)	979.4	979.4

COMPOSITION

Overall Phase	Vapour Fraction 0.0000
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	6742	0.6942	5.266e+005	0.6143	596.9	0.6019	
Propene	3.614	0.0004	152.1	0.0002	0.2920	0.0003	
Propane	343.5	0.0354	1.515e+004	0.0177	29.89	0.0301	
Cumene	2621	0.2698	3.150e+005	0.3674	364.2	0.3673	
14-iP-BZ	2.321	0.0002	376.6	0.0004	0.4381	0.0004	

Total	9712	1.0000	8.572e+005	1.0000	991.8	1.0000
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Liquid Phase

Phase Fraction 1.000

COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	6742	0.6942	5.266e+005	0.6143	596.9	0.6019
Propene	3.614	0.0004	152.1	0.0002	0.2920	0.0003
Propane	343.5	0.0354	1.515e+004	0.0177	29.89	0.0301
Cumene	2621	0.2698	3.150e+005	0.3674	364.2	0.3673
14-iP-BZ	2.321	0.0002	376.6	0.0004	0.4381	0.0004
Total	9712	1.0000	8.572e+005	1.0000	991.8	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Recycle: RCY-2	Tee: TEE-102	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Alkylation Recycled

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	50.02	50.02
Pressure: (kPa)	150.0	150.0
Molar Flow (kgmole/h)	9715	9715
Mass Flow (kg/h)	8.573e+005	8.573e+005
Std Ideal Liq VolFlow (m3/h)	991.9	991.9
Molar Enthalpy (kJ/kgmole)	2.366e+04	2.366e+04
Molar Entropy (kJ/kgmole-C)	-9.328e+01	-9.328e+01
Heat Flow (kJ/h)	2.299e+08	2.299e+08
Liq VolFlow @Std Cond (m3/h)	979.5	979.5

COMPOSITION

Overall Phase

Vapour Fraction 0.0000

COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	6750	0.6948	5.273e+005	0.6150	597.7	0.6026
Propene	3.615	0.0004	152.1	0.0002	0.2920	0.0003
Propane	343.4	0.0353	1.514e+004	0.0177	29.89	0.0301
Cumene	2616	0.2692	3.144e+005	0.3667	363.5	0.3665
14-iP-BZ	2.317	0.0002	375.9	0.0004	0.4373	0.0004
Total	9715	1.0000	8.573e+005	1.0000	991.9	1.0000
Liquid Phase					Phase Fraction	1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	6750	0.6948	5.273e+005	0.6150	597.7	0.6026
Propene	3.615	0.0004	152.1	0.0002	0.2920	0.0003
Propane	343.4	0.0353	1.514e+004	0.0177	29.89	0.0301
Cumene	2616	0.2692	3.144e+005	0.3667	363.5	0.3665
14-iP-BZ	2.317	0.0002	375.9	0.0004	0.4373	0.0004
Total	9715	1.0000	8.573e+005	1.0000	991.9	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Mixer: MIX-103	Recycle: RCY-2	
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UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Boil-Up

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH.

Vapour / Phase Fraction	1.0000	1.0000
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Temperature: (C)	219.3	219.3
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Pressure: (kPa)	1500	1500
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Molar Flow (kgmole/h)	1.300e+004	1.300e+004
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Mass Flow (kg/h)	9.915e+005	9.915e+005
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Std Ideal Liq VolFlow (m3/h)	1256	1256
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Molar Enthalpy (kJ/kgmole)	4.206e+04	4.206e+04
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Molar Entropy (kJ/kgmole-C)	6.121e+01	6.121e+01
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Heat Flow (kJ/h)	5.467e+08	5.467e+08
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Liq VolFlow @Std Cond (m3/h)	1227	1227
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COMPOSITION

Overall Phase

Vapour Fraction 1.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	7697	0.5921	6.012e+005	0.6064	681.5	0.5425	
Propene	51.87	0.0040	2183	0.0022	4.190	0.0033	
Propane	3369	0.2592	1.486e+005	0.1498	293.2	0.2334	
Cumene	1565	0.1204	1.881e+005	0.1897	217.5	0.1731	
14-iP-BZ	317.1	0.0244	5.145e+004	0.0519	59.85	0.0476	
Total	1.300e+004	1.0000	9.915e+005	1.0000	1256	1.0000	

Vapour Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	7697	0.5921	6.012e+005	0.6064	681.5	0.5425	
Propene	51.87	0.0040	2183	0.0022	4.190	0.0033	
Propane	3369	0.2592	1.486e+005	0.1498	293.2	0.2334	
Cumene	1565	0.1204	1.881e+005	0.1897	217.5	0.1731	
14-iP-BZ	317.1	0.0244	5.145e+004	0.0519	59.85	0.0476	
Total	1.300e+004	1.0000	9.915e+005	1.0000	1256	1.0000	

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	---	---	---

Propene	---	---	---
Propane	---	---	---
Cumene	---	---	---
14-iP-BZ	---	---	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Absorber: T-100	Recycle: RCY-3	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Reflux_new3	Fluid Package: Basis-1
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Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	50.02	50.02
Pressure: (kPa)	150.0	150.0
Molar Flow (kgmole/h)	1378	1378
Mass Flow (kg/h)	1.658e+005	1.658e+005

	(kgmole/h)		(kg/h)		(m3/h)	
Benzene	4.612e-003	0.0000	0.3602	0.0000	4.084e-004	0.0000
Propene	2.604e-024	0.0000	1.096e-022	0.0000	2.104e-025	0.0000
Propane	1.249e-021	0.0000	5.506e-020	0.0000	1.087e-022	0.0000
Cumene	1373	0.9963	1.650e+005	0.9950	190.8	0.9950
14-iP-BZ	5.061	0.0037	821.2	0.0050	0.9552	0.0050

Total 1378 1.0000 1.658e+005 1.0000 191.8 1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
Recycle: RCY-8	Tee: TEE-101	

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Reflux-1 @COL1 Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

(kgmole/h) (kg/h) (m³/h)

Benzene	9.534e-002	0.0000	7.447	0.0000	8.441e-003	0.0000
Propene	824.8	0.0413	3.471e+004	0.0394	66.63	0.0384
Propane	1.917e+004	0.9587	8.453e+005	0.9606	1668	0.9616
Cumene	6.047e-007	0.0000	7.268e-005	0.0000	8.405e-008	0.0000
14-iP-BZ	7.659e-011	0.0000	1.243e-008	0.0000	1.446e-011	0.0000
Total	1.999e+004	1.0000	8.800e+005	1.0000	1735	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Tower: Main Tower @COL1 Material Stream: Reflux-1

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C1-Top @COL1

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH.

Vapour / Phase Fraction	1.0000	1.0000
Temperature: (C)	41.98	41.98
Pressure: (kPa)	1450	1450
Molar Flow (kgmole/h)	2.000e+004	2.000e+004
Mass Flow (kg/h)	8.804e+005	8.804e+005
Std Ideal Liq VolFlow (m3/h)	1736	1736
Molar Enthalpy (kJ/kgmole)	-9.939e+04	-9.939e+04
Molar Entropy (kJ/kgmole-C)	1.385e+02	1.385e+02
Heat Flow (kJ/h)	-1.988e+09	-1.988e+09
Liq VolFlow @Std Cond (m3/h)	1732	1732

COMPOSITION

Overall Phase	Vapour Fraction 1.0000
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	8.936e-002	0.0000	6.980	0.0000	7.912e-003	0.0000	
Propene	810.7	0.0405	3.412e+004	0.0387	65.49	0.0377	
Propane	1.919e+004	0.9595	8.463e+005	0.9612	1670	0.9623	
Cumene	5.545e-007	0.0000	6.665e-005	0.0000	7.708e-008	0.0000	
14-iP-BZ	7.045e-011	0.0000	1.143e-008	0.0000	1.330e-011	0.0000	

Total 2.000e+004 1.0000 8.804e+005 1.0000 1736 1.0000

Vapour Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	8.936e-002	0.0000	6.980	0.0000	7.912e-003	0.0000	
Propene	810.7	0.0405	3.412e+004	0.0387	65.49	0.0377	
Propane	1.919e+004	0.9595	8.463e+005	0.9612	1670	0.9623	
Cumene	5.545e-007	0.0000	6.665e-005	0.0000	7.708e-008	0.0000	
14-iP-BZ	7.045e-011	0.0000	1.143e-008	0.0000	1.330e-011	0.0000	
Total	2.000e+004	1.0000	8.804e+005	1.0000	1736	1.0000	

K VALUE

COMPONENTS		MIXED	LIGHT	HEAVY
Benzene	---	---	---	
Propene	---	---	---	
Propane	---	---	---	
Cumene	---	---	---	
14-iP-BZ	---	---	---	

UNIT OPERATIONS

FEED TO PRODUCT FROM LOGICAL CONNECTION

Material Stream: C1-Top Tower: Main Tower @COL1

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C1-Bot @COL1

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

	OVERALL	VAPOUR PH.	LIQUID PH.
Vapour / Phase Fraction	0.0000	0.0000	1.0000
Temperature: (C)	169.5	169.5	169.5
Pressure: (kPa)	1500	1500	1500
Molar Flow (kgmole/h)	3.614e+004	0.0000	3.614e+004
Mass Flow (kg/h)	3.386e+006	0.0000	3.386e+006
Std Ideal Liq VolFlow (m3/h)	4045	0.0000	4045
Molar Enthalpy (kJ/kgmole)	1.485e+04	-2.244e+04	1.485e+04
Molar Entropy (kJ/kgmole-C)	-3.081e+00	1.043e+02	-3.081e+00
Heat Flow (kJ/h)	5.369e+08	0.000e-01	5.369e+08
Liq VolFlow @Std Cond (m3/h)	3954	0.0000	3954

COMPOSITION

Overall Phase	Vapour Fraction 0.0000
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COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.888e+004	0.5223	1.475e+006	0.4355	1672	0.4133
Propene	65.97	0.0018	2776	0.0008	5.328	0.0013
Propane	4327	0.1197	1.908e+005	0.0563	376.6	0.0931
Cumene	8802	0.2435	1.058e+006	0.3124	1223	0.3025
14-iP-BZ	4068	0.1126	6.602e+005	0.1950	767.9	0.1898
Total	3.614e+004	1.0000	3.386e+006	1.0000	4045	1.0000
Vapour Phase						Phase Fraction 0.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	0.0000	0.3397	0.0000	0.4484	0.0000	0.3341
Propene	0.0000	0.0096	0.0000	0.0068	0.0000	0.0086
Propane	0.0000	0.6073	0.0000	0.4525	0.0000	0.5871
Cumene	0.0000	0.0377	0.0000	0.0766	0.0000	0.0582
14-iP-BZ	0.0000	0.0057	0.0000	0.0156	0.0000	0.0119
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000
Liquid Phase						Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.888e+004	0.5223	1.475e+006	0.4355	1672	0.4133
Propene	65.97	0.0018	2776	0.0008	5.328	0.0013
Propane	4327	0.1197	1.908e+005	0.0563	376.6	0.0931
Cumene	8802	0.2435	1.058e+006	0.3124	1223	0.3025
14-iP-BZ	4068	0.1126	6.602e+005	0.1950	767.9	0.1898

Total 3.614e+004 1.0000 3.386e+006 1.0000 4045 1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO PRODUCT FROM LOGICAL CONNECTION

Material Stream: C1-Bot Tower: Main Tower @COL1

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C1-Feed @COL1

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

(kgmole/h) (kg/h) (m3/h)

Benzene	1.118e+004	0.4830	8.734e+005	0.3646	990.0	0.3549
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	981.2	0.0424	4.327e+004	0.0181	85.39	0.0306
Cumene	7238	0.3126	8.699e+005	0.3632	1006	0.3606
14-iP-BZ	3751	0.1620	6.087e+005	0.2541	708.0	0.2538
Total	2.315e+004	1.0000	2.395e+006	1.0000	2789	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Tower: Main Tower @COL1 Material Stream: C1-Feed

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Boil-Up @COL1

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH.

Vapour / Phase Fraction	1.0000	1.0000
Temperature: (C)	219.3	219.3
Pressure: (kPa)	1500	1500
Molar Flow (kgmole/h)	1.300e+004	1.300e+004
Mass Flow (kg/h)	9.915e+005	9.915e+005
Std Ideal Liq VolFlow (m3/h)	1256	1256
Molar Enthalpy (kJ/kgmole)	4.206e+04	4.206e+04
Molar Entropy (kJ/kgmole-C)	6.121e+01	6.121e+01
Heat Flow (kJ/h)	5.467e+08	5.467e+08
Liq VolFlow @Std Cond (m3/h)	1227	1227

COMPOSITION

Overall Phase	Vapour Fraction 1.0000
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	7697	0.5921	6.012e+005	0.6064	681.5	0.5425
Propene	51.87	0.0040	2183	0.0022	4.190	0.0033
Propane	3369	0.2592	1.486e+005	0.1498	293.2	0.2334
Cumene	1565	0.1204	1.881e+005	0.1897	217.5	0.1731
14-iP-BZ	317.1	0.0244	5.145e+004	0.0519	59.85	0.0476

Total 1.300e+004 1.0000 9.915e+005 1.0000 1256 1.0000

Vapour Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)	
Benzene	7697	0.5921	6.012e+005	0.6064	681.5	0.5425
Propene	51.87	0.0040	2183	0.0022	4.190	0.0033
Propane	3369	0.2592	1.486e+005	0.1498	293.2	0.2334
Cumene	1565	0.1204	1.881e+005	0.1897	217.5	0.1731
14-iP-BZ	317.1	0.0244	5.145e+004	0.0519	59.85	0.0476
Total	1.300e+004	1.0000	9.915e+005	1.0000	1256	1.0000

K VALUE

COMPONENTS		MIXED	LIGHT	HEAVY
Benzene	---	---	---	
Propene	---	---	---	
Propane	---	---	---	
Cumene	---	---	---	
14-iP-BZ	---	---	---	

UNIT OPERATIONS

FEED TO PRODUCT FROM LOGICAL CONNECTION

Tower: Main Tower @COL1 Material Stream: Boil-Up

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Reflux-2 @COL2

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	50.02	50.02
Pressure: (kPa)	150.0	150.0
Molar Flow (kgmole/h)	8327	8327
Mass Flow (kg/h)	7.351e+005	7.351e+005
Std Ideal Liq VolFlow (m3/h)	850.5	850.5
Molar Enthalpy (kJ/kgmole)	2.358e+04	2.358e+04
Molar Entropy (kJ/kgmole-C)	-9.317e+01	-9.317e+01
Heat Flow (kJ/h)	1.963e+08	1.963e+08
Liq VolFlow @Std Cond (m3/h)	839.9	839.9

COMPOSITION

Overall Phase	Vapour Fraction 0.0000
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COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	5778	0.6939	4.513e+005	0.6139	511.6	0.6015
Propene	3.100	0.0004	130.4	0.0002	0.2504	0.0003
Propane	294.6	0.0354	1.299e+004	0.0177	25.64	0.0301
Cumene	2249	0.2701	2.704e+005	0.3678	312.6	0.3676
14-iP-BZ	1.995	0.0002	323.8	0.0004	0.3766	0.0004
Total	8327	1.0000	7.351e+005	1.0000	850.5	1.0000
Liquid Phase					Phase Fraction	1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	5778	0.6939	4.513e+005	0.6139	511.6	0.6015
Propene	3.100	0.0004	130.4	0.0002	0.2504	0.0003
Propane	294.6	0.0354	1.299e+004	0.0177	25.64	0.0301
Cumene	2249	0.2701	2.704e+005	0.3678	312.6	0.3676
14-iP-BZ	1.995	0.0002	323.8	0.0004	0.3766	0.0004
Total	8327	1.0000	7.351e+005	1.0000	850.5	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO PRODUCT FROM LOGICAL CONNECTION

Tower: Main Tower @COL2 Material Stream: Reflux-2

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Boilup-2 @COL2 Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH.

Vapour / Phase Fraction	1.0000	1.0000
Temperature: (C)	172.7	172.7
Pressure: (kPa)	120.0	120.0
Molar Flow (kgmole/h)	8500	8500
Mass Flow (kg/h)	1.065e+006	1.065e+006
Std Ideal Liq VolFlow (m3/h)	1232	1232
Molar Enthalpy (kJ/kgmole)	2.310e+04	2.310e+04
Molar Entropy (kJ/kgmole-C)	1.339e+02	1.339e+02
Heat Flow (kJ/h)	1.963e+08	1.963e+08
Liq VolFlow @Std Cond (m3/h)	1229	1229

COMPOSITION

Overall Phase

Vapour Fraction 1.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	3.669e-002	0.0000	2.866	0.0000	3.249e-003	0.0000	
Propene	2.086e-022	0.0000	8.780e-021	0.0000	1.685e-023	0.0000	
Propane	9.137e-020	0.0000	4.029e-018	0.0000	7.952e-021	0.0000	
Cumene	7474	0.8794	8.984e+005	0.8437	1039	0.8430	
14-iP-BZ	1025	0.1206	1.664e+005	0.1563	193.5	0.1570	
Total	8500	1.0000	1.065e+006	1.0000	1232	1.0000	

Vapour Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	3.669e-002	0.0000	2.866	0.0000	3.249e-003	0.0000	
Propene	2.086e-022	0.0000	8.780e-021	0.0000	1.685e-023	0.0000	
Propane	9.137e-020	0.0000	4.029e-018	0.0000	7.952e-021	0.0000	
Cumene	7474	0.8794	8.984e+005	0.8437	1039	0.8430	
14-iP-BZ	1025	0.1206	1.664e+005	0.1563	193.5	0.1570	
Total	8500	1.0000	1.065e+006	1.0000	1232	1.0000	

K VALUE

COMPONENTS		MIXED	LIGHT	HEAVY
Benzene	---	---	---	

Propene	---	---	---
Propane	---	---	---
Cumene	---	---	---
14-iP-BZ	---	---	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Tower: Main Tower @COL2 Material Stream: Boilup-2

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C2-Top @COL2

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH.

Vapour / Phase Fraction	1.0000	1.0000
Temperature: (C)	115.6	115.6
Pressure: (kPa)	100.0	100.0
Molar Flow (kgmole/h)	2.866e+004	2.866e+004
Mass Flow (kg/h)	2.498e+006	2.498e+006

14-iP-BZ 6.632 0.0002 1076 0.0004 1.252 0.0004

Total 2.866e+004 1.0000 2.498e+006 1.0000 2914 1.0000

K VALUE

COMPONENTS		MIXED	LIGHT	HEAVY
Benzene	---	---	---	
Propene	---	---	---	
Propane	---	---	---	
Cumene	---	---	---	
14-iP-BZ	---	---	---	

UNIT OPERATIONS

FEED TO PRODUCT FROM LOGICAL CONNECTION

Material Stream: C2-Top Tower: Main Tower @COL2

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C2-Bot @COL2

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH. LIQUID PH.

Vapour / Phase Fraction	0.0000	0.0000	1.0000
Temperature: (C)	169.5	169.5	169.5
Pressure: (kPa)	120.0	120.0	120.0
Molar Flow (kgmole/h)	2.411e+004	0.0000	2.411e+004
Mass Flow (kg/h)	3.164e+006	0.0000	3.164e+006
Std Ideal Liq VolFlow (m3/h)	3666	0.0000	3666
Molar Enthalpy (kJ/kgmole)	-2.569e+04	2.474e+04	-2.569e+04
Molar Entropy (kJ/kgmole-C)	5.065e+01	1.311e+02	5.065e+01
Heat Flow (kJ/h)	-6.194e+08	0.000e-01	-6.194e+08
Liq VolFlow @Std Cond (m3/h)	3653	0.0000	3653

COMPOSITION

Overall Phase	Vapour Fraction 0.0000
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	5.063e-002	0.0000	3.954	0.0000	4.483e-003	0.0000	
Propene	2.173e-022	0.0000	9.143e-021	0.0000	1.755e-023	0.0000	
Propane	9.538e-020	0.0000	4.206e-018	0.0000	8.301e-021	0.0000	
Cumene	1.778e+004	0.7375	2.137e+006	0.6754	2471	0.6742	
14-iP-BZ	6328	0.2625	1.027e+006	0.3246	1194	0.3258	
Total	2.411e+004	1.0000	3.164e+006	1.0000	3666	1.0000	

Vapour Phase	Phase Fraction 0.0000
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW LIQVOL FRAC

(kgmole/h)	(kg/h)	(m3/h)
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Benzene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Cumene	0.0000	0.9138	0.0000	0.8871	0.0000	0.8865
14-iP-BZ	0.0000	0.0862	0.0000	0.1129	0.0000	0.1135
Total	0.0000	1.0000	0.0000	1.0000	0.0000	1.0000
Liquid Phase					Phase Fraction	1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	5.063e-002	0.0000	3.954	0.0000	4.483e-003	0.0000	
Propene	2.173e-022	0.0000	9.143e-021	0.0000	1.755e-023	0.0000	
Propane	9.538e-020	0.0000	4.206e-018	0.0000	8.301e-021	0.0000	
Cumene	1.778e+004	0.7375	2.137e+006	0.6754	2471	0.6742	
14-iP-BZ	6328	0.2625	1.027e+006	0.3246	1194	0.3258	
Total	2.411e+004	1.0000	3.164e+006	1.0000	3666	1.0000	

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO PRODUCT FROM LOGICAL CONNECTION

Material Stream: C2-Bot Tower: Main Tower @COL2

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C2-Feed-LP @COL2

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH. LIQUID PH.

Vapour / Phase Fraction	0.6057	0.6057	0.3943
Temperature: (C)	141.8	141.8	141.8
Pressure: (kPa)	120.0	120.0	120.0
Molar Flow (kgmole/h)	3.594e+004	2.177e+004	1.417e+004
Mass Flow (kg/h)	3.862e+006	2.059e+006	1.804e+006
Std Ideal Liq VolFlow (m3/h)	4496	2409	2088
Molar Enthalpy (kJ/kgmole)	2.116e+04	5.212e+04	-2.640e+04
Molar Entropy (kJ/kgmole-C)	3.975e+01	5.357e+01	1.851e+01
Heat Flow (kJ/h)	7.606e+08	1.135e+09	-3.741e+08
Liq VolFlow @Std Cond (m3/h)	4418	2368	2068

COMPOSITION

Overall Phase

Vapour Fraction 0.6057

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.373e+004	0.3819	1.072e+006	0.2776	1215	0.2703
Propene	14.15	0.0004	595.6	0.0002	1.143	0.0003
Propane	1341	0.0373	5.913e+004	0.0153	116.7	0.0260
Cumene	1.555e+004	0.4327	1.869e+006	0.4839	2161	0.4807
14-iP-BZ	5308	0.1477	8.613e+005	0.2230	1002	0.2228
Total	3.594e+004	1.0000	3.862e+006	1.0000	4496	1.0000

Vapour Phase

Phase Fraction 0.6057

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.181e+004	0.5424	9.223e+005	0.4480	1045	0.4341
Propene	13.98	0.0006	588.2	0.0003	1.129	0.0005
Propane	1323	0.0608	5.832e+004	0.0283	115.1	0.0478
Cumene	7651	0.3515	9.196e+005	0.4467	1063	0.4415
14-iP-BZ	972.6	0.0447	1.578e+005	0.0767	183.6	0.0762
Total	2.177e+004	1.0000	2.059e+006	1.0000	2409	1.0000

Liquid Phase

Phase Fraction 0.3943

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)
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Benzene	1919	0.1354	1.499e+005	0.0831	169.9	0.0814
Propene	0.1775	0.0000	7.468	0.0000	1.433e-002	0.0000
Propane	18.36	0.0013	809.7	0.0004	1.598	0.0008
Cumene	7899	0.5574	9.495e+005	0.5264	1098	0.5259
14-iP-BZ	4335	0.3059	7.034e+005	0.3900	818.2	0.3919
Total	1.417e+004	1.0000	1.804e+006	1.0000	2088	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	4.006	4.006	---
Propene	51.28	51.28	---
Propane	46.89	46.89	---
Cumene	0.6305	0.6305	---
14-iP-BZ	0.1461	0.1461	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Tower: Main Tower @COL2 Material Stream: C2-Feed-LP

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Reflux-3 @COL3

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	50.02	50.02
Pressure: (kPa)	150.0	150.0
Molar Flow (kgmole/h)	1375	1375
Mass Flow (kg/h)	1.655e+005	1.655e+005
Std Ideal Liq VolFlow (m3/h)	191.3	191.3
Molar Enthalpy (kJ/kgmole)	-3.516e+04	-3.516e+04
Molar Entropy (kJ/kgmole-C)	-3.253e+01	-3.253e+01
Heat Flow (kJ/h)	-4.833e+07	-4.833e+07
Liq VolFlow @Std Cond (m3/h)	191.0	191.0

COMPOSITION

Overall Phase	Vapour Fraction 0.0000
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COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	4.613e-003	0.0000	0.3603	0.0000	4.084e-004	0.0000	
Propene	2.613e-024	0.0000	1.099e-022	0.0000	2.110e-025	0.0000	
Propane	1.252e-021	0.0000	5.523e-020	0.0000	1.090e-022	0.0000	
Cumene	1369	0.9960	1.646e+005	0.9946	190.3	0.9945	
14-iP-BZ	5.550	0.0040	900.6	0.0054	1.048	0.0055	

Total	1375	1.0000	1.655e+005	1.0000	191.3	1.0000
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Liquid Phase	Phase Fraction	1.000
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COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)		(kg/h)		(m3/h)	
Benzene	4.613e-003	0.0000	0.3603	0.0000	4.084e-004	0.0000
Propene	2.613e-024	0.0000	1.099e-022	0.0000	2.110e-025	0.0000
Propane	1.252e-021	0.0000	5.523e-020	0.0000	1.090e-022	0.0000
Cumene	1369	0.9960	1.646e+005	0.9946	190.3	0.9945
14-iP-BZ	5.550	0.0040	900.6	0.0054	1.048	0.0055
Total	1375	1.0000	1.655e+005	1.0000	191.3	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Tower: Main Tower	@COL3	Material Stream: Reflux-3
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UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: Boilup-3 @COL3

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH.

Vapour / Phase Fraction	1.0000	1.0000
Temperature: (C)	176.6	176.6
Pressure: (kPa)	120.0	120.0
Molar Flow (kgmole/h)	5000	5000
Mass Flow (kg/h)	6.362e+005	6.362e+005
Std Ideal Liq VolFlow (m3/h)	736.6	736.6
Molar Enthalpy (kJ/kgmole)	2.082e+04	2.082e+04
Molar Entropy (kJ/kgmole-C)	1.374e+02	1.374e+02
Heat Flow (kJ/h)	1.041e+08	1.041e+08
Liq VolFlow @Std Cond (m3/h)	734.3	734.3

COMPOSITION

Overall Phase	Vapour Fraction 1.0000
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COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
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	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	2.395e-005	0.0000	1.871e-003	0.0000	2.120e-006	0.0000
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Cumene	4163	0.8325	5.003e+005	0.7864	578.6	0.7854
14-iP-BZ	837.4	0.1675	1.359e+005	0.2136	158.1	0.2146
Total	5000	1.0000	6.362e+005	1.0000	736.6	1.0000
Vapour Phase					Phase Fraction	1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	2.395e-005	0.0000	1.871e-003	0.0000	2.120e-006	0.0000
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Cumene	4163	0.8325	5.003e+005	0.7864	578.6	0.7854
14-iP-BZ	837.4	0.1675	1.359e+005	0.2136	158.1	0.2146
Total	5000	1.0000	6.362e+005	1.0000	736.6	1.0000

K VALUE

COMPONENTS		MIXED	LIGHT	HEAVY
Benzene	---	---	---	
Propene	---	---	---	
Propane	---	---	---	
Cumene	---	---	---	
14-iP-BZ	---	---	---	

UNIT OPERATIONS

FEED TO PRODUCT FROM LOGICAL CONNECTION

Tower: Main Tower @COL3 Material Stream: Boilup-3

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C3-Top @COL3

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL VAPOUR PH.

Vapour / Phase Fraction 1.0000 1.0000

Temperature: (C) 153.5 153.5

Pressure: (kPa) 100.0 100.0

Molar Flow (kgmole/h) 4594 4594

Mass Flow (kg/h) 5.528e+005 5.528e+005

Std Ideal Liq VolFlow (m3/h) 639.3 639.3

Molar Enthalpy (kJ/kgmole) 2.703e+04 2.703e+04

Molar Entropy (kJ/kgmole-C) 1.213e+02 1.213e+02

Heat Flow (kJ/h) 1.242e+08 1.242e+08

Liq VolFlow @Std Cond (m3/h) 638.1 638.1

COMPOSITION

Overall Phase

Vapour Fraction 1.0000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.537e-002	0.0000	1.201	0.0000	1.361e-003	0.0000	
Propene	8.681e-024	0.0000	3.653e-022	0.0000	7.012e-025	0.0000	
Propane	4.162e-021	0.0000	1.835e-019	0.0000	3.622e-022	0.0000	
Cumene	4577	0.9963	5.501e+005	0.9950	636.1	0.9950	
14-iP-BZ	16.87	0.0037	2737	0.0050	3.184	0.0050	
Total	4594	1.0000	5.528e+005	1.0000	639.3	1.0000	

Vapour Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.537e-002	0.0000	1.201	0.0000	1.361e-003	0.0000	
Propene	8.681e-024	0.0000	3.653e-022	0.0000	7.012e-025	0.0000	
Propane	4.162e-021	0.0000	1.835e-019	0.0000	3.622e-022	0.0000	
Cumene	4577	0.9963	5.501e+005	0.9950	636.1	0.9950	
14-iP-BZ	16.87	0.0037	2737	0.0050	3.184	0.0050	
Total	4594	1.0000	5.528e+005	1.0000	639.3	1.0000	

K VALUE

COMPONENTS		MIXED	LIGHT	HEAVY
Benzene	---	---	---	

Propene	---	---	---
Propane	---	---	---
Cumene	---	---	---
14-iP-BZ	---	---	---

UNIT OPERATIONS

FEED TO	PRODUCT FROM	LOGICAL CONNECTION
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Material Stream: C3-Top Tower: Main Tower @COL3

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C3-Bot @COL3

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

	OVERALL	VAPOUR PH.	LIQUID PH.
Vapour / Phase Fraction	0.0000	0.0000	1.0000
Temperature: (C)	173.2	173.2	173.2
Pressure: (kPa)	120.0	120.0	120.0
Molar Flow (kgmole/h)	1.739e+004	0.0000	1.739e+004
Mass Flow (kg/h)	2.348e+006	0.0000	2.348e+006

	(kgmole/h)	(kg/h)		(m3/h)		
Benzene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propene	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Propane	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Cumene	0.0000	0.8730	0.0000	0.8358	0.0000	0.8350
14-iP-BZ	0.0000	0.1270	0.0000	0.1642	0.0000	0.1650

Total 0.0000 1.0000 0.0000 1.0000 0.0000 1.0000

Liquid Phase

Phase Fraction 1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)		(kg/h)		(m3/h)		
Benzene	1.566e-005	0.0000	1.223e-003	0.0000	1.386e-006	0.0000	
Propene	1.209e-026	0.0000	5.088e-025	0.0000	9.766e-028	0.0000	
Propane	1.236e-026	0.0000	5.451e-025	0.0000	1.076e-027	0.0000	
Cumene	1.126e+004	0.6476	1.354e+006	0.5765	1565	0.5751	
14-iP-BZ	6127	0.3524	9.943e+005	0.4235	1156	0.4249	
Total	1.739e+004	1.0000	2.348e+006	1.0000	2722	1.0000	

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO PRODUCT FROM LOGICAL CONNECTION

Material Stream: C3-Bot Tower: Main Tower @COL3

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

Material Stream: C2-Bottoms @COL3

Fluid Package: Basis-1

Property Package: Peng-Robinson

CONDITIONS

OVERALL LIQUID PH.

Vapour / Phase Fraction	0.0000	1.0000
Temperature: (C)	172.7	172.7
Pressure: (kPa)	120.0	120.0
Molar Flow (kgmole/h)	1.561e+004	1.561e+004
Mass Flow (kg/h)	2.099e+006	2.099e+006
Std Ideal Liq VolFlow (m3/h)	2433	2433
Molar Enthalpy (kJ/kgmole)	-3.061e+04	-3.061e+04
Molar Entropy (kJ/kgmole-C)	5.406e+01	5.406e+01
Heat Flow (kJ/h)	-4.777e+08	-4.777e+08
Liq VolFlow @Std Cond (m3/h)	2424	2424

COMPOSITION

Overall Phase	Vapour Fraction 0.0000
---------------	------------------------

COMPONENTS	MOLE FLOW	MOLE FRAC	MASS FLOW	MASS FRAC	LIQVOL FLOW	LIQVOL FRAC
------------	-----------	-----------	-----------	-----------	-------------	-------------

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.075e-002	0.0000	0.8399	0.0000	9.520e-004	0.0000
Propene	6.081e-024	0.0000	2.559e-022	0.0000	4.912e-025	0.0000
Propane	2.910e-021	0.0000	1.283e-019	0.0000	2.532e-022	0.0000
Cumene	1.031e+004	0.6603	1.239e+006	0.5902	1433	0.5888
14-iP-BZ	5301	0.3397	8.603e+005	0.4098	1001	0.4112
Total	1.561e+004	1.0000	2.099e+006	1.0000	2433	1.0000
Liquid Phase					Phase Fraction	1.000

COMPONENTS MOLE FLOW MOLE FRAC MASS FLOW MASS FRAC LIQVOL FLOW
LIQVOL FRAC

	(kgmole/h)	(kg/h)	(m3/h)			
Benzene	1.075e-002	0.0000	0.8399	0.0000	9.520e-004	0.0000
Propene	6.081e-024	0.0000	2.559e-022	0.0000	4.912e-025	0.0000
Propane	2.910e-021	0.0000	1.283e-019	0.0000	2.532e-022	0.0000
Cumene	1.031e+004	0.6603	1.239e+006	0.5902	1433	0.5888
14-iP-BZ	5301	0.3397	8.603e+005	0.4098	1001	0.4112
Total	1.561e+004	1.0000	2.099e+006	1.0000	2433	1.0000

K VALUE

COMPONENTS	MIXED	LIGHT	HEAVY
Benzene	0.0000	0.0000	---
Propene	0.0000	0.0000	---
Propane	0.0000	0.0000	---
Cumene	0.0000	0.0000	---
14-iP-BZ	0.0000	0.0000	---

UNIT OPERATIONS

FEED TO PRODUCT FROM LOGICAL CONNECTION

Tower: Main Tower @COL3 Material Stream: C2-Bottoms

UTILITIES

(No utilities reference this stream)

PROCESS UTILITY

All Unit Operations (Case (Main)+ Templates):

Pump: P-1

CONNECTIONS

Inlet Stream

Stream Name	From Unit Operation
-------------	---------------------

Total_Benzene	MIX-103 Mixer
---------------	---------------

Outlet Stream

Stream Name	To Unit Operation
-------------	-------------------

Benzene HP	MIX-100 Mixer
------------	---------------

Energy Stream

Stream Name From Unit Operation

Q-P1

PARAMETERS

Adiabatic Efficiency (%): 75.00 Delta P: 3350 kPa Pressure Ratio: 23.33 Duty: 2689 kW

Multiphase Pump Not Active

CURVES

Delta P: 3350 kPa Duty: 2689 kW

Coefficient A: 0.0000 Coefficient B: 0.0000 Coefficient C: 0.0000 Coefficient D: 0.0000

Coefficient E: 0.0000 Coefficient F: 0.0000

Parameter Preferences Units for Head: m Flow Basis ActVolFlow Units for Flow: m3/h

User Variables

RATING

Head Offset: 0.0000 m Efficiency Offset: 0.0000

Gas Fraction Curve Collection

Gas Fraction Basis ---

CC-0

Design GVF: 0.0000 Curve GVF: --- Feed GVF: 0.0000

Characteristic Curves

Speed:

Flow Head Efficiency (%)

NPSH

NPSH Required --- NPSH Required based on Turton Method

NPSH Available 4806 m Enable NPSH Curves: No

NPSH Required Parameters

Suction Specific Speed --- Units Type ---

NPSH Curves

Speed:

NPSH required Capacity

Nozzle Paramaters

Base Elevation Relative to Ground Level 0.0000 m

Total_Benzene Benzene HP

Diameter (m) 5.000e-002 5.000e-002

Elevation (Base) (m) 0.0000 0.0000

Elevation (Ground) (m) 0.0000 0.0000

Inertia

Rotational inertia (kg-m2) --- Radius of gyration (m) 0.1000 Mass (kg) --- Friction loss factor
(kg-m2/s) -3277

Design Parameters

Synchronous Speed 1800 rpm

Full Load Speed ---

Full Load Torque ---

Full Load Power ---

Gear Ratio 1.000

Motor Inertia 1000 kg-m²

Motor Friction Factor 0.1000 kg-m²/s

Modelling Options (Dynamics)

Electric Brake Not Active Gearing Not Active

Start Up

Design Flow Typical Operating Capacity 10.00 m³/h

Design Speed ---

Design Power ---

PERFORMANCE

Results

Total Head --- Velocity Head -5.864 m

Total Fluid Head ---

Pressure Head 398.6 m Delta P excluding Static Head Results ---

Pump: P-2

CONNECTIONS

Inlet Stream

Stream Name From Unit Operation

Fresh Propylene

Outlet Stream

Stream Name	To Unit Operation
-------------	-------------------

Propylene HP	MIX-100 Mixer
--------------	---------------

Energy Stream

Stream Name	From Unit Operation
-------------	---------------------

Q-P2	
------	--

PARAMETERS

Adiabatic Efficiency (%): 75.00 Delta P: 2000 kPa Pressure Ratio: 2.333 Duty: 785.7 kW

Multiphase Pump Not Active

CURVES

Delta P: 2000 kPa Duty: 785.7 kW

Coefficient A: 0.0000 Coefficient B: 0.0000 Coefficient C: 0.0000 Coefficient D: 0.0000

Coefficient E: 0.0000 Coefficient F: 0.0000

Parameter Preferences Units for Head: m Flow Basis ActVolFlow Units for Flow: m3/h

User Variables

RATING

Head Offset: 0.0000 m Efficiency Offset: 0.0000

Gas Fraction Curve Collection

Gas Fraction Basis ---

CC-0

Design GVF: 0.0000 Curve GVF: --- Feed GVF: 0.0000

Characteristic Curves

Speed:

Flow Head Efficiency (%)

NPSH

NPSH Required --- NPSH Required based on Turton Method

NPSH Available 1220 m Enable NPSH Curves: No

NPSH Required Parameters

Suction Specific Speed --- Units Type ---

NPSH Curves

Speed:

NPSH required Capacity

Nozzle Paramaters

Base Elevation Relative to Ground Level 0.0000 m

Fresh Propylene Propylene HP

Diameter (m) 5.000e-002 5.000e-002

Elevation (Base) (m) 0.0000 0.0000

Elevation (Ground) (m) 0.0000 0.0000

Inertia

Rotational inertia (kg-m2) --- Radius of gyration (m) 0.1000 Mass (kg) --- Friction loss factor (kg-m2/s) -3277

Design Parameters

Synchronous Speed 1800 rpm

Full Load Speed ---

Full Load Torque ---

Full Load Power ---

Gear Ratio 1.000

Motor Inertia 1000 kg-m²

Motor Friction Factor 0.1000 kg-m²/s

Modelling Options (Dynamics)

Electric Brake Not Active Gearing Not Active

Start Up

Design Flow Typical Operating Capacity 10.00 m³/h

Design Speed ---

Design Power ---

PERFORMANCE

Results

Total Head --- Velocity Head -10.29 m

Total Fluid Head ---

Pressure Head 402.1 m Delta P excluding Static Head Results ---

Pump: P-100

CONNECTIONS

Inlet Stream

Stream Name	From Unit Operation
-------------	---------------------

Reflux-New	V-101 Separator
------------	-----------------

Outlet Stream

Stream Name	To Unit Operation
-------------	-------------------

Reflux-Temp	RCY-4 Recycle
-------------	---------------

Energy Stream

Stream Name	From Unit Operation
-------------	---------------------

Q-Reflux	
----------	--

PARAMETERS

Adiabatic Efficiency (%): 75.00 Delta P: 0.0000 kPa Pressure Ratio: 1.000 Duty: 0.0000 kW

Multiphase Pump Not Active

CURVES

Delta P: 0.0000 kPa Duty: 0.0000 kW

Coefficient A: 0.0000 Coefficient B: 0.0000 Coefficient C: 0.0000 Coefficient D: 0.0000

Coefficient E: 0.0000 Coefficient F: 0.0000

Parameter Preferences Units for Head: m Flow Basis ActVolFlow Units for Flow: m3/h

User Variables

RATING

Head Offset: 0.0000 m Efficiency Offset: 0.0000

Gas Fraction Curve Collection

Gas Fraction Basis ---

CC-0

Design GVF: 0.0000 Curve GVF: --- Feed GVF: 0.0000

Characteristic Curves

Speed:

Flow	Head	Efficiency (%)
------	------	----------------

NPSH

NPSH Required --- NPSH Required based on Turton Method

NPSH Available 3663 m Enable NPSH Curves: No

NPSH Required Parameters

Suction Specific Speed --- Units Type ---

NPSH Curves

Speed:

NPSH required	Capacity
---------------	----------

Nozzle Paramaters

Base Elevation Relative to Ground Level 0.0000 m

	Reflux-New	Reflux-Temp
--	------------	-------------

Diameter (m)	5.000e-002	5.000e-002
--------------	------------	------------

Elevation (Base) (m)	0.0000	0.0000
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Elevation (Ground) (m)	0.0000	0.0000
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Inertia

Rotational inertia (kg-m²) --- Radius of gyration (m) 0.1000 Mass (kg) --- Friction loss factor (kg-m²/s) -3277

Design Parameters

Synchronous Speed 1800 rpm

Full Load Speed ---

Full Load Torque ---

Full Load Power ---

Gear Ratio 1.000

Motor Inertia 1000 kg-m²

Motor Friction Factor 0.1000 kg-m²/s

Modelling Options (Dynamics)

Electric Brake Not Active Gearing Not Active

Start Up

Design Flow Typical Operating Capacity 10.00 m³/h

Design Speed ---

Design Power ---

PERFORMANCE

Results

Total Head --- Velocity Head -6.676e-002 m

Total Fluid Head ---

Pressure Head 0.0000 m Delta P excluding Static Head Results ---

Pump: P-101

CONNECTIONS

Inlet Stream

Stream Name	From Unit Operation
C2-Liquid	V-102 Separator

Outlet Stream

Stream Name	To Unit Operation
C2-PumpOut	TEE-100 Tee

Energy Stream

Stream Name	From Unit Operation
Q-Pump2	

PARAMETERS

Adiabatic Efficiency (%): 75.00 Delta P: 50.00 kPa Pressure Ratio: 1.500 Duty: 54.02 kW

Multiphase Pump Not Active

CURVES

Delta P: 50.00 kPa	Duty: 54.02 kW
--------------------	----------------

Coefficient A: 0.0000 Coefficient B: 0.0000 Coefficient C: 0.0000 Coefficient D: 0.0000
Coefficient E: 0.0000 Coefficient F: 0.0000

Parameter Preferences Units for Head: m Flow Basis ActVolFlow Units for Flow: m3/h

User Variables

RATING

Head Offset: 0.0000 m Efficiency Offset: 0.0000

Gas Fraction Curve Collection

Gas Fraction Basis ---

CC-0

Design GVF: 0.0000 Curve GVF: --- Feed GVF: 0.0000

Characteristic Curves

Speed:

Flow	Head	Efficiency (%)
------	------	----------------

NPSH

NPSH Required --- NPSH Required based on Turton Method

NPSH Available 8684 m Enable NPSH Curves: No

NPSH Required Parameters

Suction Specific Speed --- Units Type ---

NPSH Curves

Speed:

NPSH required	Capacity
---------------	----------

Nozzle Paramaters

Base Elevation Relative to Ground Level 0.0000 m

	C2-Liquid	C2-PumpOut
Diameter (m)	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000

Inertia

Rotational inertia (kg-m2) --- Radius of gyration (m) 0.1000 Mass (kg) --- Friction loss factor (kg-m2/s) -3277

Design Parameters

Synchronous Speed 1800 rpm

Full Load Speed ---

Full Load Torque ---

Full Load Power ---

Gear Ratio 1.000

Motor Inertia 1000 kg-m2

Motor Friction Factor 0.1000 kg-m2/s

Modelling Options (Dynamics)

Electric Brake Not Active Gearing Not Active

Start Up

Design Flow Typical Operating Capacity 10.00 m3/h

Design Speed ---

Design Power ---

PERFORMANCE

Results

Total Head --- Velocity Head -0.2953 m

Total Fluid Head ---

Pressure Head 6.073 m Delta P excluding Static Head Results ---

Pump: P-102

CONNECTIONS

Inlet Stream

Stream Name	From Unit Operation
-------------	---------------------

C3-Liquid	V-104 Separator
-----------	-----------------

Outlet Stream

Stream Name	To Unit Operation
-------------	-------------------

C3-PumpOut	TEE-101 Tee
------------	-------------

Energy Stream

Stream Name	From Unit Operation
-------------	---------------------

Q-Pump3	
---------	--

PARAMETERS

Adiabatic Efficiency (%): 75.00 Delta P: 50.00 kPa Pressure Ratio: 1.500 Duty: 12.23 kW

Multiphase Pump Not Active

CURVES

Delta P: 50.00 kPa Duty: 12.23 kW

Coefficient A: 0.0000 Coefficient B: 0.0000 Coefficient C: 0.0000 Coefficient D: 0.0000

Coefficient E: 0.0000 Coefficient F: 0.0000

Parameter Preferences Units for Head: m Flow Basis ActVolFlow Units for Flow: m3/h

User Variables

RATING

Head Offset: 0.0000 m Efficiency Offset: 0.0000

Gas Fraction Curve Collection

Gas Fraction Basis ---

CC-0

Design GVF: 0.0000 Curve GVF: --- Feed GVF: 0.0000

Characteristic Curves

Speed:

Flow	Head	Efficiency (%)
------	------	----------------

NPSH

NPSH Required --- NPSH Required based on Turton Method

NPSH Available 457.1 m Enable NPSH Curves: No

NPSH Required Parameters

Suction Specific Speed --- Units Type ---

NPSH Curves

Speed:

NPSH required Capacity

Nozzle Paramaters

Base Elevation Relative to Ground Level 0.0000 m

C3-Liquid C3-PumpOut

Diameter (m) 5.000e-002 5.000e-002

Elevation (Base) (m) 0.0000 0.0000

Elevation (Ground) (m) 0.0000 0.0000

Inertia

Rotational inertia (kg-m2) --- Radius of gyration (m) 0.1000 Mass (kg) --- Friction loss factor (kg-m2/s) -3277

Design Parameters

Synchronous Speed 1800 rpm

Full Load Speed ---

Full Load Torque ---

Full Load Power ---

Gear Ratio 1.000

Motor Inertia 1000 kg-m2

Motor Friction Factor 0.1000 kg-m2/s

Modelling Options (Dynamics)

Electric Brake Not Active Gearing Not Active

Start Up

Design Flow Typical Operating Capacity 10.00 m3/h

Design Speed ---

Design Power ---

PERFORMANCE

Results

Total Head --- Velocity Head -3.162e-002 m

Total Fluid Head ---

Pressure Head 6.092 m Delta P excluding Static Head Results ---

Pump: P-103

CONNECTIONS

Inlet Stream

Stream Name	From Unit Operation
-------------	---------------------

Cooled-Trans_Mixed	E-107 Cooler
--------------------	--------------

Outlet Stream

Stream Name	To Unit Operation
-------------	-------------------

P6-Out	E-108 Heater
--------	--------------

Energy Stream

Stream Name From Unit Operation

Q-P6

PARAMETERS

Adiabatic Efficiency (%): 75.00 Delta P: 2380 kPa Pressure Ratio: 20.83 Duty: 2768 kW

Multiphase Pump Not Active

CURVES

Delta P: 2380 kPa Duty: 2768 kW

Coefficient A: 0.0000 Coefficient B: 0.0000 Coefficient C: 0.0000 Coefficient D: 0.0000

Coefficient E: 0.0000 Coefficient F: 0.0000

Parameter Preferences Units for Head: m Flow Basis ActVolFlow Units for Flow: m3/h

User Variables

RATING

Head Offset: 0.0000 m Efficiency Offset: 0.0000

Gas Fraction Curve Collection

Gas Fraction Basis ---

CC-0

Design GVF: 0.0000 Curve GVF: --- Feed GVF: 0.0000

Characteristic Curves

Speed:

Flow	Head	Efficiency (%)
------	------	----------------

NPSH

NPSH Required --- NPSH Required based on Turton Method

NPSH Available 1.006e+004 m Enable NPSH Curves: No

NPSH Required Parameters

Suction Specific Speed --- Units Type ---

NPSH Curves

Speed:

NPSH required Capacity

Nozzle Paramaters

Base Elevation Relative to Ground Level 0.0000 m

Cooled-Trans_Mixed P6-Out

Diameter (m) 5.000e-002 5.000e-002

Elevation (Base) (m) 0.0000 0.0000

Elevation (Ground) (m) 0.0000 0.0000

Inertia

Rotational inertia (kg-m2) --- Radius of gyration (m) 0.1000 Mass (kg) --- Friction loss factor (kg-m2/s) -3277

Design Parameters

Synchronous Speed 1800 rpm

Full Load Speed ---

Full Load Torque ---

Full Load Power ---

Gear Ratio 1.000

Motor Inertia 1000 kg-m²

Motor Friction Factor 0.1000 kg-m²/s

Modelling Options (Dynamics)

Electric Brake Not Active Gearing Not Active

Start Up

Design Flow Typical Operating Capacity 10.00 m³/h

Design Speed ---

Design Power ---

PERFORMANCE

Results

Total Head --- Velocity Head -36.68 m

Total Fluid Head ---

Pressure Head 296.6 m Delta P excluding Static Head Results ---

Mixer: MIX-100

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
-------------	---------------------

Benzene HP	P-1 Pump
------------	----------

Propylene HP P-2 Pump

Outlet Stream

STREAM NAME TO UNIT OPERATION

Mixed Feed E-100 Heater

PARAMETERS

User Variables

NOZZLE PARAMETERS

Base Elevation Relative to Ground Level 0.0000 m

	Benzene HP	Propylene HP	Mixed Feed
Diameter (m)	5.000e-002	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000	0.0000

Mixer: MIX-101

CONNECTIONS

Inlet Stream

STREAM NAME FROM UNIT OPERATION

DIPB_To_R2 V-105 Separator

Transalkylation Recycle TEE-102 Tee

Outlet Stream

STREAM NAME	TO UNIT OPERATION
-------------	-------------------

Trans_Mixed	E-107 Cooler
-------------	--------------

PARAMETERS

User Variables

NOZZLE PARAMETERS

Base Elevation Relative to Ground Level 0.0000 m

	DIPB_To_R2	Transalkylation Recycle	Trans_Mixed
Diameter (m)	5.000e-002	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000	0.0000

Mixer: MIX-102

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
-------------	---------------------

C2-Feed	V-100 Separator
---------	-----------------

Trans_Recycled RCY-1 Recycle
Outlet Stream

STREAM NAME TO UNIT OPERATION
C2-Feed_New VLV-101 Valve
PARAMETERS

User Variables

NOZZLE PARAMETERS

Base Elevation Relative to Ground Level 0.0000 m

	C2-Feed	Trans_Recycled	C2-Feed_New
Diameter (m)	5.000e-002	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000	0.0000

Mixer: MIX-103

CONNECTIONS

Inlet Stream

STREAM NAME FROM UNIT OPERATION
Alkylation Recycled RCY-2 Recycle

Fresh Benzene

Outlet Stream

STREAM NAME	TO UNIT OPERATION
-------------	-------------------

Total_Benzene	P-1 Pump
---------------	----------

PARAMETERS

User Variables

NOZZLE PARAMETERS

Base Elevation Relative to Ground Level 0.0000 m

	Alkylation	Recycled	Fresh Benzene	Total_Benzene
Diameter (m)	5.000e-002	5.000e-002	5.000e-002	
Elevation (Base) (m)	0.0000	0.0000	0.0000	
Elevation (Ground) (m)	0.0000	0.0000	0.0000	

Heater: E-100

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
-------------	---------------------

Mixed Feed	MIX-100 Mixer
------------	---------------

Outlet Stream

STREAM NAME	TO UNIT OPERATION
-------------	-------------------

Reactor Feed	R-1 Plug Flow Reactor
--------------	-----------------------

Energy Stream

STREAM NAME	FROM UNIT OPERATION
-------------	---------------------

Q-Heater

PARAMETERS

Pressure Drop: 50.00 kPa Duty: 5.723e+008 kJ/h Volume: 0.1000 m³

Function: Not Selected Zones: 1

User Variables

RATING

NOZZLE PARAMETERS

Base Elevation Relative to Ground Level 0.0000 m

	Mixed Feed	Reactor Feed
Diameter (m)	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000

PERFORMANCE PROFILES

Zone	Pressure	Temperature	Vapour Fraction	Enthalpy
	(kPa)	(C)		(kJ/kgmole)

0.0000	0.0000	---
0.0000	0.0000	---
0.0000	0.0000	---
0.0000	0.0000	---
0.0000	0.0000	---

Vapour Phase

Mass Flow	Molecular Wt	Density	Mass Sp Heat	Viscosity	Thermal Cond
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	

---	---	---	---	---	---
---	---	---	---	---	---
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---	---	---	---	---	---

Std Gas Flow	Z Factor	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
(STD_m3/h)	(kPa)	(C)			

---	---	---	---	---	---
---	---	---	---	---	---
---	---	---	---	---	---
---	---	---	---	---	---
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---	---	---	---	---	---
---	---	---	---	---	---

Light Liquid Phase

Mass Flow (kg/h)	Density (kg/m3)	Mass Sp Heat (kJ/kg-C)	Viscosity (cP)	Thermal Cond (W/m-K)	Surface Tens (dyne/cm)
2395326.93	755.82	1.78	0.30	0.12	18.41
2395326.93	742.63	1.84	0.27	0.11	17.05
2395326.93	729.07	1.90	0.26	0.11	15.73
2395326.93	715.08	1.96	0.23	0.10	14.46
2395326.93	700.57	2.02	0.19	0.10	13.24
2395326.93	685.45	2.09	0.16	0.09	12.16
2395326.93	669.55	2.16	0.14	0.09	11.33
2395326.93	652.66	2.24	0.13	0.09	10.51
2395326.93	634.42	2.32	0.11	0.09	9.71
2395326.93	616.22	2.41	0.10	0.09	8.92
2395326.93	597.16	2.52	0.09	0.09	8.14

Molecular Wt	Sp Gravity (kPa)	Pseudo Pc (C)	Pseudo Tc	Pseudo Zc	Pseudo Omega
67.91	0.76	4673.98	221.00	0.27	0.20
67.91	0.74	4673.98	221.00	0.27	0.20
67.91	0.73	4673.98	221.00	0.27	0.20
67.91	0.72	4673.98	221.00	0.27	0.20

67.91	0.70	4673.98	221.00	0.27	0.20
67.91	0.69	4673.98	221.00	0.27	0.20
67.91	0.67	4673.98	221.00	0.27	0.20
67.91	0.65	4673.98	221.00	0.27	0.20
67.91	0.63	4673.98	221.00	0.27	0.20
67.91	0.62	4673.98	221.00	0.27	0.20
67.91	0.60	4673.98	221.00	0.27	0.20

Heavy Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)

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Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			

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Mixed Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)

2395326.93	755.82	1.78	0.30	0.12	18.41
2395326.93	742.63	1.84	0.27	0.11	17.05
2395326.93	729.07	1.90	0.26	0.11	15.73
2395326.93	715.08	1.96	0.23	0.10	14.46
2395326.93	700.57	2.02	0.19	0.10	13.24
2395326.93	685.45	2.09	0.16	0.09	12.16
2395326.93	669.55	2.16	0.14	0.09	11.33
2395326.93	652.66	2.24	0.13	0.09	10.51
2395326.93	634.42	2.32	0.11	0.09	9.71
2395326.93	616.22	2.41	0.10	0.09	8.92
2395326.93	597.16	2.52	0.09	0.09	8.14

Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			

67.91	0.76	4673.98	221.00	0.27	0.20
67.91	0.74	4673.98	221.00	0.27	0.20

67.91	0.73	4673.98	221.00	0.27	0.20
67.91	0.72	4673.98	221.00	0.27	0.20
67.91	0.70	4673.98	221.00	0.27	0.20
67.91	0.69	4673.98	221.00	0.27	0.20
67.91	0.67	4673.98	221.00	0.27	0.20
67.91	0.65	4673.98	221.00	0.27	0.20
67.91	0.63	4673.98	221.00	0.27	0.20
67.91	0.62	4673.98	221.00	0.27	0.20
67.91	0.60	4673.98	221.00	0.27	0.20

Heater: E-101

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
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C1-Bot	T-100 Absorber
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Outlet Stream

STREAM NAME	TO UNIT OPERATION
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Heated_Bot	V-100 Separator
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Energy Stream

STREAM NAME	FROM UNIT OPERATION
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Q-Reboiler

PARAMETERS

Pressure Drop: 0.0000 kPa Duty: 6.565e+008 kJ/h Volume: 0.1000 m3

Function: Not Selected Zones: 1

User Variables

RATING

NOZZLE PARAMETERS

Base Elevation Relative to Ground Level 0.0000 m

	C1-Bot	Heated_Bot
Diameter (m)	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000

PERFORMANCE PROFILES

Zone	Pressure (kPa)	Temperature (C)	Vapour Fraction	Enthalpy (kJ/kgmole)
Inlet	1500.00	169.50	0.0000	14854.33
0	1500.00	219.32	0.3597	33019.35

PERFORMANCE TABLE

Overall Phase

Temperature	Pressure	Heat Flow	Enthalpy
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(C)	(kPa)	(kJ/h)	(kJ/kgmole)
169.50	1500.00	0.00	14854.33
174.48	1500.00	51264072.91	16272.70
179.46	1500.00	104323040.88	17740.73
184.45	1500.00	159609188.87	19270.39
189.43	1500.00	217636215.70	20875.88
194.41	1500.00	279000423.13	22573.70
199.39	1500.00	344362860.74	24382.15
204.37	1500.00	414391981.48	26319.71
209.36	1500.00	489648376.75	28401.90
214.34	1500.00	570415646.14	30636.57
219.32	1500.00	656536500.11	33019.35

Vapour Fraction	Vap Phase Mass Frac	Heat of Vap
		(kJ/kgmole)

0.0000	0.0000	---
0.0211	0.0137	---
0.0438	0.0290	---
0.0685	0.0466	---
0.0960	0.0669	---
0.1269	0.0909	---
0.1622	0.1192	---
0.2026	0.1528	---
0.2488	0.1927	---
0.3013	0.2393	---
0.3597	0.2929	---

Vapour Phase

Mass Flow	Molecular Wt	Density	Mass Sp Heat	Viscosity	Thermal Cond
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	

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46306.72	60.60	28.62	2.06	0.01	0.03
98304.47	62.11	29.09	2.05	0.01	0.03
157724.55	63.70	29.61	2.05	0.01	0.03
226672.82	65.36	30.17	2.04	0.01	0.03
307662.52	67.08	30.76	2.04	0.01	0.03
403584.85	68.86	31.38	2.04	0.01	0.03
517536.86	70.69	32.02	2.04	0.01	0.03
652423.49	72.54	32.68	2.04	0.01	0.03
810322.90	74.41	33.33	2.04	0.01	0.03
991806.87	76.29	33.98	2.05	0.01	0.03

Std Gas Flow	Z Factor	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
(STD_m3/h)		(kPa)	(C)		

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18066.59	0.85	4449.04	180.61	0.28	0.18
37421.89	0.85	4460.56	187.87	0.28	0.19
58546.61	0.85	4471.48	195.41	0.28	0.19
82004.57	0.84	4481.42	203.20	0.28	0.19
108443.18	0.84	4489.86	211.15	0.27	0.20
138574.21	0.84	4496.17	219.19	0.27	0.20
173111.96	0.83	4499.53	227.21	0.27	0.21
212649.30	0.83	4499.04	235.07	0.27	0.21
257476.38	0.83	4493.73	242.63	0.27	0.21
307405.78	0.82	4482.76	249.79	0.27	0.22

Light Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)
3386418.93	686.66	2.34	0.14	0.09	10.59
3340112.22	683.60	2.36	0.14	0.09	10.29
3288114.46	680.40	2.38	0.13	0.09	9.98
3228694.38	677.09	2.40	0.13	0.09	9.66
3159746.11	673.66	2.42	0.13	0.09	9.34
3078756.41	670.16	2.44	0.12	0.09	9.02
2982834.08	666.58	2.47	0.12	0.09	8.70
2868882.07	662.94	2.49	0.12	0.09	8.39
2733995.44	659.25	2.51	0.11	0.08	8.08
2576096.03	655.53	2.54	0.11	0.08	7.79
2394612.06	651.78	2.56	0.11	0.08	7.50

Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			
93.70	0.69	4147.54	296.68	0.27	0.26
94.41	0.68	4141.03	299.19	0.27	0.26
95.14	0.68	4133.20	301.66	0.27	0.26
95.90	0.68	4123.71	304.13	0.27	0.26
96.70	0.67	4112.10	306.60	0.27	0.26
97.56	0.67	4097.78	309.11	0.27	0.27
98.50	0.67	4080.06	311.68	0.27	0.27
99.54	0.66	4058.12	314.33	0.27	0.27
100.70	0.66	4031.10	317.09	0.27	0.27
102.01	0.66	3998.26	319.98	0.27	0.28
103.48	0.65	3959.21	323.02	0.27	0.28

Heavy Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m ³)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)

[illegible]

Molecular Wt Sp Gravity Pseudo Pc Pseudo Tc Pseudo Zc Pseudo Omega
(kPa) (C)

[illegible]

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Mixed Liquid Phase

Mass Flow Density Mass Sp Heat Viscosity Thermal Cond Surface Tens
 (kg/h) (kg/m3) (kJ/kg-C) (cP) (W/m-K) (dyne/cm)

3386418.93	686.66	2.34	0.14	0.09	10.59
3340112.22	683.60	2.36	0.14	0.09	10.29
3288114.46	680.40	2.38	0.13	0.09	9.98
3228694.38	677.09	2.40	0.13	0.09	9.66
3159746.11	673.66	2.42	0.13	0.09	9.34
3078756.41	670.16	2.44	0.12	0.09	9.02
2982834.08	666.58	2.47	0.12	0.09	8.70
2868882.07	662.94	2.49	0.12	0.09	8.39
2733995.44	659.25	2.51	0.11	0.08	8.08
2576096.03	655.53	2.54	0.11	0.08	7.79
2394612.06	651.78	2.56	0.11	0.08	7.50

Molecular Wt Sp Gravity Pseudo Pc Pseudo Tc Pseudo Zc Pseudo Omega
 (kPa) (C)

93.70	0.69	4147.54	296.68	0.27	0.26
94.41	0.68	4141.03	299.19	0.27	0.26
95.14	0.68	4133.20	301.66	0.27	0.26
95.90	0.68	4123.71	304.13	0.27	0.26
96.70	0.67	4112.10	306.60	0.27	0.26
97.56	0.67	4097.78	309.11	0.27	0.27
98.50	0.67	4080.06	311.68	0.27	0.27
99.54	0.66	4058.12	314.33	0.27	0.27
100.70	0.66	4031.10	317.09	0.27	0.27

102.01	0.66	3998.26	319.98	0.27	0.28
103.48	0.65	3959.21	323.02	0.27	0.28

Heater: E-104

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
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C2-Bot	T-101 Absorber
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Outlet Stream

STREAM NAME	TO UNIT OPERATION
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C2-Heated_Bot	V-103 Separator
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Energy Stream

STREAM NAME	FROM UNIT OPERATION
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Q-Reb2

PARAMETERS

Pressure Drop: 0.0000 kPa Duty: 3.380e+008 kJ/h Volume: 0.1000 m3

Function: Not Selected Zones: 1

User Variables

RATING

NOZZLE PARAMETERS

Base Elevation Relative to Ground Level 0.0000 m

	C2-Bot	C2-Heated_Bot
Diameter (m)	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000

PERFORMANCE PROFILES

Zone	Pressure (kPa)	Temperature (C)	Vapour Fraction	Enthalpy (kJ/kgmole)
Inlet	120.00	169.49	0.0000	-25692.30
0	120.00	172.66	0.3526	-11673.35

PERFORMANCE TABLE

Overall Phase

Temperature (C)	Pressure (kPa)	Heat Flow (kJ/h)	Enthalpy (kJ/kgmole)
169.49	120.00	0.00	-25692.30
169.81	120.00	41708968.89	-23962.21
170.13	120.00	81184049.62	-22324.79
170.44	120.00	118641254.85	-20771.07
170.76	120.00	154271552.41	-19293.13
171.08	120.00	188244009.04	-17883.95

171.40	120.00	220708727.94	-16537.32
171.71	120.00	251799406.60	-15247.68
172.03	120.00	281635522.37	-14010.08
172.35	120.00	310324191.73	-12820.08
172.66	120.00	337961752.25	-11673.67

Vapour Fraction	Vap Phase Mass Frac	Heat of Vap
	(kJ/kgmole)	

0.0000	0.0000	---
0.0442	0.0418	---
0.0859	0.0812	---
0.1253	0.1186	---
0.1626	0.1541	---
0.1981	0.1880	---
0.2319	0.2203	---
0.2641	0.2512	---
0.2948	0.2808	---
0.3243	0.3092	---
0.3526	0.3366	---

Vapour Phase

Mass Flow	Molecular Wt	Density	Mass Sp Heat	Viscosity	Thermal Cond
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	
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132106.25	123.96	4.23	1.86	0.01	0.02
256968.42	124.10	4.23	1.86	0.01	0.02
375291.17	124.24	4.23	1.87	0.01	0.02
487697.71	124.39	4.23	1.87	0.01	0.02

594739.94	124.53	4.24	1.87	0.01	0.02
696908.02	124.68	4.24	1.87	0.01	0.02
794638.71	124.83	4.24	1.87	0.01	0.02
888322.44	124.98	4.24	1.87	0.01	0.02
978309.42	125.13	4.24	1.87	0.01	0.02
1064914.75	125.28	4.25	1.88	0.01	0.02

Std Gas Flow	Z Factor	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
(STD_m3/h)		(kPa)	(C)		

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25198.34	0.96	3140.92	363.18	0.26	0.34
48958.92	0.96	3138.36	363.37	0.26	0.34
71420.15	0.96	3135.78	363.57	0.26	0.34
92704.34	0.96	3133.18	363.77	0.26	0.34
112919.78	0.96	3130.56	363.97	0.26	0.34
132162.58	0.96	3127.93	364.17	0.26	0.34
150518.36	0.96	3125.27	364.37	0.26	0.34
168063.60	0.96	3122.59	364.57	0.26	0.34
184866.85	0.96	3119.89	364.78	0.26	0.34
200989.78	0.96	3117.16	364.99	0.26	0.34

Light Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)

3163940.29	730.92	2.28	0.19	0.11	14.27
3031834.03	730.74	2.28	0.19	0.11	14.26
2906971.87	730.56	2.28	0.19	0.11	14.25
2788649.12	730.39	2.28	0.19	0.11	14.24

2676242.58	730.21	2.28	0.19	0.11	14.23
2569200.35	730.02	2.28	0.19	0.11	14.21
2467032.26	729.84	2.29	0.19	0.11	14.20
2369301.58	729.66	2.29	0.19	0.11	14.19
2275617.85	729.47	2.29	0.19	0.11	14.17
2185630.87	729.28	2.29	0.19	0.11	14.16
2099025.53	729.09	2.29	0.19	0.11	14.15

Molecular Wt Sp Gravity Pseudo Pc Pseudo Tc Pseudo Zc Pseudo Omega
(kPa) (C)

131.24	0.73	3009.63	373.19	0.26	0.35
131.58	0.73	3003.55	373.65	0.26	0.35
131.91	0.73	2997.53	374.11	0.26	0.35
132.24	0.73	2991.55	374.56	0.26	0.35
132.57	0.73	2985.63	375.01	0.26	0.35
132.90	0.73	2979.75	375.46	0.26	0.35
133.22	0.73	2973.92	375.91	0.26	0.35
133.54	0.73	2968.13	376.35	0.26	0.35
133.86	0.73	2962.39	376.79	0.26	0.35
134.17	0.73	2956.70	377.22	0.26	0.36
134.49	0.73	2951.06	377.65	0.26	0.36

Heavy Liquid Phase

Mass Flow Density Mass Sp Heat Viscosity Thermal Cond Surface Tens
(kg/h) (kg/m3) (kJ/kg-C) (cP) (W/m-K) (dyne/cm)

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Molecular Wt Sp Gravity Pseudo Pc Pseudo Tc Pseudo Zc Pseudo Omega
(kPa) (C)

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Mixed Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)
3163940.29	730.92	2.28	0.19	0.11	14.27
3031834.03	730.74	2.28	0.19	0.11	14.26

2906971.87	730.56	2.28	0.19	0.11	14.25
2788649.12	730.39	2.28	0.19	0.11	14.24
2676242.58	730.21	2.28	0.19	0.11	14.23
2569200.35	730.02	2.28	0.19	0.11	14.21
2467032.26	729.84	2.29	0.19	0.11	14.20
2369301.58	729.66	2.29	0.19	0.11	14.19
2275617.85	729.47	2.29	0.19	0.11	14.17
2185630.87	729.28	2.29	0.19	0.11	14.16
2099025.53	729.09	2.29	0.19	0.11	14.15

Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			

131.24	0.73	3009.63	373.19	0.26	0.35
131.58	0.73	3003.55	373.65	0.26	0.35
131.91	0.73	2997.53	374.11	0.26	0.35
132.24	0.73	2991.55	374.56	0.26	0.35
132.57	0.73	2985.63	375.01	0.26	0.35
132.90	0.73	2979.75	375.46	0.26	0.35
133.22	0.73	2973.92	375.91	0.26	0.35
133.54	0.73	2968.13	376.35	0.26	0.35
133.86	0.73	2962.39	376.79	0.26	0.35
134.17	0.73	2956.70	377.22	0.26	0.36
134.49	0.73	2951.06	377.65	0.26	0.36

Heater: E-106

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
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C3-Bot	T-102 Absorber
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Outlet Stream

STREAM NAME	TO UNIT OPERATION
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C3-Heated_Bot	V-105 Separator
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Energy Stream

STREAM NAME	FROM UNIT OPERATION
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Q-reb3	
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PARAMETERS

Pressure Drop: 0.0000 kPa Duty: 2.041e+008 kJ/h Volume: 0.1000 m3

Function: Not Selected Zones: 1

User Variables

RATING

NOZZLE PARAMETERS

Base Elevation Relative to Ground Level 0.0000 m

	C3-Bot	C3-Heated_Bot
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Diameter (m)	5.000e-002	5.000e-002
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Elevation (Base) (m)	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000

PERFORMANCE PROFILES

Zone	Pressure (kPa)	Temperature (C)	Vapour Fraction	Enthalpy (kJ/kgmole)
Inlet	120.00	173.21	0.0000	-31404.80
0	120.00	176.59	0.2876	-19667.14

PERFORMANCE TABLE

Overall Phase

Temperature (C)	Pressure (kPa)	Heat Flow (kJ/h)	Enthalpy (kJ/kgmole)
173.21	120.00	0.00	-31404.80
173.55	120.00	23608174.65	-30047.16
173.89	120.00	46346497.36	-28739.54
174.22	120.00	68284515.27	-27477.95
174.56	120.00	89485364.29	-26258.75
174.90	120.00	110006514.15	-25078.63
175.24	120.00	129900416.21	-23934.59
175.57	120.00	149215065.75	-22823.86
175.91	120.00	167994491.57	-21743.91
176.25	120.00	186279183.85	-20692.41
176.59	120.00	204106469.07	-19667.21
Vapour Fraction	Vap Phase Mass Frac	Heat of Vap	
	(kJ/kgmole)		

0.0000	0.0000	---
0.0338	0.0315	---
0.0662	0.0617	---
0.0974	0.0909	---
0.1274	0.1191	---
0.1563	0.1463	---
0.1843	0.1727	---
0.2113	0.1983	---
0.2375	0.2232	---
0.2629	0.2474	---
0.2876	0.2710	---

Vapour Phase

Mass Flow	Molecular Wt	Density	Mass Sp Heat	Viscosity	Thermal Cond
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	
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73883.66	125.70	4.25	1.88	0.01	0.02
144965.87	125.87	4.25	1.88	0.01	0.02
213474.32	126.04	4.26	1.88	0.01	0.02
279615.82	126.20	4.26	1.89	0.01	0.02
343578.76	126.37	4.26	1.89	0.01	0.02
405535.20	126.55	4.26	1.89	0.01	0.02
465642.71	126.72	4.27	1.89	0.01	0.02
524046.00	126.89	4.27	1.89	0.01	0.02
580878.26	127.07	4.27	1.89	0.01	0.02
636262.40	127.24	4.28	1.90	0.01	0.02

Std Gas Flow Z Factor Pseudo Pc Pseudo Tc Pseudo Zc Pseudo Omega

(STD_m3/h)			(kPa)	(C)		
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13897.23	0.96		3109.45	365.57	0.26	0.34
27231.65	0.96		3106.47	365.80	0.26	0.34
40047.76	0.96		3103.45	366.03	0.26	0.34
52385.93	0.96		3100.42	366.26	0.26	0.35
64282.90	0.96		3097.36	366.50	0.26	0.35
75772.20	0.96		3094.27	366.73	0.26	0.35
86884.48	0.96		3091.16	366.97	0.26	0.35
97647.87	0.96		3088.02	367.21	0.26	0.35
108088.22	0.96		3084.86	367.45	0.26	0.35
118229.32	0.96		3081.67	367.69	0.26	0.35

Light Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)
2347912.27	728.76	2.29	0.19	0.11	14.13
2274028.61	728.56	2.29	0.19	0.11	14.11
2202946.40	728.35	2.30	0.19	0.11	14.10
2134437.95	728.14	2.30	0.19	0.11	14.09
2068296.45	727.93	2.30	0.19	0.11	14.07
2004333.51	727.72	2.30	0.19	0.11	14.06
1942377.07	727.50	2.30	0.18	0.11	14.04
1882269.55	727.29	2.30	0.18	0.11	14.03
1823866.27	727.07	2.30	0.18	0.11	14.02
1767034.01	726.86	2.30	0.18	0.11	14.00
1711649.87	726.64	2.31	0.18	0.11	13.99

Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			
135.02	0.73	2941.43	378.38	0.26	0.36
135.35	0.73	2935.55	378.83	0.26	0.36
135.67	0.73	2929.72	379.28	0.26	0.36
135.99	0.73	2923.94	379.72	0.26	0.36
136.31	0.73	2918.21	380.15	0.26	0.36
136.62	0.73	2912.53	380.59	0.26	0.36
136.94	0.73	2906.90	381.02	0.26	0.36
137.25	0.73	2901.31	381.44	0.26	0.36
137.55	0.73	2895.77	381.87	0.26	0.36
137.86	0.73	2890.27	382.28	0.26	0.36
138.16	0.73	2884.82	382.70	0.26	0.36

Heavy Liquid Phase

[illegible]

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Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			

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Mixed Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)
2347912.27	728.76	2.29	0.19	0.11	14.13
2274028.61	728.56	2.29	0.19	0.11	14.11
2202946.40	728.35	2.30	0.19	0.11	14.10
2134437.95	728.14	2.30	0.19	0.11	14.09
2068296.45	727.93	2.30	0.19	0.11	14.07
2004333.51	727.72	2.30	0.19	0.11	14.06
1942377.07	727.50	2.30	0.18	0.11	14.04
1882269.55	727.29	2.30	0.18	0.11	14.03
1823866.27	727.07	2.30	0.18	0.11	14.02

1767034.01	726.86	2.30	0.18	0.11	14.00
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1711649.87	726.64	2.31	0.18	0.11	13.99
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Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			

135.02	0.73	2941.43	378.38	0.26	0.36
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135.35	0.73	2935.55	378.83	0.26	0.36
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135.67	0.73	2929.72	379.28	0.26	0.36
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135.99	0.73	2923.94	379.72	0.26	0.36
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136.31	0.73	2918.21	380.15	0.26	0.36
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136.62	0.73	2912.53	380.59	0.26	0.36
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136.94	0.73	2906.90	381.02	0.26	0.36
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137.25	0.73	2901.31	381.44	0.26	0.36
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137.55	0.73	2895.77	381.87	0.26	0.36
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137.86	0.73	2890.27	382.28	0.26	0.36
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138.16	0.73	2884.82	382.70	0.26	0.36
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Heater: E-108

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
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P6-Out	P-103 Pump
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Outlet Stream

STREAM NAME TO UNIT OPERATION

Trans Feed CRV-100 Conversion Reactor

Energy Stream

STREAM NAME FROM UNIT OPERATION

Q-E9

PARAMETERS

Pressure Drop: 0.0000 kPa Duty: 7.134e+008 kJ/h Volume: 0.1000 m3

Function: Not Selected Zones: 1

User Variables

RATING

NOZZLE PARAMETERS

Base Elevation Relative to Ground Level 0.0000 m

	P6-Out	Trans Feed
Diameter (m)	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000

PERFORMANCE PROFILES

Zone	Pressure (kPa)	Temperature (C)	Vapour Fraction	Enthalpy (kJ/kgmole)
Inlet	2500.00	81.00	0.0000	-22846.74

0	2500.00	210.00	0.0000	9433.83
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PERFORMANCE TABLE

Overall Phase

Temperature	Pressure	Heat Flow	Enthalpy
(C)	(kPa)	(kJ/h)	(kJ/kgmole)
81.00	2500.00	0.00	-22846.74
93.90	2500.00	63091967.81	-19991.99
106.80	2500.00	127995856.69	-17055.26
119.70	2500.00	194709528.93	-14036.65
132.60	2500.00	263235033.06	-10936.05
145.50	2500.00	333579642.83	-7753.14
158.40	2500.00	405757201.47	-4487.30
171.30	2500.00	479789905.87	-1137.52
184.20	2500.00	555710736.87	2297.70
197.10	2500.00	633566862.83	5820.48
210.00	2500.00	713424555.07	9433.83

[illegible]

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Vapour Phase

Mass Flow	Molecular Wt	Density	Mass Sp Heat	Viscosity	Thermal Cond
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	

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Std Gas Flow Z Factor	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
(STD_m3/h)	(kPa)	(C)		

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Light Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)

2568918.85	819.68	1.88	0.38	0.13	21.46
2568918.85	807.69	1.93	0.34	0.12	20.17
2568918.85	795.49	1.99	0.30	0.12	18.91
2568918.85	783.06	2.04	0.27	0.12	17.66
2568918.85	770.37	2.10	0.24	0.11	16.43
2568918.85	757.38	2.15	0.22	0.11	15.21
2568918.85	744.03	2.21	0.19	0.11	14.02
2568918.85	730.28	2.26	0.18	0.10	12.84
2568918.85	716.05	2.32	0.16	0.10	11.69
2568918.85	701.38	2.38	0.14	0.10	10.55
2568918.85	686.18	2.44	0.13	0.09	9.45

Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			

116.24	0.82	3566.96	346.68	0.27	0.31
116.24	0.81	3566.96	346.68	0.27	0.31
116.24	0.80	3566.96	346.68	0.27	0.31
116.24	0.78	3566.96	346.68	0.27	0.31
116.24	0.77	3566.96	346.68	0.27	0.31

116.24	0.76	3566.96	346.68	0.27	0.31
116.24	0.74	3566.96	346.68	0.27	0.31
116.24	0.73	3566.96	346.68	0.27	0.31
116.24	0.72	3566.96	346.68	0.27	0.31
116.24	0.70	3566.96	346.68	0.27	0.31
116.24	0.69	3566.96	346.68	0.27	0.31

Heavy Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)

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Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			

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Mixed Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)
2568918.85	819.68	1.88	0.38	0.13	21.46
2568918.85	807.69	1.93	0.34	0.12	20.17
2568918.85	795.49	1.99	0.30	0.12	18.91
2568918.85	783.06	2.04	0.27	0.12	17.66
2568918.85	770.37	2.10	0.24	0.11	16.43
2568918.85	757.38	2.15	0.22	0.11	15.21
2568918.85	744.03	2.21	0.19	0.11	14.02
2568918.85	730.28	2.26	0.18	0.10	12.84
2568918.85	716.05	2.32	0.16	0.10	11.69
2568918.85	701.38	2.38	0.14	0.10	10.55
2568918.85	686.18	2.44	0.13	0.09	9.45

Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			
116.24	0.82	3566.96	346.68	0.27	0.31
116.24	0.81	3566.96	346.68	0.27	0.31
116.24	0.80	3566.96	346.68	0.27	0.31

116.24	0.78	3566.96	346.68	0.27	0.31
116.24	0.77	3566.96	346.68	0.27	0.31
116.24	0.76	3566.96	346.68	0.27	0.31
116.24	0.74	3566.96	346.68	0.27	0.31
116.24	0.73	3566.96	346.68	0.27	0.31
116.24	0.72	3566.96	346.68	0.27	0.31
116.24	0.70	3566.96	346.68	0.27	0.31
116.24	0.69	3566.96	346.68	0.27	0.31

Plug Flow Reactor: R-1

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
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Reactor Feed	E-100 Heater
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Outlet Stream

STREAM NAME	TO UNIT OPERATION
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Alkylation Effluents	VLV-100 Valve
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Energy Stream

STREAM NAME	TO UNIT OPERATION
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Q-R1

PARAMETERS

Physical Parameters

Type : User Specified Pressure Drop: 50.00 kPa

Heat Transfer

Type : Direct Q Value Energy Stream : Q-R1 Duty : 1.185e+009 kJ/h

Dimensions

Total Volume: 157.1 m³ Length: 8.000 m Diameter: 5.000 m Number of Tubes: 1

Wall Thickness: 5.000e-003 m Void Fraction: 0.4500 Void Volume: 70.69 m³

Reaction Info

Reaction Set: Set-1 Initialize From: Current

Integration Information

Number of Segments: 100 Minimum Step Fraction: 1.0e-06 Minimum Step Length: 8.0e-06 m

Catalyst Data

Particle Diameter: 1.000e-003 m Particle Sphericity: 1.0000 Solid Density: 2500 kg/m³

Bulk Density: 1375 kg/m³ Solid Heat Capacity: 250.0 kJ/kg-C

User Variables

RATING

Sizing

Tube Dimensions

Total Volume 157.1 m³ Length 8.000 m Diameter 5.000 m Number of Tubes 1 Wall Thickness 5.000e-003 m

Tube Packing

Void Fraction 0.4500 Void Volume 70.69 m³

Nozzle Parameters

Base Elevation Relative to Ground Level 0.0000 m Diameter 5.000 m Length 8.000

Alkylation Effluents Reactor Feed

Diameter (m) 5.000e-002 5.000e-002

Elevation (Base) (m) 0.0000 0.0000

Elevation (Ground) (m) 0.0000 0.0000

Conditions

Length	Temperature	Pressure	Vapour Fraction Duty	
(m)	(C)	(kPa)		(kJ/h)
0.040	150.0	3449.50	0.0000	1184552178
0.120	150.0	3449.00	0.0000	1063
0.200	150.0	3448.50	0.0000	595
0.280	150.0	3448.00	0.0000	601
0.360	150.0	3447.50	0.0000	601
0.440	150.0	3447.00	0.0000	597
0.520	150.0	3446.50	0.0000	601
0.600	150.0	3446.00	0.0000	601
0.680	150.0	3445.50	0.0000	597
0.760	150.0	3445.00	0.0000	601

0.840	150.0	3444.50	0.0000	601
0.920	150.0	3444.00	0.0000	597
1.000	150.0	3443.50	0.0000	601
1.080	150.0	3443.00	0.0000	601
1.160	150.0	3442.50	0.0000	597
1.240	150.0	3442.00	0.0000	601
1.320	150.0	3441.50	0.0000	601
1.400	150.0	3441.00	0.0000	597
1.480	150.0	3440.50	0.0000	601
1.560	150.0	3440.00	0.0000	601
1.640	150.0	3439.50	0.0000	597
1.720	150.0	3439.00	0.0000	601
1.800	150.0	3438.50	0.0000	601
1.880	150.0	3438.00	0.0000	597
1.960	150.0	3437.50	0.0000	601
2.040	150.0	3437.00	0.0000	601
2.120	150.0	3436.50	0.0000	597
2.200	150.0	3436.00	0.0000	601
2.280	150.0	3435.50	0.0000	601
2.360	150.0	3435.00	0.0000	597
2.440	150.0	3434.50	0.0000	601
2.520	150.0	3434.00	0.0000	601
2.600	150.0	3433.50	0.0000	597
2.680	150.0	3433.00	0.0000	601
2.760	150.0	3432.50	0.0000	601
2.840	150.0	3432.00	0.0000	597
2.920	150.0	3431.50	0.0000	600

3.000	150.0	3431.00	0.0000	600
3.080	150.0	3430.50	0.0000	597
3.160	150.0	3430.00	0.0000	600
3.240	150.0	3429.50	0.0000	600
3.320	150.0	3429.00	0.0000	597
3.400	150.0	3428.50	0.0000	600
3.480	150.0	3428.00	0.0000	600
3.560	150.0	3427.50	0.0000	597
3.640	150.0	3427.00	0.0000	600
3.720	150.0	3426.50	0.0000	600
3.800	150.0	3426.00	0.0000	597
3.880	150.0	3425.50	0.0000	600
3.960	150.0	3425.00	0.0000	600
4.040	150.0	3424.50	0.0000	597
4.120	150.0	3424.00	0.0000	600
4.200	150.0	3423.50	0.0000	600
4.280	150.0	3423.00	0.0000	597
4.360	150.0	3422.50	0.0000	600
4.440	150.0	3422.00	0.0000	600
4.520	150.0	3421.50	0.0000	596
4.600	150.0	3421.00	0.0000	600
4.680	150.0	3420.50	0.0000	600
4.760	150.0	3420.00	0.0000	596
4.840	150.0	3419.50	0.0000	600
4.920	150.0	3419.00	0.0000	600
5.000	150.0	3418.50	0.0000	596
5.080	150.0	3418.00	0.0000	600

5.160	150.0	3417.50	0.0000	600
5.240	150.0	3417.00	0.0000	596
5.320	150.0	3416.50	0.0000	600
5.400	150.0	3416.00	0.0000	600
5.480	150.0	3415.50	0.0000	596
5.560	150.0	3415.00	0.0000	600
5.640	150.0	3414.50	0.0000	600
5.720	150.0	3414.00	0.0000	596
5.800	150.0	3413.50	0.0000	600
5.880	150.0	3413.00	0.0000	600
5.960	150.0	3412.50	0.0000	596
6.040	150.0	3412.00	0.0000	600
6.120	150.0	3411.50	0.0000	600
6.200	150.0	3411.00	0.0000	596
6.280	150.0	3410.50	0.0000	600
6.360	150.0	3410.00	0.0000	600
6.440	150.0	3409.50	0.0000	596
6.520	150.0	3409.00	0.0000	600
6.600	150.0	3408.50	0.0000	600
6.680	150.0	3408.00	0.0000	596
6.760	150.0	3407.50	0.0000	600
6.840	150.0	3407.00	0.0000	600
6.920	150.0	3406.50	0.0000	596
7.000	150.0	3406.00	0.0000	600
7.080	150.0	3405.50	0.0000	600
7.160	150.0	3405.00	0.0000	596
7.240	150.0	3404.50	0.0000	600

7.320	150.0	3404.00	0.0000	600
7.400	150.0	3403.50	0.0000	596
7.480	150.0	3403.00	0.0000	600
7.560	150.0	3402.50	0.0000	600
7.640	150.0	3402.00	0.0000	596
7.720	150.0	3401.50	0.0000	600
7.800	150.0	3401.00	0.0000	600
7.880	150.0	3400.50	0.0000	596
7.960	150.0	3400.00	0.0000	600

Length	Enthalpy	Entropy	Inside HTC	Overall HTC
(m)	(kJ/kgmole)	(kJ/kgmole-C)	(kJ/h-m ² -C)	(kJ/h-m ² -C)
0.040	11005	-16.06	---	---
0.120	11005	-16.06	---	---
0.200	11005	-16.06	---	---
0.280	11005	-16.06	---	---
0.360	11005	-16.05	---	---
0.440	11005	-16.05	---	---
0.520	11005	-16.05	---	---
0.600	11005	-16.05	---	---
0.680	11005	-16.05	---	---
0.760	11005	-16.05	---	---
0.840	11005	-16.05	---	---
0.920	11005	-16.05	---	---
1.000	11005	-16.05	---	---
1.080	11005	-16.05	---	---
1.160	11005	-16.05	---	---
1.240	11005	-16.05	---	---

1.320	11005	-16.05	---	---
1.400	11005	-16.05	---	---
1.480	11005	-16.05	---	---
1.560	11005	-16.05	---	---
1.640	11005	-16.05	---	---
1.720	11005	-16.05	---	---
1.800	11005	-16.05	---	---
1.880	11005	-16.05	---	---
1.960	11005	-16.05	---	---
2.040	11005	-16.05	---	---
2.120	11005	-16.05	---	---
2.200	11005	-16.05	---	---
2.280	11005	-16.05	---	---
2.360	11005	-16.05	---	---
2.440	11005	-16.05	---	---
2.520	11005	-16.05	---	---
2.600	11005	-16.05	---	---
2.680	11005	-16.05	---	---
2.760	11004	-16.05	---	---
2.840	11004	-16.05	---	---
2.920	11004	-16.05	---	---
3.000	11004	-16.05	---	---
3.080	11004	-16.05	---	---
3.160	11004	-16.05	---	---
3.240	11004	-16.05	---	---
3.320	11004	-16.05	---	---
3.400	11004	-16.05	---	---

3.480	11004	-16.05	---	---
3.560	11004	-16.05	---	---
3.640	11004	-16.05	---	---
3.720	11004	-16.05	---	---
3.800	11004	-16.05	---	---
3.880	11004	-16.05	---	---
3.960	11004	-16.05	---	---
4.040	11004	-16.05	---	---
4.120	11004	-16.05	---	---
4.200	11004	-16.05	---	---
4.280	11004	-16.05	---	---
4.360	11004	-16.05	---	---
4.440	11004	-16.05	---	---
4.520	11004	-16.05	---	---
4.600	11004	-16.05	---	---
4.680	11004	-16.05	---	---
4.760	11004	-16.05	---	---
4.840	11004	-16.05	---	---
4.920	11004	-16.05	---	---
5.000	11004	-16.05	---	---
5.080	11004	-16.05	---	---
5.160	11004	-16.05	---	---
5.240	11004	-16.05	---	---
5.320	11004	-16.05	---	---
5.400	11004	-16.05	---	---
5.480	11004	-16.05	---	---
5.560	11004	-16.05	---	---

5.640	11004	-16.05	---	---
5.720	11004	-16.05	---	---
5.800	11004	-16.05	---	---
5.880	11003	-16.05	---	---
5.960	11003	-16.05	---	---
6.040	11003	-16.05	---	---
6.120	11003	-16.05	---	---
6.200	11003	-16.05	---	---
6.280	11003	-16.05	---	---
6.360	11003	-16.05	---	---
6.440	11003	-16.05	---	---
6.520	11003	-16.05	---	---
6.600	11003	-16.05	---	---
6.680	11003	-16.05	---	---
6.760	11003	-16.05	---	---
6.840	11003	-16.05	---	---
6.920	11003	-16.05	---	---
7.000	11003	-16.05	---	---
7.080	11003	-16.05	---	---
7.160	11003	-16.05	---	---
7.240	11003	-16.05	---	---
7.320	11003	-16.05	---	---
7.400	11003	-16.05	---	---
7.480	11003	-16.05	---	---
7.560	11003	-16.05	---	---
7.640	11003	-16.05	---	---
7.720	11003	-16.05	---	---

7.800	11003	-16.05	---	---
7.880	11003	-16.05	---	---
7.960	11003	-16.05	---	---

Flows

Length	Molar Flow	Mass Flow	Volumetric Flow	Heat Flow
(m)	(kgmole/h)	(kg/h)	(m3/h)	(kJ/h)
0.040	23151.979	2395353.14	3222.401	2.548e+008
0.120	23151.974	2395353.14	3222.405	2.548e+008
0.200	23151.974	2395353.14	3222.409	2.548e+008
0.280	23151.974	2395353.14	3222.413	2.548e+008
0.360	23151.974	2395353.14	3222.416	2.548e+008
0.440	23151.974	2395353.14	3222.420	2.548e+008
0.520	23151.974	2395353.14	3222.423	2.548e+008
0.600	23151.974	2395353.14	3222.427	2.548e+008
0.680	23151.974	2395353.14	3222.430	2.548e+008
0.760	23151.974	2395353.14	3222.434	2.548e+008
0.840	23151.974	2395353.14	3222.437	2.548e+008
0.920	23151.974	2395353.14	3222.441	2.548e+008
1.000	23151.974	2395353.14	3222.445	2.548e+008
1.080	23151.974	2395353.14	3222.448	2.548e+008
1.160	23151.974	2395353.14	3222.452	2.548e+008
1.240	23151.974	2395353.14	3222.455	2.548e+008
1.320	23151.974	2395353.14	3222.459	2.548e+008
1.400	23151.974	2395353.14	3222.462	2.548e+008
1.480	23151.974	2395353.14	3222.466	2.548e+008
1.560	23151.974	2395353.14	3222.470	2.548e+008

1.640	23151.974	2395353.14	3222.473	2.548e+008
1.720	23151.974	2395353.14	3222.477	2.548e+008
1.800	23151.974	2395353.14	3222.480	2.548e+008
1.880	23151.974	2395353.14	3222.484	2.548e+008
1.960	23151.974	2395353.14	3222.487	2.548e+008
2.040	23151.974	2395353.14	3222.491	2.548e+008
2.120	23151.974	2395353.14	3222.494	2.548e+008
2.200	23151.974	2395353.14	3222.498	2.548e+008
2.280	23151.974	2395353.14	3222.502	2.548e+008
2.360	23151.974	2395353.14	3222.505	2.548e+008
2.440	23151.974	2395353.14	3222.509	2.548e+008
2.520	23151.974	2395353.14	3222.512	2.548e+008
2.600	23151.974	2395353.14	3222.516	2.548e+008
2.680	23151.974	2395353.14	3222.519	2.548e+008
2.760	23151.974	2395353.14	3222.523	2.548e+008
2.840	23151.974	2395353.14	3222.526	2.548e+008
2.920	23151.974	2395353.14	3222.530	2.548e+008
3.000	23151.974	2395353.14	3222.534	2.548e+008
3.080	23151.974	2395353.14	3222.537	2.548e+008
3.160	23151.974	2395353.14	3222.541	2.548e+008
3.240	23151.974	2395353.14	3222.544	2.548e+008
3.320	23151.974	2395353.14	3222.548	2.548e+008
3.400	23151.974	2395353.14	3222.551	2.548e+008
3.480	23151.974	2395353.14	3222.555	2.548e+008
3.560	23151.974	2395353.14	3222.558	2.548e+008
3.640	23151.974	2395353.14	3222.562	2.548e+008
3.720	23151.974	2395353.14	3222.566	2.548e+008

3.800	23151.974	2395353.14	3222.569	2.548e+008
3.880	23151.974	2395353.14	3222.573	2.548e+008
3.960	23151.974	2395353.14	3222.576	2.548e+008
4.040	23151.974	2395353.14	3222.580	2.548e+008
4.120	23151.974	2395353.14	3222.583	2.548e+008
4.200	23151.974	2395353.14	3222.587	2.548e+008
4.280	23151.974	2395353.14	3222.590	2.548e+008
4.360	23151.974	2395353.14	3222.594	2.548e+008
4.440	23151.974	2395353.14	3222.598	2.548e+008
4.520	23151.974	2395353.14	3222.601	2.548e+008
4.600	23151.974	2395353.14	3222.605	2.548e+008
4.680	23151.974	2395353.14	3222.608	2.548e+008
4.760	23151.974	2395353.14	3222.612	2.548e+008
4.840	23151.974	2395353.14	3222.615	2.548e+008
4.920	23151.974	2395353.14	3222.619	2.548e+008
5.000	23151.974	2395353.14	3222.623	2.548e+008
5.080	23151.974	2395353.14	3222.626	2.548e+008
5.160	23151.974	2395353.14	3222.630	2.548e+008
5.240	23151.974	2395353.14	3222.633	2.548e+008
5.320	23151.974	2395353.14	3222.637	2.548e+008
5.400	23151.974	2395353.14	3222.640	2.548e+008
5.480	23151.974	2395353.14	3222.644	2.548e+008
5.560	23151.974	2395353.14	3222.647	2.548e+008
5.640	23151.974	2395353.14	3222.651	2.548e+008
5.720	23151.974	2395353.14	3222.655	2.548e+008
5.800	23151.974	2395353.14	3222.658	2.548e+008
5.880	23151.974	2395353.14	3222.662	2.548e+008

5.960	23151.974	2395353.14	3222.665	2.548e+008
6.040	23151.974	2395353.14	3222.669	2.548e+008
6.120	23151.974	2395353.14	3222.672	2.548e+008
6.200	23151.974	2395353.14	3222.676	2.547e+008
6.280	23151.974	2395353.14	3222.679	2.547e+008
6.360	23151.974	2395353.14	3222.683	2.547e+008
6.440	23151.974	2395353.14	3222.687	2.547e+008
6.520	23151.974	2395353.14	3222.690	2.547e+008
6.600	23151.974	2395353.14	3222.694	2.547e+008
6.680	23151.974	2395353.14	3222.697	2.547e+008
6.760	23151.974	2395353.14	3222.701	2.547e+008
6.840	23151.974	2395353.14	3222.704	2.547e+008
6.920	23151.974	2395353.14	3222.708	2.547e+008
7.000	23151.974	2395353.14	3222.712	2.547e+008
7.080	23151.974	2395353.14	3222.715	2.547e+008
7.160	23151.974	2395353.14	3222.719	2.547e+008
7.240	23151.974	2395353.14	3222.722	2.547e+008
7.320	23151.974	2395353.14	3222.726	2.547e+008
7.400	23151.974	2395353.14	3222.729	2.547e+008
7.480	23151.974	2395353.14	3222.733	2.547e+008
7.560	23151.974	2395353.14	3222.736	2.547e+008
7.640	23151.974	2395353.14	3222.740	2.547e+008
7.720	23151.974	2395353.14	3222.744	2.547e+008
7.800	23151.974	2395353.14	3222.747	2.547e+008
7.880	23151.974	2395353.14	3222.751	2.547e+008
7.960	23151.974	2395353.14	3222.754	2.547e+008

Segment Overall Reaction Rates (kgmole/m³-s)

Length (m)	Rxn-1	Rxn-2
4.000e-002	3.290	1.473
0.1200	3.158e-006	1.414e-006
0.2000	4.687e-012	2.099e-012
0.2800	0.0000	0.0000
0.3600	1.943e-006	8.700e-007
0.4400	2.883e-012	1.291e-012
0.5200	0.0000	0.0000
0.6000	1.943e-006	8.700e-007
0.6800	2.883e-012	1.291e-012
0.7600	0.0000	0.0000
0.8400	1.943e-006	8.700e-007
0.9200	2.883e-012	1.291e-012
1.000	0.0000	0.0000
1.080	1.943e-006	8.700e-007
1.160	2.883e-012	1.291e-012
1.240	0.0000	0.0000
1.320	1.943e-006	8.700e-007
1.400	2.883e-012	1.291e-012
1.480	0.0000	0.0000
1.560	1.943e-006	8.699e-007
1.640	2.883e-012	1.291e-012
1.720	0.0000	0.0000
1.800	1.943e-006	8.699e-007
1.880	2.883e-012	1.291e-012
1.960	0.0000	0.0000

2.040	1.943e-006	8.699e-007
2.120	2.883e-012	1.291e-012
2.200	0.0000	0.0000
2.280	1.943e-006	8.699e-007
2.360	2.883e-012	1.291e-012
2.440	0.0000	0.0000
2.520	1.943e-006	8.699e-007
2.600	2.883e-012	1.291e-012
2.680	0.0000	0.0000
2.760	1.943e-006	8.699e-007
2.840	2.883e-012	1.291e-012
2.920	0.0000	0.0000
3.000	1.942e-006	8.699e-007
3.080	2.883e-012	1.291e-012
3.160	0.0000	0.0000
3.240	1.942e-006	8.699e-007
3.320	2.883e-012	1.291e-012
3.400	0.0000	0.0000
3.480	1.942e-006	8.699e-007
3.560	2.883e-012	1.291e-012
3.640	0.0000	0.0000
3.720	1.942e-006	8.699e-007
3.800	2.883e-012	1.291e-012
3.880	0.0000	0.0000
3.960	1.942e-006	8.699e-007
4.040	2.883e-012	1.291e-012
4.120	0.0000	0.0000

4.200	1.942e-006	8.699e-007
4.280	2.883e-012	1.291e-012
4.360	0.0000	0.0000
4.440	1.942e-006	8.699e-007
4.520	2.883e-012	1.291e-012
4.600	0.0000	0.0000
4.680	1.942e-006	8.698e-007
4.760	2.883e-012	1.291e-012
4.840	0.0000	0.0000
4.920	1.942e-006	8.698e-007
5.000	2.883e-012	1.291e-012
5.080	0.0000	0.0000
5.160	1.942e-006	8.698e-007
5.240	2.883e-012	1.291e-012
5.320	0.0000	0.0000
5.400	1.942e-006	8.698e-007
5.480	2.883e-012	1.291e-012
5.560	0.0000	0.0000
5.640	1.942e-006	8.698e-007
5.720	2.883e-012	1.291e-012
5.800	0.0000	0.0000
5.880	1.942e-006	8.698e-007
5.960	2.883e-012	1.291e-012
6.040	0.0000	0.0000
6.120	1.942e-006	8.698e-007
6.200	2.883e-012	1.291e-012
6.280	0.0000	0.0000

6.360	1.942e-006	8.698e-007
6.440	2.883e-012	1.291e-012
6.520	0.0000	0.0000
6.600	1.942e-006	8.698e-007
6.680	2.883e-012	1.291e-012
6.760	0.0000	0.0000
6.840	1.942e-006	8.698e-007
6.920	2.883e-012	1.291e-012
7.000	0.0000	0.0000
7.080	1.942e-006	8.698e-007
7.160	2.883e-012	1.291e-012
7.240	0.0000	0.0000
7.320	1.942e-006	8.698e-007
7.400	2.882e-012	1.291e-012
7.480	0.0000	0.0000
7.560	1.942e-006	8.698e-007
7.640	2.882e-012	1.291e-012
7.720	0.0000	0.0000
7.800	1.942e-006	8.697e-007
7.880	2.882e-012	1.291e-012
7.960	0.0000	0.0000

Component Production Rates (kgmole/m³-s)

Length (m)	Benzene	Propene	Propane	Cumene
4.000e-002	-3.290	-4.763	0.0000	1.816
0.1200	-3.158e-006	-4.572e-006	0.0000	1.744e-006
0.2000	-4.687e-012	-6.786e-012	0.0000	2.588e-012

0.2800	0.0000	0.0000	0.0000	0.0000
0.3600	-1.943e-006	-2.813e-006	0.0000	1.073e-006
0.4400	-2.883e-012	-4.174e-012	0.0000	1.592e-012
0.5200	0.0000	0.0000	0.0000	0.0000
0.6000	-1.943e-006	-2.813e-006	0.0000	1.073e-006
0.6800	-2.883e-012	-4.174e-012	0.0000	1.592e-012
0.7600	0.0000	0.0000	0.0000	0.0000
0.8400	-1.943e-006	-2.813e-006	0.0000	1.073e-006
0.9200	-2.883e-012	-4.174e-012	0.0000	1.592e-012
1.000	0.0000	0.0000	0.0000	0.0000
1.080	-1.943e-006	-2.813e-006	0.0000	1.073e-006
1.160	-2.883e-012	-4.174e-012	0.0000	1.592e-012
1.240	0.0000	0.0000	0.0000	0.0000
1.320	-1.943e-006	-2.813e-006	0.0000	1.073e-006
1.400	-2.883e-012	-4.174e-012	0.0000	1.592e-012
1.480	0.0000	0.0000	0.0000	0.0000
1.560	-1.943e-006	-2.813e-006	0.0000	1.073e-006
1.640	-2.883e-012	-4.174e-012	0.0000	1.592e-012
1.720	0.0000	0.0000	0.0000	0.0000
1.800	-1.943e-006	-2.813e-006	0.0000	1.073e-006
1.880	-2.883e-012	-4.174e-012	0.0000	1.592e-012
1.960	0.0000	0.0000	0.0000	0.0000
2.040	-1.943e-006	-2.812e-006	0.0000	1.073e-006
2.120	-2.883e-012	-4.174e-012	0.0000	1.592e-012
2.200	0.0000	0.0000	0.0000	0.0000
2.280	-1.943e-006	-2.812e-006	0.0000	1.073e-006
2.360	-2.883e-012	-4.174e-012	0.0000	1.592e-012

2.440	0.0000	0.0000	0.0000	0.0000
2.520	-1.943e-006	-2.812e-006	0.0000	1.073e-006
2.600	-2.883e-012	-4.174e-012	0.0000	1.592e-012
2.680	0.0000	0.0000	0.0000	0.0000
2.760	-1.943e-006	-2.812e-006	0.0000	1.073e-006
2.840	-2.883e-012	-4.174e-012	0.0000	1.592e-012
2.920	0.0000	0.0000	0.0000	0.0000
3.000	-1.942e-006	-2.812e-006	0.0000	1.073e-006
3.080	-2.883e-012	-4.174e-012	0.0000	1.592e-012
3.160	0.0000	0.0000	0.0000	0.0000
3.240	-1.942e-006	-2.812e-006	0.0000	1.073e-006
3.320	-2.883e-012	-4.174e-012	0.0000	1.592e-012
3.400	0.0000	0.0000	0.0000	0.0000
3.480	-1.942e-006	-2.812e-006	0.0000	1.073e-006
3.560	-2.883e-012	-4.174e-012	0.0000	1.592e-012
3.640	0.0000	0.0000	0.0000	0.0000
3.720	-1.942e-006	-2.812e-006	0.0000	1.073e-006
3.800	-2.883e-012	-4.174e-012	0.0000	1.592e-012
3.880	0.0000	0.0000	0.0000	0.0000
3.960	-1.942e-006	-2.812e-006	0.0000	1.073e-006
4.040	-2.883e-012	-4.174e-012	0.0000	1.592e-012
4.120	0.0000	0.0000	0.0000	0.0000
4.200	-1.942e-006	-2.812e-006	0.0000	1.073e-006
4.280	-2.883e-012	-4.174e-012	0.0000	1.592e-012
4.360	0.0000	0.0000	0.0000	0.0000
4.440	-1.942e-006	-2.812e-006	0.0000	1.073e-006
4.520	-2.883e-012	-4.174e-012	0.0000	1.592e-012

4.600	0.0000	0.0000	0.0000	0.0000
4.680	-1.942e-006	-2.812e-006	0.0000	1.073e-006
4.760	-2.883e-012	-4.174e-012	0.0000	1.592e-012
4.840	0.0000	0.0000	0.0000	0.0000
4.920	-1.942e-006	-2.812e-006	0.0000	1.073e-006
5.000	-2.883e-012	-4.174e-012	0.0000	1.592e-012
5.080	0.0000	0.0000	0.0000	0.0000
5.160	-1.942e-006	-2.812e-006	0.0000	1.073e-006
5.240	-2.883e-012	-4.174e-012	0.0000	1.592e-012
5.320	0.0000	0.0000	0.0000	0.0000
5.400	-1.942e-006	-2.812e-006	0.0000	1.073e-006
5.480	-2.883e-012	-4.174e-012	0.0000	1.592e-012
5.560	0.0000	0.0000	0.0000	0.0000
5.640	-1.942e-006	-2.812e-006	0.0000	1.073e-006
5.720	-2.883e-012	-4.174e-012	0.0000	1.592e-012
5.800	0.0000	0.0000	0.0000	0.0000
5.880	-1.942e-006	-2.812e-006	0.0000	1.072e-006
5.960	-2.883e-012	-4.174e-012	0.0000	1.592e-012
6.040	0.0000	0.0000	0.0000	0.0000
6.120	-1.942e-006	-2.812e-006	0.0000	1.072e-006
6.200	-2.883e-012	-4.173e-012	0.0000	1.592e-012
6.280	0.0000	0.0000	0.0000	0.0000
6.360	-1.942e-006	-2.812e-006	0.0000	1.072e-006
6.440	-2.883e-012	-4.173e-012	0.0000	1.592e-012
6.520	0.0000	0.0000	0.0000	0.0000
6.600	-1.942e-006	-2.812e-006	0.0000	1.072e-006
6.680	-2.883e-012	-4.173e-012	0.0000	1.592e-012

6.760	0.0000	0.0000	0.0000	0.0000
6.840	-1.942e-006	-2.812e-006	0.0000	1.072e-006
6.920	-2.883e-012	-4.173e-012	0.0000	1.592e-012
7.000	0.0000	0.0000	0.0000	0.0000
7.080	-1.942e-006	-2.812e-006	0.0000	1.072e-006
7.160	-2.883e-012	-4.173e-012	0.0000	1.592e-012
7.240	0.0000	0.0000	0.0000	0.0000
7.320	-1.942e-006	-2.812e-006	0.0000	1.072e-006
7.400	-2.882e-012	-4.173e-012	0.0000	1.592e-012
7.480	0.0000	0.0000	0.0000	0.0000
7.560	-1.942e-006	-2.812e-006	0.0000	1.072e-006
7.640	-2.882e-012	-4.173e-012	0.0000	1.592e-012
7.720	0.0000	0.0000	0.0000	0.0000
7.800	-1.942e-006	-2.812e-006	0.0000	1.072e-006
7.880	-2.882e-012	-4.173e-012	0.0000	1.592e-012
7.960	0.0000	0.0000	0.0000	0.0000

Length (m) 14-iP-BZ

4.000e-002	1.473
0.1200	1.414e-006
0.2000	2.099e-012
0.2800	0.0000
0.3600	8.700e-007
0.4400	1.291e-012
0.5200	0.0000
0.6000	8.700e-007
0.6800	1.291e-012
0.7600	0.0000

0.8400	8.700e-007
0.9200	1.291e-012
1.000	0.0000
1.080	8.700e-007
1.160	1.291e-012
1.240	0.0000
1.320	8.700e-007
1.400	1.291e-012
1.480	0.0000
1.560	8.699e-007
1.640	1.291e-012
1.720	0.0000
1.800	8.699e-007
1.880	1.291e-012
1.960	0.0000
2.040	8.699e-007
2.120	1.291e-012
2.200	0.0000
2.280	8.699e-007
2.360	1.291e-012
2.440	0.0000
2.520	8.699e-007
2.600	1.291e-012
2.680	0.0000
2.760	8.699e-007
2.840	1.291e-012
2.920	0.0000

3.000	8.699e-007
3.080	1.291e-012
3.160	0.0000
3.240	8.699e-007
3.320	1.291e-012
3.400	0.0000
3.480	8.699e-007
3.560	1.291e-012
3.640	0.0000
3.720	8.699e-007
3.800	1.291e-012
3.880	0.0000
3.960	8.699e-007
4.040	1.291e-012
4.120	0.0000
4.200	8.699e-007
4.280	1.291e-012
4.360	0.0000
4.440	8.699e-007
4.520	1.291e-012
4.600	0.0000
4.680	8.698e-007
4.760	1.291e-012
4.840	0.0000
4.920	8.698e-007
5.000	1.291e-012
5.080	0.0000

5.160	8.698e-007
5.240	1.291e-012
5.320	0.0000
5.400	8.698e-007
5.480	1.291e-012
5.560	0.0000
5.640	8.698e-007
5.720	1.291e-012
5.800	0.0000
5.880	8.698e-007
5.960	1.291e-012
6.040	0.0000
6.120	8.698e-007
6.200	1.291e-012
6.280	0.0000
6.360	8.698e-007
6.440	1.291e-012
6.520	0.0000
6.600	8.698e-007
6.680	1.291e-012
6.760	0.0000
6.840	8.698e-007
6.920	1.291e-012
7.000	0.0000
7.080	8.698e-007
7.160	1.291e-012
7.240	0.0000

7.320	8.698e-007
7.400	1.291e-012
7.480	0.0000
7.560	8.698e-007
7.640	1.291e-012
7.720	0.0000
7.800	8.697e-007
7.880	1.291e-012
7.960	0.0000

Transport

Length Viscosity Molecular Weight Mass Density Heat Capacity Surface Tension Z Factor

(m)	(cP)	(kg/m3)	(kJ/kgmole-C)	(dyne/cm)		
4.000e-002	0.1907	103.5	743.3	223.932	13.73	0.1365
0.1200	0.1907	103.5	743.3	223.933	13.73	0.1364
0.2000	0.1907	103.5	743.3	223.933	13.73	0.1364
0.2800	0.1907	103.5	743.3	223.934	13.73	0.1364
0.3600	0.1907	103.5	743.3	223.934	13.73	0.1364
0.4400	0.1907	103.5	743.3	223.934	13.73	0.1364
0.5200	0.1907	103.5	743.3	223.935	13.73	0.1363
0.6000	0.1907	103.5	743.3	223.935	13.73	0.1363
0.6800	0.1907	103.5	743.3	223.935	13.73	0.1363
0.7600	0.1907	103.5	743.3	223.936	13.73	0.1363
0.8400	0.1907	103.5	743.3	223.936	13.73	0.1363
0.9200	0.1907	103.5	743.3	223.936	13.73	0.1363
1.000	0.1907	103.5	743.3	223.937	13.73	0.1362
1.080	0.1907	103.5	743.3	223.937	13.73	0.1362

1.160	0.1907	103.5	743.3	223.937	13.73	0.1362
1.240	0.1907	103.5	743.3	223.938	13.73	0.1362
1.320	0.1907	103.5	743.3	223.938	13.73	0.1362
1.400	0.1907	103.5	743.3	223.938	13.73	0.1361
1.480	0.1907	103.5	743.3	223.939	13.73	0.1361
1.560	0.1907	103.5	743.3	223.939	13.73	0.1361
1.640	0.1907	103.5	743.3	223.939	13.73	0.1361
1.720	0.1907	103.5	743.3	223.940	13.73	0.1361
1.800	0.1907	103.5	743.3	223.940	13.73	0.1360
1.880	0.1907	103.5	743.3	223.940	13.73	0.1360
1.960	0.1907	103.5	743.3	223.941	13.73	0.1360
2.040	0.1907	103.5	743.3	223.941	13.73	0.1360
2.120	0.1907	103.5	743.3	223.941	13.73	0.1360
2.200	0.1907	103.5	743.3	223.942	13.73	0.1359
2.280	0.1907	103.5	743.3	223.942	13.73	0.1359
2.360	0.1907	103.5	743.3	223.942	13.73	0.1359
2.440	0.1907	103.5	743.3	223.943	13.73	0.1359
2.520	0.1907	103.5	743.3	223.943	13.73	0.1359
2.600	0.1907	103.5	743.3	223.943	13.73	0.1358
2.680	0.1907	103.5	743.3	223.944	13.73	0.1358
2.760	0.1907	103.5	743.3	223.944	13.73	0.1358
2.840	0.1907	103.5	743.3	223.944	13.73	0.1358
2.920	0.1907	103.5	743.3	223.945	13.73	0.1358
3.000	0.1907	103.5	743.3	223.945	13.73	0.1357
3.080	0.1907	103.5	743.3	223.945	13.73	0.1357
3.160	0.1907	103.5	743.3	223.946	13.73	0.1357
3.240	0.1907	103.5	743.3	223.946	13.73	0.1357

3.320	0.1907	103.5	743.3	223.946	13.73	0.1357
3.400	0.1907	103.5	743.3	223.947	13.73	0.1356
3.480	0.1907	103.5	743.3	223.947	13.73	0.1356
3.560	0.1907	103.5	743.3	223.947	13.73	0.1356
3.640	0.1907	103.5	743.3	223.948	13.73	0.1356
3.720	0.1907	103.5	743.3	223.948	13.73	0.1356
3.800	0.1907	103.5	743.3	223.948	13.73	0.1355
3.880	0.1907	103.5	743.3	223.949	13.73	0.1355
3.960	0.1907	103.5	743.3	223.949	13.73	0.1355
4.040	0.1907	103.5	743.3	223.949	13.73	0.1355
4.120	0.1907	103.5	743.3	223.950	13.73	0.1355
4.200	0.1907	103.5	743.3	223.950	13.73	0.1354
4.280	0.1907	103.5	743.3	223.950	13.73	0.1354
4.360	0.1907	103.5	743.3	223.951	13.73	0.1354
4.440	0.1907	103.5	743.3	223.951	13.73	0.1354
4.520	0.1907	103.5	743.3	223.951	13.73	0.1354
4.600	0.1907	103.5	743.3	223.952	13.73	0.1353
4.680	0.1907	103.5	743.3	223.952	13.73	0.1353
4.760	0.1907	103.5	743.3	223.952	13.73	0.1353
4.840	0.1907	103.5	743.3	223.953	13.73	0.1353
4.920	0.1907	103.5	743.3	223.953	13.73	0.1353
5.000	0.1907	103.5	743.3	223.953	13.73	0.1352
5.080	0.1907	103.5	743.3	223.954	13.73	0.1352
5.160	0.1907	103.5	743.3	223.954	13.73	0.1352
5.240	0.1907	103.5	743.3	223.955	13.73	0.1352
5.320	0.1907	103.5	743.3	223.955	13.73	0.1352
5.400	0.1907	103.5	743.3	223.955	13.73	0.1352

5.480	0.1907	103.5	743.3	223.956	13.73	0.1351
5.560	0.1907	103.5	743.3	223.956	13.73	0.1351
5.640	0.1907	103.5	743.3	223.956	13.73	0.1351
5.720	0.1907	103.5	743.3	223.957	13.73	0.1351
5.800	0.1907	103.5	743.3	223.957	13.73	0.1351
5.880	0.1907	103.5	743.3	223.957	13.73	0.1350
5.960	0.1907	103.5	743.3	223.958	13.73	0.1350
6.040	0.1907	103.5	743.3	223.958	13.73	0.1350
6.120	0.1907	103.5	743.3	223.958	13.73	0.1350
6.200	0.1907	103.5	743.3	223.959	13.73	0.1350
6.280	0.1907	103.5	743.3	223.959	13.73	0.1349
6.360	0.1907	103.5	743.3	223.959	13.73	0.1349
6.440	0.1907	103.5	743.3	223.960	13.73	0.1349
6.520	0.1907	103.5	743.3	223.960	13.73	0.1349
6.600	0.1907	103.5	743.3	223.960	13.73	0.1349
6.680	0.1907	103.5	743.3	223.961	13.73	0.1348
6.760	0.1907	103.5	743.3	223.961	13.73	0.1348
6.840	0.1907	103.5	743.3	223.961	13.73	0.1348
6.920	0.1907	103.5	743.3	223.962	13.73	0.1348
7.000	0.1907	103.5	743.3	223.962	13.73	0.1348
7.080	0.1907	103.5	743.3	223.962	13.73	0.1347
7.160	0.1907	103.5	743.3	223.963	13.73	0.1347
7.240	0.1907	103.5	743.3	223.963	13.73	0.1347
7.320	0.1907	103.5	743.3	223.963	13.73	0.1347
7.400	0.1907	103.5	743.3	223.964	13.73	0.1347
7.480	0.1907	103.5	743.3	223.964	13.73	0.1346
7.560	0.1907	103.5	743.3	223.964	13.73	0.1346

7.640	0.1907	103.5	743.3	223.965	13.73	0.1346
7.720	0.1907	103.5	743.3	223.965	13.73	0.1346
7.800	0.1907	103.5	743.3	223.965	13.73	0.1346
7.880	0.1907	103.5	743.3	223.966	13.73	0.1345
7.960	0.1907	103.5	743.3	223.966	13.73	0.1345

Component Molar Flowrates (kgmole/h)

Length (m)	Benzene	Propene	Propane	Cumene
4.000e-002	11181.5740	0.0048	981.1578	7237.9799
0.1200	11181.5707	0.0000	981.1578	7237.9817
0.2000	11181.5739	0.0000	981.1578	7237.9754
0.2800	11181.5739	0.0000	981.1578	7237.9754
0.3600	11181.5739	0.0000	981.1578	7237.9754
0.4400	11181.5758	0.0000	981.1578	7237.9716
0.5200	11181.5758	0.0000	981.1578	7237.9716
0.6000	11181.5758	0.0000	981.1578	7237.9716
0.6800	11181.5777	0.0000	981.1578	7237.9678
0.7600	11181.5777	0.0000	981.1578	7237.9678
0.8400	11181.5777	0.0000	981.1578	7237.9678
0.9200	11181.5796	0.0000	981.1578	7237.9639
1.000	11181.5796	0.0000	981.1578	7237.9639
1.080	11181.5796	0.0000	981.1578	7237.9639
1.160	11181.5815	0.0000	981.1578	7237.9601
1.240	11181.5815	0.0000	981.1578	7237.9601
1.320	11181.5815	0.0000	981.1578	7237.9601
1.400	11181.5835	0.0000	981.1578	7237.9562
1.480	11181.5835	0.0000	981.1578	7237.9562

1.560	11181.5835	0.0000	981.1578	7237.9562
1.640	11181.5854	0.0000	981.1578	7237.9524
1.720	11181.5854	0.0000	981.1578	7237.9524
1.800	11181.5854	0.0000	981.1578	7237.9524
1.880	11181.5873	0.0000	981.1578	7237.9486
1.960	11181.5873	0.0000	981.1578	7237.9486
2.040	11181.5873	0.0000	981.1578	7237.9486
2.120	11181.5892	0.0000	981.1578	7237.9447
2.200	11181.5892	0.0000	981.1578	7237.9447
2.280	11181.5892	0.0000	981.1578	7237.9447
2.360	11181.5911	0.0000	981.1578	7237.9409
2.440	11181.5911	0.0000	981.1578	7237.9409
2.520	11181.5911	0.0000	981.1578	7237.9409
2.600	11181.5931	0.0000	981.1578	7237.9370
2.680	11181.5931	0.0000	981.1578	7237.9370
2.760	11181.5931	0.0000	981.1578	7237.9370
2.840	11181.5950	0.0000	981.1578	7237.9332
2.920	11181.5950	0.0000	981.1578	7237.9332
3.000	11181.5950	0.0000	981.1578	7237.9332
3.080	11181.5969	0.0000	981.1578	7237.9294
3.160	11181.5969	0.0000	981.1578	7237.9294
3.240	11181.5969	0.0000	981.1578	7237.9294
3.320	11181.5988	0.0000	981.1578	7237.9255
3.400	11181.5988	0.0000	981.1578	7237.9255
3.480	11181.5988	0.0000	981.1578	7237.9255
3.560	11181.6007	0.0000	981.1578	7237.9217
3.640	11181.6007	0.0000	981.1578	7237.9217

3.720	11181.6007	0.0000	981.1578	7237.9217
3.800	11181.6027	0.0000	981.1578	7237.9179
3.880	11181.6027	0.0000	981.1578	7237.9179
3.960	11181.6027	0.0000	981.1578	7237.9179
4.040	11181.6046	0.0000	981.1578	7237.9140
4.120	11181.6046	0.0000	981.1578	7237.9140
4.200	11181.6046	0.0000	981.1578	7237.9140
4.280	11181.6065	0.0000	981.1578	7237.9102
4.360	11181.6065	0.0000	981.1578	7237.9102
4.440	11181.6065	0.0000	981.1578	7237.9102
4.520	11181.6084	0.0000	981.1578	7237.9063
4.600	11181.6084	0.0000	981.1578	7237.9063
4.680	11181.6084	0.0000	981.1578	7237.9063
4.760	11181.6103	0.0000	981.1578	7237.9025
4.840	11181.6103	0.0000	981.1578	7237.9025
4.920	11181.6103	0.0000	981.1578	7237.9025
5.000	11181.6123	0.0000	981.1578	7237.8987
5.080	11181.6123	0.0000	981.1578	7237.8987
5.160	11181.6123	0.0000	981.1578	7237.8987
5.240	11181.6142	0.0000	981.1578	7237.8948
5.320	11181.6142	0.0000	981.1578	7237.8948
5.400	11181.6142	0.0000	981.1578	7237.8948
5.480	11181.6161	0.0000	981.1578	7237.8910
5.560	11181.6161	0.0000	981.1578	7237.8910
5.640	11181.6161	0.0000	981.1578	7237.8910
5.720	11181.6180	0.0000	981.1578	7237.8872
5.800	11181.6180	0.0000	981.1578	7237.8872

5.880	11181.6180	0.0000	981.1578	7237.8872
5.960	11181.6199	0.0000	981.1578	7237.8833
6.040	11181.6199	0.0000	981.1578	7237.8833
6.120	11181.6199	0.0000	981.1578	7237.8833
6.200	11181.6219	0.0000	981.1578	7237.8795
6.280	11181.6219	0.0000	981.1578	7237.8795
6.360	11181.6219	0.0000	981.1578	7237.8795
6.440	11181.6238	0.0000	981.1578	7237.8756
6.520	11181.6238	0.0000	981.1578	7237.8756
6.600	11181.6238	0.0000	981.1578	7237.8756
6.680	11181.6257	0.0000	981.1578	7237.8718
6.760	11181.6257	0.0000	981.1578	7237.8718
6.840	11181.6257	0.0000	981.1578	7237.8718
6.920	11181.6276	0.0000	981.1578	7237.8680
7.000	11181.6276	0.0000	981.1578	7237.8680
7.080	11181.6276	0.0000	981.1578	7237.8680
7.160	11181.6295	0.0000	981.1578	7237.8641
7.240	11181.6295	0.0000	981.1578	7237.8641
7.320	11181.6295	0.0000	981.1578	7237.8641
7.400	11181.6314	0.0000	981.1578	7237.8603
7.480	11181.6314	0.0000	981.1578	7237.8603
7.560	11181.6314	0.0000	981.1578	7237.8603
7.640	11181.6334	0.0000	981.1578	7237.8564
7.720	11181.6334	0.0000	981.1578	7237.8564
7.800	11181.6334	0.0000	981.1578	7237.8564
7.880	11181.6353	0.0000	981.1578	7237.8526
7.960	11181.6353	0.0000	981.1578	7237.8526

Length (m) 14-iP-BZ

4.000e-002 3751.2621

0.1200 3751.2636

0.2000 3751.2667

0.2800 3751.2667

0.3600 3751.2667

0.4400 3751.2686

0.5200 3751.2686

0.6000 3751.2686

0.6800 3751.2705

0.7600 3751.2705

0.8400 3751.2705

0.9200 3751.2725

1.000 3751.2725

1.080 3751.2725

1.160 3751.2744

1.240 3751.2744

1.320 3751.2744

1.400 3751.2763

1.480 3751.2763

1.560 3751.2763

1.640 3751.2782

1.720 3751.2782

1.800 3751.2782

1.880 3751.2801

1.960 3751.2801

2.040 3751.2801

2.120	3751.2821
2.200	3751.2821
2.280	3751.2821
2.360	3751.2840
2.440	3751.2840
2.520	3751.2840
2.600	3751.2859
2.680	3751.2859
2.760	3751.2859
2.840	3751.2878
2.920	3751.2878
3.000	3751.2878
3.080	3751.2897
3.160	3751.2897
3.240	3751.2897
3.320	3751.2917
3.400	3751.2917
3.480	3751.2917
3.560	3751.2936
3.640	3751.2936
3.720	3751.2936
3.800	3751.2955
3.880	3751.2955
3.960	3751.2955
4.040	3751.2974
4.120	3751.2974
4.200	3751.2974

4.280	3751.2993
4.360	3751.2993
4.440	3751.2993
4.520	3751.3013
4.600	3751.3013
4.680	3751.3013
4.760	3751.3032
4.840	3751.3032
4.920	3751.3032
5.000	3751.3051
5.080	3751.3051
5.160	3751.3051
5.240	3751.3070
5.320	3751.3070
5.400	3751.3070
5.480	3751.3089
5.560	3751.3089
5.640	3751.3089
5.720	3751.3108
5.800	3751.3108
5.880	3751.3108
5.960	3751.3128
6.040	3751.3128
6.120	3751.3128
6.200	3751.3147
6.280	3751.3147
6.360	3751.3147

6.440	3751.3166
6.520	3751.3166
6.600	3751.3166
6.680	3751.3185
6.760	3751.3185
6.840	3751.3185
6.920	3751.3204
7.000	3751.3204
7.080	3751.3204
7.160	3751.3224
7.240	3751.3224
7.320	3751.3224
7.400	3751.3243
7.480	3751.3243
7.560	3751.3243
7.640	3751.3262
7.720	3751.3262
7.800	3751.3262
7.880	3751.3281
7.960	3751.3281

Component Mole Fractions

Length (m)	Benzene	Propene	Propane	Cumene
4.000e-002	0.4830	0.0000	0.0424	0.3126
0.1200	0.4830	0.0000	0.0424	0.3126
0.2000	0.4830	0.0000	0.0424	0.3126
0.2800	0.4830	0.0000	0.0424	0.3126

0.3600	0.4830	0.0000	0.0424	0.3126
0.4400	0.4830	0.0000	0.0424	0.3126
0.5200	0.4830	0.0000	0.0424	0.3126
0.6000	0.4830	0.0000	0.0424	0.3126
0.6800	0.4830	0.0000	0.0424	0.3126
0.7600	0.4830	0.0000	0.0424	0.3126
0.8400	0.4830	0.0000	0.0424	0.3126
0.9200	0.4830	0.0000	0.0424	0.3126
1.000	0.4830	0.0000	0.0424	0.3126
1.080	0.4830	0.0000	0.0424	0.3126
1.160	0.4830	0.0000	0.0424	0.3126
1.240	0.4830	0.0000	0.0424	0.3126
1.320	0.4830	0.0000	0.0424	0.3126
1.400	0.4830	0.0000	0.0424	0.3126
1.480	0.4830	0.0000	0.0424	0.3126
1.560	0.4830	0.0000	0.0424	0.3126
1.640	0.4830	0.0000	0.0424	0.3126
1.720	0.4830	0.0000	0.0424	0.3126
1.800	0.4830	0.0000	0.0424	0.3126
1.880	0.4830	0.0000	0.0424	0.3126
1.960	0.4830	0.0000	0.0424	0.3126
2.040	0.4830	0.0000	0.0424	0.3126
2.120	0.4830	0.0000	0.0424	0.3126
2.200	0.4830	0.0000	0.0424	0.3126
2.280	0.4830	0.0000	0.0424	0.3126
2.360	0.4830	0.0000	0.0424	0.3126
2.440	0.4830	0.0000	0.0424	0.3126

2.520	0.4830	0.0000	0.0424	0.3126
2.600	0.4830	0.0000	0.0424	0.3126
2.680	0.4830	0.0000	0.0424	0.3126
2.760	0.4830	0.0000	0.0424	0.3126
2.840	0.4830	0.0000	0.0424	0.3126
2.920	0.4830	0.0000	0.0424	0.3126
3.000	0.4830	0.0000	0.0424	0.3126
3.080	0.4830	0.0000	0.0424	0.3126
3.160	0.4830	0.0000	0.0424	0.3126
3.240	0.4830	0.0000	0.0424	0.3126
3.320	0.4830	0.0000	0.0424	0.3126
3.400	0.4830	0.0000	0.0424	0.3126
3.480	0.4830	0.0000	0.0424	0.3126
3.560	0.4830	0.0000	0.0424	0.3126
3.640	0.4830	0.0000	0.0424	0.3126
3.720	0.4830	0.0000	0.0424	0.3126
3.800	0.4830	0.0000	0.0424	0.3126
3.880	0.4830	0.0000	0.0424	0.3126
3.960	0.4830	0.0000	0.0424	0.3126
4.040	0.4830	0.0000	0.0424	0.3126
4.120	0.4830	0.0000	0.0424	0.3126
4.200	0.4830	0.0000	0.0424	0.3126
4.280	0.4830	0.0000	0.0424	0.3126
4.360	0.4830	0.0000	0.0424	0.3126
4.440	0.4830	0.0000	0.0424	0.3126
4.520	0.4830	0.0000	0.0424	0.3126
4.600	0.4830	0.0000	0.0424	0.3126

4.680	0.4830	0.0000	0.0424	0.3126
4.760	0.4830	0.0000	0.0424	0.3126
4.840	0.4830	0.0000	0.0424	0.3126
4.920	0.4830	0.0000	0.0424	0.3126
5.000	0.4830	0.0000	0.0424	0.3126
5.080	0.4830	0.0000	0.0424	0.3126
5.160	0.4830	0.0000	0.0424	0.3126
5.240	0.4830	0.0000	0.0424	0.3126
5.320	0.4830	0.0000	0.0424	0.3126
5.400	0.4830	0.0000	0.0424	0.3126
5.480	0.4830	0.0000	0.0424	0.3126
5.560	0.4830	0.0000	0.0424	0.3126
5.640	0.4830	0.0000	0.0424	0.3126
5.720	0.4830	0.0000	0.0424	0.3126
5.800	0.4830	0.0000	0.0424	0.3126
5.880	0.4830	0.0000	0.0424	0.3126
5.960	0.4830	0.0000	0.0424	0.3126
6.040	0.4830	0.0000	0.0424	0.3126
6.120	0.4830	0.0000	0.0424	0.3126
6.200	0.4830	0.0000	0.0424	0.3126
6.280	0.4830	0.0000	0.0424	0.3126
6.360	0.4830	0.0000	0.0424	0.3126
6.440	0.4830	0.0000	0.0424	0.3126
6.520	0.4830	0.0000	0.0424	0.3126
6.600	0.4830	0.0000	0.0424	0.3126
6.680	0.4830	0.0000	0.0424	0.3126
6.760	0.4830	0.0000	0.0424	0.3126

6.840	0.4830	0.0000	0.0424	0.3126
6.920	0.4830	0.0000	0.0424	0.3126
7.000	0.4830	0.0000	0.0424	0.3126
7.080	0.4830	0.0000	0.0424	0.3126
7.160	0.4830	0.0000	0.0424	0.3126
7.240	0.4830	0.0000	0.0424	0.3126
7.320	0.4830	0.0000	0.0424	0.3126
7.400	0.4830	0.0000	0.0424	0.3126
7.480	0.4830	0.0000	0.0424	0.3126
7.560	0.4830	0.0000	0.0424	0.3126
7.640	0.4830	0.0000	0.0424	0.3126
7.720	0.4830	0.0000	0.0424	0.3126
7.800	0.4830	0.0000	0.0424	0.3126
7.880	0.4830	0.0000	0.0424	0.3126
7.960	0.4830	0.0000	0.0424	0.3126

Length (m) 14-iP-BZ

4.000e-002 0.1620

0.1200 0.1620

0.2000 0.1620

0.2800 0.1620

0.3600 0.1620

0.4400 0.1620

0.5200 0.1620

0.6000 0.1620

0.6800 0.1620

0.7600 0.1620

0.8400 0.1620

0.9200	0.1620
1.000	0.1620
1.080	0.1620
1.160	0.1620
1.240	0.1620
1.320	0.1620
1.400	0.1620
1.480	0.1620
1.560	0.1620
1.640	0.1620
1.720	0.1620
1.800	0.1620
1.880	0.1620
1.960	0.1620
2.040	0.1620
2.120	0.1620
2.200	0.1620
2.280	0.1620
2.360	0.1620
2.440	0.1620
2.520	0.1620
2.600	0.1620
2.680	0.1620
2.760	0.1620
2.840	0.1620
2.920	0.1620
3.000	0.1620

3.080	0.1620
3.160	0.1620
3.240	0.1620
3.320	0.1620
3.400	0.1620
3.480	0.1620
3.560	0.1620
3.640	0.1620
3.720	0.1620
3.800	0.1620
3.880	0.1620
3.960	0.1620
4.040	0.1620
4.120	0.1620
4.200	0.1620
4.280	0.1620
4.360	0.1620
4.440	0.1620
4.520	0.1620
4.600	0.1620
4.680	0.1620
4.760	0.1620
4.840	0.1620
4.920	0.1620
5.000	0.1620
5.080	0.1620
5.160	0.1620

5.240	0.1620
5.320	0.1620
5.400	0.1620
5.480	0.1620
5.560	0.1620
5.640	0.1620
5.720	0.1620
5.800	0.1620
5.880	0.1620
5.960	0.1620
6.040	0.1620
6.120	0.1620
6.200	0.1620
6.280	0.1620
6.360	0.1620
6.440	0.1620
6.520	0.1620
6.600	0.1620
6.680	0.1620
6.760	0.1620
6.840	0.1620
6.920	0.1620
7.000	0.1620
7.080	0.1620
7.160	0.1620
7.240	0.1620
7.320	0.1620

7.400	0.1620
7.480	0.1620
7.560	0.1620
7.640	0.1620
7.720	0.1620
7.800	0.1620
7.880	0.1620
7.960	0.1620

Valve: VLV-100

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
Alkylation Effluents	R-1 Plug Flow Reactor
Outlet Stream	

STREAM NAME	TO UNIT OPERATION
C1-Feed	T-100 Absorber

PARAMETERS

Physical Properties

Pressure Drop: 1900 kPa

User Variables

RATING

Sizing

Sizing Conditions

Inlet Pressure 3400 kPa Molecular Weight 103.5 Current

Valve Opening 50.00 % Delta P 1900 kPa Flow Rate 2.395e+006 kg/h

Valve Sizing Method and Type

Sizing Method: ANSI/ISA

Valve Operating Characteristic and Sizing Information

Linear Sized Coefficient: Cv (standard) cal/min.sqrt(psi)

F1 0.9000 Cv 943.6 USGPM(60F,1psi) Cg 3.158e+004 Fp 1.000 Xt 0.7000 Rigorous Cp/Cv Method

Nozzle Parameters

Base Elevation Relative to Ground Level 0.0000 m

Alkylation Effluents C1-Feed

Diameter (m)	5.000e-002	5.000e-002
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Elevation (Base) (m)	0.0000	0.0000
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Elevation (Ground) (m)	0.0000	0.0000
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Elevation (% of Height) (%)

Valve: VLV-101

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
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C2-Feed_New	MIX-102 Mixer
-------------	---------------

Outlet Stream

STREAM NAME	TO UNIT OPERATION
-------------	-------------------

C2-Feed-LP	T-101 Absorber
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PARAMETERS

Physical Properties

Pressure Drop: 1380 kPa

User Variables

RATING

Sizing

Sizing Conditions

Inlet Pressure 1500 kPa Molecular Weight 107.5 Current

Valve Opening 50.00 % Delta P 1380 kPa Flow Rate 3.862e+006 kg/h

Valve Sizing Method and Type

Sizing Method: ANSI/ISA

Valve Operating Characteristic and Sizing Information

Linear Sized Coefficient: Cv (standard) cal/min.sqrt(psi)

F1 0.9000 Cv 2958 USGPM(60F,1psi) Cg 9.900e+004 Fp 1.000 Xt 0.7000 Rigorous Cp/Cv Method

Nozzle Parameters

Base Elevation Relative to Ground Level 0.0000 m

	C2-Feed_New	C2-Feed-LP
Diameter (m)	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000
Elevation (% of Height) (%)		

Absorber: T-100

CONNECTIONS

Inlet Stream

Stream Name	Stage	From Unit Operation
Reflux-1	1__Main Tower	RCY-4 Recycle
C1-Feed	5__Main Tower	VLV-100 Valve
Boil-Up	10__Main Tower	RCY-3 Recycle

Outlet Stream

Stream Name	Stage	To Unit Operation
-------------	-------	-------------------

C1-Bot 10__Main Tower E-101 Heater

C1-Top 1__Main Tower E-102 Cooler

MONITOR

Specifications Summary

Specified Value	Current Value	Wt. Error
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Wt. Tol.	Abs. Tol.	Active Estimate Used
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SPECS

Column Specification Parameters

Fix/Rang:	Prim/Alter:	Lower Bnd:	Upper Bnd:
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SUBCOOLING

Degrees of Subcooling

Subcool to

User Variables

PROFILES

General Parameters

Sub-Flow Sheet: T-100 (COL1) Number of Stages: 10

Profile Estimates

	Temperature	Net Liquid	Net Vapour
	(C)	(kgmole/h)	(kgmole/h)
1__Main Tower	41.98	1.998e+004	2.000e+004
2__Main Tower	42.25	1.994e+004	1.999e+004
3__Main Tower	42.91	1.908e+004	1.995e+004
4__Main Tower	49.12	1.218e+004	1.909e+004
5__Main Tower	119.1	3.129e+004	1.219e+004
6__Main Tower	119.4	3.133e+004	8148
7__Main Tower	120.1	3.138e+004	8188
8__Main Tower	123.4	3.154e+004	8241
9__Main Tower	135.8	3.246e+004	8394
10__Main Tower	169.5	3.614e+004	9317

EFFICIENCIES

Stage Efficiencies

Benzene	Propene	Propane	Cumene
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000

14-iP-BZ

1.000

1.000

1.000

1.000

1.000

1.000

1.000

1.000

1.000

1.000

SOLVER

Column Solving Algorithm: HYSIM Inside-Out

Solving Options Acceleration Parameters

Maximum Iterations: 10000 Accelerate K Value & H Model Parameters: Off

Equilibrium Error Tolerance: 1.000e-05

Heat/Spec Error Tolerance: 5.000e-004

Save Solutions as Initial Estimate: On

Super Critical Handling Model: Simple K

Trace Level: Low

Init from Ideal K's: Off Damping Parameters

Initial Estimate Generator Parameters Azeotrope Check: Off

Iterative IEG (Good for Chemicals): Off Fixed Damping Factor: 0

SIDE STRIPPERS

SIDE RECTIFIERS

PUMP AROUNDS

VAP BYPASSES

RATING

Tray Sections

Tray Section	Main Tower @COL1
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Tray Diameter (m)	1.500
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Weir Height (m)	5.000e-002
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Weir Length (m)	1.200
-----------------	-------

Tray Space (m)	0.5000
----------------	--------

Tray Volume (m3)	0.8836
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Disable Heat Loss Calculations No

Heat Model	None
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Rating Calculations	No
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Tray Hold Up (m3)	8.836e-002
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Vessels

Vessel

Diameter

Length

Volume

Orientation

Vessel has a Boot

Boot Diameter

Boot Length

Hold Up

Other Equipment In Column Flowsheet

Pressure Profile

	Pressure (kPa)	Pressure Drop (kPa)
1__Main Tower	1450 kPa	5.556 kPa
2__Main Tower	1456 kPa	5.556 kPa
3__Main Tower	1461 kPa	5.556 kPa
4__Main Tower	1467 kPa	5.556 kPa
5__Main Tower	1472 kPa	5.556 kPa
6__Main Tower	1478 kPa	5.556 kPa
7__Main Tower	1483 kPa	5.556 kPa
8__Main Tower	1489 kPa	5.556 kPa
9__Main Tower	1494 kPa	5.556 kPa
10__Main Tower	1500 kPa	---

Pressure Solving Options

Pressure Tolerance 1.000e-004 Pressure Drop Tolerance 1.000e-004

Damping Factor 1.000 Max Press Iterations 100

SUMMARY

Flow Basis: Molar The composition option is selected

Feed Composition

	Reflux-1	C1-Feed	Boil-Up
Flow Rate (kgmole/h)	1.999406e+04	2.315197e+04	1.300000e+04
	---	---	---
Benzene	0.0000	0.4830	0.5921
Propene	0.0413	0.0000	0.0040
Propane	0.9587	0.0424	0.2592
Cumene	0.0000	0.3126	0.1204
14-iP-BZ	0.0000	0.1620	0.0244

Flow Basis: Molar The composition option is selected

Feed Flows

	Reflux-1	C1-Feed	Boil-Up
Flow Rate (kgmole/h)	1.999406e+04	2.315197e+04	1.300000e+04
	---	---	---
Benzene (kgmole/h)	0.0953	1.118164e+04	7.697129e+03
Propene (kgmole/h)	824.8153	0.0000	51.8737
Propane (kgmole/h)	1.916915e+04	981.1578	3.369283e+03
Cumene (kgmole/h)	0.0000	7.237853e+03	1.564631e+03
14-iP-BZ (kgmole/h)	0.0000	3.751328e+03	317.0844

Products

Flow Basis: Molar The composition option is selected

Product Compositions

	C1-Top	C1-Bot
Flow Rate (kgmole/h)	2.000314e+04	3.614289e+04
	---	---
Benzene	0.0000	0.5223
Propene	0.0405	0.0018

Propane	0.9595	0.1197
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Cumene	0.0000	0.2435
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14-iP-BZ	0.0000	0.1126
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Flow Basis: Molar The composition option is selected

Product Flows

	C1-Top	C1-Bot
Flow Rate (kgmole/h)	2.000314e+04	3.614289e+04

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Benzene (kgmole/h)	0.0894	1.887877e+04
Propene (kgmole/h)	810.7227	65.9663
Propane (kgmole/h)	1.919233e+04	4.327257e+03
Cumene (kgmole/h)	0.0000	8.802483e+03
14-iP-BZ (kgmole/h)	0.0000	4.068413e+03

Flow Basis: Molar The composition option is selected

Product Recoveries

	C1-Top	C1-Bot
Flow Rate (kgmole/h)	2.000314e+04	3.614289e+04

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Benzene (%)	0.0005	99.9995
Propene (%)	92.4755	7.5245
Propane (%)	81.6015	18.3985
Cumene (%)	0.0000	100.0000
14-iP-BZ (%)	0.0000	100.0000

COLUMN PROFILES

Reflux Ratio: 0.9989 Reboil Ratio: --- The Flows Option is Selected Flow Basis: Molar

Column Profiles Flows

	Temp	Pres	Net Liq	Net Vap	Net Feed	Net Draws	Duty
	(C)	(kPa)	(kgmole/h)	(kgmole/h)	(kgmole/h)	(kgmole/h)	(kJ/h)
1__Main Tower	42.0	1450	1.998e+004	---	1.999e+004	2.000e+004	---
2__Main Tower	42.2	1456	1.994e+004	1.999e+004	---	---	---
3__Main Tower	42.9	1461	1.908e+004	1.995e+004	---	---	---
4__Main Tower	49.1	1467	1.218e+004	1.909e+004	---	---	---
5__Main Tower	119.1	1472	3.129e+004	1.219e+004	2.315e+004	---	---
6__Main Tower	119.4	1478	3.133e+004	8148	---	---	---
7__Main Tower	120.1	1483	3.138e+004	8188	---	---	---
8__Main Tower	123.4	1489	3.154e+004	8241	---	---	---
9__Main Tower	135.8	1494	3.246e+004	8394	---	---	---
10__Main Tower	169.5	1500	---	9317	1.300e+004	3.614e+004	---

Column Profiles Energy

	Temperature	Liq Enthalpy	Vap Enthalpy	Heat Loss
	(C)	(kJ/kgmole)	(kJ/kgmole)	(kJ/h)
1__Main Tower	41.98	-1.135e+005	-9.939e+004	---
2__Main Tower	42.25	-1.141e+005	-1.002e+005	---
3__Main Tower	42.91	-1.133e+005	-1.008e+005	---
4__Main Tower	49.12	-9.564e+004	-9.943e+004	---
5__Main Tower	119.1	-1.954e+004	-7.386e+004	---
6__Main Tower	119.4	-1.951e+004	-7.384e+004	---
7__Main Tower	120.1	-1.902e+004	-7.344e+004	---
8__Main Tower	123.4	-1.651e+004	-7.122e+004	---
9__Main Tower	135.8	-6744	-6.086e+004	---
10__Main Tower	169.5	1.485e+004	-2.244e+004	---

FEEDS / PRODUCTS

Flow Basis: Molar

Stream	Type	Duty	State	Flows	Enthalpy	Temp
		(kJ/h)		(kgmole/h)	(kJ/kgmole)	(C)
1__ Main Tower Reflux-1	Feed	---	Liquid	2.00e+004	-1.1e+005	41.90
C1-Top	Draw	---	Vapour	2.00e+004	-9.9e+004	41.98
2__ Main Tower						
3__ Main Tower						
4__ Main Tower						
5__ Main Tower C1-Feed	Feed	---	Liquid	2.32e+004	1.1e+004	150.42
6__ Main Tower						
7__ Main Tower						
8__ Main Tower						
9__ Main Tower						
10__ Main Tower Boil-Up	Feed	---	Vapour	1.30e+004	4.2e+004	219.26
C1-Bot	Draw	---	Liquid	3.61e+004	1.5e+004	169.50

SETUP

Sub-Flowsheet

Internal Feed Stream	External Feed Stream	Transition	Transfer Basis
Reflux-1	Reflux-1	Fluid Pkg Transition P-H Flash	
C1-Feed	C1-Feed	Fluid Pkg Transition P-H Flash	
Boil-Up	Boil-Up	Fluid Pkg Transition P-H Flash	
Internal Prod Stream	External Prod Stream	Transition	Transfer Basis
C1-Bot	C1-Bot	Fluid Pkg Transition P-H Flash	
C1-Top	C1-Top	Fluid Pkg Transition P-H Flash	

VARIABLES

Column Flowsheet Vars Available as Parameters

Data Source	Variable	Component	Description
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COMPONENT MAPS

Feed Streams

Feed Name	In to SubFlowSheet	Out of SubFlowSheet
-----------	--------------------	---------------------

Reflux-1

C1-Feed

Product Stream

Product Name	In to SubFlowSheet	Out of SubFlowSheet
--------------	--------------------	---------------------

C1-Bot

TRAY by TRAY PROPERTIES TABLES

Column Temperature / Pressure Profile

Column Stage	Temperature	Pressure
	(C)	(kPa)
1__Main Tower	41.98	1450
2__Main Tower	42.25	1456
3__Main Tower	42.91	1461
4__Main Tower	49.12	1467
5__Main Tower	119.1	1472
6__Main Tower	119.4	1478

7__Main Tower	120.1	1483
8__Main Tower	123.4	1489
9__Main Tower	135.8	1494
10__Main Tower	169.5	1500

Column Flow Profile

Options Selected

Molar flow is selected Net is selected as flow basis

Tray Number Vapour Bulk Liquid

(kgmole/h) (kgmole/h)

1__Main Tower	2.000e+004	1.998e+004
2__Main Tower	1.999e+004	1.994e+004
3__Main Tower	1.995e+004	1.908e+004
4__Main Tower	1.909e+004	1.218e+004
5__Main Tower	1.219e+004	3.129e+004
6__Main Tower	8148	3.133e+004
7__Main Tower	8188	3.138e+004
8__Main Tower	8241	3.154e+004
9__Main Tower	8394	3.246e+004
10__Main Tower	9317	3.614e+004

Column Properties Profile

Options Selected

Mass basis is selected

Stage Surf Tens Mol Wt Dens Visc Therm Con Heat Cap

L-Liq L-Liq L-Liq L-Liq L-Liq

(dyne/cm) (kg/m3) (cP) (W/m-K) (kJ/kg-C)

1__Main Tower	4.95	44.0	465	8.38e-002	8.54e-002	3.17
2__Main Tower	4.94	44.1	464	8.38e-002	8.52e-002	3.17
3__Main Tower	5.02	44.3	467	8.49e-002	8.50e-002	3.14
4__Main Tower	6.62	48.7	516	0.104	8.57e-002	2.75
5__Main Tower	12.9	89.2	709	0.183	9.47e-002	2.17
6__Main Tower	12.9	89.2	709	0.183	9.46e-002	2.17
7__Main Tower	12.8	89.2	708	0.183	9.46e-002	2.17
8__Main Tower	12.7	89.6	707	0.186	9.47e-002	2.18
9__Main Tower	12.4	90.8	704	0.174	9.46e-002	2.22
10__Main Tower	10.6	93.7	687	0.142	9.15e-002	2.34

Column Composition Profile

Options Selected

Fraction is selected as the composition basis Net is selected as flow basis

Molar basis is selected

Stage	Benzene	Propene	Propane	Cumene
	L-Liq	L-Liq	L-Liq	L-Liq
1__Main Tower	0.0001	0.0345	0.9655	0.0000
2__Main Tower	0.0007	0.0286	0.9707	0.0000
3__Main Tower	0.0083	0.0236	0.9681	0.0001
4__Main Tower	0.1075	0.0177	0.8626	0.0108
5__Main Tower	0.3854	0.0046	0.2556	0.2341
6__Main Tower	0.3856	0.0045	0.2559	0.2339
7__Main Tower	0.3877	0.0043	0.2542	0.2338
8__Main Tower	0.3988	0.0040	0.2437	0.2339

9__Main Tower	0.4420	0.0032	0.2038	0.2338
10__Main Tower	0.5223	0.0018	0.1197	0.2435

Heavy/Light Key Component Ratios

Options Selected

Molar basis is selected

Key Components

Light Key	Chosen	Heavy Key	Chosen
Benzene	On	Benzene	Off
Propene	On	Propene	Off
Propane	On	Propane	Off
Cumene	On	Cumene	Off
14-iP-BZ	Off	14-iP-BZ	On

Stage Light Liq

1__Main Tower	3.527e+011
2__Main Tower	4.455e+008
3__Main Tower	5.682e+005
4__Main Tower	704.8
5__Main Tower	7.315
6__Main Tower	7.325
7__Main Tower	7.337
8__Main Tower	7.363
9__Main Tower	7.532
10__Main Tower	7.884

Column K-Values Profile

Tray Number	Benzene	Propene	Propane	Cumene
1__Main Tower	8.068e-002	1.176	0.9938	7.108e-003
2__Main Tower	8.132e-002	1.178	0.9954	7.198e-003
3__Main Tower	8.148e-002	1.185	1.003	7.242e-003
4__Main Tower	7.715e-002	1.290	1.123	6.939e-003
5__Main Tower	0.2788	3.579	3.380	4.593e-002
6__Main Tower	0.2795	3.574	3.376	4.616e-002
7__Main Tower	0.2831	3.590	3.392	4.710e-002
8__Main Tower	0.3004	3.693	3.496	5.132e-002
9__Main Tower	0.3764	4.128	3.933	7.086e-002
10__Main Tower	0.6504	5.270	5.072	0.1548

Absorber: T-101

CONNECTIONS

Inlet Stream

Stream Name	Stage	From Unit Operation
Reflux-2	1__Main Tower	RCY-6 Recycle
Boilup-2	22__Main Tower	RCY-5 Recycle
C2-Feed-LP	6__Main Tower	VLV-101 Valve

Outlet Stream

Stream Name	Stage	To Unit Operation
C2-Bot	22__Main Tower	E-104 Heater
C2-Top	1__Main Tower	E-103 Cooler

MONITOR

Specifications Summary

	Specified Value	Current Value	Wt. Error
Max_Benzene	1.000e-003	---	---
Reflux Ratio	1.500	0.2898	-0.8068
	Wt. Tol.	Abs. Tol.	Active Estimate Used
Max_Benzene	1.000e-002	1.000e-003	Off On Off
Reflux Ratio	1.000e-002	1.000e-002	Off On Off

SPECS

Column Specification Parameters

Max_Benzene

Fix/Rang: Fixed Prim/Alter: Primary Lower Bnd: --- Upper Bnd: ---

Stage: 22__Main Tower Flow Basis: Mole Fraction Phase: Liquid

Components:

Reflux Ratio

Fix/Rang: Fixed Prim/Alter: Primary Lower Bnd: --- Upper Bnd: ---

Stage: 1__Main Tower Flow Basis: Molar Liquid Spec: ---

SUBCOOLING

Degrees of Subcooling

Subcool to

User Variables

PROFILES

General Parameters

Sub-Flow Sheet: T-101 (COL2) Number of Stages: 22

Profile Estimates

	Temperature (C)	Net Liquid (kgmole/h)	Net Vapour (kgmole/h)
1__Main Tower	115.6	8306	2.866e+004
2__Main Tower	126.3	8333	2.864e+004
3__Main Tower	128.8	8222	2.866e+004
4__Main Tower	130.4	7947	2.855e+004
5__Main Tower	132.9	7473	2.828e+004
6__Main Tower	137.1	2.076e+004	2.780e+004
7__Main Tower	141.5	2.121e+004	5155
8__Main Tower	146.5	2.163e+004	5601
9__Main Tower	152.3	2.219e+004	6023
10__Main Tower	157.6	2.277e+004	6580
11__Main Tower	161.4	2.323e+004	7161
12__Main Tower	163.8	2.353e+004	7622
13__Main Tower	165.1	2.371e+004	7921
14__Main Tower	165.9	2.382e+004	8099
15__Main Tower	166.4	2.389e+004	8209
16__Main Tower	166.9	2.395e+004	8286
17__Main Tower	167.2	2.401e+004	8346
18__Main Tower	167.6	2.406e+004	8399
19__Main Tower	167.9	2.410e+004	8449

1.000	1.000	1.000	1.000
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1.000	1.000	1.000	1.000
-------	-------	-------	-------

14-iP-BZ

1.000

1.000

1.000

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SOLVER

Column Solving Algorithm: HYSIM Inside-Out

Solving Options Acceleration Parameters

Maximum Iterations: 10000 Accelerate K Value & H Model Parameters: Off

Equilibrium Error Tolerance: 1.000e-05

Heat/Spec Error Tolerance: 5.000e-004

Save Solutions as Initial Estimate: On

Super Critical Handling Model: Simple K

Trace Level: Low

Init from Ideal K's: Off Damping Parameters

Initial Estimate Generator Parameters Azeotrope Check: Off

Iterative IEG (Good for Chemicals): Off Fixed Damping Factor: 0

SIDE STRIPPERS

SIDE RECTIFIERS

PUMP AROUNDS

VAP BYPASSES

RATING

Tray Sections

Tray Section Main Tower @COL2

Tray Diameter (m) 1.500

Weir Height (m)	5.000e-002
Weir Length (m)	1.200
Tray Space (m)	0.5000
Tray Volume (m3)	0.8836
Disable Heat Loss Calculations	No
Heat Model	None
Rating Calculations	No
Tray Hold Up (m3)	8.836e-002

Vessels

Vessel

Diameter

Length

Volume

Orientation

Vessel has a Boot

Boot Diameter

Boot Length

Hold Up

Other Equipment In Column Flowsheet

Pressure Profile

	Pressure (kPa)	Pressure Drop (kPa)
1__Main Tower	100.0 kPa	0.9524 kPa
2__Main Tower	101.0 kPa	0.9524 kPa

3__Main Tower	101.9 kPa	0.9524 kPa
4__Main Tower	102.9 kPa	0.9524 kPa
5__Main Tower	103.8 kPa	0.9524 kPa
6__Main Tower	104.8 kPa	0.9524 kPa
7__Main Tower	105.7 kPa	0.9524 kPa
8__Main Tower	106.7 kPa	0.9524 kPa
9__Main Tower	107.6 kPa	0.9524 kPa
10__Main Tower	108.6 kPa	0.9524 kPa
11__Main Tower	109.5 kPa	0.9524 kPa
12__Main Tower	110.5 kPa	0.9524 kPa
13__Main Tower	111.4 kPa	0.9524 kPa
14__Main Tower	112.4 kPa	0.9524 kPa
15__Main Tower	113.3 kPa	0.9524 kPa
16__Main Tower	114.3 kPa	0.9524 kPa
17__Main Tower	115.2 kPa	0.9524 kPa
18__Main Tower	116.2 kPa	0.9524 kPa
19__Main Tower	117.1 kPa	0.9524 kPa
20__Main Tower	118.1 kPa	0.9524 kPa
21__Main Tower	119.0 kPa	0.9524 kPa
22__Main Tower	120.0 kPa	---

Pressure Solving Options

Pressure Tolerance 1.000e-004 Pressure Drop Tolerance 1.000e-004

Damping Factor 1.000 Max Press Iterations 100

SUMMARY

Flow Basis: Molar The composition option is selected

Feed Composition

	Reflux-2	C2-Feed-LP	Boilup-2
Flow Rate (kgmole/h)	8.327239e+03	3.593936e+04	8.500000e+03

	---	---	---
Benzene	0.6939	0.3819	0.0000
Propene	0.0004	0.0004	0.0000
Propane	0.0354	0.0373	0.0000
Cumene	0.2701	0.4327	0.8794
14-iP-BZ	0.0002	0.1477	0.1206

Flow Basis: Molar The composition option is selected

Feed Flows

	Reflux-2	C2-Feed-LP	Boilup-2
Flow Rate (kgmole/h)	8.327239e+03	3.593936e+04	8.500000e+03

	---	---	---
Benzene (kgmole/h)	5.778204e+03	1.372673e+04	0.0367
Propene (kgmole/h)	3.0996	14.1548	0.0000
Propane (kgmole/h)	294.6017	1.340984e+03	0.0000
Cumene (kgmole/h)	2.249338e+03	1.554994e+04	7.474488e+03
14-iP-BZ (kgmole/h)	1.9954	5.307552e+03	1.025476e+03

Products

Flow Basis: Molar The composition option is selected

Product Compositions

	C2-Top	C2-Bot
Flow Rate (kgmole/h)	2.865854e+04	2.410806e+04

	---	---
Benzene	0.6806	0.0000
Propene	0.0006	0.0000

Propane	0.0571	0.0000
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Cumene	0.2615	0.7375
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14-iP-BZ	0.0002	0.2625
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Flow Basis: Molar The composition option is selected

Product Flows

	C2-Top	C2-Bot
Flow Rate (kgmole/h)	2.865854e+04	2.410806e+04

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Benzene (kgmole/h)	1.950492e+04	0.0506
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Propene (kgmole/h)	17.2544	0.0000
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Propane (kgmole/h)	1.635586e+03	0.0000
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Cumene (kgmole/h)	7.494150e+03	1.777962e+04
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14-iP-BZ (kgmole/h)	6.6323	6.328390e+03
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Flow Basis: Molar The composition option is selected

Product Recoveries

	C2-Top	C2-Bot
Flow Rate (kgmole/h)	2.865854e+04	2.410806e+04

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Benzene (%)	99.9997	0.0003
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Propene (%)	100.0000	0.0000
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Propane (%)	100.0000	0.0000
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Cumene (%)	29.6519	70.3481
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14-iP-BZ (%)	0.1047	99.8953
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COLUMN PROFILES

Reflux Ratio: 0.2898 Reboil Ratio: --- The Flows Option is Selected Flow Basis: Molar

Column Profiles Flows

	Temp	Pres	Net Liq	Net Vap	Net Feed	Net Draws	Duty	
	(C)	(kPa)	(kgmole/h)	(kgmole/h)	(kgmole/h)	(kgmole/h)	(kJ/h)	
1__Main Tower	115.6	100.0	8306	---	8327	2.866e+004	---	
2__Main Tower	126.3	101.0	8333	2.864e+004	---	---	---	
3__Main Tower	128.8	101.9	8222	2.866e+004	---	---	---	
4__Main Tower	130.4	102.9	7947	2.855e+004	---	---	---	
5__Main Tower	132.9	103.8	7473	2.828e+004	---	---	---	
6__Main Tower	137.1	104.8	2.076e+004	2.780e+004	3.594e+004	---	---	
7__Main Tower	141.5	105.7	2.121e+004	5155	---	---	---	
8__Main Tower	146.5	106.7	2.163e+004	5601	---	---	---	
9__Main Tower	152.3	107.6	2.219e+004	6023	---	---	---	
10__Main Tower	157.6	108.6	2.277e+004	6580	---	---	---	
11__Main Tower	161.4	109.5	2.323e+004	7161	---	---	---	
12__Main Tower	163.8	110.5	2.353e+004	7622	---	---	---	
13__Main Tower	165.1	111.4	2.371e+004	7921	---	---	---	
14__Main Tower	165.9	112.4	2.382e+004	8099	---	---	---	
15__Main Tower	166.4	113.3	2.389e+004	8209	---	---	---	
16__Main Tower	166.9	114.3	2.395e+004	8286	---	---	---	
17__Main Tower	167.2	115.2	2.401e+004	8346	---	---	---	
18__Main Tower	167.6	116.2	2.406e+004	8399	---	---	---	
19__Main Tower	167.9	117.1	2.410e+004	8449	---	---	---	
20__Main Tower	168.3	118.1	2.415e+004	8496	---	---	---	
21__Main Tower	168.7	119.0	2.417e+004	8540	---	---	---	
22__Main Tower	169.5	120.0	---	8566	8500	2.411e+004	---	

Column Profiles Energy

	Temperature	Liq Enthalpy	Vap Enthalpy	Heat Loss
(C)	(kJ/kgmole)	(kJ/kgmole)	(kJ/h)	

1__Main Tower	115.6	942.3	6.185e+004	---
2__Main Tower	126.3	-4544	5.532e+004	---
3__Main Tower	128.8	-6441	5.367e+004	---
4__Main Tower	130.4	-9435	5.335e+004	---
5__Main Tower	132.9	-1.548e+004	5.309e+004	---
6__Main Tower	137.1	-2.514e+004	5.253e+004	---
7__Main Tower	141.5	-2.469e+004	5.697e+004	---
8__Main Tower	146.5	-2.525e+004	5.214e+004	---
9__Main Tower	152.3	-2.577e+004	4.473e+004	---
10__Main Tower	157.6	-2.600e+004	3.706e+004	---
11__Main Tower	161.4	-2.601e+004	3.123e+004	---
12__Main Tower	163.8	-2.592e+004	2.775e+004	---
13__Main Tower	165.1	-2.580e+004	2.599e+004	---
14__Main Tower	165.9	-2.568e+004	2.519e+004	---
15__Main Tower	166.4	-2.556e+004	2.486e+004	---
16__Main Tower	166.9	-2.543e+004	2.475e+004	---
17__Main Tower	167.2	-2.531e+004	2.475e+004	---
18__Main Tower	167.6	-2.519e+004	2.478e+004	---
19__Main Tower	167.9	-2.507e+004	2.484e+004	---
20__Main Tower	168.3	-2.497e+004	2.489e+004	---
21__Main Tower	168.7	-2.498e+004	2.492e+004	---
22__Main Tower	169.5	-2.569e+004	2.474e+004	---

FEEDS / PRODUCTS

Flow Basis: Molar

Stream	Type	Duty	State	Flows	Enthalpy	Temp
		(kJ/h)		(kgmole/h)	(kJ/kgmole)	(C)

1__ Main Tower Reflux-2 Feed	---	Liquid	8.33e+003	2.4e+004	50.02
C2-Top Draw	---	Vapour	2.87e+004	6.2e+004	115.65
2__ Main Tower					
3__ Main Tower					
4__ Main Tower					
5__ Main Tower					
6__ Main Tower C2-Feed-LP Feed	---	Mixed	3.59e+004	2.1e+004	141.81
7__ Main Tower					
8__ Main Tower					
9__ Main Tower					
10__ Main Tower					
11__ Main Tower					
12__ Main Tower					
13__ Main Tower					
14__ Main Tower					
15__ Main Tower					
16__ Main Tower					
17__ Main Tower					
18__ Main Tower					
19__ Main Tower					
20__ Main Tower					
21__ Main Tower					
22__ Main Tower Boilup-2 Feed	---	Vapour	8.50e+003	2.3e+004	172.65
C2-Bot Draw	---	Liquid	2.41e+004	-2.6e+004	169.49

SETUP

Sub-Flowsheet

Internal Feed Stream	External Feed Stream	Transition	Transfer Basis
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Reflux-2	Reflux-2	Fluid Pkg Transition	P-H Flash
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Boilup-2	Boilup-2	Fluid Pkg Transition	P-H Flash
----------	----------	----------------------	-----------

C2-Feed-LP	C2-Feed-LP	Fluid Pkg Transition	P-H Flash
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Internal Prod Stream	External Prod Stream	Transition	Transfer Basis
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C2-Bot	C2-Bot	Fluid Pkg Transition	P-H Flash
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C2-Top	C2-Top	Fluid Pkg Transition	P-H Flash
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VARIABLES

Column Flowsheet Vars Available as Parameters

Data Source	Variable	Component	Description
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COMPONENT MAPS

Feed Streams

Feed Name	In to SubFlowSheet	Out of SubFlowSheet
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Reflux-2

Boilup-2

Product Stream

Product Name	In to SubFlowSheet	Out of SubFlowSheet
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C2-Bot

TRAY by TRAY PROPERTIES TABLES

Column Temperature / Pressure Profile

Column Stage	Temperature	Pressure
	(C)	(kPa)
1__Main Tower	115.6	100.0
2__Main Tower	126.3	101.0
3__Main Tower	128.8	101.9
4__Main Tower	130.4	102.9
5__Main Tower	132.9	103.8
6__Main Tower	137.1	104.8
7__Main Tower	141.5	105.7
8__Main Tower	146.5	106.7
9__Main Tower	152.3	107.6
10__Main Tower	157.6	108.6
11__Main Tower	161.4	109.5
12__Main Tower	163.8	110.5
13__Main Tower	165.1	111.4
14__Main Tower	165.9	112.4
15__Main Tower	166.4	113.3
16__Main Tower	166.9	114.3
17__Main Tower	167.2	115.2
18__Main Tower	167.6	116.2
19__Main Tower	167.9	117.1
20__Main Tower	168.3	118.1
21__Main Tower	168.7	119.0
22__Main Tower	169.5	120.0

Column Flow Profile

Options Selected

Molar flow is selected Net is selected as flow basis

Tray Number Vapour Bulk Liquid

(kgmole/h) (kgmole/h)

1__Main Tower	2.866e+004	8306
2__Main Tower	2.864e+004	8333
3__Main Tower	2.866e+004	8222
4__Main Tower	2.855e+004	7947
5__Main Tower	2.828e+004	7473
6__Main Tower	2.780e+004	2.076e+004
7__Main Tower	5155	2.121e+004
8__Main Tower	5601	2.163e+004
9__Main Tower	6023	2.219e+004
10__Main Tower	6580	2.277e+004
11__Main Tower	7161	2.323e+004
12__Main Tower	7622	2.353e+004
13__Main Tower	7921	2.371e+004
14__Main Tower	8099	2.382e+004
15__Main Tower	8209	2.389e+004
16__Main Tower	8286	2.395e+004
17__Main Tower	8346	2.401e+004
18__Main Tower	8399	2.406e+004
19__Main Tower	8449	2.410e+004
20__Main Tower	8496	2.415e+004
21__Main Tower	8540	2.417e+004
22__Main Tower	8566	2.411e+004

Column Properties Profile

Options Selected

Mass basis is selected

Stage	Surf Tens	Mol Wt	Dens	Visc	Therm Con	Heat Cap
	L-Liq	L-Liq	L-Liq	L-Liq	L-Liq	
	(dyne/cm)	(kg/m3)	(cP)	(W/m-K)	(kJ/kg-C)	
1__Main Tower	18.1	110	781	0.270	0.114	2.03
2__Main Tower	17.3	114	770	0.250	0.113	2.08
3__Main Tower	17.1	115	768	0.246	0.113	2.10
4__Main Tower	17.1	117	767	0.245	0.113	2.10
5__Main Tower	16.9	120	766	0.243	0.114	2.11
6__Main Tower	16.8	126	764	0.240	0.115	2.13
7__Main Tower	16.4	126	760	0.232	0.114	2.15
8__Main Tower	16.0	127	754	0.224	0.113	2.17
9__Main Tower	15.6	129	748	0.214	0.112	2.20
10__Main Tower	15.2	130	743	0.206	0.111	2.23
11__Main Tower	14.9	130	739	0.200	0.110	2.24
12__Main Tower	14.7	130	736	0.197	0.110	2.25
13__Main Tower	14.6	131	735	0.195	0.109	2.26
14__Main Tower	14.6	131	734	0.194	0.109	2.26
15__Main Tower	14.5	131	734	0.193	0.109	2.27
16__Main Tower	14.5	131	733	0.193	0.109	2.27
17__Main Tower	14.4	131	733	0.192	0.109	2.27
18__Main Tower	14.4	131	733	0.191	0.109	2.27
19__Main Tower	14.4	131	732	0.191	0.109	2.27
20__Main Tower	14.4	131	732	0.190	0.109	2.27
21__Main Tower	14.3	131	732	0.190	0.109	2.27

22__Main Tower 14.3 131 731 0.189 0.109 2.28

Column Composition Profile

Options Selected

Fraction is selected as the composition basis Net is selected as flow basis

Molar basis is selected

Stage	Benzene	Propene	Propane	Cumene
	L-Liq	L-Liq	L-Liq	L-Liq
1__Main Tower	0.2544	0.0000	0.0013	0.7413
2__Main Tower	0.1638	0.0000	0.0010	0.8255
3__Main Tower	0.1491	0.0000	0.0009	0.8241
4__Main Tower	0.1443	0.0000	0.0009	0.7914
5__Main Tower	0.1378	0.0000	0.0009	0.7204
6__Main Tower	0.1275	0.0000	0.0009	0.6055
7__Main Tower	0.1142	0.0000	0.0001	0.6227
8__Main Tower	0.0885	0.0000	0.0000	0.6503
9__Main Tower	0.0590	0.0000	0.0000	0.6823
10__Main Tower	0.0340	0.0000	0.0000	0.7103
11__Main Tower	0.0175	0.0000	0.0000	0.7292
12__Main Tower	0.0083	0.0000	0.0000	0.7399
13__Main Tower	0.0038	0.0000	0.0000	0.7454
14__Main Tower	0.0017	0.0000	0.0000	0.7482
15__Main Tower	0.0008	0.0000	0.0000	0.7496
16__Main Tower	0.0003	0.0000	0.0000	0.7505
17__Main Tower	0.0002	0.0000	0.0000	0.7510

18__Main Tower	0.0001	0.0000	0.0000	0.7514
19__Main Tower	0.0000	0.0000	0.0000	0.7517
20__Main Tower	0.0000	0.0000	0.0000	0.7518
21__Main Tower	0.0000	0.0000	0.0000	0.7501
22__Main Tower	0.0000	0.0000	0.0000	0.7375

Heavy/Light Key Component Ratios

Options Selected

Molar basis is selected

Key Components

Light Key	Chosen	Heavy Key	Chosen
Benzene	On	Benzene	Off
Propene	On	Propene	Off
Propane	On	Propane	Off
Cumene	On	Cumene	Off
14-iP-BZ	Off	14-iP-BZ	On

Stage Light Liq

1__Main Tower	325.5
2__Main Tower	101.7
3__Main Tower	37.65
4__Main Tower	14.76
5__Main Tower	6.098
6__Main Tower	2.758
7__Main Tower	2.801
8__Main Tower	2.828
9__Main Tower	2.866

10__Main Tower 2.911
11__Main Tower 2.947
12__Main Tower 2.972
13__Main Tower 2.988
14__Main Tower 2.999
15__Main Tower 3.006
16__Main Tower 3.013
17__Main Tower 3.019
18__Main Tower 3.024
19__Main Tower 3.029
20__Main Tower 3.029
21__Main Tower 3.002
22__Main Tower 2.810

Column K-Values Profile

Tray Number	Benzene	Propene	Propane	Cumene
1__Main Tower	2.676	49.11	45.38	0.3528
2__Main Tower	3.376	53.92	49.49	0.4823
3__Main Tower	3.531	54.58	50.04	0.5132
4__Main Tower	3.630	54.72	50.13	0.5320
5__Main Tower	3.818	55.22	50.51	0.5662
6__Main Tower	4.163	56.47	51.56	0.6313
7__Main Tower	4.496	58.10	53.07	0.7061
8__Main Tower	4.888	59.91	54.71	0.7965
9__Main Tower	5.382	62.09	56.69	0.9152
10__Main Tower	5.854	63.98	58.42	1.034
11__Main Tower	6.189	65.12	59.47	1.124

12__Main Tower	6.376	65.56	59.89	1.176
13__Main Tower	6.460	65.58	59.91	1.203
14__Main Tower	6.490	65.36	59.72	1.216
15__Main Tower	6.493	65.04	59.44	1.221
16__Main Tower	6.484	64.68	59.12	1.223
17__Main Tower	6.470	64.30	58.78	1.224
18__Main Tower	6.453	63.92	58.43	1.224
19__Main Tower	6.436	63.54	58.09	1.224
20__Main Tower	6.419	63.16	57.75	1.224
21__Main Tower	6.410	62.82	57.44	1.225
22__Main Tower	6.450	62.63	57.28	1.239

Absorber: T-102

CONNECTIONS

Inlet Stream

Stream Name	Stage	From Unit Operation
Reflux-3	1__Main Tower	RCY-8 Recycle
Boilup-3	20__Main Tower	RCY-7 Recycle
C2-Bottoms	10__Main Tower	V-103 Separator

Outlet Stream

Stream Name	Stage	To Unit Operation
C3-Bot	20__Main Tower	E-106 Heater
C3-Top	1__Main Tower	E-105 Cooler

MONITOR

Specifications Summary

Specified Value	Current Value	Wt. Error
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Wt. Tol.	Abs. Tol.	Active Estimate Used
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SPECS

Column Specification Parameters

Fix/Rang:	Prim/Alter:	Lower Bnd:	Upper Bnd:
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SUBCOOLING

Degrees of Subcooling

Subcool to

User Variables

PROFILES

General Parameters

Sub-Flow Sheet: T-102 (COL3)	Number of Stages: 20
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Profile Estimates

Temperature	Net Liquid	Net Vapour
(C)	(kgmole/h)	(kgmole/h)

1__Main Tower	153.5	2255	4594
2__Main Tower	154.5	2227	5474
3__Main Tower	155.7	2188	5445
4__Main Tower	157.2	2139	5407
5__Main Tower	159.1	2084	5358
6__Main Tower	161.3	2028	5303
7__Main Tower	163.5	1978	5247
8__Main Tower	165.7	1938	5197
9__Main Tower	167.5	1908	5157
10__Main Tower	169.0	1.705e+004	5126
11__Main Tower	169.3	1.709e+004	4663
12__Main Tower	169.7	1.714e+004	4705
13__Main Tower	170.1	1.718e+004	4747
14__Main Tower	170.4	1.722e+004	4788
15__Main Tower	170.8	1.726e+004	4830
16__Main Tower	171.2	1.730e+004	4871
17__Main Tower	171.5	1.734e+004	4912
18__Main Tower	171.9	1.738e+004	4953
19__Main Tower	172.3	1.741e+004	4992
20__Main Tower	173.2	1.739e+004	5021

EFFICIENCIES

Stage Efficiencies

Benzene	Propene	Propane	Cumene
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000

1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000
1.000	1.000	1.000	1.000

14-iP-BZ

1.000
1.000
1.000
1.000
1.000
1.000
1.000
1.000
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SOLVER

Column Solving Algorithm: HYSIM Inside-Out

Solving Options Acceleration Parameters

Maximum Iterations: 10000 Accelerate K Value & H Model Parameters: Off

Equilibrium Error Tolerance: 1.000e-05

Heat/Spec Error Tolerance: 5.000e-004

Save Solutions as Initial Estimate: On

Super Critical Handling Model: Simple K

Trace Level: Low

Init from Ideal K's: Off Damping Parameters

Initial Estimate Generator Parameters Azeotrope Check: Off

Iterative IEG (Good for Chemicals): Off Fixed Damping Factor: 0

SIDE STRIPPERS

SIDE RECTIFIERS

PUMP AROUNDS

VAP BYPASSES

RATING

Tray Sections

Tray Section	Main Tower @COL3
--------------	------------------

Tray Diameter (m)	1.500
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Weir Height (m)	5.000e-002
-----------------	------------

Weir Length (m)	1.200
-----------------	-------

Tray Space (m)	0.5000
----------------	--------

Tray Volume (m3)	0.8836
------------------	--------

Disable Heat Loss Calculations No

Heat Model	None
------------	------

Rating Calculations	No
---------------------	----

Tray Hold Up (m3)	8.836e-002
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Vessels

Vessel

Diameter

Length

Volume

Orientation

Vessel has a Boot

Boot Diameter

Boot Length

Hold Up

Other Equipment In Column Flowsheet

Pressure Profile

	Pressure (kPa)	Pressure Drop (kPa)
1__ Main Tower	100.0 kPa	1.053 kPa
2__ Main Tower	101.1 kPa	1.053 kPa
3__ Main Tower	102.1 kPa	1.053 kPa
4__ Main Tower	103.2 kPa	1.053 kPa
5__ Main Tower	104.2 kPa	1.053 kPa
6__ Main Tower	105.3 kPa	1.053 kPa
7__ Main Tower	106.3 kPa	1.053 kPa
8__ Main Tower	107.4 kPa	1.053 kPa
9__ Main Tower	108.4 kPa	1.053 kPa
10__ Main Tower	109.5 kPa	1.053 kPa
11__ Main Tower	110.5 kPa	1.053 kPa
12__ Main Tower	111.6 kPa	1.053 kPa
13__ Main Tower	112.6 kPa	1.053 kPa
14__ Main Tower	113.7 kPa	1.053 kPa
15__ Main Tower	114.7 kPa	1.053 kPa
16__ Main Tower	115.8 kPa	1.053 kPa

17__Main Tower	116.8 kPa	1.053 kPa
18__Main Tower	117.9 kPa	1.053 kPa
19__Main Tower	118.9 kPa	1.053 kPa
20__Main Tower	120.0 kPa	---

Pressure Solving Options

Pressure Tolerance 1.000e-004 Pressure Drop Tolerance 1.000e-004

Damping Factor 1.000 Max Press Iterations 100

SUMMARY

Flow Basis: Molar The composition option is selected

Feed Composition

	Reflux-3	C2-Bottoms	Boilup-3
Flow Rate (kgmole/h)	1.374674e+03	1.560806e+04	5.000000e+03
	---	---	---
Benzene	0.0000	0.0000	0.0000
Propene	0.0000	0.0000	0.0000
Propane	0.0000	0.0000	0.0000
Cumene	0.9960	0.6603	0.8325
14-iP-BZ	0.0040	0.3397	0.1675

Flow Basis: Molar The composition option is selected

Feed Flows

	Reflux-3	C2-Bottoms	Boilup-3
Flow Rate (kgmole/h)	1.374674e+03	1.560806e+04	5.000000e+03
	---	---	---
Benzene (kgmole/h)	0.0046	0.0108	0.0000
Propene (kgmole/h)	0.0000	0.0000	0.0000

Propane (kgmole/h)	0.0000	0.0000	0.0000
Cumene (kgmole/h)	1.369119e+03	1.030671e+04	4.162584e+03
14-iP-BZ (kgmole/h)	5.5500	5.301338e+03	837.4157

Products

Flow Basis: Molar The composition option is selected

Product Compositions

	C3-Top	C3-Bot
Flow Rate (kgmole/h)	4.593595e+03	1.738914e+04

	---	---
Benzene	0.0000	0.0000
Propene	0.0000	0.0000
Propane	0.0000	0.0000
Cumene	0.9963	0.6476
14-iP-BZ	0.0037	0.3524

Flow Basis: Molar The composition option is selected

Product Flows

	C3-Top	C3-Bot
Flow Rate (kgmole/h)	4.593595e+03	1.738914e+04

	---	---
Benzene (kgmole/h)	0.0154	0.0000
Propene (kgmole/h)	0.0000	0.0000
Propane (kgmole/h)	0.0000	0.0000
Cumene (kgmole/h)	4.576710e+03	1.126170e+04
14-iP-BZ (kgmole/h)	16.8691	6.127435e+03

Flow Basis: Molar The composition option is selected

Product Recoveries

C3-Top	C3-Bot
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Flow Rate (kgmole/h) 4.593595e+03 1.738914e+04

	---	---
Benzene (%)	99.8983	0.1017
Propene (%)	99.8609	0.1391
Propane (%)	99.9997	0.0003
Cumene (%)	28.8963	71.1037
14-iP-BZ (%)	0.2745	99.7255

COLUMN PROFILES

Reflux Ratio: 0.4910 Reboil Ratio: --- The Flows Option is Selected Flow Basis: Molar

Column Profiles Flows

	Temp (C)	Pres (kPa)	Net Liq (kgmole/h)	Net Vap (kgmole/h)	Net Feed (kgmole/h)	Net Draws (kgmole/h)	Duty (kJ/h)
1__Main Tower	153.5	100.0	2255	---	1375	4594	---
2__Main Tower	154.5	101.1	2227	5474	---	---	---
3__Main Tower	155.7	102.1	2188	5445	---	---	---
4__Main Tower	157.2	103.2	2139	5407	---	---	---
5__Main Tower	159.1	104.2	2084	5358	---	---	---
6__Main Tower	161.3	105.3	2028	5303	---	---	---
7__Main Tower	163.5	106.3	1978	5247	---	---	---
8__Main Tower	165.7	107.4	1938	5197	---	---	---
9__Main Tower	167.5	108.4	1908	5157	---	---	---
10__Main Tower	169.0	109.5	1.705e+004	5126	1.561e+004	---	---
11__Main Tower	169.3	110.5	1.709e+004	4663	---	---	---
12__Main Tower	169.7	111.6	1.714e+004	4705	---	---	---
13__Main Tower	170.1	112.6	1.718e+004	4747	---	---	---
14__Main Tower	170.4	113.7	1.722e+004	4788	---	---	---

15__Main Tower	170.8	114.7	1.726e+004	4830	---	---	---
16__Main Tower	171.2	115.8	1.730e+004	4871	---	---	---
17__Main Tower	171.5	116.8	1.734e+004	4912	---	---	---
18__Main Tower	171.9	117.9	1.738e+004	4953	---	---	---
19__Main Tower	172.3	118.9	1.741e+004	4992	---	---	---
20__Main Tower	173.2	120.0	---	5021	5000	1.739e+004	---

Column Profiles Energy

	Temperature (C)	Liq Enthalpy (kJ/kgmole)	Vap Enthalpy (kJ/kgmole)	Heat Loss (kJ/h)
1__Main Tower	153.5	-1.114e+004	2.703e+004	---
2__Main Tower	154.5	-1.218e+004	2.692e+004	---
3__Main Tower	155.7	-1.378e+004	2.669e+004	---
4__Main Tower	157.2	-1.606e+004	2.632e+004	---
5__Main Tower	159.1	-1.897e+004	2.578e+004	---
6__Main Tower	161.3	-2.226e+004	2.507e+004	---
7__Main Tower	163.5	-2.552e+004	2.427e+004	---
8__Main Tower	165.7	-2.833e+004	2.348e+004	---
9__Main Tower	167.5	-3.050e+004	2.280e+004	---
10__Main Tower	169.0	-3.201e+004	2.230e+004	---
11__Main Tower	169.3	-3.185e+004	2.238e+004	---
12__Main Tower	169.7	-3.170e+004	2.247e+004	---
13__Main Tower	170.1	-3.154e+004	2.256e+004	---
14__Main Tower	170.4	-3.139e+004	2.264e+004	---
15__Main Tower	170.8	-3.123e+004	2.272e+004	---
16__Main Tower	171.2	-3.108e+004	2.281e+004	---
17__Main Tower	171.5	-3.093e+004	2.289e+004	---
18__Main Tower	171.9	-3.080e+004	2.297e+004	---

19__Main Tower 172.3 -3.077e+004 2.300e+004 ---

20__Main Tower 173.2 -3.140e+004 2.279e+004 ---

FEEDS / PRODUCTS

Flow Basis: Molar

Stream	Type	Duty	State	Flows	Enthalpy	Temp
		(kJ/h)	(kgmole/h)	(kJ/kgmole)	(C)	
1__Main Tower	Reflux-3	Feed	---	Liquid	1.37e+003	-3.5e+004 50.02
C3-Top	Draw	---	Vapour	4.59e+003	2.7e+004	153.55
2__Main Tower						
3__Main Tower						
4__Main Tower						
5__Main Tower						
6__Main Tower						
7__Main Tower						
8__Main Tower						
9__Main Tower						
10__Main Tower	C2-Bottoms	Feed	---	Liquid	1.56e+004	-3.1e+004 172.66
11__Main Tower						
12__Main Tower						
13__Main Tower						
14__Main Tower						
15__Main Tower						
16__Main Tower						
17__Main Tower						
18__Main Tower						
19__Main Tower						

20__Main Tower Boilup-3 Feed --- Vapour 5.00e+003 2.1e+004 176.58

C3-Bot Draw --- Liquid 1.74e+004 -3.1e+004 173.21

SETUP

Sub-Flowsheet

Internal Feed Stream External Feed Stream Transition Transfer Basis

Reflux-3 Reflux-3 Fluid Pkg Transition P-H Flash

Boilup-3 Boilup-3 Fluid Pkg Transition P-H Flash

C2-Bottoms C2-Bottoms Fluid Pkg Transition P-H Flash

Internal Prod Stream External Prod Stream Transition Transfer Basis

C3-Bot C3-Bot Fluid Pkg Transition P-H Flash

C3-Top C3-Top Fluid Pkg Transition P-H Flash

VARIABLES

Column Flowsheet Vars Available as Parameters

Data Source	Variable	Component	Description
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COMPONENT MAPS

Feed Streams

Feed Name	In to SubFlowSheet	Out of SubFlowSheet
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Reflux-3

Boilup-3

Product Stream

Product Name	In to SubFlowSheet	Out of SubFlowSheet
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C3-Bot

TRAY by TRAY PROPERTIES TABLES

Column Temperature / Pressure Profile

Column Stage	Temperature	Pressure
(C)	(kPa)	
1__Main Tower	153.5	100.0
2__Main Tower	154.5	101.1
3__Main Tower	155.7	102.1
4__Main Tower	157.2	103.2
5__Main Tower	159.1	104.2
6__Main Tower	161.3	105.3
7__Main Tower	163.5	106.3
8__Main Tower	165.7	107.4
9__Main Tower	167.5	108.4
10__Main Tower	169.0	109.5
11__Main Tower	169.3	110.5
12__Main Tower	169.7	111.6
13__Main Tower	170.1	112.6
14__Main Tower	170.4	113.7
15__Main Tower	170.8	114.7
16__Main Tower	171.2	115.8
17__Main Tower	171.5	116.8
18__Main Tower	171.9	117.9
19__Main Tower	172.3	118.9

20__Main Tower 173.2 120.0

Column Flow Profile

Options Selected

Molar flow is selected Net is selected as flow basis

Tray Number Vapour Bulk Liquid

(kgmole/h) (kgmole/h)

1__Main Tower	4594	2255
2__Main Tower	5474	2227
3__Main Tower	5445	2188
4__Main Tower	5407	2139
5__Main Tower	5358	2084
6__Main Tower	5303	2028
7__Main Tower	5247	1978
8__Main Tower	5197	1938
9__Main Tower	5157	1908
10__Main Tower	5126	1.705e+004
11__Main Tower	4663	1.709e+004
12__Main Tower	4705	1.714e+004
13__Main Tower	4747	1.718e+004
14__Main Tower	4788	1.722e+004
15__Main Tower	4830	1.726e+004
16__Main Tower	4871	1.730e+004
17__Main Tower	4912	1.734e+004
18__Main Tower	4953	1.738e+004
19__Main Tower	4992	1.741e+004
20__Main Tower	5021	1.739e+004

Column Properties Profile

Options Selected

Mass basis is selected

Stage	Surf Tens	Mol Wt	Dens	Visc	Therm Con	Heat Cap
	L-Liq	L-Liq	L-Liq	L-Liq	L-Liq	
	(dyne/cm)	(kg/m3)	(cP)	(W/m-K)	(kJ/kg-C)	
1__Main Tower	15.2	121	742	0.206	0.108	2.21
2__Main Tower	15.2	121	741	0.205	0.108	2.22
3__Main Tower	15.1	123	741	0.204	0.108	2.22
4__Main Tower	15.0	124	740	0.202	0.109	2.23
5__Main Tower	15.0	126	739	0.201	0.109	2.24
6__Main Tower	14.9	128	738	0.199	0.109	2.24
7__Main Tower	14.7	130	736	0.197	0.110	2.25
8__Main Tower	14.6	132	735	0.195	0.110	2.26
9__Main Tower	14.5	134	734	0.194	0.110	2.27
10__Main Tower	14.5	135	733	0.192	0.110	2.27
11__Main Tower	14.4	135	732	0.192	0.110	2.28
12__Main Tower	14.4	135	732	0.191	0.110	2.28
13__Main Tower	14.4	135	732	0.191	0.110	2.28
14__Main Tower	14.3	135	731	0.190	0.110	2.28
15__Main Tower	14.3	135	731	0.189	0.110	2.28
16__Main Tower	14.3	134	731	0.189	0.109	2.28
17__Main Tower	14.2	134	730	0.188	0.109	2.29
18__Main Tower	14.2	134	730	0.188	0.109	2.29
19__Main Tower	14.2	135	729	0.187	0.109	2.29
20__Main Tower	14.1	135	729	0.186	0.109	2.29

Column Composition Profile

Options Selected

Fraction is selected as the composition basis Net is selected as flow basis

Molar basis is selected

Stage	Benzene	Propene	Propane	Cumene
	L-Liq	L-Liq	L-Liq	L-Liq
1__Main Tower	0.0000	0.0000	0.0000	0.9856
2__Main Tower	0.0000	0.0000	0.0000	0.9690
3__Main Tower	0.0000	0.0000	0.0000	0.9441
4__Main Tower	0.0000	0.0000	0.0000	0.9092
5__Main Tower	0.0000	0.0000	0.0000	0.8646
6__Main Tower	0.0000	0.0000	0.0000	0.8138
7__Main Tower	0.0000	0.0000	0.0000	0.7629
8__Main Tower	0.0000	0.0000	0.0000	0.7180
9__Main Tower	0.0000	0.0000	0.0000	0.6824
10__Main Tower	0.0000	0.0000	0.0000	0.6569
11__Main Tower	0.0000	0.0000	0.0000	0.6574
12__Main Tower	0.0000	0.0000	0.0000	0.6580
13__Main Tower	0.0000	0.0000	0.0000	0.6585
14__Main Tower	0.0000	0.0000	0.0000	0.6591
15__Main Tower	0.0000	0.0000	0.0000	0.6596
16__Main Tower	0.0000	0.0000	0.0000	0.6602
17__Main Tower	0.0000	0.0000	0.0000	0.6607
18__Main Tower	0.0000	0.0000	0.0000	0.6609

19__Main Tower 0.0000 0.0000 0.0000 0.6595

20__Main Tower 0.0000 0.0000 0.0000 0.6476

Heavy/Light Key Component Ratios

Options Selected

Molar basis is selected

Key Components

Light Key	Chosen	Heavy Key	Chosen
Benzene	On	Benzene	Off
Propene	On	Propene	Off
Propane	On	Propane	Off
Cumene	On	Cumene	Off
14-iP-BZ	Off	14-iP-BZ	On

Stage Light Liq

1__Main Tower 68.30

2__Main Tower 31.24

3__Main Tower 16.89

4__Main Tower 10.01

5__Main Tower 6.387

6__Main Tower 4.372

7__Main Tower 3.218

8__Main Tower 2.546

9__Main Tower 2.149

10__Main Tower 1.914

11__Main Tower 1.919

12__Main Tower 1.924

13__Main Tower 1.929

14__Main Tower 1.933

15__Main Tower 1.938

16__Main Tower 1.943

17__Main Tower 1.947

18__Main Tower 1.949

19__Main Tower 1.937

20__Main Tower 1.838

Column K-Values Profile

Tray Number	Benzene	Propene	Propane	Cumene
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1__Main Tower	5.805	67.97	62.03	1.011
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2__Main Tower	5.847	67.67	61.75	1.024
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3__Main Tower	5.921	67.47	61.57	1.044
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4__Main Tower	6.035	67.41	61.52	1.073
---------------	-------	-------	-------	-------

5__Main Tower	6.192	67.52	61.62	1.112
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6__Main Tower	6.384	67.75	61.84	1.161
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7__Main Tower	6.588	68.03	62.11	1.214
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8__Main Tower	6.778	68.25	62.34	1.264
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9__Main Tower	6.930	68.36	62.45	1.307
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10__Main Tower	7.038	68.32	62.43	1.339
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11__Main Tower	7.013	61.12	61.99	1.338
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12__Main Tower	6.989	1.062	61.56	1.337
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13__Main Tower	6.964	1.000	61.13	1.336
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14__Main Tower	6.940	1.000	4.004	1.335
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15__Main Tower	6.916	1.000	1.000	1.334
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16__Main Tower	6.893	1.000	1.000	1.333
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17__Main Tower	6.870	1.000	1.000	1.332
18__Main Tower	6.848	1.000	1.000	1.331
19__Main Tower	6.835	1.000	1.000	1.333
20__Main Tower	6.874	1.000	1.000	1.348

Separator: V-100

CONNECTIONS

Inlet Stream

Stream Name	From Unit Operation
-------------	---------------------

Heated_Bot	Heater: E-101
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Outlet Stream

Stream Name	To Unit Operation
-------------	-------------------

Boilup-New	Recycle: RCY-3
------------	----------------

C2-Feed	Mixer: MIX-102
---------	----------------

Energy Stream

Stream Name	From Unit Operation
-------------	---------------------

PARAMETERS

Vessel Volume: --- Level SP: 50.00 % Liquid Volume: ---

Vessel Pressure: 1500 kPa Pressure Drop: 0.0000 kPa Duty: 0.0000 kJ/h Heat Transfer Mode: Heating

User Variables

RATING

Sizing

Cylinder Vertical Separator has a Boot: No
Volume: --- Diameter: --- Height: --- Head Height: ---

Nozzles

Base Elevation Relative to Ground Level 0.0000 m Diameter --- Height ---

	Heated_Bot	Boilup-New	C2-Feed
Diameter (m)	5.000e-002	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000	0.0000
Elevation (% of Height) (%)	---	---	---

Level Taps: Level Tap Specification

Level Tap	PV High	PV Low	OP High	OP Low
-----------	---------	--------	---------	--------

Level Taps: Calculated Level Tap Values

Level Tap	Liquid Level	Aqueous Level
-----------	--------------	---------------

Options

PV Work Term Contribution (%) 100.00

Separator: V-101

CONNECTIONS

Inlet Stream

Stream Name	From Unit Operation
-------------	---------------------

Cooled-New	Cooler: E-102
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Outlet Stream

Stream Name	To Unit Operation
-------------	-------------------

Fuel-Gas	
----------	--

Reflux-New	Pump: P-100
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Energy Stream

Stream Name	From Unit Operation
-------------	---------------------

PARAMETERS

Vessel Volume: --- Level SP: 50.00 % Liquid Volume: ---

Vessel Pressure: 1450 kPa Pressure Drop: 0.0000 kPa Duty: 0.0000 kJ/h Heat Transfer Mode:
Heating

User Variables

RATING

Sizing

Cylinder Vertical Separator has a Boot: No
Volume: --- Diameter: --- Height: --- Head Height: ---

Nozzles

Base Elevation Relative to Ground Level 0.0000 m Diameter --- Height ---

	Cooled-New	Fuel-Gas	Reflux-New
Diameter (m)	5.000e-002	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000	0.0000
Elevation (% of Height) (%)	---	---	---

Level Taps: Level Tap Specification

Level Tap	PV High	PV Low	OP High	OP Low
-----------	---------	--------	---------	--------

Level Taps: Calculated Level Tap Values

Level Tap	Liquid Level	Aqueous Level
-----------	--------------	---------------

Options

PV Work Term Contribution (%) 100.00

Separator: V-102

CONNECTIONS

Inlet Stream

Stream Name	From Unit Operation
-------------	---------------------

C2-Cooled	Cooler: E-103
-----------	---------------

Outlet Stream

Stream Name	To Unit Operation
-------------	-------------------

C2-Gas	
--------	--

C2-Liquid	Pump: P-101
-----------	-------------

Energy Stream

Stream Name	From Unit Operation
-------------	---------------------

PARAMETERS

Vessel Volume: --- Level SP: 50.00 % Liquid Volume: ---

Vessel Pressure: 100.0 kPa Pressure Drop: 0.0000 kPa Duty: 0.0000 kJ/h Heat Transfer Mode:
Heating

User Variables

RATING

Sizing

Cylinder Vertical Separator has a Boot: No
Volume: --- Diameter: --- Height: --- Head Height: ---
Nozzles

Base Elevation Relative to Ground Level 0.0000 m Diameter --- Height ---

	C2-Cooled	C2-Gas	C2-Liquid
Diameter (m)	5.000e-002	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000	0.0000
Elevation (% of Height) (%)	---	---	---

Level Taps: Level Tap Specification

Level Tap	PV High	PV Low	OP High	OP Low
-----------	---------	--------	---------	--------

Level Taps: Calculated Level Tap Values

Level Tap	Liquid Level	Aqueous Level
-----------	--------------	---------------

Options

PV Work Term Contribution (%) 100.00

Separator: V-103

CONNECTIONS

Inlet Stream

Stream Name From Unit Operation

C2-Heated_Bot Heater: E-104

Outlet Stream

Stream Name To Unit Operation

Boilup-New2 Recycle: RCY-5

C2-Bottoms Absorber: T-102

Energy Stream

Stream Name From Unit Operation

PARAMETERS

Vessel Volume: --- Level SP: 50.00 % Liquid Volume: ---

Vessel Pressure: 120.0 kPa Pressure Drop: 0.0000 kPa Duty: 0.0000 kJ/h Heat Transfer Mode:
Heating

User Variables

RATING

Sizing

Cylinder Vertical Separator has a Boot: No

Volume: --- Diameter: --- Height: --- Head Height: ---

Nozzles

Base Elevation Relative to Ground Level 0.0000 m Diameter --- Height ---

C2-Heated_Bot Boilup-New2 C2-Bottoms

Diameter (m) 5.000e-002 5.000e-002 5.000e-002

Elevation (Base) (m) 0.0000 0.0000 0.0000

Elevation (Ground) (m) 0.0000 0.0000 0.0000

Elevation (% of Height) (%) --- --- ---

Level Taps: Level Tap Specification

Level Tap PV High PV Low OP High OP Low

Level Taps: Calculated Level Tap Values

Level Tap Liquid Level Aqueous Level

Options

PV Work Term Contribution (%) 100.00

Separator: V-104

CONNECTIONS

Inlet Stream

Stream Name From Unit Operation

C3-Cooled Cooler: E-105

Outlet Stream

Stream Name To Unit Operation

C3-Gas

C3-Liquid Pump: P-102

Energy Stream

Stream Name From Unit Operation

PARAMETERS

Vessel Volume: --- Level SP: 50.00 % Liquid Volume: ---

Vessel Pressure: 100.0 kPa Pressure Drop: 0.0000 kPa Duty: 0.0000 kJ/h Heat Transfer Mode:
Heating

User Variables

RATING

Sizing

Cylinder Vertical Separator has a Boot: No

Volume: --- Diameter: --- Height: --- Head Height: ---

Nozzles

Base Elevation Relative to Ground Level 0.0000 m Diameter --- Height ---

	C3-Cooled	C3-Gas	C3-Liquid
Diameter (m)	5.000e-002	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000	0.0000

Elevation (Ground) (m)	0.0000	0.0000	0.0000
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Elevation (% of Height) (%)	---	---	---
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Level Taps: Level Tap Specification

Level Tap	PV High	PV Low	OP High	OP Low
-----------	---------	--------	---------	--------

Level Taps: Calculated Level Tap Values

Level Tap	Liquid Level	Aqueous Level
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Options

PV Work Term Contribution (%) 100.00

Separator: V-105

CONNECTIONS

Inlet Stream

Stream Name	From Unit Operation
-------------	---------------------

C3-Heated_Bot	Heater: E-106
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Outlet Stream

Stream Name	To Unit Operation
-------------	-------------------

Boilup-New3	Recycle: RCY-7
-------------	----------------

DIPB_To_R2	Mixer: MIX-101
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Energy Stream

Stream Name From Unit Operation

PARAMETERS

Vessel Volume: --- Level SP: 50.00 % Liquid Volume: ---

Vessel Pressure: 120.0 kPa Pressure Drop: 0.0000 kPa Duty: 0.0000 kJ/h Heat Transfer Mode:
Heating

User Variables

RATING

Sizing

Cylinder Vertical Separator has a Boot: No

Volume: --- Diameter: --- Height: --- Head Height: ---

Nozzles

Base Elevation Relative to Ground Level 0.0000 m Diameter --- Height ---

C3-Heated_Bot Boilup-New3 DIPB_To_R2

Diameter (m) 5.000e-002 5.000e-002 5.000e-002

Elevation (Base) (m) 0.0000 0.0000 0.0000

Elevation (Ground) (m) 0.0000 0.0000 0.0000

Elevation (% of Height) (%) --- --- ---

Level Taps: Level Tap Specification

Level Tap PV High PV Low OP High OP Low

Level Taps: Calculated Level Tap Values

Level Tap Liquid Level Aqueous Level

Options

PV Work Term Contribution (%) 100.00

Cooler: E-102

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
-------------	---------------------

C1-Top	T-100 Absorber
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Outlet Stream

STREAM NAME	TO UNIT OPERATION
-------------	-------------------

Cooled-New	V-101 Separator
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Energy Stream

STREAM NAME	TO UNIT OPERATION
-------------	-------------------

Q-Condenser	
-------------	--

PARAMETERS

Pressure Drop: 0.0000 kPa Duty: 2.674e+008 kJ/h Volume: 0.1000 m3

Function: Not Selected Zones: 1

User Variables

NOZZLE PARAMETERS

Base Elevation Relative to Ground Level 0.0000 m

	C1-Top	Cooled-New
Diameter (m)	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000

PERFORMANCE PROFILES

Zone	Pressure (kPa)	Temperature (C)	Vapour Fraction	Enthalpy (kJ/kgmole)
Inlet	1450.00	41.98	1.0000	-99390.35
0	1450.00	41.91	0.0000	-112759.40

PERFORMANCE TABLE

Overall Phase

Temperature (C)	Pressure (kPa)	Heat Flow (kJ/h)	Enthalpy (kJ/kgmole)
41.98	1450.00	0.00	-99390.35
41.97	1450.00	-24726213.54	-100626.47
41.96	1450.00	-53321197.74	-102056.00

41.96	1450.00	-82337057.16	-103506.56
41.95	1450.00	-110921981.34	-104935.58
41.94	1450.00	-138836235.15	-106331.08
41.94	1450.00	-166009463.10	-107689.52
41.93	1450.00	-192429479.92	-109010.32
41.92	1450.00	-218107121.05	-110294.00
41.91	1450.00	-243063011.89	-111541.60
41.91	1450.00	-267321939.91	-112754.35

Vapour Fraction	Vap Phase Mass Frac	Heat of Vap
	(kJ/kgmole)	

1.0000	1.0000	---
0.9075	0.9075	---
0.8006	0.8006	---
0.6921	0.6920	---
0.5852	0.5851	---
0.4808	0.4807	---
0.3792	0.3791	---
0.2804	0.2803	---
0.1844	0.1843	---
0.0911	0.0910	---
0.0004	0.0004	---

Vapour Phase

Mass Flow	Molecular Wt	Density	Mass Sp Heat	Viscosity	Thermal Cond
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	
880446.88	44.02	32.19	2.10	0.01	0.02
799022.62	44.01	32.19	2.10	0.01	0.02

704855.48	44.01	32.19	2.10	0.01	0.02
609306.50	44.01	32.19	2.10	0.01	0.02
515182.22	44.01	32.19	2.10	0.01	0.02
423272.38	44.01	32.19	2.10	0.01	0.02
333808.61	44.01	32.19	2.10	0.01	0.02
246830.85	44.01	32.19	2.10	0.01	0.02
162303.24	44.00	32.19	2.10	0.01	0.02
80157.63	44.00	32.18	2.10	0.01	0.02
312.20	44.00	32.18	2.10	0.01	0.02

Std Gas Flow	Z Factor	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
(STD_m3/h)		(kPa)	(C)		

472963.04	0.76	4271.41	96.55	0.28	0.15
429235.07	0.76	4271.61	96.55	0.28	0.15
378660.46	0.76	4271.86	96.54	0.28	0.15
327340.63	0.76	4272.11	96.54	0.28	0.15
276782.99	0.76	4272.38	96.54	0.28	0.15
227411.78	0.76	4272.64	96.53	0.28	0.15
179351.54	0.76	4272.91	96.53	0.28	0.15
132623.84	0.76	4273.17	96.53	0.28	0.15
87209.53	0.76	4273.44	96.52	0.28	0.15
43072.12	0.76	4273.71	96.52	0.28	0.15
167.77	0.76	4273.97	96.51	0.28	0.15

Light Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)

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-----	-----	-----	-----	-----	-----

81424.26	464.53	3.17	0.08	0.09	4.95
175591.40	464.54	3.17	0.08	0.09	4.95
271140.39	464.56	3.17	0.08	0.09	4.95
365264.66	464.58	3.17	0.08	0.09	4.95
457174.50	464.60	3.17	0.08	0.09	4.95
546638.28	464.62	3.17	0.08	0.09	4.95
633616.03	464.64	3.17	0.08	0.09	4.96
718143.65	464.66	3.17	0.08	0.09	4.96
800289.25	464.68	3.17	0.08	0.09	4.96
880134.68	464.70	3.17	0.08	0.09	4.96

Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			

---	---	---	---	---	---
44.03	0.46	4269.39	96.58	0.28	0.15
44.03	0.46	4269.59	96.58	0.28	0.15
44.02	0.46	4269.81	96.57	0.28	0.15
44.02	0.46	4270.04	96.57	0.28	0.15
44.02	0.46	4270.26	96.57	0.28	0.15
44.02	0.46	4270.49	96.56	0.28	0.15
44.02	0.46	4270.72	96.56	0.28	0.15
44.02	0.46	4270.95	96.56	0.28	0.15
44.02	0.46	4271.18	96.55	0.28	0.15
44.02	0.46	4271.40	96.55	0.28	0.15

Heavy Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)

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---	---	---	---	---	---

Molecular Wt Sp Gravity Pseudo Pc Pseudo Tc Pseudo Zc Pseudo Omega
(kPa) (C)

---	---	---	---	---	---
---	---	---	---	---	---
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---	---	---	---	---	---
---	---	---	---	---	---

Mixed Liquid Phase

Mass Flow Density Mass Sp Heat Viscosity Thermal Cond Surface Tens

(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)
---	---	---	---	---	---
81424.26	464.53	3.17	0.08	0.09	4.95
175591.40	464.54	3.17	0.08	0.09	4.95
271140.39	464.56	3.17	0.08	0.09	4.95
365264.66	464.58	3.17	0.08	0.09	4.95
457174.50	464.60	3.17	0.08	0.09	4.95
546638.28	464.62	3.17	0.08	0.09	4.95
633616.03	464.64	3.17	0.08	0.09	4.96
718143.65	464.66	3.17	0.08	0.09	4.96
800289.25	464.68	3.17	0.08	0.09	4.96
880134.68	464.70	3.17	0.08	0.09	4.96
Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			
---	---	---	---	---	---
44.03	0.46	4269.39	96.58	0.28	0.15
44.03	0.46	4269.59	96.58	0.28	0.15
44.02	0.46	4269.81	96.57	0.28	0.15
44.02	0.46	4270.04	96.57	0.28	0.15
44.02	0.46	4270.26	96.57	0.28	0.15
44.02	0.46	4270.49	96.56	0.28	0.15
44.02	0.46	4270.72	96.56	0.28	0.15
44.02	0.46	4270.95	96.56	0.28	0.15
44.02	0.46	4271.18	96.55	0.28	0.15
44.02	0.46	4271.40	96.55	0.28	0.15

Cooler: E-103

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
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C2-Top	T-101 Absorber
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Outlet Stream

STREAM NAME	TO UNIT OPERATION
-------------	-------------------

C2-Cooled	V-102 Separator
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Energy Stream

STREAM NAME	TO UNIT OPERATION
-------------	-------------------

Q-Cond2	
---------	--

PARAMETERS

Pressure Drop: 0.0000 kPa Duty: 1.164e+009 kJ/h Volume: 0.1000 m3

Function: Not Selected Zones: 1

User Variables

NOZZLE PARAMETERS

Base Elevation Relative to Ground Level 0.0000 m

C2-Top	C2-Cooled
--------	-----------

Diameter (m)	5.000e-002	5.000e-002
--------------	------------	------------

Elevation (Base) (m)	0.0000	0.0000
----------------------	--------	--------

Elevation (Ground) (m)	0.0000	0.0000
------------------------	--------	--------

PERFORMANCE PROFILES

Zone	Pressure (kPa)	Temperature (C)	Vapour Fraction	Enthalpy (kJ/kgmole)
------	-------------------	--------------------	-----------------	-------------------------

Inlet	100.00	115.65	1.0000	61852.50
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0	100.00	50.00	0.0318	21241.99
---	--------	-------	--------	----------

PERFORMANCE TABLE

Overall Phase

Temperature (C)	Pressure (kPa)	Heat Flow (kJ/h)	Enthalpy (kJ/kgmole)
--------------------	-------------------	---------------------	-------------------------

115.65	100.00	0.00	61852.50
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109.08	100.00	-171844187.71	55856.24
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102.52	100.00	-327411002.54	50427.95
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95.95	100.00	-489035748.70	44788.28
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89.39	100.00	-675454084.17	38283.47
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82.82	100.00	-852542737.32	32104.20
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76.26	100.00	-963645504.06	28227.43
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69.69	100.00	-1032476927.52	25825.65
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63.13	100.00	-1083292105.40	24052.52
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56.56	100.00	-1125744173.00	22571.22
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50.00	100.00	-1163837989.99	21241.99
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Vapour Fraction	Vap Phase Mass Frac	Heat of Vap
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		(kJ/kgmole)
1.0000	1.0000	---
0.8574	0.8258	---
0.7257	0.6760	---
0.5817	0.5250	---
0.4051	0.3529	---
0.2361	0.1954	---
0.1411	0.1095	---
0.0935	0.0679	---
0.0657	0.0449	---
0.0467	0.0302	---
0.0318	0.0196	---

Vapour Phase

Mass Flow	Molecular Wt	Density	Mass Sp Heat	Viscosity	Thermal Cond
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	
2498207.65	87.17	2.77	1.52	0.01	0.02
2062968.70	83.96	2.71	1.47	0.01	0.02
1688691.71	81.19	2.67	1.44	0.01	0.02
1311591.67	78.68	2.63	1.41	0.01	0.02
881578.02	75.94	2.59	1.39	0.01	0.02
488256.24	72.15	2.50	1.39	0.01	0.02
273456.55	67.63	2.38	1.42	0.01	0.02
169693.74	63.34	2.27	1.46	0.01	0.02
112203.54	59.58	2.17	1.49	0.01	0.02
75469.83	56.39	2.10	1.53	0.01	0.02
48956.43	53.73	2.04	1.55	0.01	0.02

Std Gas Flow	Z Factor	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
(STD_m3/h)		(kPa)	(C)		

677614.99	0.97	4436.91	295.95	0.27	0.24
580976.14	0.97	4548.60	289.40	0.27	0.23
491777.79	0.97	4637.31	283.26	0.27	0.23
394156.35	0.97	4701.61	276.61	0.27	0.22
274483.61	0.97	4732.30	266.73	0.27	0.21
160005.60	0.98	4710.50	248.79	0.27	0.21
95599.65	0.98	4651.31	225.20	0.28	0.20
63349.27	0.98	4586.12	202.16	0.28	0.19
44531.01	0.98	4526.14	181.81	0.28	0.18
31645.66	0.98	4473.91	164.47	0.28	0.18
21542.26	0.98	4429.59	149.97	0.28	0.17

Light Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)

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435238.95	788.36	1.99	0.28	0.12	18.65
809515.94	795.24	1.96	0.30	0.12	19.16
1186615.98	801.90	1.92	0.31	0.12	19.67
1616629.63	808.05	1.88	0.32	0.12	20.19
2009951.41	813.68	1.84	0.33	0.12	20.76
2224751.10	819.13	1.81	0.35	0.12	21.38
2328513.91	824.50	1.78	0.38	0.12	22.01
2386004.11	829.73	1.75	0.40	0.12	22.64
2422737.82	834.78	1.73	0.43	0.12	23.27

Molecular Wt Sp Gravity Pseudo Pc Pseudo Tc Pseudo Zc Pseudo Omega
(kPa) (C)

Heavy Liquid Phase

[illegible]

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Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			
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Mixed Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)
---	---	---	---	---	---
435238.95	788.36	1.99	0.28	0.12	18.65
809515.94	795.24	1.96	0.30	0.12	19.16
1186615.98	801.90	1.92	0.31	0.12	19.67
1616629.63	808.05	1.88	0.32	0.12	20.19
2009951.41	813.68	1.84	0.33	0.12	20.76
2224751.10	819.13	1.81	0.35	0.12	21.38
2328513.91	824.50	1.78	0.38	0.12	22.01

2386004.11	829.73	1.75	0.40	0.12	22.64
2422737.82	834.78	1.73	0.43	0.12	23.27
2449251.22	839.60	1.70	0.46	0.12	23.89

Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			

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106.49	0.79	3765.43	335.29	0.27	0.30
103.00	0.80	3906.59	329.51	0.27	0.29
98.98	0.80	4068.83	322.84	0.27	0.28
94.82	0.81	4235.78	315.84	0.27	0.26
91.81	0.81	4352.33	310.52	0.27	0.26
90.38	0.82	4401.69	307.57	0.27	0.25
89.63	0.82	4421.52	305.62	0.27	0.25
89.11	0.83	4430.63	303.97	0.27	0.25
88.68	0.83	4435.09	302.39	0.27	0.25
88.27	0.84	4437.15	300.74	0.27	0.25

Cooler: E-105

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
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C3-Top	T-102 Absorber
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Outlet Stream

STREAM NAME	TO UNIT OPERATION
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C3-Cooled	V-104 Separator
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Energy Stream

STREAM NAME	TO UNIT OPERATION
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Q-Cond3	
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PARAMETERS

Pressure Drop: 0.0000 kPa Duty: 2.855e+008 kJ/h Volume: 0.1000 m3

Function: Not Selected Zones: 1

User Variables

NOZZLE PARAMETERS

Base Elevation Relative to Ground Level 0.0000 m

	C3-Top	C3-Cooled
Diameter (m)	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000

PERFORMANCE PROFILES

Zone	Pressure (kPa)	Temperature (C)	Vapour Fraction	Enthalpy (kJ/kgmole)
Inlet	100.00	153.55	1.0000	27027.35
0	100.00	50.00	0.0000	-35134.58

Overall Phase

[illegible]

0.0000	0.0000	---
0.0000	0.0000	---
0.0000	0.0000	---

Vapour Phase

Mass Flow	Molecular Wt	Density	Mass Sp Heat	Viscosity	Thermal Cond
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	

552831.70	120.35	3.53	1.79	0.01	0.02
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Std Gas Flow	Z Factor	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
(STD_m3/h)	(kPa)	(C)			

108612.95	0.96	3206.03	358.21	0.26	0.34
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Light Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)

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552831.70	752.09	2.17	0.22	0.11	16.17
552831.70	762.12	2.12	0.24	0.11	17.13
552831.70	771.98	2.08	0.27	0.12	18.10
552831.70	781.66	2.04	0.29	0.12	19.07
552831.70	791.19	1.99	0.32	0.12	20.06
552831.70	800.58	1.95	0.35	0.12	21.05
552831.70	809.84	1.90	0.39	0.13	22.06
552831.70	818.97	1.86	0.43	0.13	23.07
552831.70	828.00	1.82	0.48	0.13	24.09
552831.70	836.93	1.77	0.54	0.13	25.12

Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			

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120.35	0.75	3206.03	358.21	0.26	0.34
120.35	0.76	3206.03	358.21	0.26	0.34
120.35	0.77	3206.03	358.21	0.26	0.34
120.35	0.78	3206.03	358.21	0.26	0.34
120.35	0.79	3206.03	358.21	0.26	0.34

120.35	0.80	3206.03	358.21	0.26	0.34
120.35	0.81	3206.03	358.21	0.26	0.34
120.35	0.82	3206.03	358.21	0.26	0.34
120.35	0.83	3206.03	358.21	0.26	0.34
120.35	0.84	3206.03	358.21	0.26	0.34

Heavy Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)

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Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			

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Mixed Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)

---	---	---	---	---	---
552831.70	752.09	2.17	0.22	0.11	16.17
552831.70	762.12	2.12	0.24	0.11	17.13
552831.70	771.98	2.08	0.27	0.12	18.10
552831.70	781.66	2.04	0.29	0.12	19.07
552831.70	791.19	1.99	0.32	0.12	20.06
552831.70	800.58	1.95	0.35	0.12	21.05
552831.70	809.84	1.90	0.39	0.13	22.06
552831.70	818.97	1.86	0.43	0.13	23.07
552831.70	828.00	1.82	0.48	0.13	24.09
552831.70	836.93	1.77	0.54	0.13	25.12

Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			

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120.35	0.75	3206.03	358.21	0.26	0.34
120.35	0.76	3206.03	358.21	0.26	0.34
120.35	0.77	3206.03	358.21	0.26	0.34

120.35	0.78	3206.03	358.21	0.26	0.34
120.35	0.79	3206.03	358.21	0.26	0.34
120.35	0.80	3206.03	358.21	0.26	0.34
120.35	0.81	3206.03	358.21	0.26	0.34
120.35	0.82	3206.03	358.21	0.26	0.34
120.35	0.83	3206.03	358.21	0.26	0.34
120.35	0.84	3206.03	358.21	0.26	0.34

Cooler: E-107

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
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Trans_Mixed	MIX-101 Mixer
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Outlet Stream

STREAM NAME	TO UNIT OPERATION
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Cooled-Trans_Mixed	P-103 Pump
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Energy Stream

STREAM NAME	TO UNIT OPERATION
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Q-107

PARAMETERS

Pressure Drop: 0.0000 kPa Duty: 2.980e+008 kJ/h Volume: 0.1000 m3

Function: Not Selected Zones: 1

User Variables

NOZZLE PARAMETERS

Base Elevation Relative to Ground Level 0.0000 m

	Trans_Mixed	Cooled-Trans_Mixed
Diameter (m)	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000

PERFORMANCE PROFILES

Zone	Pressure (kPa)	Temperature (C)	Vapour Fraction	Enthalpy (kJ/kgmole)
Inlet	120.00	123.52	0.1226	-9812.73
0	120.00	80.00	0.0000	-23297.56

PERFORMANCE TABLE

Overall Phase

Temperature (C)	Pressure (kPa)	Heat Flow (kJ/h)	Enthalpy (kJ/kgmole)
123.52	120.00	0.00	-9812.76
119.17	120.00	-50708769.83	-12107.20
114.82	120.00	-93217562.93	-14030.61

110.47	120.00	-128495178.85	-15626.82
106.11	120.00	-158933028.13	-17004.06
101.76	120.00	-186443703.73	-18248.84
97.41	120.00	-212166410.11	-19412.73
93.06	120.00	-234088410.33	-20404.64
88.70	120.00	-255610379.70	-21378.45
84.35	120.00	-276922316.07	-22342.76
80.00	120.00	-298024208.96	-23297.56

Vapour Fraction	Vap Phase Mass Frac	Heat of Vap
	(kJ/kgmole)	

0.1226	0.0883	---
0.0809	0.0564	---
0.0506	0.0339	---
0.0304	0.0195	---
0.0170	0.0104	---
0.0075	0.0044	---
0.0004	0.0002	---
0.0000	0.0000	---
0.0000	0.0000	---
0.0000	0.0000	---
0.0000	0.0000	---

Vapour Phase

Mass Flow	Molecular Wt	Density	Mass Sp Heat	Viscosity	Thermal Cond
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	
226905.09	83.77	3.14	1.55	0.01	0.02
144791.93	80.96	3.06	1.54	0.01	0.02

87071.28	77.88	2.98	1.54	0.01	0.02
50182.45	74.67	2.88	1.55	0.01	0.02
26843.69	71.51	2.79	1.56	0.01	0.02
11405.55	68.51	2.70	1.58	0.01	0.02
544.72	65.72	2.62	1.59	0.01	0.02

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Std Gas Flow	Z Factor	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
(STD_m3/h)		(kPa)	(C)		

64043.94	0.97	4491.26	283.16	0.27	0.23
42284.31	0.97	4517.72	272.99	0.27	0.23
26433.42	0.97	4525.93	260.37	0.27	0.22
15889.73	0.97	4520.77	246.23	0.27	0.21
8875.80	0.98	4508.09	231.77	0.27	0.21
3936.55	0.98	4491.79	217.75	0.27	0.20
195.99	0.98	4474.01	204.52	0.28	0.20

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Light Liquid Phase

Mass Flow	Density	Mass Sp Heat	Viscosity	Thermal Cond	Surface Tens
(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)

2342013.76	778.79	2.07	0.26	0.12	17.69
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2424126.93	782.91	2.05	0.27	0.12	18.03
2481847.58	786.87	2.03	0.28	0.12	18.37
2518736.40	790.71	2.01	0.29	0.12	18.73
2542075.16	794.45	1.99	0.30	0.12	19.10
2557513.30	798.12	1.97	0.31	0.12	19.46
2568374.14	801.73	1.95	0.32	0.12	19.83
2568918.85	805.85	1.93	0.34	0.12	20.26
2568918.85	809.98	1.92	0.35	0.12	20.69
2568918.85	814.09	1.90	0.37	0.12	21.13
2568918.85	818.18	1.88	0.38	0.13	21.57

Molecular Wt Sp Gravity Pseudo Pc Pseudo Tc Pseudo Zc Pseudo Omega
(kPa) (C)

120.77	0.78	3437.86	355.56	0.27	0.32
119.34	0.78	3483.25	353.17	0.27	0.32
118.28	0.79	3515.87	351.28	0.27	0.31
117.54	0.79	3537.05	349.83	0.27	0.31
117.01	0.79	3550.70	348.67	0.27	0.31
116.60	0.80	3559.94	347.66	0.27	0.31
116.26	0.80	3566.62	346.74	0.27	0.31
116.24	0.81	3566.96	346.68	0.27	0.31
116.24	0.81	3566.96	346.68	0.27	0.31
116.24	0.81	3566.96	346.68	0.27	0.31
116.24	0.82	3566.96	346.68	0.27	0.31

Heavy Liquid Phase

Mass Flow Density Mass Sp Heat Viscosity Thermal Cond Surface Tens
(kg/h) (kg/m3) (kJ/kg-C) (cP) (W/m-K) (dyne/cm)

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Molecular Wt Sp Gravity Pseudo Pc Pseudo Tc Pseudo Zc Pseudo Omega
(kPa) (C)

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Mixed Liquid Phase

Mass Flow Density Mass Sp Heat Viscosity Thermal Cond Surface Tens

(kg/h)	(kg/m3)	(kJ/kg-C)	(cP)	(W/m-K)	(dyne/cm)
2342013.76	778.79	2.07	0.26	0.12	17.69
2424126.93	782.91	2.05	0.27	0.12	18.03
2481847.58	786.87	2.03	0.28	0.12	18.37
2518736.40	790.71	2.01	0.29	0.12	18.73
2542075.16	794.45	1.99	0.30	0.12	19.10
2557513.30	798.12	1.97	0.31	0.12	19.46
2568374.14	801.73	1.95	0.32	0.12	19.83
2568918.85	805.85	1.93	0.34	0.12	20.26
2568918.85	809.98	1.92	0.35	0.12	20.69
2568918.85	814.09	1.90	0.37	0.12	21.13
2568918.85	818.18	1.88	0.38	0.13	21.57

Molecular Wt	Sp Gravity	Pseudo Pc	Pseudo Tc	Pseudo Zc	Pseudo Omega
	(kPa)	(C)			
120.77	0.78	3437.86	355.56	0.27	0.32
119.34	0.78	3483.25	353.17	0.27	0.32
118.28	0.79	3515.87	351.28	0.27	0.31
117.54	0.79	3537.05	349.83	0.27	0.31
117.01	0.79	3550.70	348.67	0.27	0.31
116.60	0.80	3559.94	347.66	0.27	0.31
116.26	0.80	3566.62	346.74	0.27	0.31
116.24	0.81	3566.96	346.68	0.27	0.31
116.24	0.81	3566.96	346.68	0.27	0.31
116.24	0.81	3566.96	346.68	0.27	0.31
116.24	0.82	3566.96	346.68	0.27	0.31

Tee: TEE-100

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
-------------	---------------------

C2-PumpOut	P-101 Pump
------------	------------

Outlet Stream

STREAM NAME	TO UNIT OPERATION
-------------	-------------------

Reflux-New2	RCY-6 Recycle
-------------	---------------

Benzene Ovrhd	TEE-102 Tee
---------------	-------------

PARAMETERS

	Flow Ratios	Dynamic Valve Openings
Reflux-New2	0.3000	30.00
Benzene Ovrhd	0.7000	70.00

Valve Control: Multiple Stream

User Variables

RATING

Nozzle Parameters

Base Elevation Relative to Ground Level 0.0000 m

	C2-PumpOut	Reflux-New2	Benzene Ovrhd
Diameter (m)	5.000e-002	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000	0.0000

Tee: TEE-101

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
-------------	---------------------

C3-PumpOut	P-102 Pump
------------	------------

Outlet Stream

STREAM NAME	TO UNIT OPERATION
-------------	-------------------

Cumene

Reflux_new3	RCY-8 Recycle
-------------	---------------

PARAMETERS

	Flow Ratios	Dynamic Valve Openings
Cumene	0.7000	70.00
Reflux_new3	0.3000	30.00

Valve Control: Multiple Stream

User Variables

RATING

Nozzle Parameters

Base Elevation Relative to Ground Level 0.0000 m

	C3-PumpOut	Cumene	Reflux_new3
Diameter (m)	5.000e-002	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000	0.0000

Tee: TEE-102

CONNECTIONS

Inlet Stream

STREAM NAME	FROM UNIT OPERATION
-------------	---------------------

Benzene Ovrhd	TEE-100 Tee
---------------	-------------

Outlet Stream

STREAM NAME	TO UNIT OPERATION
-------------	-------------------

Transalkylation Recycle	MIX-101 Mixer
-------------------------	---------------

Alkylation Recycle RCY-2 Recycle

PARAMETERS

	Flow Ratios	Dynamic Valve Openings
Transalkylation Recycle	0.5000	50.00
Alkylation Recycle	0.5000	50.00

Valve Control: Multiple Stream

User Variables

RATING

Nozzle Parameters

Base Elevation Relative to Ground Level 0.0000 m

	Benzene Ovrhd	Transalkylation Recycle	Alkylation Recycle
Diameter (m)	5.000e-002	5.000e-002	5.000e-002
Elevation (Base) (m)	0.0000	0.0000	0.0000
Elevation (Ground) (m)	0.0000	0.0000	0.0000

Conversion Reactor: CRV-100

CONNECTIONS

Inlet Stream Connections

Stream Name	From Unit Operation
-------------	---------------------

Trans Feed	E-108 Heater
------------	--------------

Outlet Stream Connections

Stream Name	To Unit Operation
-------------	-------------------

Trans_Vap	
-----------	--

Trans_Effluents	Recycle: RCY-1
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Energy Stream Connections

Stream Name	From Unit Operation
-------------	---------------------

PARAMETERS

Physical Parameters	Optional Heat Transfer
---------------------	------------------------

Delta P	Vessel Volume	Duty	Energy Stream
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0.0000 kPa	18.85 m3	0.0000 kJ/h	
------------	----------	-------------	--

User Variables

REACTION DETAILS

Reaction: Rxn-4

Component	Mole Weight	Stoichiometric Coeff.
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Benzene	78.11	-1.000
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14-iP-BZ	162.3	-1.000
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Cumene	120.2	2.000
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REACTION RESULTS FOR : Set-2

Extents

Name	Rank	Specified Use	Default	Actual	Base	Reaction Extent
		% Conversion		% Conversion	Component	
Rxn-4	0	60.00	Yes	60.00	14-iP-BZ	0.8820

Balance

Components	Total Inflow	Total Reaction	Total Outflow
Benzene	1.873	-0.8820	0.9907
Propene	1.004e-003	0.0000	1.004e-003
Propane	9.541e-002	0.0000	9.541e-002
Cumene	2.700	1.764	4.464
14-iP-BZ	1.470	-0.8820	0.5880

RATING

Sizing

Cylinder	Vertical	Reactor has a Boot: No
Volume 18.85 m3	Diameter 2.000 m	Height 6.000

Nozzles

Base Elevation Relative to Ground Level 0.0000 m Diameter 2.000 m Height 6.000

	Trans_Feed	Trans_Vap	Trans_Effluents
Diameter (m)	0.3000	0.3000	0.3000

Elevation (Base) (m)	3.000	6.000	0.0000
Elevation (Ground) (m)	3.000	6.000	0.0000
Elevation (% of Height) (%)	50.00	100.00	0.00

Recycle: RCY-1 (Ignored)

CONNECTIONS

Inlet Stream

Stream Name	From Unit Operation
Trans_Effluents	CRV-100 Conversion Reactor
Outlet Stream	

Stream Name	To Unit Operation
Trans_Recycled	MIX-102 Mixer

TOLERANCE

Vapour Fraction: 10.00 Temperature: 10.00 Pressure: 10.00
Flow: 100.0 Enthalpy: 10.00 Composition: 10.00

NUMERICAL

Acceleration Type: Dominant Eigenvalue Iteration Type: Nested

Maximum Iterations: 10 Iteration Count: 0

Iteration History

Iteration Variable	Outlet Value	Inlet Value
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User Variables

Recycle: RCY-2

CONNECTIONS

Inlet Stream

Stream Name	From Unit Operation
-------------	---------------------

Alkylation Recycle	TEE-102 Tee
--------------------	-------------

Outlet Stream

Stream Name	To Unit Operation
-------------	-------------------

Alkylation Recycled	MIX-103 Mixer
---------------------	---------------

TOLERANCE

Vapour Fraction: 10.00 Temperature: 10.00 Pressure: 10.00

Flow: 100.0 Enthalpy: 10.00 Composition: 10.00

NUMERICAL

Acceleration Type: Dominant Eigenvalue Iteration Type: Nested

Maximum Iterations: 10 Iteration Count: 0

Iteration History

Iteration	Variable	Outlet Value	Inlet Value
0	Converged	---	---

User Variables

Recycle: RCY-3

CONNECTIONS

Inlet Stream

Stream Name	From Unit Operation
Boilup-New	V-100 Separator

Outlet Stream

Stream Name	To Unit Operation
Boil-Up	T-100 Absorber

TOLERANCE

Vapour Fraction: 10.00 Temperature: 10.00 Pressure: 10.00

Flow: 10.00 Enthalpy: 10.00 Composition: 10.00

NUMERICAL

Acceleration Type: Wegstein Iteration Type: Nested

Maximum Iterations: 10 Iteration Count: 0

Wegstein Count: 3 Q Minimum: -20.00 Q Maximum: 0.0000

Iteration History

Iteration	Variable	Outlet Value	Inlet Value
0	Converged	---	---

User Variables

Recycle: RCY-4

CONNECTIONS

Inlet Stream

Stream Name	From Unit Operation
-------------	---------------------

Reflux-Temp	P-100 Pump
-------------	------------

Outlet Stream

Stream Name	To Unit Operation
-------------	-------------------

Reflux-1	T-100 Absorber
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TOLERANCE

Vapour Fraction: 10.00 Temperature: 10.00 Pressure: 10.00

Flow: 10.00 Enthalpy: 10.00 Composition: 10.00

NUMERICAL

Acceleration Type: Wegstein Iteration Type: Nested

Maximum Iterations: 10 Iteration Count: 0

Wegstein Count: 3 Q Minimum: -20.00 Q Maximum: 0.0000

Iteration History

Iteration	Variable	Outlet Value	Inlet Value
0	Converged	---	---

User Variables

Recycle: RCY-5

CONNECTIONS

Inlet Stream

Stream Name	From Unit Operation
-------------	---------------------

Boilup-New2	V-103 Separator
-------------	-----------------

Outlet Stream

Stream Name	To Unit Operation
-------------	-------------------

Boilup-2	T-101 Absorber
----------	----------------

TOLERANCE

Vapour Fraction: 10.00 Temperature: 10.00 Pressure: 10.00

Flow: 10.00 Enthalpy: 10.00 Composition: 10.00

NUMERICAL

Acceleration Type: Wegstein Iteration Type: Nested

Maximum Iterations: 10 Iteration Count: 0

Wegstein Count: 3 Q Minimum: -20.00 Q Maximum: 0.0000

Iteration History

Iteration Variable	Outlet Value	Inlet Value
0 Converged	---	---

User Variables

Recycle: RCY-6

CONNECTIONS

Inlet Stream

Stream Name	From Unit Operation
-------------	---------------------

Reflux-New2	TEE-100 Tee
-------------	-------------

Outlet Stream

Stream Name To Unit Operation

Reflux-2 T-101 Absorber

TOLERANCE

Vapour Fraction: 10.00 Temperature: 10.00 Pressure: 10.00

Flow: 10.00 Enthalpy: 10.00 Composition: 10.00

NUMERICAL

Acceleration Type: Wegstein Iteration Type: Nested

Maximum Iterations: 10 Iteration Count: 0

Wegstein Count: 3 Q Minimum: -20.00 Q Maximum: 0.0000

Iteration History

Iteration Variable	Outlet Value	Inlet Value
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0 Converged	---	---
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User Variables

Recycle: RCY-7

CONNECTIONS

Inlet Stream

Stream Name From Unit Operation

Boilup-New3 V-105 Separator

Outlet Stream

Stream Name To Unit Operation

Boilup-3 T-102 Absorber

TOLERANCE

Vapour Fraction: 10.00 Temperature: 10.00 Pressure: 10.00

Flow: 10.00 Enthalpy: 10.00 Composition: 10.00

NUMERICAL

Acceleration Type: Wegstein Iteration Type: Nested

Maximum Iterations: 10 Iteration Count: 0

Wegstein Count: 3 Q Minimum: -20.00 Q Maximum: 0.0000

Iteration History

Iteration Variable	Outlet Value	Inlet Value
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0 Converged	---	---
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User Variables

Recycle: RCY-8

CONNECTIONS

Inlet Stream

Stream Name From Unit Operation

Reflux_new3 TEE-101 Tee

Outlet Stream

Stream Name To Unit Operation

Reflux-3 T-102 Absorber

TOLERANCE

Vapour Fraction: 10.00 Temperature: 10.00 Pressure: 10.00

Flow: 10.00 Enthalpy: 10.00 Composition: 10.00

NUMERICAL

Acceleration Type: Wegstein Iteration Type: Nested

Maximum Iterations: 10 Iteration Count: 0

Wegstein Count: 3 Q Minimum: -20.00 Q Maximum: 0.0000

Iteration History

Iteration Variable	Outlet Value	Inlet Value
--------------------	--------------	-------------

0 Converged	---	---
------------------	-----	-----

User Variables

Tower: Main Tower @COL1

Vapour Draws Summary

Name:	Name:	Name:
Tray Number		
Temperature (C)		
Pressure (kPa)		
Mass Flow (kg/h)		
Molar Flow (kgmole/h)		
Ideal Liquid Volume Flow (m3/h)		
Molar Enthalpy (kJ/kgmole)		
Mass Enthalpy (kJ/kg)		
Heat Flow (kJ/h)		
Molecular Weight		
Molar Entropy (kJ/kgmole-C)		
Mass Entropy (kJ/kg-C)		
Molar Density (kgmole/m3)		
Mass Density (kg/m3)		
Std Liq Mass Den (kg/m3)		
Molar Heat Cap (kJ/kgmole-C)		
Mass Heat Cap (kJ/kg-C)		
Thermal Cond (W/m-K)		
Viscosity (cP)		
Surface Tension (dyne/cm) ---	---	---
Z Factor		
Liquid Draws Summary		

Name: Name: Name:

Tray Number

Temperature (C)

Pressure (kPa)

Mass Flow (kg/h)

Molar Flow (kgmole/h)

Ideal Liquid Volume Flow (m³/h)

Molar Enthalpy (kJ/kgmole)

Mass Enthalpy (kJ/kg)

Heat Flow (kJ/h)

Molecular Weight

Molar Entropy (kJ/kgmole-C)

Mass Entropy (kJ/kg-C)

Molar Density (kgmole/m³)

Mass Density (kg/m³)

Std Liq Mass Den (kg/m³)

Molar Heat Cap (kJ/kgmole-C)

Mass Heat Cap (kJ/kg-C)

Thermal Cond (W/m-K)

Viscosity (cP)

Surface Tension (dyne/cm)

Z Factor

Water Draws Summary

Name: Name: Name:

Tray Number

Temperature (C)

Pressure (kPa)

Mass Flow (kg/h)

Molar Flow (kgmole/h)

Water Volume Flow (m³/h)

Molar Enthalpy (kJ/kgmole)

Mass Enthalpy (kJ/kg)

Heat Flow (kJ/h)

Molecular Weight

Molar Entropy (kJ/kgmole-C)

Mass Entropy (kJ/kg-C)

Molar Density (kgmole/m³)

Mass Density (kg/m³)

Std Liq Mass Den (kg/m³)

Molar Heat Cap (kJ/kgmole-C)

Mass Heat Cap (kJ/kg-C)

Thermal Cond (W/m-K)

Viscosity (cP)

Surface Tension (dyne/cm)

Z Factor

User Variables

RATING

Sizing

Weir Height (m)

5.000e-002

Weir Length (m)	1.200
Tray Space (m)	0.5000
Tray Volume (m3)	0.8836
DC Volume (m3)	8.836e-002
Diameter (m)	1.500
Active Area (m2)	1.264
Flow Paths	1

Internal Type: Sieve

Nozzles

Tray Section Elevation Relative to Ground Level 0.0000 m

	HoldupRG	VtoAbove (m)	Diameter (m)	L toBelow (m)	Diameter (m)
1__Main Tower	4.500 m	---	---	0.0000 m	3.000e-003 m
2__Main Tower	4.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
3__Main Tower	3.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
4__Main Tower	3.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
5__Main Tower	2.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
6__Main Tower	2.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
7__Main Tower	1.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
8__Main Tower	1.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
9__Main Tower	0.5000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
10__Main Tower	0.0000 m	0.5000 m	3.000e-003 m	---	---

	Feed Tray	Elevation RH (m)	Diameter (m)
Reflux-1	1__Main Tower	0.3750 m	5.000e-002 m
C1-Feed	5__Main Tower	0.3750 m	5.000e-002 m
Boil-Up	10__Main Tower	0.0000 m	5.000e-002 m

Prod Tray	Elevation RH (m)	Diameter (m)
-----------	------------------	--------------

C1-Top	1__Main Tower	0.5000 m	5.000e-002 m
C1-Bot	10__Main Tower	0.0000 m	3.000e-003 m

Heat Loss

Efficiency

Overall Efficiency

Tray	Overall
1__Main Tower	1.000
2__Main Tower	1.000
3__Main Tower	1.000
4__Main Tower	1.000
5__Main Tower	1.000
6__Main Tower	1.000
7__Main Tower	1.000
8__Main Tower	1.000
9__Main Tower	1.000
10__Main Tower	1.000

Component Efficiency

Tray	Benzene	Propene	Propane	Cumene
1__Main Tower	1.000	1.000	1.000	1.000
2__Main Tower	1.000	1.000	1.000	1.000
3__Main Tower	1.000	1.000	1.000	1.000
4__Main Tower	1.000	1.000	1.000	1.000
5__Main Tower	1.000	1.000	1.000	1.000
6__Main Tower	1.000	1.000	1.000	1.000
7__Main Tower	1.000	1.000	1.000	1.000

8__ Main Tower	1.000	1.000	1.000	1.000
9__ Main Tower	1.000	1.000	1.000	1.000
10__ Main Tower	1.000	1.000	1.000	1.000

Tray 14-iP-BZ

1__ Main Tower	1.000
2__ Main Tower	1.000
3__ Main Tower	1.000
4__ Main Tower	1.000
5__ Main Tower	1.000
6__ Main Tower	1.000
7__ Main Tower	1.000
8__ Main Tower	1.000
9__ Main Tower	1.000
10__ Main Tower	1.000

Pressure Drop

	Pressure (kPa)	Pressure Drop (kPa)
1__ Main Tower	1450 kPa	5.556 kPa
2__ Main Tower	1456 kPa	5.556 kPa
3__ Main Tower	1461 kPa	5.556 kPa
4__ Main Tower	1467 kPa	5.556 kPa
5__ Main Tower	1472 kPa	5.556 kPa
6__ Main Tower	1478 kPa	5.556 kPa
7__ Main Tower	1483 kPa	5.556 kPa
8__ Main Tower	1489 kPa	5.556 kPa
9__ Main Tower	1494 kPa	5.556 kPa
10__ Main Tower	1500 kPa	---

Rating Enabled: No Top Tray Fixed For Update Tray Section Pressure Drop 50.00 kPa

Vapour Mole Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__Main Tower	0.0000	0.0405	0.9595	0.0000	0.0000
2__Main Tower	0.0001	0.0337	0.9662	0.0000	0.0000
3__Main Tower	0.0007	0.0279	0.9714	0.0000	0.0000
4__Main Tower	0.0083	0.0228	0.9688	0.0001	0.0000
5__Main Tower	0.1075	0.0165	0.8639	0.0108	0.0014
6__Main Tower	0.1078	0.0160	0.8640	0.0108	0.0014
7__Main Tower	0.1098	0.0154	0.8624	0.0110	0.0015
8__Main Tower	0.1198	0.0146	0.8520	0.0120	0.0016
9__Main Tower	0.1664	0.0132	0.8015	0.0166	0.0023
10__Main Tower	0.3397	0.0096	0.6073	0.0377	0.0057

Liquid Mole Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__Main Tower	0.0001	0.0345	0.9655	0.0000	0.0000
2__Main Tower	0.0007	0.0286	0.9707	0.0000	0.0000
3__Main Tower	0.0083	0.0236	0.9681	0.0001	0.0000
4__Main Tower	0.1075	0.0177	0.8626	0.0108	0.0014
5__Main Tower	0.3854	0.0046	0.2556	0.2341	0.1203
6__Main Tower	0.3856	0.0045	0.2559	0.2339	0.1201
7__Main Tower	0.3877	0.0043	0.2542	0.2338	0.1200
8__Main Tower	0.3988	0.0040	0.2437	0.2339	0.1196
9__Main Tower	0.4420	0.0032	0.2038	0.2338	0.1172
10__Main Tower	0.5223	0.0018	0.1197	0.2435	0.1126

Vapour Mass Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__Main Tower	0.0000	0.0387	0.9612	0.0000	0.0000
2__Main Tower	0.0001	0.0322	0.9677	0.0000	0.0000
3__Main Tower	0.0012	0.0267	0.9721	0.0000	0.0000
4__Main Tower	0.0146	0.0216	0.9635	0.0002	0.0000
5__Main Tower	0.1723	0.0143	0.7822	0.0265	0.0047
6__Main Tower	0.1728	0.0138	0.7821	0.0266	0.0047
7__Main Tower	0.1757	0.0133	0.7791	0.0271	0.0048
8__Main Tower	0.1900	0.0125	0.7629	0.0293	0.0053
9__Main Tower	0.2535	0.0108	0.6895	0.0388	0.0074
10__Main Tower	0.4484	0.0068	0.4525	0.0766	0.0156

Liquid Mass Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__Main Tower	0.0001	0.0329	0.9670	0.0000	0.0000
2__Main Tower	0.0012	0.0273	0.9714	0.0000	0.0000
3__Main Tower	0.0146	0.0224	0.9628	0.0002	0.0000
4__Main Tower	0.1725	0.0153	0.7810	0.0266	0.0047
5__Main Tower	0.3374	0.0022	0.1263	0.3154	0.2187
6__Main Tower	0.3377	0.0021	0.1265	0.3152	0.2185
7__Main Tower	0.3394	0.0020	0.1256	0.3149	0.2181
8__Main Tower	0.3478	0.0019	0.1200	0.3138	0.2166
9__Main Tower	0.3804	0.0015	0.0990	0.3096	0.2095
10__Main Tower	0.4355	0.0008	0.0563	0.3124	0.1950

Vapour LiqVolume Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__Main Tower	0.0000	0.0377	0.9623	0.0000	0.0000
2__Main Tower	0.0001	0.0314	0.9686	0.0000	0.0000
3__Main Tower	0.0007	0.0260	0.9733	0.0000	0.0000
4__Main Tower	0.0085	0.0212	0.9702	0.0001	0.0000
5__Main Tower	0.1084	0.0152	0.8564	0.0170	0.0030
6__Main Tower	0.1087	0.0147	0.8565	0.0171	0.0031
7__Main Tower	0.1107	0.0142	0.8546	0.0174	0.0031
8__Main Tower	0.1207	0.0134	0.8435	0.0190	0.0035
9__Main Tower	0.1668	0.0121	0.7900	0.0261	0.0050
10__Main Tower	0.3341	0.0086	0.5871	0.0582	0.0119

Liquid LiqVolume Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__Main Tower	0.0001	0.0321	0.9679	0.0000	0.0000
2__Main Tower	0.0007	0.0266	0.9727	0.0000	0.0000
3__Main Tower	0.0085	0.0219	0.9695	0.0001	0.0000
4__Main Tower	0.1085	0.0163	0.8552	0.0170	0.0030
5__Main Tower	0.3047	0.0033	0.1987	0.2906	0.2027
6__Main Tower	0.3049	0.0032	0.1990	0.2904	0.2025
7__Main Tower	0.3067	0.0031	0.1977	0.2903	0.2023
8__Main Tower	0.3155	0.0029	0.1895	0.2905	0.2016
9__Main Tower	0.3502	0.0023	0.1587	0.2908	0.1980
10__Main Tower	0.4133	0.0013	0.0931	0.3025	0.1898

Tower: Main Tower @COL2

Vapour Draws Summary

	Name:	Name:	Name:
Tray Number			
Temperature (C)			
Pressure (kPa)			
Mass Flow (kg/h)			
Molar Flow (kgmole/h)			
Ideal Liquid Volume Flow (m3/h)			
Molar Enthalpy (kJ/kgmole)			
Mass Enthalpy (kJ/kg)			
Heat Flow (kJ/h)			
Molecular Weight			
Molar Entropy (kJ/kgmole-C)			
Mass Entropy (kJ/kg-C)			
Molar Density (kgmole/m3)			
Mass Density (kg/m3)			
Std Liq Mass Den (kg/m3)			
Molar Heat Cap (kJ/kgmole-C)			
Mass Heat Cap (kJ/kg-C)			
Thermal Cond (W/m-K)			
Viscosity (cP)			
Surface Tension (dyne/cm) ---	---	---	---
Z Factor			

Liquid Draws Summary

Name: Name: Name:

Tray Number

Temperature (C)

Pressure (kPa)

Mass Flow (kg/h)

Molar Flow (kgmole/h)

Ideal Liquid Volume Flow (m³/h)

Molar Enthalpy (kJ/kgmole)

Mass Enthalpy (kJ/kg)

Heat Flow (kJ/h)

Molecular Weight

Molar Entropy (kJ/kgmole-C)

Mass Entropy (kJ/kg-C)

Molar Density (kgmole/m³)

Mass Density (kg/m³)

Std Liq Mass Den (kg/m³)

Molar Heat Cap (kJ/kgmole-C)

Mass Heat Cap (kJ/kg-C)

Thermal Cond (W/m-K)

Viscosity (cP)

Surface Tension (dyne/cm)

Z Factor

Water Draws Summary

Name: Name: Name:

Tray Number

Temperature (C)

Pressure (kPa)

Mass Flow (kg/h)

Molar Flow (kgmole/h)

Water Volume Flow (m³/h)

Molar Enthalpy (kJ/kgmole)

Mass Enthalpy (kJ/kg)

Heat Flow (kJ/h)

Molecular Weight

Molar Entropy (kJ/kgmole-C)

Mass Entropy (kJ/kg-C)

Molar Density (kgmole/m³)

Mass Density (kg/m³)

Std Liq Mass Den (kg/m³)

Molar Heat Cap (kJ/kgmole-C)

Mass Heat Cap (kJ/kg-C)

Thermal Cond (W/m-K)

Viscosity (cP)

Surface Tension (dyne/cm)

Z Factor

User Variables

RATING

Sizing

Weir Height (m)	5.000e-002
Weir Length (m)	1.200
Tray Space (m)	0.5000
Tray Volume (m3)	0.8836
DC Volume (m3)	8.836e-002
Diameter (m)	1.500
Active Area (m2)	1.264
Flow Paths	1

Internal Type: Sieve

Nozzles

Tray Section Elevation Relative to Ground Level 0.0000 m

	HoldupRG	VtoAbove (m)	Diameter (m)	L toBelow (m)	Diameter (m)
1__Main Tower	10.50 m	---	---	0.0000 m	3.000e-003 m
2__Main Tower	10.00 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
3__Main Tower	9.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
4__Main Tower	9.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
5__Main Tower	8.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
6__Main Tower	8.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
7__Main Tower	7.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
8__Main Tower	7.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
9__Main Tower	6.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
10__Main Tower	6.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
11__Main Tower	5.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
12__Main Tower	5.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
13__Main Tower	4.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
14__Main Tower	4.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m

15__	Main Tower	3.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
16__	Main Tower	3.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
17__	Main Tower	2.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
18__	Main Tower	2.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
19__	Main Tower	1.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
20__	Main Tower	1.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
21__	Main Tower	0.5000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
22__	Main Tower	0.0000 m	0.5000 m	3.000e-003 m	---	---

	Feed Tray	Elevation	RH (m)	Diameter (m)
Reflux-2	1__ Main Tower	0.3750 m		5.000e-002 m
C2-Feed-LP	6__ Main Tower	0.3750 m		5.000e-002 m
Boilup-2	22__ Main Tower	0.0000 m		5.000e-002 m

	Prod Tray	Elevation	RH (m)	Diameter (m)
C2-Top	1__ Main Tower	0.5000 m		5.000e-002 m
C2-Bot	22__ Main Tower	0.0000 m		3.000e-003 m

Heat Loss

Efficiency

Overall Efficiency

Tray	Overall
1__ Main Tower	1.000
2__ Main Tower	1.000
3__ Main Tower	1.000
4__ Main Tower	1.000
5__ Main Tower	1.000
6__ Main Tower	1.000

7__Main Tower	1.000
8__Main Tower	1.000
9__Main Tower	1.000
10__Main Tower	1.000
11__Main Tower	1.000
12__Main Tower	1.000
13__Main Tower	1.000
14__Main Tower	1.000
15__Main Tower	1.000
16__Main Tower	1.000
17__Main Tower	1.000
18__Main Tower	1.000
19__Main Tower	1.000
20__Main Tower	1.000
21__Main Tower	1.000
22__Main Tower	1.000

Component Efficiency

Tray	Benzene	Propene	Propane	Cumene
1__Main Tower	1.000	1.000	1.000	1.000
2__Main Tower	1.000	1.000	1.000	1.000
3__Main Tower	1.000	1.000	1.000	1.000
4__Main Tower	1.000	1.000	1.000	1.000
5__Main Tower	1.000	1.000	1.000	1.000
6__Main Tower	1.000	1.000	1.000	1.000
7__Main Tower	1.000	1.000	1.000	1.000
8__Main Tower	1.000	1.000	1.000	1.000
9__Main Tower	1.000	1.000	1.000	1.000

10__Main Tower	1.000	1.000	1.000
11__Main Tower	1.000	1.000	1.000
12__Main Tower	1.000	1.000	1.000
13__Main Tower	1.000	1.000	1.000
14__Main Tower	1.000	1.000	1.000
15__Main Tower	1.000	1.000	1.000
16__Main Tower	1.000	1.000	1.000
17__Main Tower	1.000	1.000	1.000
18__Main Tower	1.000	1.000	1.000
19__Main Tower	1.000	1.000	1.000
20__Main Tower	1.000	1.000	1.000
21__Main Tower	1.000	1.000	1.000
22__Main Tower	1.000	1.000	1.000

Tray 14-iP-BZ

1__Main Tower	1.000
2__Main Tower	1.000
3__Main Tower	1.000
4__Main Tower	1.000
5__Main Tower	1.000
6__Main Tower	1.000
7__Main Tower	1.000
8__Main Tower	1.000
9__Main Tower	1.000
10__Main Tower	1.000
11__Main Tower	1.000
12__Main Tower	1.000
13__Main Tower	1.000

14__Main Tower 1.000
15__Main Tower 1.000
16__Main Tower 1.000
17__Main Tower 1.000
18__Main Tower 1.000
19__Main Tower 1.000
20__Main Tower 1.000
21__Main Tower 1.000
22__Main Tower 1.000

Pressure Drop

	Pressure (kPa)	Pressure Drop (kPa)
1__Main Tower	100.0 kPa	0.9524 kPa
2__Main Tower	101.0 kPa	0.9524 kPa
3__Main Tower	101.9 kPa	0.9524 kPa
4__Main Tower	102.9 kPa	0.9524 kPa
5__Main Tower	103.8 kPa	0.9524 kPa
6__Main Tower	104.8 kPa	0.9524 kPa
7__Main Tower	105.7 kPa	0.9524 kPa
8__Main Tower	106.7 kPa	0.9524 kPa
9__Main Tower	107.6 kPa	0.9524 kPa
10__Main Tower	108.6 kPa	0.9524 kPa
11__Main Tower	109.5 kPa	0.9524 kPa
12__Main Tower	110.5 kPa	0.9524 kPa
13__Main Tower	111.4 kPa	0.9524 kPa
14__Main Tower	112.4 kPa	0.9524 kPa
15__Main Tower	113.3 kPa	0.9524 kPa

16__Main Tower	114.3 kPa	0.9524 kPa
17__Main Tower	115.2 kPa	0.9524 kPa
18__Main Tower	116.2 kPa	0.9524 kPa
19__Main Tower	117.1 kPa	0.9524 kPa
20__Main Tower	118.1 kPa	0.9524 kPa
21__Main Tower	119.0 kPa	0.9524 kPa
22__Main Tower	120.0 kPa	---

Rating Enabled: No Top Tray Fixed For Update Tray Section Pressure Drop 20.00 kPa

Vapour Mole Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__Main Tower	0.6806	0.0006	0.0571	0.2615	0.0002
2__Main Tower	0.5531	0.0005	0.0472	0.3981	0.0011
3__Main Tower	0.5265	0.0005	0.0471	0.4229	0.0030
4__Main Tower	0.5237	0.0005	0.0472	0.4210	0.0076
5__Main Tower	0.5260	0.0005	0.0477	0.4079	0.0180
6__Main Tower	0.5307	0.0005	0.0485	0.3822	0.0380
7__Main Tower	0.5134	0.0000	0.0038	0.4397	0.0431
8__Main Tower	0.4324	0.0000	0.0003	0.5179	0.0494
9__Main Tower	0.3177	0.0000	0.0000	0.6245	0.0578
10__Main Tower	0.1991	0.0000	0.0000	0.7347	0.0662
11__Main Tower	0.1081	0.0000	0.0000	0.8193	0.0726
12__Main Tower	0.0532	0.0000	0.0000	0.8704	0.0763
13__Main Tower	0.0248	0.0000	0.0000	0.8969	0.0783
14__Main Tower	0.0112	0.0000	0.0000	0.9096	0.0792
15__Main Tower	0.0050	0.0000	0.0000	0.9154	0.0796
16__Main Tower	0.0022	0.0000	0.0000	0.9180	0.0798

17__Main Tower	0.0010	0.0000	0.0000	0.9191	0.0799
18__Main Tower	0.0004	0.0000	0.0000	0.9197	0.0799
19__Main Tower	0.0002	0.0000	0.0000	0.9199	0.0799
20__Main Tower	0.0001	0.0000	0.0000	0.9199	0.0801
21__Main Tower	0.0000	0.0000	0.0000	0.9191	0.0809
22__Main Tower	0.0000	0.0000	0.0000	0.9138	0.0862

Liquid Mole Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__Main Tower	0.2544	0.0000	0.0013	0.7413	0.0031
2__Main Tower	0.1638	0.0000	0.0010	0.8255	0.0097
3__Main Tower	0.1491	0.0000	0.0009	0.8241	0.0259
4__Main Tower	0.1443	0.0000	0.0009	0.7914	0.0634
5__Main Tower	0.1378	0.0000	0.0009	0.7204	0.1409
6__Main Tower	0.1275	0.0000	0.0009	0.6055	0.2661
7__Main Tower	0.1142	0.0000	0.0001	0.6227	0.2631
8__Main Tower	0.0885	0.0000	0.0000	0.6503	0.2612
9__Main Tower	0.0590	0.0000	0.0000	0.6823	0.2586
10__Main Tower	0.0340	0.0000	0.0000	0.7103	0.2557
11__Main Tower	0.0175	0.0000	0.0000	0.7292	0.2533
12__Main Tower	0.0083	0.0000	0.0000	0.7399	0.2517
13__Main Tower	0.0038	0.0000	0.0000	0.7454	0.2507
14__Main Tower	0.0017	0.0000	0.0000	0.7482	0.2501
15__Main Tower	0.0008	0.0000	0.0000	0.7496	0.2496
16__Main Tower	0.0003	0.0000	0.0000	0.7505	0.2492
17__Main Tower	0.0002	0.0000	0.0000	0.7510	0.2488
18__Main Tower	0.0001	0.0000	0.0000	0.7514	0.2485

19__Main Tower	0.0000	0.0000	0.0000	0.7517	0.2482
20__Main Tower	0.0000	0.0000	0.0000	0.7518	0.2482
21__Main Tower	0.0000	0.0000	0.0000	0.7501	0.2499
22__Main Tower	0.0000	0.0000	0.0000	0.7375	0.2625

Vapour Mass Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__Main Tower	0.6098	0.0003	0.0289	0.3606	0.0004
2__Main Tower	0.4629	0.0002	0.0223	0.5127	0.0018
3__Main Tower	0.4350	0.0002	0.0219	0.5377	0.0051
4__Main Tower	0.4313	0.0002	0.0220	0.5335	0.0130
5__Main Tower	0.4318	0.0002	0.0221	0.5152	0.0307
6__Main Tower	0.4330	0.0002	0.0223	0.4799	0.0645
7__Main Tower	0.4006	0.0000	0.0017	0.5279	0.0698
8__Main Tower	0.3246	0.0000	0.0001	0.5983	0.0770
9__Main Tower	0.2272	0.0000	0.0000	0.6870	0.0858
10__Main Tower	0.1357	0.0000	0.0000	0.7705	0.0938
11__Main Tower	0.0711	0.0000	0.0000	0.8296	0.0992
12__Main Tower	0.0343	0.0000	0.0000	0.8634	0.1022
13__Main Tower	0.0158	0.0000	0.0000	0.8804	0.1037
14__Main Tower	0.0071	0.0000	0.0000	0.8884	0.1044
15__Main Tower	0.0032	0.0000	0.0000	0.8921	0.1047
16__Main Tower	0.0014	0.0000	0.0000	0.8937	0.1049
17__Main Tower	0.0006	0.0000	0.0000	0.8944	0.1049
18__Main Tower	0.0003	0.0000	0.0000	0.8948	0.1050
19__Main Tower	0.0001	0.0000	0.0000	0.8949	0.1050
20__Main Tower	0.0001	0.0000	0.0000	0.8948	0.1052

21__Main Tower	0.0000	0.0000	0.0000	0.8938	0.1062
22__Main Tower	0.0000	0.0000	0.0000	0.8871	0.1129

Liquid Mass Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__Main Tower	0.1814	0.0000	0.0005	0.8135	0.0045
2__Main Tower	0.1126	0.0000	0.0004	0.8731	0.0139
3__Main Tower	0.1013	0.0000	0.0004	0.8618	0.0365
4__Main Tower	0.0965	0.0000	0.0004	0.8149	0.0882
5__Main Tower	0.0895	0.0000	0.0003	0.7200	0.1901
6__Main Tower	0.0791	0.0000	0.0003	0.5778	0.3428
7__Main Tower	0.0705	0.0000	0.0000	0.5918	0.3376
8__Main Tower	0.0542	0.0000	0.0000	0.6132	0.3326
9__Main Tower	0.0359	0.0000	0.0000	0.6378	0.3264
10__Main Tower	0.0205	0.0000	0.0000	0.6591	0.3204
11__Main Tower	0.0105	0.0000	0.0000	0.6736	0.3159
12__Main Tower	0.0050	0.0000	0.0000	0.6818	0.3132
13__Main Tower	0.0023	0.0000	0.0000	0.6861	0.3116
14__Main Tower	0.0010	0.0000	0.0000	0.6883	0.3106
15__Main Tower	0.0005	0.0000	0.0000	0.6895	0.3100
16__Main Tower	0.0002	0.0000	0.0000	0.6903	0.3095
17__Main Tower	0.0001	0.0000	0.0000	0.6909	0.3091
18__Main Tower	0.0000	0.0000	0.0000	0.6913	0.3087
19__Main Tower	0.0000	0.0000	0.0000	0.6916	0.3083
20__Main Tower	0.0000	0.0000	0.0000	0.6917	0.3083
21__Main Tower	0.0000	0.0000	0.0000	0.6898	0.3102
22__Main Tower	0.0000	0.0000	0.0000	0.6754	0.3246

Vapour LiqVolume Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__Main Tower	0.5927	0.0005	0.0489	0.3575	0.0004
2__Main Tower	0.4507	0.0004	0.0378	0.5093	0.0018
3__Main Tower	0.4234	0.0004	0.0372	0.5339	0.0051
4__Main Tower	0.4197	0.0004	0.0372	0.5297	0.0130
5__Main Tower	0.4201	0.0004	0.0374	0.5114	0.0306
6__Main Tower	0.4212	0.0004	0.0378	0.4762	0.0643
7__Main Tower	0.3952	0.0000	0.0029	0.5313	0.0706
8__Main Tower	0.3201	0.0000	0.0002	0.6018	0.0779
9__Main Tower	0.2236	0.0000	0.0000	0.6898	0.0867
10__Main Tower	0.1333	0.0000	0.0000	0.7722	0.0945
11__Main Tower	0.0698	0.0000	0.0000	0.8303	0.0999
12__Main Tower	0.0336	0.0000	0.0000	0.8635	0.1028
13__Main Tower	0.0155	0.0000	0.0000	0.8802	0.1043
14__Main Tower	0.0070	0.0000	0.0000	0.8880	0.1050
15__Main Tower	0.0031	0.0000	0.0000	0.8916	0.1053
16__Main Tower	0.0014	0.0000	0.0000	0.8932	0.1054
17__Main Tower	0.0006	0.0000	0.0000	0.8939	0.1055
18__Main Tower	0.0003	0.0000	0.0000	0.8942	0.1055
19__Main Tower	0.0001	0.0000	0.0000	0.8943	0.1055
20__Main Tower	0.0001	0.0000	0.0000	0.8943	0.1057
21__Main Tower	0.0000	0.0000	0.0000	0.8932	0.1067
22__Main Tower	0.0000	0.0000	0.0000	0.8865	0.1135

Liquid LiqVolume Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__	Main Tower 0.1784	0.0000	0.0009	0.8161	0.0046
2__	Main Tower 0.1106	0.0000	0.0006	0.8747	0.0140
3__	Main Tower 0.0995	0.0000	0.0006	0.8631	0.0368
4__	Main Tower 0.0947	0.0000	0.0006	0.8158	0.0888
5__	Main Tower 0.0878	0.0000	0.0006	0.7203	0.1913
6__	Main Tower 0.0774	0.0000	0.0006	0.5774	0.3446
7__	Main Tower 0.0691	0.0000	0.0000	0.5915	0.3394
8__	Main Tower 0.0531	0.0000	0.0000	0.6127	0.3342
9__	Main Tower 0.0351	0.0000	0.0000	0.6370	0.3279
10__	Main Tower 0.0201	0.0000	0.0000	0.6582	0.3218
11__	Main Tower 0.0103	0.0000	0.0000	0.6725	0.3172
12__	Main Tower 0.0049	0.0000	0.0000	0.6807	0.3145
13__	Main Tower 0.0022	0.0000	0.0000	0.6849	0.3128
14__	Main Tower 0.0010	0.0000	0.0000	0.6871	0.3119
15__	Main Tower 0.0005	0.0000	0.0000	0.6883	0.3112
16__	Main Tower 0.0002	0.0000	0.0000	0.6891	0.3107
17__	Main Tower 0.0001	0.0000	0.0000	0.6896	0.3103
18__	Main Tower 0.0000	0.0000	0.0000	0.6901	0.3099
19__	Main Tower 0.0000	0.0000	0.0000	0.6904	0.3096
20__	Main Tower 0.0000	0.0000	0.0000	0.6904	0.3095
21__	Main Tower 0.0000	0.0000	0.0000	0.6885	0.3115
22__	Main Tower 0.0000	0.0000	0.0000	0.6742	0.3258

Tower: Main Tower @COL3

Vapour Draws Summary

	Name:	Name:	Name:
Tray Number			
Temperature (C)			
Pressure (kPa)			
Mass Flow (kg/h)			
Molar Flow (kgmole/h)			
Ideal Liquid Volume Flow (m3/h)			
Molar Enthalpy (kJ/kgmole)			
Mass Enthalpy (kJ/kg)			
Heat Flow (kJ/h)			
Molecular Weight			
Molar Entropy (kJ/kgmole-C)			
Mass Entropy (kJ/kg-C)			
Molar Density (kgmole/m3)			
Mass Density (kg/m3)			
Std Liq Mass Den (kg/m3)			
Molar Heat Cap (kJ/kgmole-C)			
Mass Heat Cap (kJ/kg-C)			
Thermal Cond (W/m-K)			
Viscosity (cP)			
Surface Tension (dyne/cm) ---	---	---	---
Z Factor			
Liquid Draws Summary			

Name: Name: Name:

Tray Number

Temperature (C)

Pressure (kPa)

Mass Flow (kg/h)

Molar Flow (kgmole/h)

Ideal Liquid Volume Flow (m³/h)

Molar Enthalpy (kJ/kgmole)

Mass Enthalpy (kJ/kg)

Heat Flow (kJ/h)

Molecular Weight

Molar Entropy (kJ/kgmole-C)

Mass Entropy (kJ/kg-C)

Molar Density (kgmole/m³)

Mass Density (kg/m³)

Std Liq Mass Den (kg/m³)

Molar Heat Cap (kJ/kgmole-C)

Mass Heat Cap (kJ/kg-C)

Thermal Cond (W/m-K)

Viscosity (cP)

Surface Tension (dyne/cm)

Z Factor

Water Draws Summary

Name: Name: Name:

Tray Number

Temperature (C)

Pressure (kPa)
Mass Flow (kg/h)
Molar Flow (kgmole/h)
Water Volume Flow (m³/h)
Molar Enthalpy (kJ/kgmole)
Mass Enthalpy (kJ/kg)
Heat Flow (kJ/h)
Molecular Weight
Molar Entropy (kJ/kgmole-C)
Mass Entropy (kJ/kg-C)
Molar Density (kgmole/m³)
Mass Density (kg/m³)
Std Liq Mass Den (kg/m³)
Molar Heat Cap (kJ/kgmole-C)
Mass Heat Cap (kJ/kg-C)
Thermal Cond (W/m-K)
Viscosity (cP)
Surface Tension (dyne/cm)
Z Factor
User Variables

RATING

Sizing

Weir Height (m)	5.000e-002
Weir Length (m)	1.200

Tray Space (m)	0.5000
Tray Volume (m3)	0.8836
DC Volume (m3)	8.836e-002
Diameter (m)	1.500
Active Area (m2)	1.264
Flow Paths	1

Internal Type: Sieve

Nozzles

Tray Section Elevation Relative to Ground Level 0.0000 m

	HoldupRG	VtoAbove (m)	Diameter (m)	L toBelow (m)	Diameter (m)
1__Main Tower	9.500 m	---	---	0.0000 m	3.000e-003 m
2__Main Tower	9.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
3__Main Tower	8.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
4__Main Tower	8.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
5__Main Tower	7.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
6__Main Tower	7.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
7__Main Tower	6.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
8__Main Tower	6.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
9__Main Tower	5.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
10__Main Tower	5.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
11__Main Tower	4.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
12__Main Tower	4.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
13__Main Tower	3.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
14__Main Tower	3.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
15__Main Tower	2.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
16__Main Tower	2.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m

17__Main Tower	1.500 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
18__Main Tower	1.000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
19__Main Tower	0.5000 m	0.5000 m	3.000e-003 m	0.0000 m	3.000e-003 m
20__Main Tower	0.0000 m	0.5000 m	3.000e-003 m	---	---

	Feed Tray	Elevation	RH (m)	Diameter (m)
Reflux-3	1__Main Tower	0.3750 m		5.000e-002 m
C2-Bottoms	10__Main Tower	0.3750 m		5.000e-002 m
Boilup-3	20__Main Tower	0.0000 m		5.000e-002 m

	Prod Tray	Elevation	RH (m)	Diameter (m)
C3-Top	1__Main Tower	0.5000 m		5.000e-002 m
C3-Bot	20__Main Tower	0.0000 m		3.000e-003 m

Heat Loss

Efficiency

Overall Efficiency

Tray	Overall
1__Main Tower	1.000
2__Main Tower	1.000
3__Main Tower	1.000
4__Main Tower	1.000
5__Main Tower	1.000
6__Main Tower	1.000
7__Main Tower	1.000
8__Main Tower	1.000
9__Main Tower	1.000
10__Main Tower	1.000

11__Main Tower	1.000
12__Main Tower	1.000
13__Main Tower	1.000
14__Main Tower	1.000
15__Main Tower	1.000
16__Main Tower	1.000
17__Main Tower	1.000
18__Main Tower	1.000
19__Main Tower	1.000
20__Main Tower	1.000

Component Efficiency

Tray	Benzene	Propene	Propane	Cumene
1__Main Tower	1.000	1.000	1.000	1.000
2__Main Tower	1.000	1.000	1.000	1.000
3__Main Tower	1.000	1.000	1.000	1.000
4__Main Tower	1.000	1.000	1.000	1.000
5__Main Tower	1.000	1.000	1.000	1.000
6__Main Tower	1.000	1.000	1.000	1.000
7__Main Tower	1.000	1.000	1.000	1.000
8__Main Tower	1.000	1.000	1.000	1.000
9__Main Tower	1.000	1.000	1.000	1.000
10__Main Tower	1.000	1.000	1.000	1.000
11__Main Tower	1.000	1.000	1.000	1.000
12__Main Tower	1.000	1.000	1.000	1.000
13__Main Tower	1.000	1.000	1.000	1.000
14__Main Tower	1.000	1.000	1.000	1.000
15__Main Tower	1.000	1.000	1.000	1.000

16__ Main Tower	1.000	1.000	1.000	1.000
17__ Main Tower	1.000	1.000	1.000	1.000
18__ Main Tower	1.000	1.000	1.000	1.000
19__ Main Tower	1.000	1.000	1.000	1.000
20__ Main Tower	1.000	1.000	1.000	1.000

Tray 14-iP-BZ

1__ Main Tower	1.000
2__ Main Tower	1.000
3__ Main Tower	1.000
4__ Main Tower	1.000
5__ Main Tower	1.000
6__ Main Tower	1.000
7__ Main Tower	1.000
8__ Main Tower	1.000
9__ Main Tower	1.000
10__ Main Tower	1.000
11__ Main Tower	1.000
12__ Main Tower	1.000
13__ Main Tower	1.000
14__ Main Tower	1.000
15__ Main Tower	1.000
16__ Main Tower	1.000
17__ Main Tower	1.000
18__ Main Tower	1.000
19__ Main Tower	1.000
20__ Main Tower	1.000

Pressure Drop

	Pressure (kPa)	Pressure Drop (kPa)
1__ Main Tower	100.0 kPa	1.053 kPa
2__ Main Tower	101.1 kPa	1.053 kPa
3__ Main Tower	102.1 kPa	1.053 kPa
4__ Main Tower	103.2 kPa	1.053 kPa
5__ Main Tower	104.2 kPa	1.053 kPa
6__ Main Tower	105.3 kPa	1.053 kPa
7__ Main Tower	106.3 kPa	1.053 kPa
8__ Main Tower	107.4 kPa	1.053 kPa
9__ Main Tower	108.4 kPa	1.053 kPa
10__ Main Tower	109.5 kPa	1.053 kPa
11__ Main Tower	110.5 kPa	1.053 kPa
12__ Main Tower	111.6 kPa	1.053 kPa
13__ Main Tower	112.6 kPa	1.053 kPa
14__ Main Tower	113.7 kPa	1.053 kPa
15__ Main Tower	114.7 kPa	1.053 kPa
16__ Main Tower	115.8 kPa	1.053 kPa
17__ Main Tower	116.8 kPa	1.053 kPa
18__ Main Tower	117.9 kPa	1.053 kPa
19__ Main Tower	118.9 kPa	1.053 kPa
20__ Main Tower	120.0 kPa	---

Rating Enabled: No Top Tray Fixed For Update Tray Section Pressure Drop 20.00 kPa

Vapour Mole Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__ Main Tower	0.0000	0.0000	0.0000	0.9963	0.0037

2__Main Tower	0.0000	0.0000	0.0000	0.9920	0.0080
3__Main Tower	0.0000	0.0000	0.0000	0.9852	0.0148
4__Main Tower	0.0000	0.0000	0.0000	0.9753	0.0247
5__Main Tower	0.0000	0.0000	0.0000	0.9616	0.0384
6__Main Tower	0.0000	0.0000	0.0000	0.9447	0.0553
7__Main Tower	0.0000	0.0000	0.0000	0.9259	0.0741
8__Main Tower	0.0000	0.0000	0.0000	0.9076	0.0924
9__Main Tower	0.0000	0.0000	0.0000	0.8918	0.1082
10__Main Tower	0.0000	0.0000	0.0000	0.8796	0.1204
11__Main Tower	0.0000	0.0000	0.0000	0.8797	0.1203
12__Main Tower	0.0000	0.0000	0.0000	0.8797	0.1203
13__Main Tower	0.0000	0.0000	0.0000	0.8798	0.1202
14__Main Tower	0.0000	0.0000	0.0000	0.8798	0.1202
15__Main Tower	0.0000	0.0000	0.0000	0.8799	0.1201
16__Main Tower	0.0000	0.0000	0.0000	0.8800	0.1200
17__Main Tower	0.0000	0.0000	0.0000	0.8800	0.1200
18__Main Tower	0.0000	0.0000	0.0000	0.8799	0.1201
19__Main Tower	0.0000	0.0000	0.0000	0.8790	0.1210
20__Main Tower	0.0000	0.0000	0.0000	0.8730	0.1270

Liquid Mole Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__Main Tower	0.0000	0.0000	0.0000	0.9856	0.0144
2__Main Tower	0.0000	0.0000	0.0000	0.9690	0.0310
3__Main Tower	0.0000	0.0000	0.0000	0.9441	0.0559
4__Main Tower	0.0000	0.0000	0.0000	0.9092	0.0908
5__Main Tower	0.0000	0.0000	0.0000	0.8646	0.1354

6__Main Tower	0.0000	0.0000	0.0000	0.8138	0.1862
7__Main Tower	0.0000	0.0000	0.0000	0.7629	0.2371
8__Main Tower	0.0000	0.0000	0.0000	0.7180	0.2820
9__Main Tower	0.0000	0.0000	0.0000	0.6824	0.3176
10__Main Tower	0.0000	0.0000	0.0000	0.6569	0.3431
11__Main Tower	0.0000	0.0000	0.0000	0.6574	0.3426
12__Main Tower	0.0000	0.0000	0.0000	0.6580	0.3420
13__Main Tower	0.0000	0.0000	0.0000	0.6585	0.3415
14__Main Tower	0.0000	0.0000	0.0000	0.6591	0.3409
15__Main Tower	0.0000	0.0000	0.0000	0.6596	0.3404
16__Main Tower	0.0000	0.0000	0.0000	0.6602	0.3398
17__Main Tower	0.0000	0.0000	0.0000	0.6607	0.3393
18__Main Tower	0.0000	0.0000	0.0000	0.6609	0.3391
19__Main Tower	0.0000	0.0000	0.0000	0.6595	0.3405
20__Main Tower	0.0000	0.0000	0.0000	0.6476	0.3524

Vapour Mass Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__Main Tower	0.0000	0.0000	0.0000	0.9950	0.0050
2__Main Tower	0.0000	0.0000	0.0000	0.9892	0.0108
3__Main Tower	0.0000	0.0000	0.0000	0.9802	0.0198
4__Main Tower	0.0000	0.0000	0.0000	0.9669	0.0331
5__Main Tower	0.0000	0.0000	0.0000	0.9489	0.0511
6__Main Tower	0.0000	0.0000	0.0000	0.9267	0.0733
7__Main Tower	0.0000	0.0000	0.0000	0.9025	0.0975
8__Main Tower	0.0000	0.0000	0.0000	0.8791	0.1209
9__Main Tower	0.0000	0.0000	0.0000	0.8593	0.1407

10__Main Tower	0.0000	0.0000	0.0000	0.8441	0.1559
11__Main Tower	0.0000	0.0000	0.0000	0.8441	0.1559
12__Main Tower	0.0000	0.0000	0.0000	0.8442	0.1558
13__Main Tower	0.0000	0.0000	0.0000	0.8443	0.1557
14__Main Tower	0.0000	0.0000	0.0000	0.8443	0.1557
15__Main Tower	0.0000	0.0000	0.0000	0.8444	0.1556
16__Main Tower	0.0000	0.0000	0.0000	0.8445	0.1555
17__Main Tower	0.0000	0.0000	0.0000	0.8445	0.1555
18__Main Tower	0.0000	0.0000	0.0000	0.8444	0.1556
19__Main Tower	0.0000	0.0000	0.0000	0.8433	0.1567
20__Main Tower	0.0000	0.0000	0.0000	0.8358	0.1642

Liquid Mass Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__Main Tower	0.0000	0.0000	0.0000	0.9806	0.0194
2__Main Tower	0.0000	0.0000	0.0000	0.9586	0.0414
3__Main Tower	0.0000	0.0000	0.0000	0.9260	0.0740
4__Main Tower	0.0000	0.0000	0.0000	0.8812	0.1188
5__Main Tower	0.0000	0.0000	0.0000	0.8255	0.1745
6__Main Tower	0.0000	0.0000	0.0000	0.7640	0.2360
7__Main Tower	0.0000	0.0000	0.0000	0.7045	0.2955
8__Main Tower	0.0000	0.0000	0.0000	0.6534	0.3466
9__Main Tower	0.0000	0.0000	0.0000	0.6142	0.3858
10__Main Tower	0.0000	0.0000	0.0000	0.5864	0.4136
11__Main Tower	0.0000	0.0000	0.0000	0.5870	0.4130
12__Main Tower	0.0000	0.0000	0.0000	0.5876	0.4124
13__Main Tower	0.0000	0.0000	0.0000	0.5882	0.4118

14__Main Tower	0.0000	0.0000	0.0000	0.5888	0.4112
15__Main Tower	0.0000	0.0000	0.0000	0.5894	0.4106
16__Main Tower	0.0000	0.0000	0.0000	0.5900	0.4100
17__Main Tower	0.0000	0.0000	0.0000	0.5905	0.4095
18__Main Tower	0.0000	0.0000	0.0000	0.5908	0.4092
19__Main Tower	0.0000	0.0000	0.0000	0.5893	0.4107
20__Main Tower	0.0000	0.0000	0.0000	0.5765	0.4235

Vapour LiqVolume Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__Main Tower	0.0000	0.0000	0.0000	0.9950	0.0050
2__Main Tower	0.0000	0.0000	0.0000	0.9891	0.0108
3__Main Tower	0.0000	0.0000	0.0000	0.9801	0.0199
4__Main Tower	0.0000	0.0000	0.0000	0.9667	0.0333
5__Main Tower	0.0000	0.0000	0.0000	0.9486	0.0514
6__Main Tower	0.0000	0.0000	0.0000	0.9263	0.0737
7__Main Tower	0.0000	0.0000	0.0000	0.9020	0.0980
8__Main Tower	0.0000	0.0000	0.0000	0.8785	0.1215
9__Main Tower	0.0000	0.0000	0.0000	0.8586	0.1414
10__Main Tower	0.0000	0.0000	0.0000	0.8433	0.1567
11__Main Tower	0.0000	0.0000	0.0000	0.8434	0.1566
12__Main Tower	0.0000	0.0000	0.0000	0.8434	0.1566
13__Main Tower	0.0000	0.0000	0.0000	0.8435	0.1565
14__Main Tower	0.0000	0.0000	0.0000	0.8436	0.1564
15__Main Tower	0.0000	0.0000	0.0000	0.8436	0.1564
16__Main Tower	0.0000	0.0000	0.0000	0.8437	0.1563
17__Main Tower	0.0000	0.0000	0.0000	0.8437	0.1563

18__Main Tower	0.0000	0.0000	0.0000	0.8436	0.1564
19__Main Tower	0.0000	0.0000	0.0000	0.8425	0.1575
20__Main Tower	0.0000	0.0000	0.0000	0.8350	0.1650

Liquid LiqVolume Fractions

Tray Number	Benzene	Propene	Propane	Cumene	14-iP-BZ
1__Main Tower	0.0000	0.0000	0.0000	0.9805	0.0195
2__Main Tower	0.0000	0.0000	0.0000	0.9583	0.0417
3__Main Tower	0.0000	0.0000	0.0000	0.9256	0.0744
4__Main Tower	0.0000	0.0000	0.0000	0.8806	0.1194
5__Main Tower	0.0000	0.0000	0.0000	0.8247	0.1753
6__Main Tower	0.0000	0.0000	0.0000	0.7630	0.2370
7__Main Tower	0.0000	0.0000	0.0000	0.7033	0.2967
8__Main Tower	0.0000	0.0000	0.0000	0.6521	0.3479
9__Main Tower	0.0000	0.0000	0.0000	0.6128	0.3872
10__Main Tower	0.0000	0.0000	0.0000	0.5850	0.4150
11__Main Tower	0.0000	0.0000	0.0000	0.5856	0.4144
12__Main Tower	0.0000	0.0000	0.0000	0.5862	0.4138
13__Main Tower	0.0000	0.0000	0.0000	0.5868	0.4132
14__Main Tower	0.0000	0.0000	0.0000	0.5874	0.4126
15__Main Tower	0.0000	0.0000	0.0000	0.5880	0.4120
16__Main Tower	0.0000	0.0000	0.0000	0.5886	0.4114
17__Main Tower	0.0000	0.0000	0.0000	0.5891	0.4109
18__Main Tower	0.0000	0.0000	0.0000	0.5894	0.4106
19__Main Tower	0.0000	0.0000	0.0000	0.5879	0.4121
20__Main Tower	0.0000	0.0000	0.0000	0.5751	0.4249

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